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CHAPTER 1

BINOMIAL THEOREM

1. INTRODUCTION

Given two number a and b, if we are required to find $(a+b)^2$, we can just add a and b then multiply the sum by itself. Another way of doing it is to find $a^2 + 2ab + b^2$. We also know that $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, Now we may need to use the expression for $(a+b)^5$, $(a+b)^7$, etc. But we cannot remember all the expressions. Binomial theorem helps to find these expressions.

An algebraic expression containing two term is called a binomial expression for example,

$(2x + 3)$, $(x^2 - \frac{1}{x})$, $(x + a)$, etc are binomial expression.

Binomial theorem was first given by Sir Issac Newton.

PASCALS TRIANGLE

A Triangular arrangement of numbers as show. The Numbers give the coefficients for the expansion of $(x+y)^n$. The first row is for n =0, the second for n =1 etc Each row has 1 as its first and last number other numbers are generated by adding the two numbers immediately to the left and right in the row above.

```
                1
              1   1
            1   2   1
          1   3   3   1
        1   4   6   4   4  1
      1   5  10  10   5   1
    1   6 15  20  15  6  1
                          etc
```

2. BINOMIAL THEOREM FOR POSITIVE INTEGRAL INDEX

The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as Binomial theorem. If x and a are real number, then for all $n \in N$,

$(x+a)^n = {}^nC_0\, x^n\, a^0 + {}^nC_1\, x^{n-1}\, a^1 + {}^nC_2\, x^{n-2}\, a^2 + \ldots + {}^nC_r\, x^{n-r}\, a^r + \ldots + {}^nC_{n-1}\, x^1\, a^{n-1} + {}^nC_n\, x^0\, a^n$

i.e. $(x+a)^n = \sum_{r=0}^{n} {}^nC_r\, x^{n-r}\, a^r$

This theorem can be proved by Induction.

Properties

(i) There are $(n+1)$ terms in the expansion.

(ii) The sum of the exponents of x and y in any term of the expansion is equal to n.

(iii) The binomial coefficients of terms equidistant from the beginning and the end are equal, since ${}^nC_r = {}^nC_{n-r}$.

(iv) The term ${}^nC_r\, x^{n-r}\, y^r$ is the $(r+1)$th term from the beginning of the expansion. It is usually denoted by T_{r+1} and is called the general term of the expansion.

(v) The rth term from the end is equal to the $(n-r+2)$th term from the beginning, i.e., ${}^nC_{n-r+1}\, x^{r-1}\, y^{n-r+1}$.

(vi) Term independent of x contains no x, Hence find the value of r for which the exponent of x is zero,

General Term

In the expansion of $(x + a)^n$, $(r + 1)^{th}$ term is called the general term which can be represented by T_{r+1}.

$T_{r+1} = {}^nC_r\, x^{n-r}\, a^r = {}^nC_r$ (first term)$^{n-r}$ (second term)^r

TERM INDEPENDENT OF X

Term independent of x does not contain x, hence find the value of r for which the exponent of x is Zero.

To find a term from the end in the expansion of $(x + a)^n$

It can be easily seen that in the expansion of $(x + a)^n$, $(r + 1)^{th}$ term from end = $(n - r + 1)^{th}$ term from beginning.

i.e. $T_{r+1}(E) = T_{n-r+1}(B)$

$T_r(E) = T_{n-r+2}(B)$

Some deduction of binomial expansion

(i) Expansion of $(x-a)^n$

$(x - a)^n = {}^nC_0x^na^0 - {}^nC_1x^{n-1}a^1 + {}^nC_2x^{n-2}\,a^2 - {}^nC_3x^{n-3}a^3 + \ldots\ldots+(-1)^r\,{}^nC_rx^{n-r}a^r + \ldots..+ (-1)^n\,{}^nC_n\,x^0a^n$

This expansion can be obtained by putting $(-a)$ in place of a in the expansion of $(x + a)^n$.

General term = $(r + 1)^{th}$ term

$T_{r+1} = {}^nC_r(-1)^r.x^{n-r}\,a^r$

(ii) By putting $x = 1$ and $a = x$ in the expansion of $(x + a)^n$, we get the following result

$(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots\ldots+ {}^nC_rx^r + \ldots..+ {}^nC_n\,x^n$

which is the standard form of binomial expansion

General term = $(r + 1)^{th}$ term

$$T_{r+1} = {}^nC_rx^r = \frac{n(n-1)(n-2).....(n-r+1)}{r!}\,x^r$$

(iii) By putting (–x) in place of x in the expansion of $(1 + x)^n$

$(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - {}^nC_3x^3 + \ldots.. + (-1)^r\,{}^nC_rx^r + \ldots..+ {}^nC_nx^n.$

General term = $(r + 1)^{th}$ term

$$T_{r+1} = (-1)^r.{}^nC_rx^r = (-1)^r\,\frac{n(n-1)(n-2).....(n-r+1)}{r!}\,.\,x^r$$

Important Formulae

If C_0, C_1, C_2, C_3 ..... represent ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3$ ..... in the expansion of $(1 + x)^n$. Then

(i) $C_0 + C_1 + C_2 + C_3 + \ldots.. + C_n = 2^n$

(ii) $C_0 - C_1 + C_2 - C_3 + \ldots.. +(-1)^nC_n = 0$

(iii) $C_0 + C_2 + C_4 + \ldots.. = C_1 + C_3 + C_5 + \ldots.. = 2^{n-1}$

Number of terms in the expansion of $(x + y + z)^n$

$(x + y + z)^n$ can be expanded as –

$(x + y + z)^n = \{(x + y) + z\}^n = (x + y)^n + {}^nC_1(x + y)^{n-1}.\,z + {}^nC_2(x + y)^{n-2}\,z^2 + \ldots.. + {}^nC_nZ^n.$

= $(n + 1)$ terms + n terms + $(n - 1)$ terms +..... + 1 term

$\therefore$ Total number of terms = $(n + 1) + n + (n - 1)+ \ldots.. + 1 = \frac{(n+1)(n+2)}{2}$

The number of distinct terms in the expansion of $(x_1 + x_2 + x_3 +.....x_r)^n$ is given by ${}^{n+r-1}C_{r-1}$

Integral and fractional part

If $(\sqrt{A}+B)^n = I + f$, where I and n are positive integers, n being odd and $0 < f < 1$, then

$(I + f)\,.\,f = K^n$ where $A - B^2 = K > 0$ and $\sqrt{A} - B < 1$

If n is an even integer, then $(I + f)\,(1 - f) = K^n$.

Illustration 1: Find the 7th term from the end in the expansion of $\left(x - \frac{2}{x^2}\right)^{10}$.

(a) ${}^{10}C_4 2^4 \left(\frac{1}{x^2}\right)$ (b) ${}^{10}C_4 2^4$ (c) $-{}^{10}C_3 2^3 x$ (d) None of these

Solution : **(a)** The 7th term from the end = 5th term from beginning

$$T_5 = {}^{10}C_4 x^6 \left(-\frac{2}{x^2}\right)^4 = {}^{10}C_4 \cdot 2^4 \left(\frac{1}{x^2}\right)$$

Illustration 2

Find the coefficient of x^{-8} in the expansion of $\left(x - \frac{1}{2x^2}\right)^{10}$

Solution : We have, $T_{r+1} = {}^{10}C_r (x)^{10-r} \left(\frac{-1}{2x^2}\right)^r = {}^{10}C_r (-1)^r 2^{-r} x^{10-3r}$

To find the coefficient of x^{-8}, we have

$10 - 3r = -8$ i.e., $r = 6$.

Thus, the 7th term has x^{-8} and its coefficient is

$${}^{10}C_6 (-1)^6 2^{-6} = \frac{105}{32}.$$

Illustration 3: Show that $\left[\left(3+\sqrt{5}\right)^{2n}\right] + 1$ is divisible by 2^{n+1} for every $n \in N$, where $[x]$ denotes the integral part of x.

Solution : Let $\left(3+\sqrt{5}\right)^{2n} = I + f$, where I and f are the integral and the fractional pats of $\left(3+\sqrt{5}\right)^{2n}$ respectively.

Let $\left(3-\sqrt{5}\right)^{2n} = g$, where g is a fraction.

we have, $I + f + g = \left(3+\sqrt{5}\right)^{2n} + \left(3-\sqrt{5}\right)^{2n}$

$$= \left(14+6\sqrt{5}\right)^n + \left(14-6\sqrt{5}\right)^n = 2^n \left[\left(7+3\sqrt{5}\right)^n + \left(7-3\sqrt{5}\right)^n\right]$$

$$= 2^n . 2\left[C_0 . 7^n + C_2 (7)^{n-2} . \left(3\sqrt{5}\right)^2 + \ldots\right]$$

i.e. $I + f + g = (2^{n+1})k, k \in N$ .... (i)

i.e. $f + g = (2^{n+1})k - I$ = a positive integer .... (ii)

Since $0 < f, g < 1$, therefore we have

$0 < f + g < 2$ .... (iii)

Thus, using results (ii) and (iii), we have

$f + g = 1$ [Since the only integral value in (0, 2) is 1] .... (iv)

Putting result (iv) in equation (i), we have

$I + 1 = (2^{n+1})k$

which proves that $\left[\left(3+\sqrt{5}\right)^{2n}\right] + 1$ is divisible by 2^{n+1}

Illustration 4: Find the number of terms in the expansion of $(x + y + 2z)^8$

Solution :

$\because$ $n = 8$ and from the above given formula we have

Number of terms = $\frac{(8+1)(8+2)}{2} = 45$

Student Task

Q.1 If coefficient of $(2r + 3)^{th}$ and $(r - 1)^{th}$ terms in the expansion of $(1 + x)^{15}$ are equal, then find the value of r.

Q.2 If p and q be positive, then prove that the coefficients of x^p and x^q in the expansion of $(1 + x)^{p+q}$ will be equal.

Q.3 Find the term independent of y in the expansion of $(y^{-1/6} - y^{1/3})^9$.

Q.4 Find the coefficient of x^5 in the expansion of the product $(1 + 2x)^6 (1 - x)^7$.

Q.5 Find the number of integral terms in the expansion of $\left(\sqrt{2} + \sqrt[3]{3}\right)^{87}$.

Q.6 Find the value of $\left(\sqrt{2}+1\right)^6 - \left(\sqrt{2}-1\right)^6$.

Q.7 If the 4th term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is 5/2, then find the values of a and n.

Q.8 Find the greatest integer less than or equal to $(\sqrt{2} + 1)^6$.

Q.9 Show that the integer just greater than $(\sqrt{3} + 1)^{2m}$ contains 2^{m+1} as a factor.

ANSWERS KEY

1.	5	**3.**	– 84	**4.**	171	**5.**	15
6.	$140\sqrt{2}$	**7.**	1/2, 6	**8.**	197		

3. MIDDLE TERM IN THE EXPANSION OF $(X + A)^N$

There are $(n + 1)$ terms in the expansion of $(x + a)^n$. The middle term depends upon the value of n.

(a) When n is even, then only one middle term exists and it is $\left(\frac{n}{2}+1\right)^{th}$ term

Middle term = ${}^nC_{\frac{n}{2}} x^{\frac{n}{2}}.a^{\frac{n}{2}}$

(b) When n is odd, there will be two middle terms and they are $\frac{(n+1)^{th}}{2}$ and $\frac{(n+3)^{th}}{2}$ terms.

The first middle term = ${}^nC_{\frac{n-1}{2}} x^{\frac{n+1}{2}}.a^{\frac{n-1}{2}}$; The second middle term = ${}^nC_{\frac{n+1}{2}} x^{\frac{n-1}{2}}.a^{\frac{n+1}{2}}$

Illustration 5: Find the middle term in the expansion of $\left(ax - \frac{b}{x}\right)^9$.

Solution : The expansion has two middle terms, viz $\left(\frac{9+1}{2}\right)^{th} = 5^{th}$ term and $\left(\frac{9+3}{2}\right)^{th} = 6^{th}$ term.

Hence, the middle terms are $T_5 = {}^9C_4\,(ax)^{9-4}\left(\frac{-b}{x}\right)^4 = \frac{9.8.7.6}{1.2.3.4}\left(a^5b^4x\right) = 126a^5b^4x$

and $T_6 = {}^9C_5\,(ax)^{9-5}\left(\frac{-b}{x}\right)^5 = \frac{9.8.7.6}{1.2.3.4}\left(-a^4b^5x^{-1}\right) = 126a^4b^5x^{-1}$

Illustration 6: If the middle term in the expansion of $\left(x^2+\frac{1}{x}\right)^n$ is 924 x^6,n is even then find *n*.

Solution : Since *n* is even therefore $\left(\frac{n}{2}+1\right)^{th}$ term is middle term,

hence ${}^nC_{n/2}\,(x^2)^{n/2}\left(\frac{1}{x}\right)^{n/2} = 924\,x^6 \Rightarrow x^{n/2} = x^6 \Rightarrow n = 12$

4. GREATEST BINOMIAL COEFFICIENTS

In a binomial expansion binomial coefficients of the middle terms are called as greatest binomial coefficients.

(i) If *n* is even : When $r = \frac{n}{2}$ i.e. ${}^nC_{n/2}$ takes maximum value.

(ii) If *n* is odd : $r = \frac{n-1}{2}$ or $\frac{n+1}{2}$ i.e. ${}^nC_{\frac{n-1}{2}} = {}^nC_{\frac{n+1}{2}}$ and take maximum value.

5. NUMERICALLY GREATEST TERM IN THE EXPANSION OF $(X+A)^N$

Let T_{r+1} be the greatest term in $(x + a)^n$

Then $\frac{T_{r+1}}{T_r} \geq 1$ and $\frac{T_{r+2}}{T_{r+1}} \leq 1$

$$\left|\frac{T_{r+1}}{T_r}\right| = \left|\frac{{}^nC_r x^{n-r}a^r}{{}^nC_{r-1}x^{n-r+1}a^{r+1}}\right| = \left(\frac{n-r+1}{r}\right)\left|\frac{a}{x}\right|$$

Now $\left(\frac{n-r+1}{r}\right)\left|\frac{a}{x}\right| \geq 1 \Rightarrow \frac{n+1}{r} \geq \left|\frac{x}{a}\right| + 1 \Rightarrow r \leq \frac{n+1}{\left|\frac{x}{a}\right|+1}$

Algorithm

step I : Calculate $\frac{n+1}{\left|\frac{x}{a}\right|+1} = k$ (say)

step II :

(a) If *k* is an integer than T_k and T_{k+1} are the numerically greatest term.

(b) If *k* is not an integer. Let m is its integral part than T_{m+1} is the numerically greatest term.

Illustration 7: Find the term which has the greatest binomial coefficient in the expansion of $(x^2 + 2/x)^6$

Solution : We know that Binomial Coefficient of middle term is the greatest Binomial coefficient.
Since $n = 6$ is even, So the middle term is $T_{n/2+1}$
$\therefore$ middle term = $n/2 + 1 = 3 + 1 \Rightarrow 4^{th}$ term

Illustration 8: Given that the 4th term in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ has the maximum numerical value, find the range of values of x for which this will be true.

Solution : According to the question, $|t_4| \geq |t_3|$, $|t_4| \geq |t_5|$.

Now $t_{r+1} = {}^{10}C_r . 2^{10-r}\left(\frac{3x}{8}\right)^r$

$\therefore \quad t_4 = {}^{10}C_3 . 2^7 . \left(\frac{3x}{8}\right)^3$, $t_3 = {}^{10}C_2 . 2^8 . \left(\frac{3x}{8}\right)^2$ and $t_5 = {}^{10}C_4 . 2^6 . \left(\frac{3x}{8}\right)^4$

Now, $|t_4| \geq |t_3| \Rightarrow {}^{10}C_3 . 2^7 . \left(\frac{3}{8}\right)^3 . |x|^3 \geq {}^{10}C_2 . 2^8 . \left(\frac{3}{8}\right)^2 . |x|^2$

....(i)

and $|t_4| \geq |t_5| \Rightarrow {}^{10}C_3 . 2^7 . \left(\frac{3}{8}\right)^3 . |x|^3 \geq {}^{10}C_4 . 2^6 . \left(\frac{3}{8}\right)^4 . |x|^4$

....(ii)

from (i) $\frac{10.9.8}{6}.\frac{3}{8}.|x|^3 \geq \frac{10.9}{2}.2|x|^2$

or $45|x|^3 \geq 90|x|^2$ or $|x|^3 - 2|x|^2 \geq 0$ or $|x| \geq 2$, as x cannot be zero.

from (ii) $\frac{10.9.8}{6}.2|x|^3 \geq \frac{10.9.8.7}{24}.\frac{3}{8}.|x|^4$ or $|x|^3 \geq \frac{7}{8}.\frac{3}{8}|x|^4$

or $|x|^3\left\{1-\frac{21}{64}|x|\right\} \geq 0 \Rightarrow 1-\frac{21}{64}|x| \geq 0 \qquad \therefore |x| \leq \frac{64}{21}$

Thus, we get $|x| \geq 2$ and $|x| \leq \frac{64}{21} = 3\frac{1}{21}$. So $x \in \left[-\frac{64}{21}, -2\right] \cup \left[2, \frac{64}{21}\right]$

Student Task

Q.1 Find numerically greatest term in the expansion of $(3-5x)^{11}$ when $x = \frac{1}{5}$

Q.2 Given T_3 in the expansion of $(1-3x)^6$ has maximum numerical value. Find the range of x

Q.3 Find the numerically greatest term in the expansian of $(3-2x)^9$ when x =1

Q.4 In the expansion of $\left(\frac{1}{2}+\frac{2x}{3}\right)^n$ when $x = -\frac{1}{2}$ it is knows that 3rd term is the greatest term. Find the possible integral values of n.

6. TO DETERMINE A PARTICULAR TERM IN THE EXPANSION

In the expansion of $\left(x^{\alpha} \pm \frac{1}{x^{\beta}}\right)^n$, if x^m occurs in T_{r+1}, then r is given by

$$n\alpha - r(\alpha + \beta) = m \Rightarrow r = \frac{n\alpha - m}{\alpha + \beta}$$

Thus in above expansion if constant term i.e. the term which is independent of x, occurs in T_{r+1}

then r is determined by $n\alpha - r(\alpha + \beta) = 0 \Rightarrow r = \frac{n\alpha}{\alpha+\beta}$

Illustration 9: Find the term independent of x in the expansion of $\left(\frac{4}{3}x^2 - \frac{3}{2x}\right)^9$.

Solution : Here comparing $\left(\frac{4}{3}x^2 - \frac{3}{2x}\right)^9$ with $\left(x^{\alpha} \pm \frac{1}{x^{\beta}}\right)^n$

We get $\alpha = 2$, $\beta = 1$, $n = 9$ and $r = \frac{9(2)}{2+1} = 6$

$\therefore$ (6 + 1) = 7th term is independent of x.

Illustration 10: Find the coefficient of x^{39} in the expansion of $(x^4 - 1/x^3)^{15}$.

Solution : From above formula

$$r = \frac{15(4)-39}{4+3} = 3$$

$\therefore$ The required term = $T_4 = {}^{15}C_3\,(x^4)^{12}\,(-1/x^3)^3 = -455\,x^{39}$

$\therefore$ Coefficient of $x^{39} = -455$

7. BINOMIAL COEFFICIENTS & THEIR PROPERTIES

In the expansion of $(1 + x)^n$; i.e. $(1 + x)^n = {}^nC_0 + {}^nC_1x + ...+ {}^nC_rx^r + ...+ {}^nC_nx^n$

The coefficients ${}^nC_0, {}^nC_1, {}^nC_n$ of various powers of x, are called binomial coefficients and they are written as $C_0, C_1, C_2, ...C_n$.

Hence $(1 + x)^n = C_0 + C_1x + C_2x^2 + ... + C_rx^r + ... + C_nx^n$ ....(1)

Where $C_0 = 1$, $C_1 = n$, $C_2 = \frac{n(n-1)}{2!}$

$$C_r = \frac{n(n-1).....(n-r+1)}{r!},\ C_n = 1$$

Now, we shall obtain some important expressions involving binomial coefficients-

(a) Sum of coefficient : putting x = 1 in (1), we get

$C_0 + C_1 + C_2 + ...+ C^n = 2^n$ ....(2)

i.e. $\sum_{r=0}^{n} {}^nC_r = 2^n$

Generally we use the method of differentiation when the coefficient of binomial expansion C_k is a polynomial in k.

***Illustration 11*:** Find the sum of the series : $1^2.\ C_1 + 2^2.\ C_2 + 3^2.\ C_3 + ... + n^2.\ C_n$

Solution :

We have $(1 + x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$ ....(i)

Differentiating equation (i) w.r.t. x, we have

$n(1 + x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + ... + nC_nx^{n-1}$ ....(ii)

Multiplying equation (ii) throughout by x, we have

$nx(1 + x)^{n-1} = C_1x + 2C_2x^2 + 3C_3x^3 + ... + nC_nx^n$ ....(iii)

Differentiating equation (iii) w.r.t. x, we have

$n(1 + x)^{n-1} + n(n-1)(1 + x)^{n-2} = C_1 + 2^2.\ C_2x + 3^2.\ C_3x^2 + ... + n^2.\ C_n\,x^{n-1}$ ....(iv)

Putting x = 1 in equation (iii), we have

$1^2 . C_1 + 2^2 . C_2 + 3^2 . C_3 + ... + n^2 . C_n$

$= n . 2^{n-1} + n(n-1) . 2^{n-2} = (n^2 + n)\, 2^{n-2} = n(n+1)2^{n-2}$.

(b) **Sum of coefficients with alternate signs:**

Putting $x = -1$ in (1) We get

$C_0 - C_1 + C_2 - C_3 + ... = 0$

....(3)

i.e. $\sum_{r=0}^{n} (-1)^r \, {}^nC_r = 0$

(c) **Sum of coefficient of even and odd terms**

From (3), we have

$C_0 + C_2 + C_4 + ... = C_1 + C_3 + C_5 + ...$

....(4)

i.e. Sum of coefficient of even and odd terms are equals

from (2) and (4) we have

$C_0 + C_2 + ... = C_1 + C_3 + ... = 2^{n-1}$

(d) **Sum of products of coefficients**

Replacing x by $1/x$ in (1) We get

$$\left(1+\frac{1}{x}\right)^n = C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + ... + \frac{C_n}{x^n} + ...$$

....(5)

Multiplying (1) by (5), we get

$$\frac{(1+x)^{2n}}{x^n} = (C_0 + C_1 x + C_2 x^2 + ...)\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + ...\right)$$

Now, comparing coefficients of x^r on both the sides, we get

$$C_0 C_r + C_1 C_{r+1} + ... + C_{n-r}\, C_n = {}^{2n}C_{n-r} = \frac{2n!}{(n+r)!(n-r)!}$$

....(6)

(e) **Sum of squares of coefficients**

Putting $r = 0$ in (6), we get

$$C_0^2 + C_1^2 + C_2^2 + ... + C_n^2 = \frac{2n!}{n!n!}$$

(f) Putting $r = 1$ in (6) we get

$$C_0 C_1 + C_1 C_2 + C_2 C_3 + ... + C_{n-1}\, C_n = {}^{2n}C_{n-1} = \frac{2n!}{(n+1)!(n-1)!} \quad .....(7)$$

(g) Putting $r = 2$ in (6), we get

$$C_0 C_2 + C_1 C_3 + C_2 C_4 + ... + C_{n-2}\, C_n = {}^{2n}C_{n-2} = \frac{2n!}{(n+2)!(n-2)!} \quad .....(8)$$

(h) Differentiating both sides of (1) w.r.t. x, we get

$n(1 + x)^{n-1} = C_1 + 2C_2 x + 3C_3 x^2 + ... + {}^nC_n x^{n-1}$

Now putting $x = 1$ and $x = -1$ respectively

$C_1 + 2C_2 + 3C_3 + ... + {}^nC_n = n.2^{n-1}$

.....(9)

and $C_1 - 2C_2 + 3C_3 ..... = 0$ .....(10)

i.e. $\sum_{r=0}^{n}(-1)^r r.\,{}^nC_r = 0$ similarly $\sum_{r=0}^{n}(-1)^r r^2.\,{}^nC_r = 0$, $\sum_{r=0}^{n}(-1)^r r^3.\,{}^nC_r = 0$

(i) Adding (2) and (9)

$C_0 + 2C_1 + 3C_2 + ...+ {}^{(n+1)}C_n = 2^{n-1}(n+2)$ ....(11)

(j) Integrating (1) w.r.t. x between the limits 0 to 1, we get,

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \left[C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + ... + \frac{C_nx^{n+1}}{n+1}\right]_0^1$$

$$\Rightarrow \quad C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ...+ \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1} \qquad ....(12)$$

Integrating (1) w.r.t. x between the limits –1 to 0, we get

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_{-1}^0 = \left[C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + ... + \frac{C_nx^{n+1}}{n+1}\right]_{-1}^0$$

$$\Rightarrow \quad C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + ...+ \frac{(-1)^n.C_n}{n+1} = \frac{1}{(n+1)} \qquad .....(13)$$

Note :-

(i) ${}^nC_r = n.\,{}^{n-1}C_{r-1}$ (ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (iii) $\frac{1}{r+1}\,{}^nC_r = \frac{1}{n+1}\,{}^{n+1}C_{r+1}$

(iv) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$ (v)

$${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1} = \frac{n}{r}.\frac{n-1}{r-1}\,{}^{n-2}C_{r-2} = ... = \frac{n(n-1)(n-2)(n-3)....(n-r+1)}{r(r-1)(r-2)...1}$$

Illustration 12: Find the sum of the series $C_1 - \frac{C_2}{2} + \frac{C_3}{3} - \frac{C_4}{4} + \ldots + (-1)^{n+1}\frac{C_n}{n}$.

Solution : We have $(1-x)^n = C_0 - C_1x + C_2x^2 - .... + C_n(-1)^n x^n$

i.e. $1-(1-x)^n = C_1x - C_2x^2 + C_3x^3 - \ldots + C_n(-1)^{n+1}x^n$

i.e. $\frac{1-(1-x)^n}{x} = C_1 - C_2x + C_3x^2 - \ldots + C_n(-1)^{n+1}x^{n-1}$

.... (i)

Integrating equation (i) w.r.t. x from 0 to 1, we have

$$\int_0^1 \frac{1-(1-x)^n}{x}dx = \int_0^1 \left[C_1 - C_2x + C_3x^2 + \ldots\right]dx$$

Now, L.H.S. = $\int_0^1 \frac{1-x^n}{1-x}dx = \int_0^1 \left(1 + x + x^2 + \ldots + x^{n-1}\right)dx = \left[x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots + \frac{x^n}{n}\right]_0^1$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$$

and R.H.S.= $\left[C_1 x - C_2 \frac{x^2}{2} + C_3 \frac{x^3}{3} - \ldots \right]_0^1 = C_1 - \frac{C_2}{2} + \frac{C_3}{3} - \ldots + (-1)^{n+1} \frac{C_n}{n+1}$

Hence, we have , $C_1 - \frac{C_2}{2} + \frac{C_3}{3} + \ldots + (-1)^{n+1} \frac{C_n}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$.

Illustration 13: Find the value of ${}^{4n}C_0 = {}^{4n}C_4 + {}^{4n}C_8 + \ldots + {}^{4n}C_{4n}$

Solution : $(1 + x)^{4n} = {}^{4n}C_0 + {}^{4n}C_1 x + {}^{4n}C_2 x^2 + {}^{4n}C_3 x^3 + {}^{4n}C_4 x^4 + \ldots + {}^{4n}C_n x^{4n}$

Put $x = 1$ and $x = -1$, then adding.

$2^{4n-1} = {}^{4n}C_0 + {}^{4n}C_2 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n}$ ..... (1)

Now put, $x = i$

$(1 + i)^{4n} = {}^{4n}C_0 + {}^{4n}C_1 i - {}^{4n}C_2 - {}^{4n}C_3 i + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n}$

Compare real and imaginary part, we get

$(-1)^n (2)^{2n} = {}^{4n}C_0 - {}^{4n}C_2 + {}^{4n}C_2 + {}^{4n}C_4 - {}^{4n}C_6 + \ldots + {}^{4n}C_{4n}$ .....(2)

Adding (i) and (2), we get

$\Rightarrow \quad {}^{4n}C_0 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n} = (-1)^n (2)^{2n-1} + 2^{4n-2}$

Illustration 14: If $(1 + x)^n = C_0 + C_1 x + C_2 x^2 + \ldots + C_n x^n$, then find the value of $3C_0 - 5C_1 + 7C_2 + \ldots + (-1)^n (2n+3) C_n$.

Solution : We have, $3C_0 - 5C_1 + 7C_2 + \ldots + (-1)^n (2n + 3) C_n$

$= 3C_0 - 3C_1 + 3C_2 + \ldots + (-1)^n 3C_n - 2C_1 + 4C_2 + \ldots + (-1)^n 2n\, C_n$

$= 3 (C_0 - C_1 + C_2 + \ldots + (-1)^n C_n) - 2 (C_1 - 2C_2 + \ldots (-1)^n n C_n) = 3 \times 0 - 2 \times 0 = 0$

Illustration 15(a): If $C_0, C_1, C_2 \ldots C_n$ are binomial coefficients, then find $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots \frac{C_n}{n+1}$.

Solution : $(1 + x)^n = C_0 + C_1 x + C_2 x^2 + \ldots + C_n x^n$

Integrate it with respect to x

$$\frac{(1+x)^{n+1}}{n+1} C_0 x + \frac{C_1 x^2}{2} + C_2 \frac{x^3}{3} + \ldots + C_n \frac{x^{n+1}}{n+1} + k$$

put $x = 0$ then, $k = \frac{1}{n+1}$, Now put $x = 1$

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{(n+1)}$$

Illustration 15(b): *If* $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots + C_n x^n$ *then prove that*

$$\sum\sum_{0 \le i,\ j \le n} (C_i + C_j)^2 = (n-1) 2^n C_n + 2^{2n}$$

Solution: **L.H.S** $\sum\sum_{0 \le i < j \le n} (C_i + C_j)^2$

$$= (C_0 + C_1)^2 + (C_0 + C_2)^2 + \ldots + (C_0 + C_n)^2 + (C_1 + C_2)^2 + (c_1 + C_3)^2 + \ldots + (C_1 + C_n)^2 +$$

$$(C_2 + C_3)^2 + (C_2 + C_4)^2 + \ldots + (C_2 + C_n)^2 + \ldots + (C_{n-1} + C_n)^2$$

$$= n\left(C_0^2 + C_1^2 + C_2^2 + ... + C_n^2\right) + 2 \sum_{0 \le i < j \le n} C_i\, C_j$$

$$= n.\ {}^{2n}C_n + 2.\left\{2^{n-1} - \frac{2n!}{2.n!n!}\right\}$$

$$= n.\ {}^{2n}C_n + 2^{2n} - {}^{2n}C_n = (n-1)\ {}^{2n}C_n + 2^{2n} = \text{R.H.S.}$$

Student Task

Q.1 If $C_0, C_1, C_2, ... C_n$ denote the coefficient in the binomial expansion of $(1 + x)^n$, then prove that :

$$3C_0 + \frac{3^2}{2} C_1 + \frac{3^3}{3} C_2 + \frac{3^4}{4} C_3 + ...+ \frac{3^{n+1}}{n+1} C_n = \frac{4^{n+1} - 1}{n+1}.$$

Q.2 Prove that the greatest coefficient in the expansion of $(1 + x)^{2n}$ is double the greatest coefficient in the expansion $(1 + x)^{2n-1}$.

Q.3 If $(1 + x)^n = \sum_{r=0}^{n} {}^nC_r x^n$, then prove that

$$\frac{C_0}{1.2}.2^2 + \frac{C_1}{2.3}.2^3 + \frac{C_2}{3.4}.2^4 + ...+ \frac{C_n}{(n+1)(n+2)}\ 2^{n+2} = \frac{3^{n+2} - 2n - 5}{(n+1)(n+2)}$$

Q.4 Find the greatest term in the expansion of $(1 + x)^{10}$ when $x = 2/3$.

Q.5 If $C_0, C_1, C_2, \ldots C_n$ denote the coefficient in the binomial expansion of $(1 + x)^n$, then prove that : $(1.2)\ C_2 + (2.3)\ C_3 + ...+ ((n - 1).n)\ C_n = n\ (n - 1)2^{n-2}$

Q.6 Show that if the greatest term in the expansion of $(1 + x)^{2n}$ has also the greatest coefficient, then x lies between $\frac{n}{n+1}$ and $\frac{n+1}{n}$.

Q.7 If $C_0, C_1, C_2, ... C_n$ denote the coefficient in the binomial expansion of $(1 + x)^n$, then prove that : $C_0 - C_1 + C_2 - C_3 + ... + (-1)^n\ C_n = 0$.

Q.8 Prove that ${}^{25}C_{10} + {}^{24}C_{10} + ... + {}^{10}C_{10} = {}^{26}C_{11}$

Q.9 $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n} = ?$

(a) 2^{n-1} (b) ${}^{2n}C_n$ (c) 2^n (d) 2^{n+1}

Answer :

(4) $210\left(\frac{2}{3}\right)^4$

8. MULTINOMIAL THEOREM

Using binomial theorem we have

$$(x + a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r, n \in N = \sum_{r=0}^{n} \frac{n!}{(n-r)!r!} x^{n-r} a^r = \sum_{r+s=n} \frac{n!}{r!s!} x^5 a^r, \text{ where } s + r = n$$

This result can be generalized in the following form.

$$(x_1 + x_2 + ... + x_k)^n = \sum_{r_1 + r_2 + ... + r_k = n} \frac{n!}{r_1!r_2!.....r_k!} x_1^{r_1} x_2^{r_2} .....x_k^{r_k}$$

The general term in the above expansion $\frac{n!}{r_1!r_2!.....r_k!}x^{r_1}x^{r_2}.....x^{r_k}$

The number of terms in the above expansion is equal to the number of none negative integral solution of the equation

$r_1+r_2...r_k=n$ because each solution of this equation gives a term in the above expansion

The Number of each solution is ${}^{n+k-1}C_{k-1}$

Q. Find the coefficient of $x^2y^3z^4w$ in the expansion of $(x-y-z+w)^{10}$

Solution: $(x-y-z+w)^{10}=\sum_{p+q+r+s=10}\frac{n!}{p!q!r!s!}(x)^p(-y)^q(-z)^r(w)^s$

We want to get $x^2y^3z^4w$ this implies that p = 2, q=3, r=4,s=1

$\therefore$ Coefficient of $x^2y^3z^4$ is $\frac{10!}{2!3!4!1!}(-1)^3(-1)^4=-12600$

9. BINOMIAL THEOREM FOR ANY RATIONAL INDEX

When n is a negative integer or a fraction then the expansion of a binomial is possible only when

(i) Its first term is 1, and

(ii) Its second term is numerically less than 1.

when $n \notin N$ and $|x| < 1$, then it states

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 +.....+ \frac{n(n-1)(n-r+1)}{r!}x^r +.....\infty$$

General Term

$$T_{r+1} = \frac{n(n-1)(n-2).....(n-r+1)}{r!}x^r$$

(i) In this expansion the coefficient of different terms can not be expressed as ${}^nC_0, {}^nC_1, {}^nC_2$,.... because n is not a positive integer.

(ii) In this case there are infinite terms in the expansion.

Approximations

$$(1+x)^n = 1+nx+\frac{n(n-1)}{1.2}x^2+\frac{n(n-1)(n-2)}{1.2.3}x^3.....$$

If $x < 1$, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when we may neglect the terms containing higher powers of x in the expansion. Thus, if x be so small that its squares and higher powers may be neglected then $(1+x)^n = 1+nx$, approximately. This is an approximate value of $(1+x)^n$.

Important Expansions

If $|x| < 1$ and $n \in Q$ but $n \notin N$, then

(a) $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 +.....+ \frac{n(n-1).....(n-r+1)}{r!}x^r + ...$

(b) $(1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + ...+ \frac{n(n-1).....(n-r+1)}{r!}(-x)^r + ...$

(c) $(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 +....+ \frac{n(n+1).....(n+r-1)}{r!}x^r + ...$

(d) $(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + ...+ \frac{n(n+1)(n+r-1)}{r!}(-x)^r + ...$

By putting n = 1, 2, 3 in the above results (c) and (d), we get the following results

(e) $(1-x)^{-1} = 1 + x + x^2 + x^3 + ...+ x^r + ...$

General term $T_{r+1} = x^r$

(f) $(1 + x)^{-1} = 1 - x + x^2 - x^3 + ...(-x)^r + ...$

General term $T_{r+1} = (-x)^r$

(g) $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + ...+ (r + 1) x^r + ...$

General term $T_{r+1} = (r + 1) x^r$

(h) $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 +....+ (r + 1) (-x)^r + ...$

General term $T_{r+1} = (r + 1) (- x)^r$

(i) $(1 - x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + ...+ \frac{(r+1)(r+2)}{2!} x^r + ...$

General term = $\frac{(r+1)(r+2)}{2!} x^r$

(j) $(1 + x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + ...+ \frac{(r+1)(r+2)}{2!} (-x)^r + ...$

General term = $\frac{(r+1)(r+2)}{2!} (-x)^r$

(k) Multinomial Expansion ($n \in N$) :

General terms in expansion of $\left(x_1 + x_2 + x_3 + ... + x_k\right)^n$ is

$\frac{n!}{a_1!a_2!a_3!.....a_k!} .x_1^{a_1} .x_2^{a_2} .x_3^{a_3} .....x_k^{a_k}$ where

$a_1 + a_2 + a_3 + ... + a_k = n$, $0 \le a_i \le n$, i = 1, 2, 3,..... k and the number of terms in the expansion are ${}^{n+k-1}C_{k-1}$.

Number of terms in $(x + y)^n = {}^{n+1}C_1$; Number of terms in $(x + y + z)^n = {}^{n+2}C_2$

Number of terms in $(x + y + z + \omega) = {}^{n+3}C_3$

SUM OF THE SERIES

(a) By comparing the coefficients of same power of x in an expansion :

In this method we use the fact that coefficient of same power of x in an appropriate identity is the given series.

Important Formulae

If C_0, C_1, C_2, C_3, ... represent ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3$ ..... in the expansion of $(1 + x)^n$. Then

(i) ${C_0}^2 + {C_1}^2 + {C_2}^2 +..... {C_n}^2 = {}^{2n}C_n$

(ii) $C_0C_r + C_1C_{r+1} + C_{2r+2} +.....+ C_{n-r}C_n = {}^{2n}C_{n-r}$ or ${}^{2n}C_{n+r}$

Illustration 16: Find the sum of the series $C_1^2 + 2.C_2^2 + 3.C_3^2 + \ldots + n.C_n^2$

Solution : We have $(1 + x)^n = C_0 + C_1x + C_2x^2 + ... + n.C_nx^n$ ....(i)

and $\left(1+\frac{1}{x}\right)^n = C_0 + C_1\frac{1}{x} + C_2\ \frac{1}{x^2} + \ldots + C_n\frac{1}{x^n}$ ....(ii)

Differentiating equation (i) w.r.t. x, we have

$n(1 + x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + ... + nC_n\, x^{n-1}$ ....(iii)

Hence, the given series = coefficient of x^{-1} in $\left(1+\frac{1}{x}\right)^n . n(1 + x)^{n-1}$

$= n \times$ coefficient of x^{n-1} in $(1+x)^{2n-1} = n.^{2n-1}C_{n-1}$.

(b) By equating the real and imaginary parts

Illustration 17: If $(3+4x)^n = p_0 + p_1x + p_2x^2 + p_3x^3 + ... + p_nx^n$, then prove that $(p_0 - p_2 + p_4 - ....)^2 + (p_1 - p_3 + p_5 - ....)^2 = 25^n$

Solution : Consider the expansion $(3+4x)^n = p_0 + p_1x + p_2x^2 + p_3x^3 + ... + p_nx^n$

Putting $x = i$ in the above expansion we get,

$(3+4i)^n = (p_0 - p_2 + p_4 - ...) + i(p_1 - p_3 + p_5 - ....)$

Equating the square of the modulus, we get,

$(p_0 - p_2 + p_4 - ....)^2 + (p_1 - p_3 + p_5 - ....)^2 = 25^n$

10. EXPONENTIAL AND LOGARITHMIC SERIES

(1) Exponential

(i) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...\infty$; where x may be any real or complex and $e = \underset{n\to\infty}{Lim}\left(1+\frac{1}{n}\right)^n$

(ii) $a^x = 1 + x(\log_e a) + \frac{x^2}{2!}(\log_e a)^2 + \frac{x^3}{3!}(\log_e a)^3 + ...\infty$

(a) $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ...\infty$

(b) e is an irrational number lying between 2.7 and 2.8. Its value correct upto 10 place of decimal is 2.7182818284.

(c) $e + e^{-1} = 2\left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\infty\right)$

(d) $e - e^{-1} = 2\left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + ...\infty\right)$

(e) Logarithms to the base 'e' are known as the Napierian system, so named after Napier, their inventor. They are also called Natural Logarithm.

(2) Logarithmic Series

(i) $\ell n(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$ where $-1 < x \le 1$

(ii) $\ell n(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$ where $-1 \le x < 1$

(iii) $\ell n\frac{(1+x)}{(1-x)} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + ...\infty\right) |x|<1$

(a) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ...\infty = ln2$ **(b)** $e^{\ln x} = x$

(c) ln 2 = 0.693 **(d)** ln 10 = 2.303

APPROXIMATIONS

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 .....$$

If x<1, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when, we may neglect the terms containing higher power of x in the expansion

$(1+x)^n = 1 + nx$ Approximately

Q. If x is so small such that its square and higher powers may be neglected then find the approximate value of $\frac{(1-3x)^{1/2}+(1-x)^{5/3}}{(4+x)^{1/2}}$

Solution: $\frac{(1-3x)^{1/2}+(1-x)^{5/3}}{(4+x)^{1/2}} = \frac{1-\frac{3}{2}x+1-\frac{5x}{3}}{2\left(1+\frac{x}{4}\right)^{1/2}} = \frac{1}{2}\left(2-\frac{19}{6}x\right)\left(1+\frac{x}{4}\right)^{-1/2} = \frac{1}{2}\left(2-\frac{19}{6}x\right)\left(1-\frac{x}{8}\right)]$

$= \frac{1}{2}\left(2-\frac{x}{4}-\frac{19}{6}x\right) = 1-\frac{x}{8}-\frac{19}{12}x = 1-\frac{41}{24}x$

REMEMBER

1. $a^x = 1 + x(\log_e a) + \frac{x^2}{2!}(\log_e a)^2 + \frac{x^3}{3!}(\log_e a)^3 + ...\infty$

2. $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ... \infty$ when $|x| < 1$

3. $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} ..... \infty$ when $|x| < 1$

4. $\log\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + ... \infty\right)$

5. $\log\frac{x+1}{x-1} = 2\left(\frac{1}{x} + \frac{1}{3x^3} + \frac{1}{5x^5} + ...\infty\right)$

6. $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ...$

7. If put $\frac{1+x}{1-x} = \frac{m}{n}$ then $\log\left(\frac{m}{n}\right) = 2\left[\left(\frac{m-n}{m+n}\right) + \frac{1}{3}\left(\frac{m-n}{m+n}\right)^3 + \frac{1}{5}\left(\frac{m-n}{m+n}\right)^5 + ...\right]$

8. $\sum_{n=0}^{\infty}\frac{1}{n!} = \sum_{n=0}^{\infty}\frac{1}{(n-1)!} = \sum_{n=0}^{\infty}\frac{1}{(n-k)!} = e$

9. $\sum_{n=1}^{\infty}\frac{1}{n!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ...+ \infty = e - 1$

10. $\sum_{n=2}^{\infty}\frac{1}{n!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{4!} + ... + \infty = e - 2$

11. $\sum_{n=0}^{\infty}\frac{1}{(n+1)!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ... + \infty = e - 1$

12. $\sum_{n=0}^{\infty}\frac{1}{(n+2)!} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{4!} + ... + \infty = e - 2$

13. $\sum_{n=1}^{\infty}\frac{1}{(n+1)!} = \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + ... + \infty = e - 2$

14. $\sum_{n=0}^{\infty}\frac{1}{(2n)!} = 1+\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\ldots\ldots = \frac{e+e^{-1}}{2} = \sum_{n=1}^{\infty}\frac{1}{(2n-2)!}$

15. $\sum_{n=1}^{\infty}\frac{1}{(2n-1)!} = \frac{1}{1!}+\frac{1}{3!}+\frac{1}{5!}+\ldots\ldots = \frac{e-e^{-1}}{2} = \sum_{n=0}^{\infty}\frac{1}{(2n+1)!}$

16. $e^{ax} = 1+\frac{(ax)}{1!}+\frac{(ax)^2}{2!}+\frac{(ax)^3}{3!}+\ldots\ldots+\frac{(ax)^n}{n!}+\ldots\ldots\ldots\infty$

17. $\sum_{n=1}^{\infty}\frac{n^2}{n!} = \sum_{n=1}^{\infty}\frac{n}{(n-1)!} = 2e$

18. $\sum_{n=1}^{\infty}\frac{n^3}{n!} = \sum_{n=1}^{\infty}\frac{n^2}{(n-1)!} = 5e$

19. $\sum_{n=1}^{\infty}\frac{n^4}{n!} = \sum_{n=1}^{\infty}\frac{n^3}{(n-1)!} = 15e$

20. The coefficient of x^n in the expansion of $e^{a+bx} = 1+\frac{(a+bx)}{1!}+\frac{(a+bx)^2}{2!}+\ldots\ldots\ldots\infty$ is $\frac{e^a.b^n}{n!}$

21. The coefficient of x^n in the expansion of $e^{-(a+bx)} = 1-\frac{(a+bx)}{2!}+\frac{(a+bx)^2}{2!}-\frac{(a+bx)^3}{3!}+\ldots\ldots\ldots\infty$ is

$(-1)^n \cdot \frac{e^a.b^n}{n!}$

23. $1+\frac{1+a}{2!}+\frac{1+a+a^2}{3!}+\frac{1+a+a^2+a^3}{4!}+\ldots\ldots\ldots\infty = \frac{e^a-e}{a-1}$ when $a > 1$

23. If α and β are the roots of the equation $x^2 - px + q = 0$, than coefficient of x^n in the expansion of

$\log(1+px+qx^2) = (-1)^{n+1}\frac{\alpha^n+\beta^n}{n}$

24. If nth term of a series is $\frac{a}{2n-1}+\frac{b}{2n}+\frac{c}{2n+1} = (-b\log 2 - c)$

Then the sum of the series is $-b\log 2 - c$. (where $a + b + c = 0$]

USEFUL TIPS

1. If a, b, c and d are the coefficients of any four consecutive terms in the expansion of $(1+x)^n$, $n \in N$, then $\frac{a}{a+b}+\frac{c}{c+d}=\frac{2b}{b+c}$.

2. ${}^{n-1}C_r + {}^{n-2}C_r + {}^{n-3}C_r + \ldots\ldots + {}^{r}C_r = {}^{n}C_{r+1}$

3. Let I denotes the integral part and F the proper fractional part of $(3+\sqrt{5})^n$ where $n \in N$ and if ρ denotes the rational part and σ the irrational part of the same, $\rho = \frac{1}{2}(I+1)$ and $\sigma = \frac{1}{2}(I+2F-1)$.

4. $\frac{C_1}{C_0}+\frac{2C_2}{C_1}+\frac{3C_3}{C_2}+.....+\frac{n.C_n}{C_{n-1}}=\frac{n(n+1)}{2}$

5. $\frac{C_0}{1}+\frac{C_2}{3}+\frac{C_4}{5}+.....=\frac{2^n}{n+1}$

6. $\frac{C_0}{2}+\frac{C_1}{3}+\frac{C_2}{4}+\frac{C_3}{5}.....+\frac{C_n}{n+2}=\frac{1+n.2^{n+1}}{(n+1)(n+2)}$

7. $({}^{2n}C_1)^2+2.({}^{2n}C_2)^2+({}^{2n}C_3)^2+.....+2n.({}^{2n}C_n)^2=\frac{(4n-1)!}{[(2n-1)!]^2}$

MISCELLANEOUS EXAMPLES

Example 1 : If $\sum_{r=0}^{2n} a_r\,(x-1)^r=\sum_{r=0}^{2n} b_r\,(x-2)^r$ and $b_r = (-1)^{r-n}$ for all $r \geq n$, find a_n

Solution : Let, $x - 1 = t$ then $\sum_{r=0}^{2n} a_r t^r=\sum_{r=0}^{2n} b_r\,(t-1)^r$

$\therefore a_n$ = coefficient of t^n in $\sum_{r=0}^{2n} b_r\,(t-1)^r$

= coefficient of t^n in $(b_0+b_1)(t-1)+.....+b_n(t-1)^n+b_{n+1}(t-1)^{n+1}+....+b_{2n}\,(t-1)^{2n}$

$= b_n\,{}^nC_0+b_{n+1}\,{}^{n+1}C_1(-1)^1+b_{n+2}\,{}^{n+2}C_2(-1)^2+.....b_{2n}\,{}^{2n}C_n(-1)^n$

$= (-1)^n\,{}^nC_0+(-1)^{n+1-n+1}\,{}^{n+1}C_1+.....+(-1)^{2n-n+n}\,{}^{2n}C_n$

$= {}^nC_0+{}^{n+1}C_1+{}^{n+2}C_2+.....\,{}^{2n}C_n={}^{2n+1}C_{n+1}={}^{2n+1}C_n$

Example 2 : Find the sum of $1+\frac{1}{4}+\frac{1.3}{4.8}+\frac{1.3.5}{4.8.12}+.....\infty$.

Solution : Comparing with $1 + nx + \frac{n(n-1)}{2!}x^2 + ...$

$nx = 1/4$ and $\frac{n(n-1)x^2}{2!}=\frac{1.3}{4.8}$

or $\frac{nx(nx-x)}{2!}=\frac{3}{32}\Rightarrow\frac{1}{4}\left(\frac{1}{4}-x\right)=\frac{3}{16}$ .....(i)

$\Rightarrow\left(\frac{1}{4}-x\right)=\frac{3}{4}\Rightarrow x=\frac{1}{4}-\frac{3}{4}=-\frac{1}{2}$ .....(ii)

Putting the value of x in (1)

$n(-1/2) = 1/4 \Rightarrow n = -1/2$

$\therefore$ Sum of series = $(1 + x)^n = (1 - 1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$

Example 3 : Show that $\frac{2^{n+1}-x^{(2n+2)}}{(n+1)\left(2-x^2\right)}={}^nC_0.x^{2n}+\frac{{}^nC_1}{2}.x^{2n-2}.\left(2-x^2\right)+\ldots+\frac{{}^nC_n}{(n+1)}.\left(2-x^2\right)^n$.

Solution : We have $(1+y)^n = {}^nC_0 + {}^nC_1y + ... + {}^nC_ny^n$
Integrating both sides w.r.t. y, we get

$$\Rightarrow\frac{(1+y)^{n+1}}{n+1}={}^nC_0y+\frac{{}^nC_1}{2}y^2+\ldots+\frac{{}^nC_n}{(n+1)}y^{n+1}+C \qquad ....(i)$$

Putting $y = 0$ in equation (i), we get $C = \left(\frac{1}{n+1}\right)$

Putting $y = \frac{2-x^2}{x^2}$ in equation (i), we get

$$\frac{\left(1+\frac{2-x^2}{x^2}\right)^{n+1}}{n+1}-\frac{1}{n+1}={}^nC_0\left(\frac{2-x^2}{x^2}\right)+\frac{{}^nC_1}{2}\left(\frac{2-x^2}{x^2}\right)^2+\ldots+\frac{{}^nC_n}{(n+1)}\left(\frac{2-x^2}{x^2}\right)^{n+1}$$

$$\Rightarrow\ \frac{2^{n+1}}{(n+1)x^{2n+2}}-\frac{1}{n+1}={}^nC_0\left(\frac{2-x^2}{x^2}\right)+\frac{{}^nC_1}{2}.\left(\frac{2-x^2}{x^2}\right)^2+\ldots+\frac{{}^nC_n}{(n+1)}\left(\frac{2-x^2}{x^2}\right)^{n+1}$$

$$\Rightarrow\frac{2^{n+1}-x^{2n+2}}{(n+1)x^{2n+2}}={}^nC_0\left(\frac{2-x^2}{x^2}\right)+\frac{{}^nC_1}{2}\left(\frac{2-x^2}{x^2}\right)^2+\ldots+\frac{{}^nC_n}{(n+1)}\left(\frac{2-x^2}{x^2}\right)^{n+1}$$

Multiplying both sides by $\left(\frac{x^{2n+2}}{2-x^2}\right)$, we get

$$\frac{2^{n+1}-x^{2n+2}}{(n+1)\left(2-x^2\right)}=\frac{{}^nC_0}{1}x^{2n}+\frac{{}^nC_1}{2}\left(2-x^2\right).x^{(2n-2)}+\ldots+\frac{{}^nC_n}{(n+1)}\left(2-x^2\right)^n$$

Example 4 : Prove that $C_1 - C_3 + C_5 - .... = 2^{n/2}\sin\frac{n\pi}{4}$

Solution: Consider the expansion, $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$ ...(i)
Putting x = – i in equation (i) we have
$(1 - i)^n = C_0 - C_1i - C_2 + C_3i + C_4 + ... (-1)^n C_n i^n$

or $(2)^{n/2}\left(\cos\left(-\frac{n\pi}{4}\right)+i\sin\left(-\frac{n\pi}{4}\right)\right) = (C_0 - C_2 + C_4 - ....) - i.(C_1 - C_3 + C_5 - ....)$ ...(ii)

equating the imaginary part in (i), we get

$$C_1 - C_3 + C_5 - ... = 2^{n/2}\sin\frac{n\pi}{4}$$

Example 5 : Find the sum of the series : $C_0 - 3C_1 + 5C_2 + ... + (-1)^n (2n + 1)C_n$.

Solution : We have $(1 - x)^n = C_0 - C_1x + C_2x^2 - .... + C_n (-1)^nx^n$ ...(i)
Replacing x by x^2 in equation (i), we have
$(1 - x^2)^n = C_0 - C_1x^2 + C_2x^4 - ... + C_n (-1)^n x^{2n}$ ...(ii)

Multiplying throughout by x, we have

$x(1-x^2)^n = C_0x - C_1x^3 + C_2x^5 - .... + C_n(-1)^n x^{2n+1}$ ...(iii)

Differentiating equation (iii) w.r.t. x, we have

$(1-x^2)^n - 2nx^2(1-x^2)^{n-1} = C_0 - 3C_1x^2 + 5C_2x^4 - ... + (-1)^n(2n+1)C_nx^{2n}$ ...(iv)

Putting $x = 1$ in equation (iv), we have

$C_0 - 3C_1 + 5C_2 - ... + (-1)^n(2n+1)C_n = 0$.

Example 6 : In the expansion of $\left(\sqrt[3]{\frac{a}{b}} + \sqrt[3]{\frac{b}{\sqrt{a}}}\right)^{21}$ the term containing same powers of a and b is –

(a) 11th (b) 13th (c) 12th (d) 6th

Solution : (b) Given : $\left(\sqrt[3]{\frac{a}{b}} + \sqrt[3]{\frac{b}{\sqrt{a}}}\right)^{21}$

$$t_{r+1} = {}^{21}C_r\left(\frac{a}{b}\right)^{\frac{21-r}{3}}\frac{b^{r/3}}{a^{r/6}} = {}^{21}C_r\, a^{\frac{42-3r}{6}}\, b^{\frac{2r-21}{3}}$$

$\therefore 42 - 3r = 4r - 42$ i.e. $r = 12$

$\therefore 13^{th}$ term contains same power of a and b.

Example 7 : If $(1 + x + x^2)^{25} = a_0 + a_1x + a_2x^2 + ... a_{50}x^{50}$ then $a_0 + a_2 + a_4 + ...+ a_{50}$ is –

(a) even (b) odd and of the form $3n$

(c) odd and of the form $(3n - 1)$ (d) odd and of the form $(3n + 1)$

Solution : (a) Putting $x = 1$ and -1 and adding

$$a_0 + a_2 + ... + a_{50} = \frac{3^{25}+1}{2} = \frac{(1+2)^{25}+1}{2}$$

$$= \frac{{}^{25}C_0 + {}^{25}C_1.2 + {}^{25}C_2.2^2 + ... + {}^{25}C_{25}.2^{25} + 1}{2} = \frac{2[1 + {}^{25}C_1 + {}^{25}C_2.2 + ... + {}^{25}C_{25}.2^{24}]}{2}$$

$= 2[13 + {}^{25}C_2 + ... + {}^{25}C_{25} \cdot 2^{23}] \Rightarrow$ even

Example 8 : Find the sum of the series : $aC_0 + a^2\frac{C_1}{2} + a^3\frac{C_2}{3} + ... + a^{n+1}\frac{C_n}{n+1}$

Solution : We have $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$ ...(i)

Integrating equation (i) w.r.t. x, we have

$$\int_0^a (1+x)^n\,dx = \int_0^a \left[C_0 + C_1x + C_2x^2 + ... + C_nx^n\right]dx \quad ...(ii)$$

Now, L.H.S. $= \left[\frac{(1+x)^{n+1}}{(n+1)}\right]_0^a = \frac{(1+a)^{n+1}-1}{(n+1)}$

and R.H.S $= \left[C_0 + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + ...\right]_0^a = aC_0 + a^2\frac{C_1}{2} + a^3\frac{C_2}{3} + ...$

Hence, we have, $aC_0 + a^2\frac{C_1}{2} + a^3\frac{C_2}{3} + ... = \frac{(1+a)^{n+1}-1}{(n+1)}$

Example 9 : If $\frac{(1-3x)^{1/2}+(1-x)^{5/3}}{\sqrt{4-x}}$ is approximately equal to $a + bx$ for small values of x, then (a,b) = ?

Solution :

$$\frac{(1-3x)^{1/2}+(1-x)^{5/3}}{2\left[1-\frac{x}{4}\right]^{1/2}} = \frac{\left[1+\frac{1}{2}(-3x)+\frac{1}{2}\left(-\frac{1}{2}\right)\frac{1}{2}(-3x)^2+.....\right]+\left[1+\frac{5}{3}(-x)+\frac{5}{3}\frac{2}{3}\frac{1}{2}(-x)^2+....\right]}{2\left[1+\frac{1}{2}\left(-\frac{x}{4}\right)+\frac{1}{2}\left(-\frac{1}{2}\right)\frac{1}{2}\left(-\frac{x}{4}\right)^2+....\right]}$$

$$= \frac{\left[1-\frac{19}{12}x+\frac{53}{144}x^2-.....\right]}{\left[1-\frac{x}{2}-\frac{1}{8}x^2-.....\right]} = 1-\frac{35}{24}x+.....$$

Neglecting higher powers of x, then $a + bx = 1 - \frac{35}{24}x \Rightarrow a = 1, b = -\frac{35}{24}$

Example 10 : Find the coefficient of x^{50} in the expression : $(1 + x)^{1000} + 2x(1 + x)^{999} + 3x^2 (1 + x)^{998} + ...+ 1001. x^{1000}$.

Solution : Let $S = (1 + x)^{1000} + 2x (1 + x)^{999} + 3x^2(1 + x)^{998} + ... 1000. x^{999} (1 + x)+ 1001. x^{1000}$ ..(i)

Above series is a arithmetic-geometric series with common ratio of G.P. equal to $\frac{x}{(1+x)}$

So multiplying both sides by $\frac{x}{(1+x)}$, we get

$$\frac{kS}{(1+x)} = x(1+x)^{999}+2x^2(1+x)^{998}+\ldots+\frac{1001.x^{1001}}{(1+x)} \quad \ldots(ii)$$

from (i) – (ii), we get

$$\left(1-\frac{x}{1+x}\right)S = (1+x)^{1000}+x(1+x)^{999}+x^2(1+x)^{998}+\ldots+x^{1000}-\frac{1001.x^{1001}}{(1+x)}$$

$$\Rightarrow \frac{S}{(1+x)} = (1+x)^{1000}\left[\frac{1-\left(\frac{x}{1+x}\right)^{1001}}{1-\left(\frac{x}{1+x}\right)}\right]-1001.\frac{x^{1001}}{(1+x)}$$

$$\Rightarrow \frac{S}{(1+x)} = (1+x)^{1000}\left[1-\frac{x^{1001}}{(1+x)^{1001}}\right].(1+x)-1001.\frac{x^{1001}}{(1+x)}$$

$\Rightarrow S = (1 + x)^{1002} – x^{1001} (1 + x) – 1001. x^{1001}$

$\therefore$ coefficient of x^{50} in above expression = $^{1002}C_{50}$

Example 11 : Find the value of $\frac{C_0}{1}+\frac{C_2}{3}+\frac{C_4}{5}+\ldots$

Solution: We have, $(1+x)^n = \sum_{r=0}^{n} C_r x^r$ ...(i)

$$(1-x)^n = \sum_{r=0}^{n} (-1)^r C_r x^r \quad ...(ii)$$

Adding equations (i) and (ii) we have,

$(1+x)^n + (1-x)^n = C_0 + C_2x^2 + C_4x^4 + \ldots$

Integrating both sides with respect to x from 0 to 1, we get

$$\frac{2^{n+1}}{n+1} = \frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \ldots$$

Example 12 : Find the coefficient of x^5 in the expansion of $(1 + x)^{21} + (1 + x)^{22} + ... + (1 + x)^{30}$.

Solution : $(1 + x)^{21} + (1 + x)^{22} + ... + (1 + x)^{30}$

$$= (1 + x)^{21} \left(\frac{(1+x)^{10} - 1}{(1+x) - 1} \right) = \frac{1}{x}\left[(1+x)^{31} - (1+x)^{21} \right]$$

$\therefore$ Coefficient of x^5 in the given expression

= Coefficient of x^5 in $\left[\frac{1}{x}\left\{ (1+x)^{31} - (1+x)^{21} \right\} \right]$ = Coefficient of x^6 in $\{(1 + x)^{31} – (1 + x)^{21}\}$ = $^{31}C_6 – {}^{21}C_6$

Example 13 : If x is numerically very small as compared with 1, then find the value of $(1 – 7x)^{1/3} (1 + 2x)^{–3/4}$.

Solution : $(1 – 7x)^{1/3} = 1 + \frac{1}{3}(– 7x) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{1.2}(– 7x)^2 + ...$ ....(i)

$(1 + 2x)^{–3/4} = 1 + (–3/4)(2x) + \frac{(-3/4)(-7/4)}{1.2}(4x^2) + ...$ ....(ii)

Multiply (i) and (ii), we get

$(1 – 7x)^{1/3} (1 + 2x)^{–3/4} = 1 - \frac{7}{3}x - \frac{3}{2}x + ... = 1 - \frac{23x}{6}$ (Neglected rest term as x small)

Example 14 : If $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nX^n$, then $\frac{(C_0 + C_1)(C_1 + C_2).....(C_{n-1} + C_n)}{C_1C_2.....C_n}$ equals.

Solution : The given expression = $\frac{C_0C_1C_2.....C_{n-1}}{C_1C_2.....C_n} \cdot \left(1 + \frac{C_1}{C_0}\right)\left(1 + \frac{C_2}{C_1}\right)\left(1 + \frac{C_3}{C_2}\right).....\left(1 + \frac{C_n}{C_{n-1}}\right)$

$= \left(1 + \frac{n}{1}\right)\left(1 + \frac{n-1}{2}\right)\left(1 + \frac{n-2}{3}\right).....\left(1 + \frac{1}{n}\right)$ (since $C_0 = C_n$) $= \frac{(n+1)^n}{n!}$

Example 15 : p, q and r are the coefficients of three consecutive terms of $(1 + x)^n$.

Prove that the index n is $\frac{2pr + q(q+r)}{q^2 - pr}$ and the term having coefficient p is $\left(\frac{p(q+r)}{q^2 - pr}\right)$th term.

Solution : General term = $^nC_k . x^k$

Let, p, q and r be the coefficients of kth, $(k + 1)^{th}$, $(k + 2)^{th}$ terms respectively.

$\therefore$ $p = {}^nC_{k-1} = \frac{n!}{(n-k+1)!(k-1)!}$ ...(i)

$q = {}^nC_k = \dfrac{n!}{(n-k)!.k!}$ ...(ii)

$r = {}^nC_{k+1} = \dfrac{n!}{(n-k-1)!.k+1}$ ...(iii)

from (i) and (ii) : $\dfrac{p}{q} = \dfrac{k}{n-k+1} \Rightarrow k = \dfrac{(n+1)p}{p+q}$ ...(iv)

from (ii) and (iii) $\dfrac{q}{r} = \dfrac{k+1}{n-k} \Rightarrow k\dfrac{(nq-r)}{q+r}$ ....(v)

from (iv) and (v) $\dfrac{(n+1)p}{p+q} = \dfrac{(nq-r)}{q+r} \Rightarrow n = \dfrac{2pr+q(p+r)}{q^2-pr}$

term having coefficient $p = \dfrac{(n+1)p^{th}}{p+q}$ term = $\dfrac{\left(\dfrac{2pr+q(p+r)}{q^2-pr}+1\right)p^{th}}{p+q}$ $term = \dfrac{p(q+r)^{th}}{q^2-pr}$ term

Example 16 : If C_r stands for ${}^nC_r$, then find the value of $2\,C_0 - \dfrac{2^2}{2}C_1 + \dfrac{2^3}{3}C_2 + ... + (-1)^{n+1}\dfrac{2^{n+1}}{(n+1)}$.

Solution : $(1 + x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$

$$\frac{(1+x)^{n+1}-1}{n+1} = C_0\,x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + ... + C_n\frac{x^{n+1}}{n+1}$$

Put $x = -2$: $\dfrac{(-1)^{n+1}-1}{n+1} = -2C_0 + C_1\dfrac{2^2}{2} - C_2\dfrac{2^3}{3} + ... + C_n\dfrac{(-2)^{n+1}}{(n+1)}$

If n is odd, then L.H.S. = 0.

Example 17 : If $x + y = 1$, then prove that $\sum_{k=0}^{n} k^2\,{}^nC_k.x^k.y^{n-k} = nxy + n^2x^2$.

Solution : We know that $(t + x)^n = \sum_{k=0}^{n} {}^nC_k.x^k.t^{n-k}$

Differentiating above equation with respect to x, we get

$\Rightarrow n.(t + x)^{n-1} = \sum_{k=0}^{n} k\,{}^nC_k.x^{k-1}.t^{n-k}$ ...(i)

now multiplying equation (i) with x and then differentiating w.r.t. x,

$\Rightarrow n[x\,(n-1)(t + x)^{n-2} + (t + x)^{n-1}.1] = \sum_{k=0}^{n} k.\,{}^nC_k.k.x^{k-1}.t^{n-k}$

$\Rightarrow nx\,[x\,(n-1)\,(t + x)^{n-2} + (t + x)^{n-1}] = \sum_{k=0}^{n} k^2.\,{}^nC_k.x^k.t^{n-k}$

Now, replacing t by y, we get

$$\Rightarrow \sum_{k=0}^{n} k^2 . {}^nC_k . x^k . y^{n-k} = nx\,[x(n-1)(y+x)^{n-2} + (y+x)^{n-1}]$$

$$\Rightarrow \sum_{k=0}^{n} k^2 . {}^nC_k . x^k . y^{n-k} = nx\,[x(n-1)+1] = nx\,[nx+1-x] \quad (\text{putting } x+y=1)$$

$$= nx\,[nx+y] = nxy + nx^2y^2$$

$$\Rightarrow \sum_{k=0}^{n} k^2 . {}^nC_k . x^k . y^{n-k} = nxy + n^2x^2.$$

Example 18 : Find the term independent of x in the expansion of $\left(\sqrt[6]{x} - \frac{1}{\sqrt[3]{x}}\right)^9$.

Solution: $T_{r+1} = {}^9C_r \left(\sqrt[6]{x}\right)^{9-r} \left(-\frac{1}{\sqrt[3]{x}}\right)^r = {}^9C_r\,(-1)^r . x^{\frac{9-r}{6} - \frac{r}{3}} = {}^9C_r . x^{\left(\frac{9-3r}{6}\right)}$

Now $\frac{9-3r}{6} = 0 \Rightarrow r = 3$; $T = -{}^9C_3$

Example 19 : Find the value of $C_0 - C_1 \frac{(1+x)}{(1+nx)} + C_2 \frac{(1+2x)}{(1+nx)^2} - C_3 . \frac{(1+3x)}{(1+nx)} + \ldots$

Solution : Consider $C_0 - C_1 \frac{(1+x)}{(1+nx)} + C_2 \frac{(1+2x)}{(1+nx)^2} - C_3 . \frac{(1+3x)}{(1+nx)} + \ldots$

$$= \underbrace{\left[C_0 - \frac{C_1}{(1+nx)} + \frac{C_2}{(1+nx)^2} - \frac{C_3}{(1+nx)^3} + \ldots\right]}_{I_1} + \frac{x}{1+nx} \underbrace{\left[-C_1 + \frac{2C_2}{1+nx} - \frac{3C_3}{(1+nx)^2} + \ldots\right]}_{I_2} \quad ..(A)$$

$\because (1-y)^n = C_0 - C_1 y + C_2 y^2 - C_3 y^3 + \ldots + (-1)^n\, y^n$ ...(i)

differentiating w.r.t y we get, $-(1-y)^{n-1} = -C_1 + 2C_2 y - 3C_3 y^2 +$ ...(ii)

Now $I_1 = \left(1 - \frac{1}{1+nx}\right)^n = \left(\frac{nx}{1+nx}\right)^n$ ...(iii)

and $I_2 = -C_1 + 2C_2 . \frac{1}{1+nx} - 3C_3 . \frac{1}{(1+nx)^2} + \ldots.$

$$= (-n)\left(1 - \frac{1}{1+nx}\right)^{n-1} = -n\left(\frac{nx}{1+nx}\right)^{n-1} \quad \ldots(iv)$$

Putting I_1 and I_2 from equation (iii) and (iv) to equation (A), we get

$$C_0 - C_1 \frac{(1+x)}{(1+nx)} + C_2 \frac{(1+x)}{(1+nx)^2} - C_3 . \frac{(1+3x)}{(1+nx)^3} + \ldots + (-1)^n\, C_n \frac{1+nx}{(1+nx)^n}$$

$$= \left(\frac{nx}{1+nx}\right)^n - \left(\frac{x}{1+nx}\right) . n . \left(\frac{nx}{1+nx}\right)^{n-1} = \left(\frac{nx}{1+nx}\right)^n - \left(\frac{nx}{1+nx}\right)^n = 0$$

Example 20 : Find the largest coefficient in the expansion of $(4 + 3x)^{25}$.

Solution : $(4 + 3x)^{25} = 4^{25}\left(1+\frac{3}{4}x\right)^{25}$

Let $(r + 1)^{th}$ term will have largest coefficient

$$\Rightarrow \frac{\text{Coefficient of } T_{r+1}}{\text{Coefficient of } T_r} \geq 1 \Rightarrow \frac{{}^{25}C_r\left(\frac{3}{4}\right)^r}{{}^{25}C_{r-1}\left(\frac{3}{4}\right)^{r-1}} \geq 1 \Rightarrow \left(\frac{25-r+1}{r}\right)\frac{3}{4} \geq 1 \Rightarrow r \leq \frac{78}{7}$$

Largest possible value of r is 11

$$\therefore \text{Coefficient of } T_{12} = 4^{25} \times {}^{25}C_{11} \times \left(\frac{3}{4}\right)^{11}$$

Example 21 : Find the sum of the series $\sum_{r=1}^{n}(-1)^{r-1} \cdot {}^nC_r(a - r)$.

Solution : $\text{Sum} = a\sum_{r=1}^{n}(-1)^{r-1} \cdot {}^nC_r - \sum_{r=1}^{n}(-1)^{r-1} \cdot r \cdot {}^nC_r$

$= a\{{}^nC_1 - {}^nC_2 + {}^nC_3 - ....\} - n\{{}^{n-1}C_0 - {}^{n-1}C_1 + ...\}$ (because $r.{}^nC_r = n.{}^{n-1}C_{r-1}$)

$= a.{}^nC_0$ (since ${}^nC_0 - {}^nC_1 + {}^nC_2 - ..... = 0$)

Example 22 : If $g(x) = \sum_{r=0}^{200}\alpha_r x^r$ and, $f(x) = \sum_{r=0}^{200}\beta_r x^r$, $\beta_r = 1$ for $r \geq 100$ and $g(x) = f(1 + x)$, show that the greatest coefficient in the expansion of $(1 + x)^{201}$ is α_{100}.

Solution : Greatest coefficient in $(1 + x)^{201}$ = coefficient of middle term

= coefficient of 101st term or coefficient of 102th term

$= {}^{201}C_{100}$ or ${}^{201}C_{101} = {}^{201}C_{100}$ ....(i)

Given $g(x) = f(1 + x)$

$$\Rightarrow \sum_{r=0}^{200}\alpha_r x^r = \sum_{r=0}^{200}\beta_r(1+x)^r$$

$$\Rightarrow \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \ldots + \alpha_{200}x^{200}$$

$$= \beta_0 + \beta_1(1+x) + \beta_2(1+x)^2 + \ldots + \beta_{100}(1+x)^{100} + \ldots + \beta_{200}(1+x)^{200}$$

$$\Rightarrow \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \ldots + \alpha_{200}x^{200}$$

$$= \beta_0 + \beta_1(1+x) + \beta_2(1+x)^2 + \ldots + (1+x)^{100} + (1+x)^{101} \ldots + (1+x)^{200}$$

$(\because \beta_{100} = \beta_{101} = \ldots = \beta_{200} = 1, \quad \therefore \quad \beta_r = 1 \forall\, r \geq 100)$

Equation coefficient of x^{100}, we get

$$\alpha_{100} = {}^{100}C_{100} + {}^{101}C_{100} + {}^{102}C_{100} + \ldots + {}^{200}C_{100}$$

$$= {}^{101}C_{101} + {}^{101}C_{100} + {}^{102}C_{100} + \ldots + {}^{200}C_{100}$$

$$= {}^{102}C_{101} + {}^{102}C_{100} + \ldots + {}^{200}C_{100}$$

. . . . . .

$= {}^{200}C_{101} + {}^{200}C_{100} = {}^{201}C_{101}$ ...(ii)

From (i) and (ii), the greatest coefficient in $(1 + x)^{201} = \alpha_{100}$

Example 23 : Find the number of irrational terms in the expansion of $\left(\sqrt[8]{5} + \sqrt[6]{2}\right)^{100}$.

Solution : $t_{r+1} = {}^{100}C_r \left(\sqrt[8]{5}\right)^{100-r} . \left(\sqrt[6]{2}\right)^r$

As 2 and 5 are coprime, t_{r+1} will be rational if $100 - r$ is a multiple of 8 and r is a multiple of 6. Also $0 \le r \le 100$

$\therefore r = 0, 6, 12, ..., 96$

$\therefore$ $100 - r = 4, 10 , 16, ..., 100$ ..... (1)

But $100 - r$ is to be a multiple of 8.

So, $100 - r = 0, 8, 16, 24, ... 96$ ..... (2)

The common terms in (1) and (2) are 16, 40, 64, and 88

$\therefore r = 84, 60, 36, 12$ give rational terms

$\therefore$ the number of irrational terms = 101 – 4 = 97

Example 24 : If $(1 + x)^n = \sum_{r=0}^{n} C_r x^r$, then evaluate : $\frac{C_0}{m!} + \frac{n.C_1}{(m+1)!} + \frac{n.(n-1).C_2}{(m+2)!} + \ldots + \frac{n(n-1)\ldots 2.1.C_n}{(n+m)!}$

Solution : Consider the expansion $\left(1 + \frac{1}{x}\right)^n . (1 + x)^{m+n}$

$$= \left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \ldots + \frac{C_n}{x^n}\right)({}^{m+n}C_0 + {}^{m+n}C_1 x + \ldots + {}^{m+n}C_m x^m + {}^{m+n}C_{m+1} + \ldots + {}^{m+n}C_{m+n} x^{m+n})$$

Equating the coefficients of x^m, we get

$C_0 {}^{m+n}C_m + C_1 {}^{m+n}C_{m+1} + \ldots + C_2 . {}^{m+n}C_{m+2} + \ldots + C_n . {}^{m+n}C_{m+n}$

= coefficent of x^m in $\frac{(1+x)^n (1+x)^{m+n}}{x^n}$

$$\Rightarrow \quad \frac{C_0.(m+n)!}{m!.n!} + C_1 \frac{(m+n)!}{(m+1)!(n-1)!} + C_2 \frac{(m+n)!}{(m+2)!(n-2)!} + \ldots + \frac{C_n.(m+n)!}{(m+n)!.1}$$

$$= {}^{m+2n}C_{m+n} = \frac{(m+2n)!}{(m+n)!.n!}$$

$$\Rightarrow \quad (m+n)! \left[\frac{C_0}{m!.n!} + \frac{C_1}{m+1!.n-1!} + \ldots + \frac{C_n}{m+n!}\right] = \frac{(m+2n)!}{(m+n)!.n!}$$

$$\Rightarrow \quad \frac{C_0}{m!} + \frac{n.C_1}{(m+1)!} + \frac{n(n-1)C_2}{(m+2)!} + \ldots + \frac{m(n-1)\ldots 21.C_n}{(m+n)!} = \frac{(m+n+1)(m+n+2)\ldots(m+2n)}{m+n!}$$

Example 25 : Prove than $C_1 + 2.3\ C_2 + 3.5\ C_3 + ... + n(2n - 1)\ C_n = n^2 . 2^{n-1}$, C_r *s* being the binomial coefficients ${}^nC_r$ *s*.

Solution : L.H.S. $= \sum_1^n r(2r-1)\ C_r = \frac{1}{2} \sum_1^n 2r\ (2r-1) C_r$

We therefor consider $\left(1 + x^2\right)^n = \sum_0^n C_r x^{2r}$

Differentiating both sides twice,

$$2xn\,(1+x^2)^{n-1} = \sum_1^n 2rC_r\,x^{2r-1}$$

$$n\left[(1+x^2)^{n-1} + 2x^2(n-1)(1+x^2)^{n-2}\right] = \sum_1^n r(2r-1)\,C_r\,x^{2r-2}$$

Substituting $x = 1$, $\sum_1^n r\,(2r-1)\,C_r = n^2.\,2^{n-1}$.

Example 26 : If $(9+4\sqrt{5})^n = m + a$, n and m being positive integers and a a proper fraction, show that $(m-1)\,m + a^2$ is an even integer.

Solution : $\left(9+4\sqrt{5}\right)^n.\left(9-4\sqrt{5}\right)^n = 1 \Rightarrow 0 < \left(9-4\sqrt{5}\right)^n < 1$

$\left(9+4\sqrt{5}\right)^n + \left(9-4\sqrt{5}\right)^n = 2\sum_0^{\left[\frac{n}{2}\right]} {}^nC_{2r}\,9^{n-2r}.\,4^{2r}.\,5^r = 2N$, where N is an interger and $\left[\frac{n}{2}\right]$ stands for the greatest integer $\le \frac{n}{2}$.

$\therefore \quad \left(9+4\sqrt{5}\right)^n = 2N - 1 + \left[1-\left(9-4\sqrt{5}\right)^n\right] = m + \alpha$, giving $m = 2N - 1$and $\alpha = 1 - \left(9-4\sqrt{5}\right)^n$.

$(m+\alpha)(1-\alpha) = 1 \Rightarrow (m-1)\,\alpha + \alpha^2 = m - 1 = 2N - 2$, an even integer.

Example 27 : In the expansion of $\left(3-\sqrt{\frac{17}{4}+3\sqrt{2}}\right)^{15}$ the 11th term is a

(A) Positive integer (B) Positive irrational number
(C) Negative integer (D) Negative irrational number

Solution : (A) $\frac{17}{4} + 3\sqrt{2} = \frac{9}{2} + 2 + 2.\frac{3}{2}.\sqrt{2} = \left(\frac{3}{2}+\sqrt{2}\right)^2$ and $\left(3-\sqrt{\frac{17}{4}+3\sqrt{2}}\right)^{15}$

$= \left(\frac{3}{2}-\sqrt{2}\right)^{15}$ and $T_{11} = {}^{15}C_{10}\left(\frac{3}{2}\right)^5.(-\sqrt{2})^{10}$ which is a positive integer.

Example 28 : The term independent of x in the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}$, $x > 0$, is a times the corresponding binomial coefficient. Then a is

(a) 3 (b) 1/3 (c) –1/3 (d) None of these

Solution : (d) $T_{r+1} = {}^{18}C_r(9x)^{18-r}\left(-\frac{1}{3\sqrt{x}}\right)^r = (-r)^r\,{}^{18}C_r\,9^{18-\frac{3r}{2}}\,x^{18-\frac{3r}{2}}$

is independent of x provided $r = 12$ and then $\alpha = 1$.

Example 29 : x^r occurs in the expansion of $\left(x^3+\frac{1}{x^4}\right)^n$ Provided

(a) $2n - r$ is divisible by 5 (b) $3n - r$ is divisible by 5
(c) $2n - r$ is divisible by 7 (d) $3n - r$ is divisible by 7

Solution : (d)

$T_{p+1} = {}^nC_p (x^3)^{n-p} (x^{-4})^p = {}^nC_p\, x^{3n-7p}$ and x^r occurs provided $p = \frac{3n-r}{7}$ is an integer.

Example 30 : In the expansion of $\left(x^2\sqrt{\frac{2}{x}}\right)^n$, the 17th term is independent of x. Find the maximum value of m such that 2^m divides the coefficient in the 15th term.

Solution : $T_{17} = {}^nC_{16} (x^2)^{n-16} \left(-\sqrt{\frac{2}{x}}\right)^{16} = {}^nC_{16}\, 2^8\, x^{2n-40}$ is independent of x which is possible in $n = 20$.

Coefficient in $T_{15} = {}^{20}C_{14}.\, 2^7 = 2^{10}.19.17.15 \Rightarrow m = 10$

Example 31 : A committee of 10 is chosen from 8 men and 7 women. The number of committees on which men are in the majority is equal to

(a) $\frac{5}{11}$ of the total (b) $\frac{6}{13}$ of the total (c) $\frac{61}{143}$ of the total (d) $\frac{117}{143}$ of the total

Solution : (c) Men can be in a majority, if the distribution is as follows.

M	6	7	8
W	4	3	2

and the number of ways of choosing them

$= {}^8C_6 \times {}^7C_4 + {}^8C_7 \times {}^7C_3 + {}^8C_8 \times {}^7C_2 = 1281$

Total number of ways of forming the committee $= {}^{15}C_{10} = 3003$

Hence number of committees $= \frac{1281}{3003}$ of the total $= \frac{61}{134}$ of the total

Example 32 : If $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + - + \ldots\ldots \infty$, then $\sum_1^\infty \frac{1}{n(2n-1)}$ is

(a) $2 - \log 2$ (b) $2 - 2\log 2$ (c) $1 - \log 2$ (d) $3 - 2\log 2$

Solution : (B) $\sum_1^\infty \frac{1}{n(2n-1)} = 2\sum_1^\infty \left(\frac{1}{2n} - \frac{1}{2n+1}\right) = 2\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + - + \ldots\text{.inf.}\right)$

$= 2\left[1 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + - + \ldots\text{inf.}\right)\right] = 2 - 2\log 2$

Note : If it is written as $\sum_1^\infty \left(\frac{1}{n} - \frac{2}{2n+1}\right) \sum_1^\infty \frac{1}{n} - 2\sum_1^\infty \frac{1}{2n+1}$

$= 1 + \frac{1}{2} + \frac{1}{3} + \ldots. - 2\left(\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \ldots.\right) = 1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} = 2 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \ldots.\right) = 2 -$ log2.

This is not correct as the steps dealing with non-convergent series $\sum_1^\infty \frac{1}{n}$ and $\sum_1^\infty \frac{1}{2n+1}$ are wrong.

Example 33 : The middle term in the expansion of $\left(1 + \frac{1}{x^2}\right)\left(1 + x^2\right)^n$ is

(a) ${}^{2n}C_n\, x^{2n}$ (b) ${}^{2n}C_n x^{-2n}$ (c) ${}^{2n}C_n$ (d) ${}^{2n}C_{n-1}$

Solution : (c)

$\left(1+\frac{1}{x^2}\right)^n (1+x^2)^n = \frac{\left(1+z^2\right)^{2n}}{x^{2n}}$, numerator has $(2n+1)$ terms.

The middle terms is $\frac{1}{x^{2n}} x^{(2n)}C_n (x^2)^n = {}^{(2n)}C_n$.

Example 34 : If in the expansion of $(2-3x^2)^{20}$ the ratio of the tenth term to the eleventh term is $\frac{45}{22}$, then the value of x is

(a) $-\frac{3}{2}$ (b) $\frac{3}{2}$ (c) $-\frac{2}{3}$ (d) $\frac{2}{3}$

Solution: (c) $\frac{T_{10}}{T_{11}} = \frac{{}^{20}C_9}{{}^{20}C_{10}} \frac{2^{11}(-3x^3)}{2^{10}(-3x^3)^{10}} = -\frac{20}{33}\frac{1}{x^3} = \frac{45}{22}$ giving $x^3 = -\frac{8}{27}$

Example 35 : Prove that the integral part of $(7+4\sqrt{3})^n$ is odd for all $n \in N$.

Solution :

As $0 < 7-4\sqrt{3} < 1$, $0 < (7-4\sqrt{3})^n < 1$

$(7+4\sqrt{3})^n + (7-4\sqrt{3})^n = 2\sum_0^m {}^nC_{2r}\ 7^{n-2r}(4\sqrt{3})^{2r}$, $m = \frac{n}{2}$ if n is even and $\frac{n-1}{2}$ otherwise.

$= 2N$, N being an integer.

$(7+4\sqrt{3})^n = 2N - 1 + \{1-(7-4\sqrt{3})^n\}$

As $0 < 1-(7-4\sqrt{3})^n < 1$, then integral part of $(7+4\sqrt{3})^n$ is $2N-1$ which is always odd.

Example 36 : The total number of selection of at least $(n+1)$ things from $(2n+1)$ different things is 256. Then the value of n is

(a)2 (b) 3 (c) 4 (d) 8

Solution : (c) ${}^{(2n+1)}C_{(n+1)} + {}^{(2n+1)}C_{(n+2)} + \ldots + {}^{(2n+1)}C_{(2n+1)} = 256$

$\frac{1}{2}\left[{}^{(2n+1)}C_0 + {}^{(2n+1)}C_1 + \ldots + {}^{(2n+1)}C_{(2n+1)}\right] = 256$

$\frac{1}{2} \cdot 2^{2n+1} = 256 \Rightarrow 2^{2n} = 256 = 2^3 \Rightarrow n = 4$

Example 37 : If ${}^{15}C_{n+3} \cdot {}^{15}C_{n+1} + 3 \cdot {}^{15}C_{n+2} + {}^{15}C_{n+3} > {}^{18}C_{n+2}$. show that the greatest value of n possible is 6.

Solution : ${}^{15}C_{n+3} \cdot {}^{15}C_{n+1} + 3 \cdot {}^{15}C_{n+2} + {}^{15}C_{n+3}$

$= \left({}^{15}C_n + {}^{15}C_{n+1}\right) + 2\left({}^{15}C_{n+1} + {}^{15}C_{n+2}\right) + \left({}^{15}C_{n+2} + {}^{15}C_{n+3}\right)$

$= {}^{16}C_{n+1} + 2x\ {}^{16}C_{n+2} + {}^{16}C_{n+3}$

$= {}^{16}C_{n+1} + {}^{16}C_{n+2} + {}^{16}C_{n+2} + {}^{16}C_{n+3}$

$= {}^{17}C_{n+1} + {}^{17}C_{n+3} = {}^{18}C_{n+3}$

Hence the given inequality becomes ${}^{18}C_{n+3} > {}^{18}C_{n+2}$

i.e $\frac{18!}{(n+3)!(15-n)!} > \frac{18!}{(n+2)!(16-n)!}$

i.e., $16-n > n+3$ i.e., $n < \frac{13}{2}$, but n is an integer $\therefore$ $n = 6$

Example 38 : Show that ${}^{(2n-2)}C_{n-2} + 2x^{(2n-2)}C_{n-1} + {}^{(2n-2)}C_n > \frac{4n}{n+1}$

Solution : ${}^{(2n-2)}C_{n-2} + 2x^{(2n-2)}C_{n-1} + {}^{(2n-2)}C_n = {}^{2n-1}C_{n-1} + {}^{(2n-1)}C_n = {}^{2n}C_n$

$$\frac{(2n)!}{(n)!(n)!} = \frac{(n+1)(n+2)(n+3).....(2n)}{1.2.3.....n}$$

$$= (n+1)\left(1+\frac{n}{2}\right)\left(1+\frac{n}{3}\right) ..... \left(1+\frac{n}{n}\right) = (n+1)k, \text{ where } k > 1 > n+1$$

and $\frac{4n}{m+1} = \frac{(n+1)^2-(n-1)^2}{n+1} = (n+1) - \frac{(n-1)^2}{n+1} < n+1$

Hence, ${}^{2n}C_n > \frac{4n}{n+1}$

Example 39 : In the expansion of $\left(x^{2/3} - \frac{1}{\sqrt{x}}\right)^{30}$, a term containing the power x^{13}

(a) Does not exist
(b) Exists and the coefficient is divisible by 29
(c) Exists and the coefficient is divisible by 63
(d) Exists and the coefficient is divisible by 65

Solution : (b, c, d)

$$T_{r+1} = {}^{30}C_r (x^{2/3})^{30-r}\left(-\frac{1}{\sqrt{x}}\right)^r = (-1)^r {}^{30}C_r \, x^{20-\frac{2r}{3}-\frac{r}{2}}$$

$\therefore 20 - \frac{7r}{6} = 13 \Rightarrow r = 6$ and the coefficient $= {}^{30}C_6 = 29.7.9.13.25.$

Example 40 : Prove that $1.3\,{}^nC_1 + 2.4\,{}^nC_2 + 3.5\,{}^nC_3 + ...+ n(n+2)\,{}^nC_n = n(n+5)2^{n-2}$

Solution : $x^2(1+x)^n = \sum_0^n {}^nC_r\, x^{r+2}$ and $\frac{d}{dx}\{x^2(1+x)^n\} = \sum_0^n (r+2)\,{}^nC_r\, x^{r+1}$

$$\therefore \sum_0^n r(r+2)\,{}^nC_r = \frac{d}{dx}\left[\frac{1}{x}\frac{d}{dx}\left\{x^2(1+x)^n\right\}\right] \text{ at } x = 1$$

$$= \frac{d}{dx}\left[\frac{1}{x}\left\{2x(1+x)^n + nx^2(1+x)^{n-1}\right\}\right] \text{ at } x = 1 = n(n+5)2^{n-2},\ n \geq 2$$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic:1 Binomial Coefficient

Q.1 Let $(1 + x^2)^2(1 + x)^n = \sum_{k=0}^{n+4} a_k.x^k$. If a_1, a_2 and a_3 are in A.P. then the value of n is
(a) 2, 3, 4 (b) 5, 6, 7
(c) 8, 9, 10 (d) –1, 4, 6

Q.2 The coefficient of x^k $(0 \le k \le n - 1)$ in the expansion of
$(x+3)^{n-1} - (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + ... + (x+2)^{n-1}$ is
(a) ${}^nC_k(3^n - 2^n)$ (b) ${}^nC_k(3^{n-k} - 2^{n-k})$
(c) ${}^nC_k(3^{n-k} - 2^{n-k})$ (d) ${}^nC_k(3^n + 2^n)$

Q.3 The coefficient of x^{1007} in the expansion $(1 + x)^{2006} + x(1 + x)^{2005} + x^2(1 + x)^{2004} + x^3(1 + x)^{2003} + ... + x^{2006}$ is
(a) ${}^{2006}C_{1007}$ (b) ${}^{2006}C_{1006}$
(c) ${}^{2007}C_{1006}$ (d) ${}^{2007}C_{1007}$

Q.4 In the expansion of $(x + a)^n$, the sum of even terms is E, the sum of odd terms is O, then $O^2 - E^2$ is equal to
(a) $(x^2 + a^2)^n$ (b) $(x^2 - a^2)^n$
(c) $(x^2 + a^2)^{2n}$ (d) $(x^2 + a^2)^{2n}$

Q.5 The sum of the last eight coefficients in the expansion of $(1 + x)^{16}$ is equal to
(a) 2^{15} (b) 2^{14}
(c) $2^{15} - \frac{1}{5}.\frac{16!}{(8!)^2}$ (d) 2^{16}

Q.6 If n is an integer greater than 1, then $a - {}^nC_1(a - 1) + {}^nC_2(a - 2) - .... + (-1)^n(a - n) =$
(a) a (b) 0
(c) a^2 (d) 2^n

Q.7 $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_{100}}{101}$ equals
(a) $\frac{2^{101}}{101}$ (b) $\frac{2^{101} - 1}{101}$
(c) $\frac{3^{101}}{101}$ (d) $\frac{3^{101} - 1}{101}$

Q.8 The value of $2C_0 + \frac{2^2C_1}{2} + \frac{2^3C_2}{3} + ... + \frac{2^{11}C_{10}}{11}$ where $C_r = {}^{10}C_r$ is
(a) $\frac{3^{11}}{11}$ (b) $\frac{3^{11} + 1}{11}$
(c) $\frac{3^{11} - 1}{11}$ (d) $\frac{3^{10} - 1}{10}$

Q.9 The coefficient of x^n in the polynomial $(x + {}^nC_0)(x + 3.{}^nC_1)(x + 5.{}^nC_2)(x + 7.{}^nC_3)....(x + (2n + 1)^nC_n)$ is
(a) $n.2^{n-1}$ (b) $n.2^n$
(c) $n.2^{n-1}$ (d) $(n + 1)2^n$

Q.10 If $P = (C_1 - C_3 + C_5 - .....)$ and $Q = C_0 - C_2 + C_4 - ....$ then $(P/Q)^{-1}$ is equal to
(a) $\cot \frac{n\pi}{4}$ (b) $\tan \frac{n\pi}{4}$
(c) $\sec \frac{n\pi}{4}$ (d) 1

Q.11 For $4 \le r \le n$, $\binom{n}{r} + 4\binom{n}{r+1} + 6\binom{n}{r+2} + 4\binom{n}{r+3} + \binom{n}{r+4}$ equals
(a) $\binom{n+4}{r+4}$ (b) $\binom{n+4}{r}$
(c) $\binom{n+3}{r-1}$ (d) $\binom{n+3}{r+3}$

Q.12 The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$ is
(a) 900 (b) 909
(c) 990 (d) 999

Q.13 The sum of the coefficient in the expansion of $(1 - 2x + 5x^n)^n$ is 'p' and the sum of the coefficients in the expansion of $(1 + x)^{2n}$ is q. Then
(a) $p = q$ (b) $p = q^2$
(c) $p^2 = q$ (d) $pq = 1$

Q.14 Which of the following is false?
(a) If the coefficients of $(4k + 3)^{th}$ and $(k + 4)^{th}$ terms in the expansion of $(1 + x)^{25}$ are same then the value of k is 4
(b) $\sum_{r=0}^{10} {}^{10}C_r 2^{10-r}3^r = 5^{10}$
(c) $\sum_{r=0}^{20} (-1)^r \, {}^{20}C_r = 0$
(d) The general term in the expansion of $(1 - x)^{-n}$ is ${}^{n+r+1}C_r x^r$

Q.15 If $(1+x)^n = \sum_{r=0}^{n} C_r.x^r$, then

$$\left(1+\frac{C_1}{C_0}\right)\left(1+\frac{C_2}{C_1}\right)....\left(1+\frac{C_n}{C_{n-1}}\right) =$$

(a) $\frac{n^{n-1}}{(n+1)!}$ (b) $\frac{(n+1)^{n+1}}{(n+1)!}$

(c) $\frac{(n+1)^n}{n!}$ (d) $\frac{(n+1)^{n+1}}{n!}$

Q.16 $\frac{C_1}{C_0}+\frac{2.C_2}{C_1}+\frac{3.C_3}{C_2}+....+\frac{20.C_{20}}{C_{19}} =$

(a) 180 (b) 210 (c) 240 (d) 280

Q.17 If in the expansion of $(1+x)^m(1-x)^n$, the coefficient of x and x^2 are 3 and –6, respectively, then the value of n is

(a) 6 (b) 9 (c) 12 (d) 24

Topic:2 Particular Term in the Expansion

Q.18 The coefficient of $x^3y^4z^5$ in the expansion of $(xy+yz+zx)^6$ is

(a) 60 (b) 120 (c) 6! (d) 0

Q.19 In the expansion of $(3x+2y-z)^8$, the coefficients of $x^2y^3z^3$ is

(a) 10084 (b) –40320

(c) 20160 (d) –43280

Q.20 If in the expansion of $\left(\frac{1}{x}+x\tan x\right)^5$ the ratio of 4th term to 2nd term is $\frac{2\pi^4}{27}$, then value of x, where $x < 0$, can be

(a) $\pi/3$ (b) $-\pi/6$

(c) $-\pi/3$ (d) $\pi/12$

Q.21 If 6th term of

$$\left[2^{\log_2\sqrt{g^{x-1}+7}}+\frac{1}{2^{\frac{1}{5}\log_2(3^{x-1}+1)}}\right]^7$$ is 84,

then x =

(a) 4 or 3 (b) 3 or 1

(c) 2 or 1 (d) 1

Q.22 If n be a positive integer and the last term in the expansion of $\left(2^{\frac{1}{3}}+2^{-\frac{1}{2}}\right)^n$ is $\left(\frac{1}{3\times 9^{1/3}}\right)^{\log_3 8}$, then the fifth term is

(a) 110 (b) 220

(c) 210 (d) 400

Q.23 The sum of all the coefficients of the expansion of $(x+y)^n$ is 1024. Then the greatest coefficient occurs in the rth term where r is equal to

(a) 4 (b) 5 (c) 6 (d) 7

Q.24 In the binomial expansion of $(a-b)^n$, $n \geq 5$, the sum of the 5th and 6th term is zero, then a/b equals

(a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$

(c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$

Q.25 The sum of the binomial coefficients of $\left(2x+\frac{1}{x}\right)^n$ is equal to 256. Then the constant term in the expansion is

(a) 1120 (b) 2110

(c) 1210 (d) 2210

Q.26 If sum of the coefficients in the expansion of $(2x+3y-2z)^n$ is 2187 then the greatest coefficient in the expansion of $(1+x)^n$

(a) 30 (b) 40 (c) 28 (d) 35

Topic:3 Greatest Terms

Q.27 The greatest term in the expansion of $(3-5x)^{15}$, when $x = 1/5$, is

(a) 6th (b) 5th

(c) 5th and 6th (d) 4th and 5th

Q.28 The greatest term of the expansion $(2x+5y)^{13}$ when $x = 10$, $y = 2$ is

(a) $^{13}C_5.20^8.10^5$ (b) $^{13}C_6.20^7.10^4$

(c) $^{13}C_4.20^9.10^4$ (d) $^{13}C_7.20^7.10^4$

Topic:4 Miscellaneous

Q.29 $\sum_{r=0}^{n}(-1)^r.{}^nC_r\left[\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{4r}}+...m \text{ terms}\right] =$

(a) $\frac{2^{mn}-1}{2^{mn}(2^n-1)}$ (b) $\frac{2^{mn}-1}{2^n-1}$

(c) $\frac{2^{mn}+1}{2^n+1}$ (d) $\frac{2^{mn}+1}{2^n-1}$

Q.30 If C_r stands for $^nC_r = \frac{n!}{r!\,n-r!}$ and $\sum_{r=1}^{n} r.C_r^2 = \lambda$ for $n > 2$, then λ is divisible by

(a) $3(n-1)$ (b) $n+1$

(c) $n(2n-1)$ (d) n^2+1

Q.31 If $k = \sum_{r=0}^{n}\frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n}\frac{r}{{}^nC_r}$ is equal to

(a) nk (b) $nk/2$
(c) $k(n-1)$ (d) $nk/3$

Q.32 $\sum_{r=1}^{n} r({}^nC_r - {}^nC_{r-1})$ is equal to
(a) $2^n + n + 1$ (b) $2^n - n + 1$
(c) $n - 2^n + 1$ (d) $n - 2^n - 1$

Q.33 If $\sum_{k=0}^{n}(k^2+k+1)k! = (2007).2007!$, then value of n is
(a) 2007 (b) 2006
(c) 2008 (d) 2005

Q.34 Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$ where [ ] denotes the greatest integer function. The value of $R \cdot f$ is
(a) 4^{2n+1} (b) 4^{2n}
(c) 4^{2n-1} (d) 4^{-2n}

Q.35 The number of term in $(x+y)^{100} + (x-y)^{100}$ is
(a) 51 (b) 101
(c) 202 (d) 50

Q.36 The value of

$$\frac{18^3+7^3+3.18.7.25}{4^6+6.243.2+15.81.4+20.27.8+15.9.16+6.3.32+64}$$

is
(a) 10 (b) 2
(c) 20 (d) 1

Q.37 If the coefficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^{2n}$, are in A.P., then
(a) $2n^2 - 9n + 7 = 0$ (b) $2n^2 + 5n + 7 = 0$
(c) $n^2 - 9n + 7 = 0$ (d) $n^2 + 9n - 7 = 0$

Q.38 $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}}$ equals

(a) $\frac{n}{2}$ (b) $\frac{n+1}{2}$

(c) $\frac{n(n+1)}{2}$ (d) $\frac{n(n-1)}{2(n+1)}$

Q.39 In the expansion of $(y^{1/5} + x^{1/10})^{55}$, the number of terms terms free of radical sign are
(a) 5 (b) 6 (c) 50 (d) 56

Q.40 Consider the following statements
S_1 : The total number of terms $(x^2 + 2x + 4)^{10}$ is 21.

S_2 : The coefficient of x^{10} in $\left(x^2 + \frac{1}{x}\right)^{20}$ is ${}^{20}C_{10}$.

S_3 : The middle term in the expansion of $(1+x)^{12}$ is ${}^{12}C_6x^6$.

S_4 : The coefficients of fifth and ninth term in the expansion of $(1+x)^n$ are same, then $n = 12$.

Now identify the correct combination of true statements.
(a) S_1, S_2, S_3, S_4 (b) S_1, S_2 only
(c) S_2, S_3 only (d) S_1, S_4 only

Q.41 The term independent of x in $(1 + x + 2x^3)\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^9$ is
(a) $6^{th}, 7^{th}$ (b) $7^{th}, 8^{th}$
(c) $6^{th}, 7^{th}, 8^{th}$ (d) $4^{th}, 5^{th}$

Q.42 For natural numbers m, n if $(1-y)^n(1+y)^n = 1 + a_1y + a_2y^2 + ...$ and $a_1 = a_2 = 10$ then (m, n) is
(a) (20, 45) (b) (35, 45)
(c) (35, 20) (d) (45, 35)

Q.43 If $(1 - x^3)^n = \sum_{r=0}^{n} a_r x^r (1-x)^{3n-2r}$, then the value of a_r, where $n \in N$ is
(a) ${}^nC_r \cdot 3^r$ (b) ${}^nC_{3r}$
(c) ${}^nC_{r-1}2^{r-1}$ (d) ${}^nC_r2^r$

Section-B
(Assertion-Reason)

Q.1 **Statement–1 :** $(\sqrt{2} - 1)^n$ can be expressed as $\sqrt{N} - \sqrt{N-1}$ for $N > 1$, n is positive integer and $n \in \mathbb{N}$.

Statement–2 : $(\sqrt{2} - 1)^n$ can be expressed as $A + B\sqrt{2}$ where A and B are integers and n is positive integer.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** If $\sum_{r=1}^{n} r^3\left(\frac{{}^nC_r}{{}^nC_{r-1}}\right)^2 = 196$, then the sum of the coefficients of power of x in the expansion of the polynomial $(x - 3x^2 + x^3)^n$ is –1.
Statement-2:

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \forall n \in N \text{ and } r \in W.$$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation

for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** The number of terms in the expansion of $\left(x+\frac{1}{x}+1\right)^n$.

Statements-2: The number of terms in the expansion of $(a_1 + a_2 + ... a_m)^n$ is ${}^{n+m-1}C_{m-1}$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The coefficient of x^{3l+2} in the expansion of $(a+x)^{\lambda}(b+x)^{\lambda+1}(c+x)^{\lambda+2}$ $\forall\lambda \in N$ is $\lambda(a+b+c)$.
Statement–2 : The coeficient of x^m in the expansion of $(a+x)^n$ is ${}^nC_m \cdot a^{n-m}$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** If n is even, then ${}^{2n}C_3 + ... + {}^{2n}C_{2n-1} = 2^{2n-1}$.
Statement–2 : ${}^{2n}C_1 + {}^{2n}C_3 + ... + {}^{2n}C_{2n-1} = 2^{2n-1}$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.6 **Statement-1:** The term independent of x in the expansion of $\left(x^{2008}+\frac{1}{x^{2008}}+2\right)^{1004}$ is ${}^{2008}C_{1004}$.
Statement-2: In a binomial expansion $\left(x+\frac{1}{x}\right)^{2n}$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.7 **Statement–1 :** The greatest coefficient of the expansion $(1 + 8x)^{16}$ is ${}^{16}C_8.8^8$.
Statement–2 : Greatest coefficient in the expansion of $(1 + x)^{2n}$ is the middle term.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.8 **Statement–1 :** In the expansion of $(1 + x)^n$ if coefficient of 21st and 22nd term are equal then $n = 41$.
Statement–2 : Middle term in txpansion of $(1 + x)^{2n}$ has greatest coefficient of $n \in N$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.9 **Statement–1 :** The number of distinct term in the expansion of $(1 + px)^{20} + (1 – px)^{20}$ is 42.
Statement–2 : Number of term in the expansion of $(1 + x)^n$ is $(n + 1)$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.10 **Statement–1 :** The coefficient of $a^3b^4c^3$ in the expansion of $(a – b + c)^{10}$ is $\frac{10!}{3!\,4!\,3!}$.
Statement–2 : The coefficient of $x^py^qz^r$ in the expansion of $(x + y + z)^n$ is $\frac{n!}{p!\,q!\,r!}$ for

all integer *n*.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 If the coefficients of rth, $(r+1)^{th}$ and $(r+2)^{m}$ are in A.P., then m and r satisfy the equation:-

(a) $m^2-m(4r-1)+4r^2+2=0$

(b) $m^2-m(4r+1)+4r^2-2=0$

(c) $m^2-m(4r+1)+4r^2+2=0$

(d) $m^2-m(4r-1)+4r^2-2=0$

Q.2 If the coefficient of x^7 in $\left[ax^2+\left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax-\left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation:-

(a) ab =1 (b) $\frac{a}{b}=1$

(c) $a+b=1$ (d) $a-b=1$

Q.3 For natural numbers m, n if $(1-y)^m(1+y)^n=1+a_1y+a_2y^2+.....$, and $a_1=a_2=10$, then (m,n) is
(a) (45, 35) (b) (35, 45)
(c) (20, 45) (d) (35, 20)

Q.4 The sum of the series
$^{20}C_0-{}^{20}C_1+{}^{20}C_2-{}^{20}C_3+.....-.....+{}^{20}C_{10}$ is:-

(a) $\frac{1}{2}\,{}^{20}C_{10}$ (b) 0

(c) $-{}^{20}C_{10}$ (d) $^{20}C_{10}$

Q.5 In the binomial expansion of $(a-b)^n$, $n\geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals

(a) $\frac{6}{n-5}$ (b) $\frac{n-5}{6}$

(c) $\frac{n-4}{5}$ (d) $\frac{5}{n-4}$

Q.6 **Statement-1 :** $\sum_{r=0}^{n}(r+1)^n C_r=(n+2)2^{n-1}$

Statement-2 :

$\sum_{r=0}^{n}(r+1)^n C_r x^r=(1+x)^n+nx(1+x)^{n-1}$

(a) Statement-1 is false, Statement - 2 is true
(b) Statement-1 is true, Statement-2 is truw; Statement-2 is a correct explanation for Statement-1
(c) Statement-1 is true, Statement-2 is truw; Statement-2 is not a correct explanation for Statement-1
(d) Statement -1 is true, Statement-2 is false

Q.7 The remainder left out when
$8^{2n}-(62)^{2n+1}$ is divided by 9 is:-
(a) 7 (b) 8 (c) 0 (d) 2

Q.8 Let $S_1=\sum_{j=1}^{10}j(j-1)^{10}C_j, S_2=\sum_{j=1}^{10}j^{10}C_j$ and

$S_3=\sum_{j=1}^{10}j^2{}^{10}C_j$

Statement-1: $S_3=55\times 2^9$.

Statement-2: $S_1=90\times 2^8$ and $S_2=10\times 2^8$
(a) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is true, Statement-2 is true; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is true, Statement-2 is false
(d) Statement-1 is false, Statement-2 is ture

Q.9 The coefficient of x^7 in the expansion of $\left(1-x-x^2+x^3\right)^6$ is :-
(a) -144 (b) 132
(c) 144 (d) -132

Q.10 If n is positive integer, then
$\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is :-
(a) a rational number other than positive integers
(b) an irrational number
(c) an odd positive integer
(d) an even positive integer

Q.11 The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is :

(a) 4 (b) 120
(c) 210 (d) 310

Q.12 If the coefficients of x^3 and x^4 in the expansion of $(1+ax+bx^2)(1-2x)^{18}$ in powers of x are both zero, then (a, b) is equal to:-

(a) $\left(16,\frac{251}{3}\right)$ (b) $\left(14,\frac{251}{3}\right)$

(c) $\left(14,\frac{272}{3}\right)$ (d) $\left(16,\frac{272}{3}\right)$

Q.13 If $(10)^9 + 2\,(11)^1\,(10)^8 + 3\,(11)^2\,(10)^7 + ... + 10\,(11)^9 = k\,(10)^9$, then k is equal to:

(a) $\frac{121}{10}$ (b) $\frac{441}{100}$
(c) 100 (d) 110

Q.14 The sum of coefficients of integeral powers of x in the binomial expansion of $\left(1-2\sqrt{x}\right)^{50}$ is

(a) $\frac{1}{2}\left(3^{50}-1\right)$ (b) $\frac{1}{2}\left(2^{50}+1\right)$

(c) $\frac{1}{2}\left(3^{50}+1\right)$ (d) $\frac{1}{2}\left(3^{50}\right)$

Q.15 If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n, x\neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is:-

(a) 729 (b) 64
(c) 2187 (d) 243

Q.16 The value of $\sum_{r=1}^{15} r^2\left(\frac{^{15}C_r}{^{15}C_{r-1}}\right)$ is equal to:-

(a) 1085 (b) 1240
(c) 680 (d) 560

Q.17 For $x\in R, x\neq -1$, if $(1+x)^{2016}+ x(1+x)^{2015} + x^2(1+x)^{2014}+...+x^{2016} = \sum_{i=0}^{2016} a_i x^i$, then a_{17} is equal to:-

(a) $\frac{2016!}{17!1999!}$ (b) $\frac{2017!}{17!2000!}$

(c) $\frac{2016!}{16!}$ (d) $\frac{2017!}{2000!}$

Q.18 If the coefficients of x^{-2} and x^{-4} in the expansion of $\left(x^{\frac{1}{3}}+\frac{1}{2x^{\frac{1}{3}}}\right)^{18}, (x>0)$, are m and n respectively, then $\frac{m}{n}$ is equal to:-

(a) $\frac{4}{5}$ (b) $\frac{5}{4}$
(c) 182 (d) 27

Q.19 The sum $\sum_{r=1}^{10}(r^2+1)\times(r!)$ is equal to:-

(a) 101×(10!) (b) (11!)
(c) 10 ×(11!) (d) 11×(11!)

Q.20 The sum of the co-efficient of all odd degree terms in the expansion of $\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, (x>1)$ is

(a) 0 (b) 1 (c) 2 (d) -1

Section-D
(Board pattern)

Q.1 The 3rd term from the end in the expansion of $\left(x-\frac{1}{x^2}\right)^7$ is .....

Q.2 If $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2 + x^2 + ...+ {}^nC_n + x^n$, then the value of ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + ... +(-1)^n\,{}^nC_n$ is _____.

Q.3 The middle term in the expansion of $\left(x^2+\frac{1}{x}\right)^{2n}$, $x\neq 0$, is _____.

Q.4 Using binomial Theorem for positive integral index, expand $(1 - x + x^2)^4$ in powers of x.

Q.5 Show that $(x+\sqrt{x^3+1})^4+(x-\sqrt{x^3+1})^4$ is a polynomial of degree 6.

Q.6 Find the term involving a^2b^5 in the expansion of $(a - 2b)^4(a + b)^3$.

Q.7 Find the remainder when 5^{99} is divided by 13.

Q.8 Find the integral part of $(\sqrt{2} + 1)^6$

Q.9 Find the greatest term in the expansion of $(2 + 3x)^{10}$ when $x = 4$.

Q.10 Find the value of r if the coefficients of $(2r + 4)^{th}$ term and $(r - 2)^{th}$ term in the expansion of $(1 + x)^{18}$ are equal.

Q.11 If T_r denotes the rth term in the expansion of $(1 + x)^n$ in ascending powers of x, then show

CHAPTER 2

PROBABILITY

1. BASIC CONCEPTS AND TERMS USED IN PROBABILITY

Experiment

An operation that has as its result some well-defined outcomes is called an experiment.

Random Experiment

An experiment whose outcome cannot be predicated with certainty is called a random experiment. That is, if one were to perform an experiment several times under the same conditions and the outcomes each time were not the same, then we would call this experiment random.

On the other hand, if the outcome of an experiment is known before hand, then that experiment is not random. Tossing of a fair coin. throwing a perfectly balanced die (unbiased die), drawing card from a well-shuffled pack of cards are all random experiment. For in no case we predict the outcome with certainly.

Note :- The term 'random'means that all the possibilities are equally likely.

Sample Space

The set of all possible outcomes of a random experiment is called the sample space for the experiment and is denoted by S.

When we toss a coin, it will land head (H) or tail (T). The sample space, written as a set, is $\{H, T\}$. Similarly when we toss two coins, the sample space, written as a set, is $\{(H, H), (H, T), (T, H), (T, T)\}$.

The ordered pair, (H, T) for example means first coin has landed heads and second coin has lended tails.

The sample space can also be written as $\{HH, HT, TH, TT\}$ for brevity.

When we throw a die, one of the numbers 1, 2, 3, 4, 5, or 6 will come up. The sample space, written as a set, is { 1, 2, 3, 4, 5, 6}.

Sample Point

Each element of the sample space is called a sample point.

For example, when we throw a dice, the sample space $S = \{1, 2, 3, 4, 5, 6\}$ and 1, 2, 3, 4, 5, 6 are called sample points of this sample space.

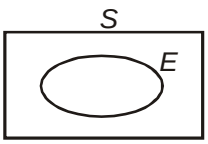
Event

A subset of the sample space S is called an event. When a coin is tossed, the sample space $S = \{H, T\}$. The subset $\{H\} \subseteq S$ denotes the event of occurrence of head in the toss of a coin, and the subset $\{T\} \subseteq S$ denotes the event of occurrence of tail.

When a die is thrown, the sample space $S = \{ 1, 2, 3, 4, 5, 6\}$. The subset $\{1, 3, 5\} \subseteq S$ denotes the event of occurrence of an odd number, and the subset {2, 4, 6} denotes the event of occurence of an even number.

Also note the following

(i) In relation to the outcomes of a random experiment, 'S' plays the same role as the universal set. 'S' is to its sample points what universal set is to its elements.



(ii) ϕ is also a subset of S called impossible event or null event.

(iii) 'S' is also a subset of S called a sure or certain event.

Simple Event

An event is called a simple event, if it is a singleton subset of the sample space S. For the toss of a coin, sample space $S = \{H, T\}$.

The occurrence of head, denoted by the singleton subset $\{H\}$ is a simple event.

For the throw of a die, sample space $S = \{1, 2, 3, 4, 5, 6\}$.

The occurrence of number 2 on the die, denoted by $\{2\}$, is a simple event.

Compound event

A subset of the sample space S, which contains more than one element is called a compound event or a mixed event.

For the toss of a coin the sample space $S = \{H, T\}$.

The occurrence of either a head or a tail, denoted by the subset $\{H, T\}$, is a compound event.

For the throw of a die the sample space $S = \{1, 2, 3. 4. 5, 6\}$.

The occurrence of a prime number, denoted by the subset $\{2, 3, 5\}$, is a compound event.

Equally likely events

Events (Outcomes) are said to be equally likely when we have no reason to believe that some outcome is more likely to occur than the other. And the sample space then is called equiprob able space. An equiprobable space allows us to assign equal probabilities to all equally likely events.

For the toss of a coin, the occurrence of each of the two sides "heads or tails" is equally likely.

Thus $P\{\text{Heads or tails}\} = \frac{1}{2}$.

For the throw of a die, the occurrence of each of the six numbers 1, 2, 3, 4, 5, or 6 is equally likely.

Thus $P\{\text{occurrence of any six numbers}\} = \frac{1}{6}$.

Mutually Exclusive Events

Two events are said to be mutually exclusive (or disjoint or incompatible)If the occurrence of one precludes (rules out) the simultaneous occurence of the other. If A & B are two mutually exclusive events then $P(A \cap B)=0$

Consider for example choosing numbers at random form the set $\{3,4,5,6,7,8,9,10,11,12\}$

If Event A is the selection of a prime number

Event B is the selection of an odd number

Event C is the selection of an Even Number

Then A and C are Mutually Exclusive as none of the numbers in this set is both prime and Even but A and B are not mutually Exclusive as some number of are both prime and odd.

Exhaustive events

For a random experiment A, a set of events is said to be exhaustive if one of them must necesarily happen every time the experiment is performed.

For the toss of a coin $\{H\}$ and $\{T\}$ make up an exhaustive set of events (cases).

For the throw of a die, $\{1\}$, $\{2\}$, $\{3\}$, $\{4\}$, $\{5\}$, $\{6\}$ make up an exhaustive set of events.

2. DEFINITION OF PROBABILITY

Let E denote the outcome of an experiment, called an event E, there are two ways to obtain the probability P of E.

Classical (a priori) definition

Suppose an event E can occur in m ways out of a total of n equally likely possible ways. Then

$$P = \frac{m}{n}.$$

Formally, letting S denote the sample space, the probability of occurrence of an event E, $P(E)$, is given by $P(E) = \frac{n(E)}{n(S)} = \frac{\text{Number of elements in the event}}{\text{Number of element in the sample space}}$

$$= \frac{\text{Number of favourable outcomes}}{\text{Number of total outcomes}}$$

For the toss of a coin, sample space $S = \{H, T\}$. Let E = the event of occurrence of head, then $E = \{H\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{1}{2}$$

For the throw of a die, sample space $S = \{1, 2, 3, 4, 5, 6\}$. Let E = the event of occurrence of an even number greater than 2, then $E = \{4, 6\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

(i) The probability of occurrence of an event is a number lying between 0 and 1, i.e., $0 \le P(E) \le 1$.

As $\phi \subseteq E \subseteq S \Rightarrow n(\phi) \le n(E) \le n(S)$

$\Rightarrow 0 \le n(E) \le n(S)$

Dividing by $n(S)$ and recalling that $P(E) = \frac{n(E)}{n(S)}$

$$0 \le \frac{n(E)}{n(S)} \le 1$$

i.e., $0 \le P(E) \le 1$

(ii) The probability of a sure or certain event is 1.

For a sure event, $n(E) = n(S)$ and thus $P(E) = \frac{n(E)}{n(S)} = \frac{n(S)}{n(S)} = 1$

(iii) The probability of an impossible event is 0.

For an impossible event, $n(E) = n(f) = 0$ and thus $P(E) = \frac{n(E)}{n(S)} = \frac{0}{n(S)} = 0 \ (n(S) \neq 0)$

Frequency (posteriori) definition

Suppose after n repetitions, where n is very large, an event occurs times. Then $P = \frac{s}{n}$

Formally, let a random experiment be performed n times under similar conditions, *i.e., n* trials of experiment be made, where n is very large, and the event E occurs r times in trials, then

$$P(E) = \lim_{n \to \infty} \frac{r}{n}$$

$P(E)$ is very often denoted by P.

Both the definitions suffer from flaws. The classical definition is essentially circular because the idea of 'equally likely' is the same as that of 'with equal probability' which has not been defined. The frequency definition is not well-defined because 'very large' has not been defined.

The current trend in probability theory is axiomatic, using set theory where we just require that probabilities of all events of an experiemnt satisfy certain axioms.

Complement of an event

The complement of an event E is denoted by E' ,E^c and $\bar{E}$. E' is nothing but the event 'not E'. That is, E' means non-occurrenced of event E.

E' occurs if and only if E doesn't occur.

As $n(E) + n(E') = n(S)$

$$\Rightarrow \quad \frac{n(E)}{n(S)} + \frac{n(E')}{n(S)} = \frac{n(S)}{n(S)} = 1$$

$$\therefore \quad P(E) + P(E') = 1$$

Odds in favour and odds against an event

Let S be a sample space, E an event and E' the complement of event E.

Odds in favour of event $E = E = \frac{n(E)}{n(E')} = \frac{\text{Number of outcomes favourable to E}}{\text{Number of outcomes against E}}$

$$= \frac{\frac{n(E)}{n(S)}}{\frac{n(E')}{n(S)}} = \frac{P(E)}{P(E')}$$

Odds against an event $E = \frac{n(E')}{n(E)} = \frac{\text{Number of outcomes against E}}{\text{Number of outcomes favourable to E}}$

$$= \frac{\frac{n(E')}{n(S)}}{\frac{n(E)}{n(S)}} = \frac{P(E')}{P(E)}$$

Of the three things "P(E), odds in favour of E, odds against E" if any one is given, the other two can be determined.

If $P(E) = \frac{4}{17}$, then odds in favour of event $E = \frac{4}{13}$ and odds against event $E = \frac{13}{4}$.

If odds in favour of $E = \frac{2}{5}$, then odds against event $E = \frac{5}{2}$ and $P(E) = \frac{2}{7}$.

If odds against $E = \frac{7}{11}$ then odds in favour of $E = \frac{11}{7}$ and $P(E) = \frac{11}{18}$.

Facts about a pack of cards

A pack of 52 playing cards have four suits having 13 cards each. The suits are called

Spades, Clubs } Black cards

Hearts, Diamonds } Red cards

The aces, kings, queens, jacks are called honour cards and king, queens, jacks are called face cards.

The game of Bridge is played by 4 people. Each gets 13 cards.

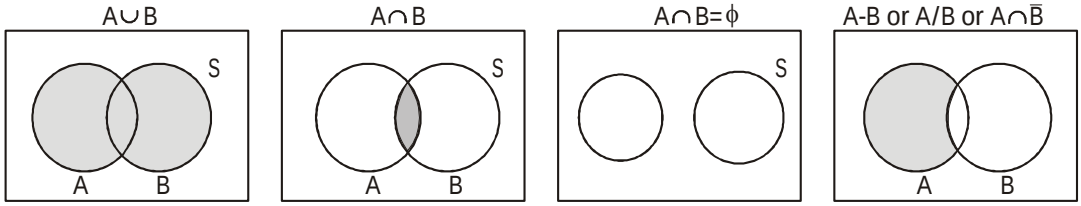
Although we have dealt all the theoretical aspects with suitable illustrations in previous classes, yet the following illustrative examples clarify the transparent concept and review of the chapter so that we shall feel accelerating understanding.

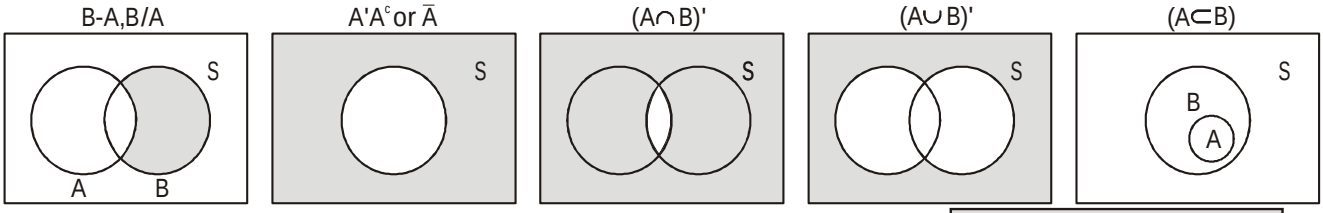
3. VENN DIAGRAMS

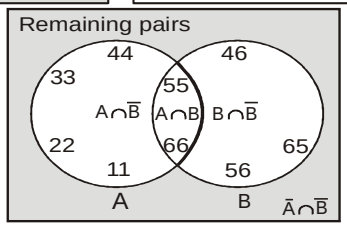
A diagram used to illustrate relationships between sets. Commonly, a rectangle represents the universal set and a circle within it represents a given set (all members of the given set are repre-

sented by points within the circle). A subset is represented by a circle within a circle and intersection is indicated by overlapping circles.

Let S is the sample space of an experiment and A, B are two events corresponding to it:





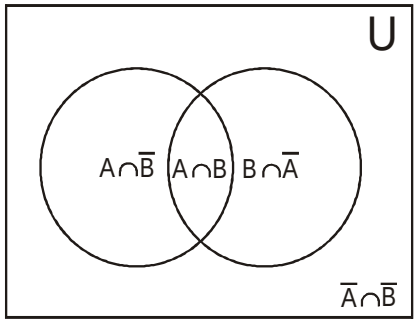


Example : Let us now conduct an experiment of tossing a pair of dice.

Two events defined on the experiment are

A: getting a doublet $\{11,22,33,44,55,66\}$

B: getting total score of 10 or more $\{64,46,55,56,65,66\}$



ADDITION THEOREM:

$A \cup B = A + B$ = A or B denotes occurrence of at least A or B.

For 2 events event A & B:

$$P(A \cup B) = P(A)P(B) - P(A \cap B)$$

Note:

$$= \begin{cases} P(A)+P(B)-P(A\cap B) \text{ (This is known as generallised addition theorem)} \\ P(A)+P(B\cap \bar{A}) \\ P(B)+P(A\cap \bar{B}) \\ P(A\cap \bar{B})+P(A\cap B)+P(B\cap \bar{A}) \\ 1-P(A^c\cap B^c) \\ 1-P(A\cup B)^c \end{cases}$$

(a) $P(A \cup B)$

$P(A + B)$

$P(A \text{ or } B)$

P(occurence of "
atleast A or B" is neither A nor B

(b) $P(A/B) = P(A-B) = P(A \cap B^c) = P(A) - P(A \cap B)$

(c) Opposite of "atleast A or B" is neither A nor B

i.e. $\overline{A+B} = 1 - (A \text{ or } B) = \bar{A} \cap \bar{B}$

Note that $P(A \cup B)+P(\overline{A} \cap \overline{B})=1$.

(c) If A & B are mutually exclusive then $P(A \cup B)=P(A)+P(B)$

(d) If A & B are mutually exclusive then $P(A \cup B)=P(A)+P(B)$

(e) For any two event A & B, P (exactly one of A,B occurrs) $=P(A)+P(B)-2P(A \cap B)$

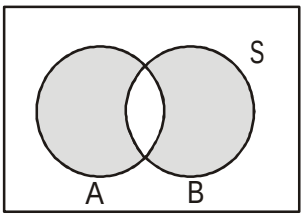
(f) $(A-B) \cup (B-A)=(A \cup B)-(A \cap B)$

(g) **De Morgan's Law :** If A & B are two subsets of a universal set U, then

(i) $(A \cup B)^c = A^c \cap B^c$ and (ii) $(A \cap B)^c = A^c \cup B^c$

(h) $(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$ & $(A \cap B \cap C)^c = A^c \cup B^c \cup C^c$

(i) $A \cup (B \cap C)=(A \cup B) \cap (A \cup C)$ & $A \cap (B \cup C)=(A \cap B) \cup (A \cap C)$



Question 1: Given two events A and B. If odds against A are as 2 : 1 and those in favour of $A \cup B$ are as 3:1, then find the range of P(B).

Solution: Clearly $P(A)=1/3, P(A \cup B)=3/4$.

Now, $P(B) \le P(A \cup B)$

$\Rightarrow \quad P(B) \le 3/4$

Also, $P(B)=P(A \cup B)-P(A)+P(A \cap B)$

$\Rightarrow P(B) \ge P(A \cup B)-P(A)+P(A \cap B)$

$\Rightarrow P(B) \ge 3/4-1/3$

$\Rightarrow P(B) \ge \frac{5}{12}$ $\qquad\qquad$ $\Rightarrow \frac{5}{12} \le P(B) \le \frac{3}{4}$

Question 2 : If A and B are two events such that $P(A \cup B)=\frac{3}{4}, P(A \cap B)=\frac{1}{4}$ and $P(A^c)=\frac{2}{3}$. THen find

(i) P(A) (ii) P(B) (iii) $P(A \cap B^c)$ (iv) $P(A^c \cap B)$

Solution: $P(A)=1-P(A^c)=1-\frac{2}{3}=\frac{1}{3}$

$P(B)=P(A \cup B)+P(A \cap B)-P(A)=\frac{3}{4}+\frac{1}{4}-\frac{1}{3}=\frac{2}{3}$

$P(A \cap B^c)=P(A)-P(A \cap B)=\frac{1}{3}-\frac{1}{4}=\frac{1}{12}$

$$P(A^c \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{1}{15}$$

PROBABILITY OF THREE EVENTS:

For any three events A,B and C we have

(a) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B)$

$-P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(b) P(at least two of A,B, C occur)

$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(c) P(exactly one of A,B,C occurs)

$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(d) P(exactly one of A,B,C occurs)

$= P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap B) + 3P(A \cap B \cap C)$

Note : If three A, B and C are pair wise mutually exclusive then they must be mutually exclusive.

I.e $P(A \cap B) = P(B \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$. However the converse of this is not true.

3. PROBLEMS HAVING GEOMETRICAL FLAVOUR

In the last section we dealt with the sample spaces that were finite, making it possible for us to count $n(E)$. But what if the sample space is uncountable?

The only uncountable sample space S that we consider here are those with some finite geoetrical measurement $m(S)$, such as length, area, or volume, and where a point in S is selected at random. Then the probability of an event A, *i,e*. the selected point belongs to E, is the ratio of $m(E)$ to $m(S)$.

$$P(E) = \frac{\text{Length of E}}{\text{Length of S}} \text{ or } P(E) = \frac{\text{area of E}}{\text{area of S}} \text{ or } P(E) = \frac{\text{Volume of E}}{\text{Volume of S}}$$

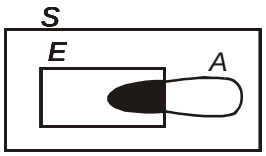
Such probability spaces are assumed to be uniform.

CONDITIONAL PROBABILITY AND MULTIPLICATION RULE

Suppose E is an event in the simple space S with $E \neq \phi$. The probability that an event A occurs once E has occurred or specifically, the conditional probability of A, given E, written $P(A/E)$ is defined as follows.

$$P(A / E) = \frac{P(A \cap E)}{P(E)}$$

As shown in the Venn diagram, $P(A/E)$ measures, in some sence, the relative probability of A with respect to the reduced space E.



Now $P(A \cap E) = \frac{n(A \cap E)}{n(S)}, P(E) = \frac{n(E)}{n(S)}$ and so $P(A / E) = \frac{P(A \cap E)}{P(S)} = \frac{n(A \cap E)}{n(E)}$

i.e. $P(A / E) = \frac{\text{number of elements in } A \cap E}{\text{number of elements in } E} = \frac{P(A \cap E)}{P(E)}$

Multiplication Theorem of Probability

Suppose A and B are events in a sample space with $B \neq \phi$.

By definition of conditional probability

$$P(A / B) = \frac{P(A \cap B)}{P(B)}$$

Which on cross - multiplying becomes

$P(A \cap B) = P(A/B).P(B)$

$P(A \cap B) = P(B).\ P(A/B)$

For three events A, B and C

$P(A \cap B \cap C) = P(A)\ P(B/A).P(C/A \cap B)$

That is, the probability that A, B and C occur is equal to the product of the following :

(a) The probability that A occurs

(b) The probability that B occurs, assuming that A has occured.

(c) The probability that C occurs, assuming that A and B have occurred.

Independence of Events

Events A and B in a probability space are said to be independent if the occurrence (or non-occurrence) of one of them doesn't influence the occurrence (or non-occurrence) of the other.

If B is independent of A, then $P(B/A) = P(B)$.

We are now ready to define the independence of two events.

Independence of two events

Events A and B are said to be independent if $P(A \cap B) = P(A).\ P(B)$

It at once follows that the relation of independence is symmatric.

Independence of three or more events

Three events A, B, C are independent if the following two conditions hold :

(i) They are pairwise independent, that is

$P(A \cap B) = P(A).P(B)$, $P(B \cap C) = P(B)\ P(C)$, $P(C \cap A) = P(C)\ P(A)$

(ii) $P(A \cap B \cap C) = P(A)P(B)P(C)$

Independence of more than three events can be defined in a similar manner. The events A_1, A_2, ..., A_n are independent if,

(i) Any proper subset of them is independent

(ii) $P(A_1 \cap A_2 .... \cap A_n) = P(A_1)P(A_2).... P(A_n)$

Tree Diagram

Sometimes we need to consider a finite sequence of experiments where each experiments has a finite number of outcomes with given probabilities.

A convenient way of representing such a process is by means of a labelled tree diagram. The multipli cation theorem can then be used to compute the probability of an event which is described by a given path of the tree

4. TOTAL PROBABILITY THEOREM

Let an event A cocurs with one of the n mutually exclusive & exhaustive events B_1, B_2, B_3,.....B_n then $A = AB_1 + AB_2 + AB_3 + ...AB_n$

$$P(A) = P(AB_1) + P(AB_2) + ... + P(AB_n) = \sum_{i=1}^{n} P(AB_1)$$

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + ... + P(B_n)P(A|B_n) = \sum_{i=1}^{n} P(B_i)P(A|B_i)$$

Question:1 A pures contains 4 copper and 3 silver cons and another purse contanins 6 copper and 2 silver coins. One coin is drawn from any one of these two purses. The probability that it is a copper coin is:-

(a) $\frac{4}{7}$ (b) $\frac{3}{4}$ (c) $\frac{2}{7}$ (d) $\frac{37}{56}$

Solution: Let A = event of selecting first purse

B = event of selecting second pure
C = envent of drawing a copper coin
Then given event has two disjoint cases: AC and BC

$$\therefore P(C) = P(AC + BC) = P(AC) + P(BC) = P(A)P(C|A) + P(B)P(C|B)$$

$$= \frac{1}{2}.\frac{4}{7} + \frac{1}{2}.\frac{6}{8} = \frac{37}{56}$$

Question:2 Three groups A, B, C are contesting for positions on the Board of directors of a company. The probabilities of their winning are 0.5, 0.3, 0.2 respectively. If the group A wins, the the probability of introducing a new product is 0.7 an the corresponding probabilities for group B and C are 0.6 and 0.5 respectively. Find the probability that the new product will be introduced.

Solution: Given P(A) = 0.5, P(B)=0.3 and P(C) = 0.2

$$\therefore P(A) + P(B) + P(C) = 1$$

then events A,B,C are exhaustive.
If P(E) = Probability of introducing a new product, then as given

$$P(E|A) = 0.7, P(E|B) = 0.6 \text{ and } P(E|C) = 0.5$$

$$\therefore P(E) = P(A).P(E|A) + P(B).P(E|B) + P(C).P(E|C)$$

$$= 0.5 \times 0.7 + 0.3 \times 0.6 + 0.2 \times 0.5 = 0.35 + 0.18 + 0.10 = 0.63$$

5. BAYE'S THEOREM

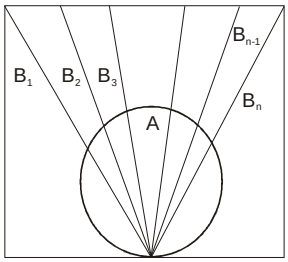
If an event A can occur only with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots B_n$ & the probabilities $P(A/B_1)$, $P(A/B_2)$.....$P(A/B_n)$ are known then,

$$P(B_1 / A) = \frac{P(B_i).P(A / B_i)}{\sum_{i=1}^{n} P(B_i).P(A / B_i)}$$

Explanation: A ≡ event what we have: B_i ≡ event what we want & remaining are alternative events.

Now, $P(AB_i) = P(A).P(B_i / A) = P(B_i)P(A / B_i)$

$$P(B_i / A) = \frac{P(B_i).P(A / B_i)}{P(A)} = \frac{P(B_i).P(A / B_i)}{\sum_{i=1}^{n} P(AB_i)}$$



$$P(B_i / A) = \frac{P(B_i).P(A / B_i)}{\sum_{i=1}^{n} P(B_i).P(A / B_i)}$$

Question: Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold?

Solution: Let E_1, E_2 and E_3 be the events that boxes I, II and III are chosen, respectively.

Then $P(E_1) = P(E_2) = P(E_2) = \frac{1}{3}$

Also, let A be the event that the coin drawn is of gold'

Then $P(A|E_1) = P(\text{a gold coin from box I}) = \frac{2}{2} = 1$

$P(A|E_2) = P$ (a gold coin from box II) = 0

$P(A|E_3) = P$ (a gold coin form box III) = $\frac{1}{2}$

Now , the probability that the other coin in the box is of gold
=the probability that gold coin is drawn from the box I.

= $P(E_1|A)$

By Baye's theorem, we know that

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)} = \frac{\frac{1}{3}\times 1}{\frac{1}{3}\times 1+\frac{1}{3}\times 0+\frac{1}{3}\times\frac{1}{2}} = \frac{2}{3}$$

6. BINOMIAL PROBABILITY DISTRIBUTION

Suppose that we have an experiment such as tossing a coin or die repeatedly or choosing a marble from an urn repeatedly. Each toss or selection is called a trial In any single trial there will be probability associated with a particular event such as head on the coin, 4 on the die, or section of a red marble. In some cases this probability will not change from one trial to the next (as in tossing a coin or die). Such trials are then said to be indendent and are often called Bernoulli trials James Bernoulli who investigated them at the end of the seventeenth century.

Let p be the probability that an event will happen in any single Bernoulli trial (called the probability of success). then q = 1-p is the probability that the event will fail to happen in any single trial(called the probability of failure). The probability that the event will happen exactly x times in n trials (i.e., successes and n - x failures will occur) is given by the probability function.

$$f(x) = P(X=x) = \binom{n}{x}p^x q^{n-x} = \frac{n!}{x!(n-x)!}p^x q^{n-x} \quad \text{.....(i)}$$

Where the random variable X denotes the number of succeses in n trials and x = 0, ...,n.

Example: The probability of getting exactly 2 heads in 6 tosses of a fair coin is

$$P(X=2) = \binom{6}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{6-2} = \frac{6!}{2!4!}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{6-2} = \frac{15}{64}$$

The discrete probability function (i) is often called the binomial distribution since for x = 0, 1,2,...n, it corresponds to successive terms in the binomial expansion

$$(q+p)^n = q^n + \binom{n}{1}q^{n-1}p + \binom{n}{2}q^{n-2}p^2 + \ldots + p^n = \sum_{x=0}^{n}\binom{n}{x}p^x q^{n-x}$$

The special case of a binomial distribution with n = 1 is also called the Bernoulli distribu tion.

Question: The repeated tosses of a coin are Bernoulli trials. Let X denotes the number of heads in an experiment of 10 trials.

Clearly, X has the binomial distribution with n = 10 and $p = \frac{1}{2}$

Therefore $P(X=x) = {}^n C_x q^{n-x}p^x, x = 0,1,2,.....,n$

Here $n=10, p=\frac{1}{2}, q=1-p=\frac{1}{2}$

Therefore $P(X=x)={}^{10}C_x\left(\frac{1}{2}\right)^{10-x}\left(\frac{1}{2}\right)^{x}={}^{10}C_x\left(\frac{1}{2}\right)^{10}$

Now (i) $P(X=6)={}^{10}C_6\left(\frac{1}{2}\right)^{10}=\frac{10!}{6!\times 4!2^{10}}=\frac{105}{512}$

(ii) P(atleast six heads)

$=P(X\geq 6)=P(X=6)+P(X=7)+P(X=8)+P(X=9)+P(X=10)$

$={}^{10}C_6\left(\frac{1}{2}\right)^{10}+{}^{10}C_7\left(\frac{1}{2}\right)^{10}+{}^{10}C_8\left(\frac{1}{2}\right)^{10}+{}^{10}C_9\left(\frac{1}{2}\right)^{10}+{}^{10}C_{10}\left(\frac{1}{2}\right)^{10}$

$=\left[\left(\frac{10!}{6!\times 4!}\right)+\left(\frac{10!}{7!\times 3!}\right)+\left(\frac{10!}{8!\times 2!}\right)+\left(\frac{10!}{9!\times 1!}\right)+\left(\frac{10!}{10!}\right)\right]\frac{1}{2^{10}}=\frac{193}{512}$

(iii) (at most six heads) $=P(X\leq 6)$

$=P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)+P(X=6)$

$=\left(\frac{1}{2}\right)^{10}+{}^{10}C_1\left(\frac{1}{2}\right)^{10}+{}^{10}C_2\left(\frac{1}{2}\right)^{10}+{}^{10}C_3\left(\frac{1}{2}\right)^{10}+{}^{10}C_4\left(\frac{1}{2}\right)^{10}+{}^{10}C_5\left(\frac{1}{2}\right)^{10}+{}^{10}C_6\left(\frac{1}{2}\right)^{10}$

$=\frac{848}{1024}=\frac{53}{64}$

MISCELLANEOUS EXAMPLES

Example 1 :

For a post three persons *A, B* and *C* appear in the interview. The probability of *A* being selected is twice that of *B* and the probability of *B* being selected is thrice that of *C*, what are the individual probabilities of *A, B, C* being selected?

Solution :

Let E_1, E_2, E_3 be the events of selection of *A, B* and *C* respectively.

Let $P(E_3) = x$

then $P(E_2) = 3P(E_3) = 3x$

and $P(E_1) = 2P(E_2) = 6x$

since E_1, E_2, E_3 are mutually exclusive and exhaustive events

$\therefore \quad P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) = 1$

$\therefore \quad P(E_1) + P(E_2) + P(E_3) = 1$

$\Rightarrow \quad 6x + 3x + x = 1$

$$\therefore \quad x = \frac{1}{10}$$

Hence $$P(E_1) = 6x = \frac{6}{10} = \frac{3}{5}$$

$$P(E_2) = 3x = \frac{3}{10}$$

and $$P(E_3) = x = \frac{1}{10}$$

Example 2 :

A student appears for test I, II and III. The student is successful if he passes either in tests I and II or test I and III. The probabilities of the student passing in test I, II, III are *p, q* and 1/2 respectively. If the probability that the student is successful is 1/2 then find the relation between *p* and *q*.

Solution :

Let *A, B* and *C* be the events that the student is successful in test I, II and III respectively. Then

P(The student is successful)

$= P[(A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A \cap B \cap C)]$

$= P(A \cap B \cap C') + P(A \cap B' \cap C) + P(A \cap B \cap C)$

$= P(A)P(B)P(C') + P(A)P(B')P(C) + P(A)P(B)P(C)$ [$\because$ *A, B* and *C* are independent]

$= P(A)P(B)\{1 - P(C)\} + P(A)\{1 - P(B)\} + P(A)P(B)P(C)$

$$= p.q\left(1 - \frac{1}{2}\right) + p(1-q)\frac{1}{2} + p.q.\frac{1}{2}$$

$$= \frac{1}{2}pq + \frac{1}{2}p(1-q) + \frac{1}{2}pq$$

$$= \frac{1}{2}(pq + p - pq + pq)$$

$$= \frac{1}{2}(pq + p)$$

$$\therefore \quad \frac{1}{2} = \frac{1}{2}(pq + p) \qquad \Rightarrow \quad 1 = pq + p$$

Hence $p(1 + q) = 1$

Example 3 :

Let A and B be two independent events. The probability that both A and B happens is $\frac{1}{12}$ and probability that neither A nor B happen is 1/2. Then find probabilities of occurrence of A and B.

Solution :

Given $P(A \cap B) = \frac{1}{12}$ and $P(A' \cap B') = \frac{1}{2}$. Since A and B are independent events,

$\therefore$ $P(A\ B) = P(A).P(B)$ and $P(A'\ B') = P(A')P(B')$

Then $P(A).P(B) = \frac{1}{2}$ and $P(A').P(B') = \frac{1}{2}$

or $P(A).P(B) = \frac{1}{2}(1 - P(A))(1 - P(B)) = \frac{1}{2}$

since $\{1 - P(A)\}\{1 - P(B)\} = \frac{1}{2}$

$\therefore$ $1 - P(A) - P(B) + P(A).P(B) = \frac{1}{2}$

$1 - P(A) - P(B) + \frac{1}{12} = \frac{1}{2}$ $\left\{\because P(A).P(B) = \frac{1}{12}\right\}$

$\therefore$ $P(A) + P(B)$ and $P(A).P(B) = \frac{1}{12}$

Hence $P(A) + P(B) = \frac{7}{12}$ and $P(A).P(B) = \frac{1}{12}$

Hencpe $P(A)$ and $P(B)$ are the roots of the quadratic equation $x^2 - \frac{7}{12}x + \frac{1}{12} = 0$

$\left(\text{since sum} = \frac{7}{12} \text{ and product} = \frac{1}{12}\right)$

$\Rightarrow$ $12x^2 - 7x + 1 = 0$

$\Rightarrow$ $(3x - 1)(4x - 1) = 0$

$\therefore$ $x = \frac{1}{2}.\frac{1}{4}$

$P(A) = \frac{1}{2}$ or $\frac{1}{4}$ and $P(B) = \frac{1}{4}$ or $\frac{1}{3}$ respectively

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic 1: Basic Concepts and Terms used in Probability

Q.1 Four digit numbers are formed by using the digits 1, 2, 3, 6 without repetition. The probability that a selected number is divisible by 3 is
(a)1/4 (b)1/3 (c)1/2 (d) 1

Q.2 In a game of six digits, two digits are already fixed for prize. If any one will find these two fixed digits then he will win the prize. Mr. A selected two numbers. The probability that Mr. A will win the prize is
(a)1/15 (b)1/6 (c)1/5 (d)1/1

Q.3 A number of four digit is selected. If the probability that number has non-repeated digits is P then the value of $125P$ is
(a)63 (b)64 (c)65 (d)66

Q.4 Three dice are thrown simultaneously. If the probabiity that numbers on them are same is P, then the value of $36P$ is
(a)3 (b)2 (c)1 (d)4

Q.5 Two cards are drawn at random from a pack of 52 playing cards. If one is red and other is black then probability is
(a)26/51 (b)1/2 (c)25/51 (d)29/51

Q.6 A bag contains 5 black and 4 white balls. Two balls are drawn. The probability that the balls are of same colour is
(a)4/9 (b)8/9 (c)2/9 (d)1/9

Q.7 A bag contains 3 red and 5 black balls. Two balls are drawn at random. The probability that one ball is red and other is black is P then the value of $28P - 15$ is
(a)1 (b)0 (c)1/2 (d)2

Q.8 If n is a positive integer then the probability that 3^n has 3 at unit place is
(a)1/4 (b)1/5 (c) 3/10 (d) 1/8

Q.9 Three distinct numbers are selected from the first 100 natural numbers. The probability that all the three numbers are divisible by 2 and 3 is
(a)1/231 (b)4/1155 (c)1/55 (d)4/25

Q.10 Three students Ram, Ali and John appear at a competitive examinations. The probability of Ram comes first is thrice that of Ali and the probability of Ali coming first is thrice of Johm. The probability of John coming first is
(a)1/13 (b)3/13 (c)12/13 (d)9/13

Q.11 A number is selected from first 100 natural numbers. The probability that number is perfect square or a perfect cube is
(a)7/50 (b)3/25 (c)2/25 (d)1/25

Q.12 Five horses are in race. Mr. A selected two of horses at random and bets on them. The probability that Mr. A selected the winning horse is
(a) 3/5 (b)2/5 (c) 1/5 (d) 4/5

Q.13 Two numbers a and b are chosen at random from the set of integers 1, 2, 3,.... 15. The probability that the equation $2a - 3b = 0$ is satisfied, is
(a)1/21 (b)2/15 (c)2/21 (d) 3/31

Q.14 5 cards are drawn from a pack of 52 cards. The probability that these 5 will contain just one king is

(a) $\frac{{}^{48}C_4}{{}^{52}C_4}$ (b) $\frac{{}^{48}C_4}{{}^{51}C_4}$

(c) $\frac{{}^{48}C_{44}}{{}^{52}C_5} \times 4$ (d) $\frac{{}^{48}C_{44}}{{}^{51}C_4} \times 4$

Q.15 Let $n = 2^3\, 4^5\, 6^8\, 5^4$. A positive factor is taken at random from the possible positive factors of n. Then the probability that the selected factor is a perfect square and divisible by 100 is
(a) 5/99 (b) 10/99 (c) 1/6 (d) 71/99

Q.16 A coin is tossed 10 times. The probability of getting at most 6 heads is
(a) 51/64 (b) 53/64
(c) 105/512 (d) 193/512

Q.17 A pair of dice is rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes before 7 is
(a) 1/5 (b) 2/5 (c) 3/5 (d) 2/3

Q.18 A bag contains 5 brown and 4 white socks. A man pulls out two socks. The probability that these are of the same colour, is
(a) 5/108 (b) 18/108
(c) 30/108 (d) f48/108

Q.19 Two numbers are selected randomly from the set $S = \{1, 2, 3, 4, 5, 6\}$ without replacement one by one. The probabiilty that minimum of the two numbers is less thatn 4, is

(a)1/15 (b)14/15 (c)1/5 (d)4/5

Q.20 If two balls are drawn from a bag containig 2 white , 4 red and 5 black balls, the chance that both are red is
(a)1/55 (b)2/55 (c)4/55 (d)6/55

Q.21 In the previous question the probabiily that one is red and the other is black is
(a)1/11 (b)2/11 (c)3/11 (d) 4/11

Q.22 Out of 20 consecutive numbers, two are chosen at random, the probability that their sum is odd is
(a) 9/10 (b) 11/20 (c) 10/19 (d) 10/21

Q.23 If the letters of "Attempt" are written down at random, then the chance that no two *t*'s are together is
(a) 1/7 (b) 2/7 (c) 3/7 (d) 4/7

Q.24 *n* men sit in a row. The probabiilty that not more than *m* men ($n > m$) sit between two particular men *A* and *B* is

(a) $\frac{(m+1)(2n-m-2)}{n(n-1)}$ (b) $\frac{m(2n-m-2)}{n(n-1)}$

(c) $\frac{(2n-m-2)}{n(n-1)}$ (d) $\frac{(2n-m(m+1)}{n(n-1)}$

Q.25 From a bag containing *a* balls, *m* balls are drawn simultaneously and replaced; then *n* balls are drawn. The change that exactly *r* balls are common to the two drawings is

(a) $\frac{{}^{m}C_{r}}{{}^{a}C_{n}}$ (b) $\frac{{}^{a-m}C_{n-r}}{{}^{a}C_{n}}$

(c) $\frac{{}^{m}C_{r}.{}^{a-m}C_{n-r}}{{}^{a}C_{n}}$ (d) $\frac{{}^{m}C_{r}.{}^{n-r}C_{a-m}}{{}^{a}C_{n}}$

Q.26 One number is selected from the set {1, 2, 3,....,100}. The probability that the number is even or a multiple of 5 is
(a) 7/10 (b) 3/5 (c) 4/5 (d) 2/5

Q.27 Out of *n* persons sitting at a round table, three, *A*, *B, C* are chosen at random. The chance that no two of these three are sitting next to one another is

(a) $\frac{(n-4)(n-5)}{2\times(n-1)}$ (b) $\frac{(n-4)(n-5)}{2\times(n-1)(n-2)(n-3)}$

(c) $\frac{(n-4)(n-5)}{(n-1)(n-2)}$ (d) $\frac{n(n-4)}{(n-1)(n-2)}$

Topic: 2 Venn Diagram Probability

Q.28 If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(C) = \frac{1}{2}$ and $P(A \cap B) = P(B \cap C) = P(C \cap A) = \frac{1}{4}$ and $P(A \cap B \cap A) = \frac{1}{8}$ then the probability that at least one of them appears is
(a) 17/24 (b) 19/24
(c) 13/24 (d) 11/24

Q.29 If $P(A) = \frac{1}{2}$, and $P(A \cap B) = \frac{1}{3}$ then the probability that *A* happens but *B* does not happen is
(a) 1/2 (b) 1/3 (c) 5/6 (d) 1/6

Q.30 A coin is tossed twice. Events *E* and *F* are defined as folows *A* = heads on first toss, *B* = heads on second toss. The probability of the occurence of the event $A \cup B$ is
(a) 1/2 (b) 3/4 (c) 1/4 (d) 1/3

Q.31 If $P(B) = \frac{3}{4}, P(A \cap B \cap \bar{C}) = \frac{1}{3}$, $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then $P(B \cap C)$
(a) 1/12 (b) 1/6 (c) 1/15 (d) 1/9

Topic: 3 Geometric Probability

Q.32 If *p* and *q* are chosen randomly from the set {1, 2, 3,..., 10} with replacement, then the probability that the roots of the equation $x^2 + px + q = 0$ are real is
(a) 0.62 (b) 0.32 (c) 0.44 (d) 0.55

Q.33 12 members of a committee are to sit down at random round a table. Probability that there are 3 members between the two particular members *A* and *B*, is
(a) 2/11 (b) 7/11 (c) 4/11 (d) 5/11

Q.34 The rod of length 10 cm is broken into three parts, so that each part is having a length as an integral multiple of 1 cm. The probability that the parts are forming a triangle, is
(a) 1/4 (b) 1/2 (c) 3/4 (d) 1/3

Q.35 If two squares of size 1 × 1 are chosen at random on a chess board, the chance that they share a common side is
(a) 1/2 (b) 1/3 (c) 1/6 (d) 1/18

Topic: 4 Miscellaneous

Q.36 Two integers *x* and *y* are chosen with replacement out of the set {0, 1, 2,...., 10}. The probability that $|x - y|$ doesn't exceed 5 is
(a) 5/11 (b) 81/121
(c) 3/11 (d)91/121

Q.37 Dailing a telephone number, an old man forgets the last two digits remembering only that they are different, dialled at random. The probability that the number is dialled correctly
(a) 1/90 (b) 1/45 (c) 2/45 (d) 2/100

Q.38 There are four balls of different colour and four boxes of colours same as of the balls the number of ways in which the balls, one in each box, could be placed such that a ball does not go to a box of its own colour is
(a) 5/8 (b) 3/8 (c) 1/8 (d) 7/8

Q.39 Four digit number is formed from all possible ways. The probability that a number $xyzt$ chosen from the number satisfy $x < y = z > t$ is
(a) 2/75 (b) 3/75 (c) 2/25 (d) 3/25

Q.40 Out of $(2n + 1)$ tickets consecutively numbered, are drawn at random. The chance that the numbers on them are in A.P. is

(a) $\frac{3n}{4n^2-1}$ (b) $\frac{2n}{4n^2-1}$

(c) $\frac{3n}{4n^2+1}$ (d) $\frac{2n}{4n^2+1}$

Q.41 Given two events A and B. If odds against A are 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then

(a) $\frac{1}{2} \le P(B) \le \frac{3}{9}$ (b) $\frac{5}{12} \le P(B) \le \frac{3}{4}$

(c) $\frac{1}{2} \le P(B) \le \frac{3}{5}$ (d) $\frac{1}{3} \le P(B) \le \frac{3}{4}$

Q.42 The decimal part of logarithm of two numbers taken at random are found to seven places. The chance that second can be subtracted from the first
(a) $(0.5)^7$ (b) $(0.2)^7$
(c) $(0.55)^7$ (d) $(0.66)^7$

Q.43 A and B play with two dice on the condition that A wins if he throws 6 before B throws 7, then the probability that A wins is

(a) $\frac{36}{1141}$ (b) $\frac{180}{1141}$

(c) $\frac{172}{1141}$ (d) $\frac{191}{1141}$

Q.44 A draws a card from a pack of n cards marked 1, 2, 3,....,n; the card is replaced in the pack and then B draws a card. The chance that A draws a higher card is

(a) $\frac{1}{n}\left(\frac{1}{n}+\frac{2}{n}+.....+\frac{n-1}{n}\right)$

(b) $\left(\frac{1}{n}+\frac{2}{n}+.....+\frac{n-1}{n}\right)$

(c) $\frac{1}{n}\left(\frac{1}{n}+\frac{2}{n}+.....+\frac{n}{n}\right)$

(d) $\left(\frac{1}{n}+\frac{2}{n}+.....+\frac{n}{n}\right)$

Q.45 In a singing competition a group of 10 people, participate, each person has exactly one friend and rest are strangers, the probabilty that both the singers of the duet are complete stragers is
(a) 1/15 (b) 1/45 (c) 8/9 (d) 22/45

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** The fair coins fall to the floor, the probability that there are five heads and five tails is 63/256.
Statement – 2 : Four far dice are rolled, the probability that the four numbers coming will be all different is 5/18.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1 :** If the odds against an event is 2/3, then probability of occuring of the event is 3/5.
Statement-2 : For two events A and B, $P(A' \cap B') = 1 - P(A \cup B)$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1 :** A lottery agent sells 100 tickets and awards 10 prizes, if Ramesh buys 10 tickets the probability of his winning a prize is $\left(1-\frac{^{90}C_{10}}{^{100}C_{10}}\right)$.

Statements-2 : If $P(A) = 0.8$, $P(B) = 0.5$, the $P(A \cap B)$ is 0.4.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** In case of mutually exclusive events A, B and C, $(A \cup B \cup C) = P(A) + P(B) + P(C)$.

Statement–2 : Independent events will never be mutualy exclusive.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** Two fair dice are thrown at a time. The probability that the difference of numbers on their face is 2, is 7/9.

Statement–2 : If $P(A) = 1/4$, $P(B) = 1/13$ events A and B are independent events then $P(A' \cap B')$ will be equal to 9/13.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.6 Let for two events A and B, $P(A) = p$ and $P(B) = q$.

Statement–1 : The probability that exactly one of the event A and B occurs is $p + q - 2pq$.

Statement–2 : The probability that the system has a solution is 1.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.7 Consider the system of linear equations $ax + by = 0$, $cx + dy = 0$, $a, b, c, d \in \{0, 1\}$.

Statement–1 : The probability that the system of equations has a unique solution is 3/8.

Statement–2 : The probability that the system has a solution is 1.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.8 **Statement–1 :**

$$P(B)=\frac{3}{4};P(A\cap B\cap \bar{C})=\frac{1}{3};P(\bar{A}\cap B\cap \bar{C})=\frac{1}{3},$$

then $P(B \cap C) = \frac{1}{12}$.

Statement–2 :

$$P(B\cap C)=P(B)-P(A\cap B\cap \bar{C})-(\bar{A}\cap \bar{B}\cap \bar{C}).$$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.9 **Statement–1 :** If $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$ and $\frac{1}{6} \le (A \cap B) \le \frac{1}{2}$.

Statement–2 : $P(A \cup B) \geq$ max. $\{P(A), P(B)\}$ and $P\{A \cap B\} \leq$ min.$\{P(A), P(B)\}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.10 **Statement–1 :** Six different balls are put in three different boxes, no box being empty, the probability of putting balls in boxes in equal numbers is 1/6.

Statement–2 : Six letters are posted in 3 letter boxes. The probability that no letter box remains empty is 20/27.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Years Question)

Q.1 Let A nd B be two events such that $P\left(\overline{A \cup B}\right)=\frac{1}{6}, P(A \cap B)=\frac{1}{4}$ and $P\left(\overline{A}\right)=\frac{1}{4}$, where $\overline{A}$ stands for complement of event A. Then events A and B are:-

(a) mutually exclusive and independent

(b) independent but not equally likely

(c) equally likely but not independent

(d) equally likely and mutually exclusive

Q.2 Three houses are avilable in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is:-

(a) 7/9 (b) 8/9 (c) 1/9 (d) 2/9

Q.3 Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the targert is hit by the second plane is:

(a) 0.14 (b) 0.2 (c) 0.7 (d) 0.06

Q.4 It is given taht the events A and B are such that $P(A)=\frac{1}{4}, P\left(\frac{A}{B}\right)=\frac{1}{2}$ and $P\left(\frac{B}{A}\right)=\frac{2}{3}$. Then P(B) is

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{2}$

Q.5 A die is thrown. Let A be the event that thenumber obtained is greater than 3 Let B the event that the number obtained is less than 5. Then $P(A \cup B)$ is

(a) $\frac{3}{5}$ (b) 0 (c) 1 (d) $\frac{2}{5}$

Q.6 One ticket is selected at random from 50 tickets numbered 00, 01, 02, ...49. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals

(a) 5 /14 (b) 1/50 (c) 1/14 (d) 1/7

Q.7 In a binomial distribution $\left(n, p=\frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then n is greater than

(a) $\frac{9}{\log_{10} 4-\log_{10} 3}$ (b) $\frac{4}{\log_{10} 4-\log_{10} 3}$

(c) $\frac{1}{\log_{10} 4-\log_{10} 3}$ (d) $\frac{1}{\log_{10} 4+\log_{10} 3}$

Q.8 An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random without replacement from the urn. The probability that three balls have differnce colours is:-

(a) $\frac{1}{3}$ (b) $\frac{2}{7}$ (c) $\frac{1}{21}$ (d) $\frac{2}{23}$

Q.9 Four numbers are chosen at random (without repllacement) from the set (1, 2, 3,.....,20)

Statement-1: The probability that the chosen numbers when when arranged in some order will form an AP is $\frac{1}{85}$

Statement-2: In the four chosen number form an AP, then the set of all possible values of common differnce is $\{\pm1,\pm2,\pm3,\pm4,\pm5\}$.

(a) Statement-1 is true, Statement -2 is true; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is true, Statement-2 is true; Statement-2is not a correct explanation for statement-1
(c) Statement-1is true, Statement-2 is false.
(d) Statement-1 is false, Statement-2 is true.

Q.10 If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is:-

(a) $P(C|D) < P(C)$ (b) $P(C|D) = \frac{P(D)}{P(C)}$

(c) $P(C|D) = P(C)$ (d) $P(C|D) \geq P(C)$

Q.11 Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval:-

(a) $\left[0,\frac{1}{2}\right]$ (b) $\left(\frac{11}{12},1\right]$

(c) $\left(\frac{1}{2},\frac{3}{4}\right]$ (d) $\left(\frac{3}{4},\frac{11}{12}\right]$

Q.12 Let A,B,C be pairwise independent events with $P(C) > 0$ and $(A \cap B \cap C) = 0$. Then $P(A^c \cap B^c | C)$ is

(a) $P(A^c) - P(B)$ (b) $P(A) - P(B^c)$

(c) $P(A^c) + P(B^c)$ (d) $P(A^c) - P(B^c)$

Q.13 Three numbers are chosen at random without replacement from {1,2,3,.....,8}. The probability that their minimum is 3, given that their maximum is 6, is :

(a) $\frac{2}{5}$ (b) $\frac{3}{8}$ (c) $\frac{1}{5}$ (d) $\frac{1}{4}$

Q.14 A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is:

(a) $\frac{17}{3^5}$ (b) $\frac{13}{3^5}$ (c) $\frac{11}{3^5}$ (d) $\frac{10}{3^5}$

Q.15 Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for the complement of the event A. Then the events A and B are:
(a) mutually exclusive and independent
(b) equally likely but not independent
(c) independent but not equally likely
(d) independent and equally likely.

Q.16 If 12 identical balls are to be placed in 3 identical boxes, then the probability that one of the boxes contains exactly 3 balls is:

(a) $220\left(\frac{1}{3}\right)^{12}$ (b) $22\left(\frac{1}{3}\right)^{11}$

(c) $\frac{55}{3}\left(\frac{2}{3}\right)^{11}$ (d) $55\left(\frac{2}{3}\right)^{10}$

Q.17 Let X be a set contaning 10 elements and P(X) be its power set. If A and B are picked up at random form P(X) , with replacement, then the probability that A and B have equal number of elements, is

(a) $\frac{^{20}C_{10}}{2^{20}}$ (b) $\frac{\left(2^{10}-1\right)}{2^{20}}$

(c) $\frac{^{20}C_{10}}{2^{20}}$ (d) $\frac{\left(2^{10}-1\right)}{2^{20}}$

Q.18 Let two fair six-faced dice A and B be thrown simultaneously. If E_1 is the event that die A shows up four, E_2 is the event that dice is odd, the which of the following statements is **NOT true?**
(a)E_1,E_2 and E_3 are independent.
(b) E_1 and E_2 are independent
(c) E_2 and E_3 are independent.
(d) E_1 and E_3 are independent.

Q.19 If A and B are any two events such that $P(A) = \frac{2}{5}$ and $P(A \cap B) = \frac{3}{20}$, then the conditional probability, $P(A|A' \cup B'))$, where A' denotes the complement of A is equal to:-
(a) 8 /17 (b) 1/4 (c) 11/20 (d) 5/17

Q.20 An experiment succeeds twice as often as it fails. The probability of at least 5 successes in the six trials of this experiment is:-

(a) $\frac{256}{729}$ (b) $\frac{192}{729}$ (c) $\frac{240}{729}$ (d) $\frac{496}{729}$

Q.21 A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is:

(a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{3}{4}$ (d) $\frac{3}{10}$

Section-D
(Board pattern)

Q.1 If E_1 and E_2 be the events assocciated with a random experiment and if E_1 implies E_2, then prove that $P(E_1) < P(E_2)$.

Q.2 If a dice is thrown once, find the chance that 'six' shows up.

Q.3 A pair of coins is tossed once. Find the probability of showing exactly one head.

Q.4 Four digit number are formed by the digits 1, 2, 3, 4, and 5 without repeating any digit. Find the probability that a number chosen at random is an odd number.

Q.5 Six boys and six girls sit in a row at random. Find the probability that (i) the six girls sit together (ii) the boys and girls sit alternately.

Q.6 From a deck of 52 cards, four cards are drawn simultaneously. Find the chance that they will be the four honours of the same suit.

Q.7 A party of 10 take their seats at a round table. What are the odds against two specified persons sitting together.

Q.8 The probability that at least one of A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, find $P(\bar{A}) + P(\bar{B})$.

Q.9 A dice is rolled twice. Find the probability that at least one of two throws shows up the number 5.

Q.10 A has 3 shares in a lottery in which there are 3 prizes and 8 blanks. B has 2 shares in a lottery in which there are 4 prizes and 6 blanks. Compare their chances of success.

Q.11 A pack of 50 tickets numbered 1 to 50 is shuffled the two tickets are drawn. Find the probability that

(i) Both the tickets drawn bear prime numbers.

(ii) Neither of the ticket drawn bear a prime number.

Q.12 A five digit number is formed at a random by using the digits 1, 2, 3, 4, 5, 6 and 7. Find the probability that the number formed has none of its digit repeated.

Q.13 A box contains 10 red marbles, 20 blue marbles and 30 green marbles. 5 marbles are drawn from the box, what is the probability that

(i) All will be blue.

(ii) At least one will be green.

Q.14 4 cards are drawn from a well shuffled deck of 52 cards. What is the probability of getting 3 diamonds and one spade?

Q.15 Three letters are dictated to three persons and an enevelope is addressed to each of them, the letters are inserted into the envelopes at random, so that each envelope contain exactly one letter. Find the probability that at least one letter is in its proper envelop.

CHAPTER 3 SEQUENCE AND SERIES

Informally, a sequence is an ordered list of objects, but in this lesson the object will usually be numbers.

e.g., 2, 0, –2, 2, 0, –2, 2, 0, –2, ...;

1, 2, 4, 8, 16, ...;

$x, x-3, x-6, x$ ....., *etc*.

You can observe that there is a pattern in each of these list of numbers/expressions. This pattern may be easy to observe and sometimes difficult. But with the help of this pattern we can tell the number at any position in the sequence.

It is obvious that there is no limit to the kinds of patterens we can form and thus infinite types of sequences can be generated. In this lesson, will focus our attention to some basic sequences. Before that let us define certain terms.

Sequence : A set of numbers arranged in some definite order and according to some definite rule is called a sequence.

Alternatively, a real sequence is a function whose domain is the set of natural numbers *N* or a subset of *N*, and range is a set of real numbers.

Series : Terms of a sequence when added form a series.

Progression : When terms of a sequences are written under specific conditions, then the sequence is called a progression.

A progression is represented as $t_1, t_2, \ldots\ldots, t_n, \ldots\ldots$or $a_1, a_2, \ldots\ldots, a_n, \ldots\ldots$ where t_1 (or a_1) means first term, and t_n (or a_n) means n^{th} term. t_k(or a_k) is called the general term of the progression. The number of terms can be finite or infinite.

$S_n = t_1 + t_2 + \ldots\ldots + t_n$, denotes the sum of corresponding series upto *n* terms.

Clearly, $t_n = S_n - S_{n-1}$.

1. ARITHMETIC PROGRESSION

(i) A succession of terms is said to be in Arithmetic Progression if the difference between any term and its preceding term is independent of the position of the term.

Arithmetic Progression is abbreviated as A.P.

e.g., 5, 9, 13, 17, 21, ...;

t, t – 5, t – 10, t – 15,.....; *etc*.

We define,

Common difference = value of a term – value of the preceding term.

In the above examples, common difference (c. d.) is 4 and –5 respectively.

The first term is usually denoted by '*a*' and the common difference by '*d*'. So , an A.P. can be written as

$a, a + d, a + 2d, a + 3d, \ldots\ldots\ldots\ldots$

If follows that the kth term of an A.P., t_k (or a_k) is given by

$t_k = a + (k-1)d$ (General term)

If an A.P. consists of n terms, its nth term is called its last term denoted by l and is given by $l = t_n = a + (n-1)d$.

(ii) N^{th} Term of an AP from End: N^{th} from end $t_n' = l-(n-1)d, \forall n \in N$ l = last term

Illustration 1: If p times the qth term of an A.P. is equal to q times p^{th} term, show that $t_{p-q} : t_{p+q} = p-q : p+q$.

Solution : Let us proceed backwards.

We require $t_{p-q} : t_{p+q}$ in terms of p and q only.

If we assume the first term as 'a' and common difference as 'd' , then we must obtain a relation between them to get the required ration in that form.

Now, $\frac{t_{p-q}}{t_{p+q}} = \frac{a+(p-q-1)d}{a+(p+q-1)d}$

But from the given condition,

$pt_q = qt_p$,

$\Rightarrow p[a + (q-1)d] = q[a + (p-1)d]$

$\Rightarrow (p-q)\, a = (pq - q - pq + p)d]$

$\Rightarrow a = d$ $(\because p \neq q)$

Hence, $\frac{t_{p-q}}{t_{p+q}} = \frac{a+(p-q-1)a}{a+(p+q-1)a} = \frac{p-q}{p+q}$.

Student Task

(a) If the mth term of an A.P is $\frac{1}{n}$ and the nth term is $\frac{1}{m}$, then find mn^{th} term of an A.P

(b) Find first negative term of the sequance 20, $19\frac{1}{4}$, $18\frac{1}{20}$, $17\frac{3}{4}$,.....

1.1 SUM OF AN ARITHMETIC PROGRESSION

If an A.P. contains n terms, the corresponding series can be summed.

Sum to n terms of an A.P. is given by

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

Proof : Let a be the first tearm, d be the commen difference, l be the last term of an A.P having n terms and S_n be the sum of n terms then

$S_n = a + (a+d) + (a+2d) + ... + (l-2d) + (l-d) + l$.....(i)

$S_n = l + (l-d) + (l-2d) + ... + (a+2d) + (a+d) + a$.....(il)

$2S_n = (a+l) + (a+l) + (a+l) + ... + (a+l) + (a+l)$..... n times

$S_n = \frac{n}{2}[a+l]$

Illustration 2: The sum of the first 13 terms of an A.P. is 21 and the sum of first 21 terms is 13. Find the sum of first 34 terms.

Solution : Given $S_{13} = 21$ and $S_{21} = 13$

$$i.e., \ \frac{13}{2}[2a+(13-1)d] = 21$$

$$\Rightarrow a + 6d = \frac{21}{13} \quad \text{....(i)}$$

$$\text{and } \frac{21}{2}[a+(21-1)d] = 13$$

$$\Rightarrow \ a + 10d = \frac{13}{21} \quad \text{....(ii)}$$

From equation (i) & (ii),

$$a = \frac{849}{273} \text{ and } d = -\frac{68}{273}$$

$$\therefore S_{34} = \frac{34}{2}[2a+(34-1)d] = -34$$

Illustration 3: Find the arithmetic progression, consisting of 10 terms if the sum of the terms occupying the even places is equal to 15 and the sum of those occupying the odd places is equal to $12\frac{1}{2}$.

Solution : Let the successive terms of the A.P. be $a_1, a_2, ..., a_9, a_{10}$.

By hypothesis $a_2 + a_4 + a_6 + a_8 + a_{10} = 15$

i.e., $(a + d) + (a + 3d) + ...+ (a + 9d) = 15$

i.e., $5a + 25d = 15$ ....(i)

and $a_1 + a_3 + a_5 + a_7 + a_9 = 12\frac{1}{2}$

$5a + 20d = 12\frac{1}{2}$ ....(ii)

From (i) and (ii), we get $5d = 12\frac{1}{2}$ or $d = \frac{1}{2}$ and $a = \frac{1}{2}$

Hence the A.P. is $\frac{1}{2}, 1, 1\frac{1}{2}, 2, ...$

Student Task

(a)Then Sum of the products of the ten numbers $\pm1, \pm2, \pm3, \pm4, \pm5$, *taking two at a time is*

(b) The 6th term of an A>P is equal to 2, the value of the common difference ofthe A.P which makes the product $a_1 a_4 a_5$ *least is given by*

1.2 IMPORTANT FACTS

(i) If a constant is added to (or subtracted from) every term of an A.P., the resulting sequences will also be an A.P. with the same common difference;

(ii) If every term of an A.P. is multiplied by (or divided by) a fixed constant, the resulting sequence will also be an A.P.

(iii) It is convenient in problems dealing with only three terms to take the three terms in A.P. as $a-d$, a, $a+d$, with a as the middle term and d as the common difference.

(iv) Four terms in A.P. can be taken as $a-3d$, $a-d$, $a+d$, $a+3d$. Note that $2d$ is the common difference.

(v) A sequence is an AP if and Only if the sum of its n terms is of the form An^2+Bn, where A and B are constants.

(vi) In a finite AP, sum of the terms equation distant from the beginning and end is always same and is equal to the sum of the first and last term.

n^{th} term from beginning = $t_n = a+(n-1)d$

nth term from end $t_n' = l-(n-1)d$

where a = First term, l = last term

$$tn+tn'=a+(n-1)d+l-(n-1)d$$

$$tn+tn'=a+l$$

for n =1 $t_1+t_1'=a+l$

for n =2 $t_2+t_2'=a+l$

Illustration 4: Find four numbers in A.P. such that their sum is 50 and greatest of them is 4 times the least.

Solution : Let the four numbers in *A.P.* be $\alpha-3\beta$, $\alpha-\beta$, $\alpha+\beta$, $\alpha+3\beta$.

Given $\alpha-3\beta+\alpha-\beta+\alpha+\beta+\alpha+3\beta=50$

or $4\alpha=50 \quad \therefore \alpha=\frac{25}{2}$

and $\alpha+3\beta=4(\alpha-3\beta)$ or $3\alpha=15\beta$

or $\beta=\frac{\alpha}{5}=\frac{25}{5\times2}=\frac{5}{2}$

Hence, the four numbers are 5, 10, 15, 20.

Illustration 5: If the numbers $\frac{1}{b+c},\frac{1}{c+a},\frac{1}{a+b}$ form an A.P., then prove that a^2, b^2, c^2 are also in A.P.

Solution : $\frac{1}{b+c},\frac{1}{c+a},\frac{1}{a+b}$ are in A.P.

$\Rightarrow (c+a)(a+b), (b+c)(a+b), (b+c)(c+a)$ are in A.P.

[by multiplying by $(b+c)(c+a)(a+b)$]

$\Rightarrow (a^2+ab+bc+ca), (b^2+ba+bc+ac), (c^2+ca+cb+ab)$ are in A.P.

$\Rightarrow a^2, b^2, c^2$ are iun A.P. (subtracting ab + bc + ca from all the three terms)

2. GEOMETRIC PROGRESSION

A succession of terms is said to be in Geometric Progression if the ratio of a term to its preceding term is independent of the position of the term. Geometric Progression is abbreviated as G.P.

e.g., 3, 15, 75, 375,.....;

t, $t/3$, $t/9$, $t/27$, ...; *etc*.

We define

$$\text{Common ratio} = \frac{\text{value of a term}}{\text{value of its preceding term}}$$

In the above examples, common ratio (c.r.) is 5 and 1/3 respectively.

The first term is usally denoted by 'a' and the common ratio by 'r' .So , a G.P. can be written as a, ar, ar^2, ar^3, ...

So, k^{th} term of a G.P. is given by

$t_k = ar^{k-1}$ (General term)

For a G.P. consisting of n terms, the last term is

$l = t_n = ar^{n-1}$

Illustration 6: If the fifth term of a G.P. is 81 and second term is 24, find the G.P.

Solution : Here $t_5 = 81$ and $t_2 = 24$

$\therefore t_n = ar^{n-1}$ $\therefore$ $81 = ar^4$ ....(i)

and $24 = ar^{2-1}$ or $24 = ar$ ....(ii)

dividing (i) by (ii), we get

$$\frac{ar^4}{ar} = \frac{81}{24} = \frac{27}{8} \quad \text{or} \quad r^3 = \left(\frac{3}{2}\right)^3$$

$$\therefore \quad r = \frac{3}{2}$$

Putting the value of r in (ii), we get $a = \frac{24}{r} = 24.\frac{2}{3} = 16$

Hence the required G.P. is 16, 24, 36, 54,.....

2.1 SUM OF TERM IN G.P.

Sum to n terms of a G.P. is given by

$$S_n = \frac{a(1-r^n)}{1-r} \text{ provided } r \neq 1$$

Student Task

Show that the sequence (f_n)defined by tn = 2.3^n+1 is not a G.P

The sum formula shows that if the magnitude of r is less than 1, r^n decreases as n increase. If n becomes very large, *i.e.*, an $n \to \infty$, $r^n \to 0$. So, if the number of terms is very large, we can get the sum of a G.P. to infinite terms, as a finite quantity.

$$\therefore \qquad S_\infty = \frac{a}{1-r} \qquad \text{if} \quad |r| < 1$$

Proof :

Let $S_n = a + ar + ar^2 + \ldots\ldots\ldots\ldots\ldots + ar^{n-1}$ ...(i)

Consider two cases : $r = 1$, $r \neq 1$

If $r = 1$, $S_n = \underbrace{a + a + a + \ldots\ldots\ldots + a}_{n\,times} = na$

If $r \neq 1$, multiplying (i) by r, we have

$rS_n = ar + ar^2 + ar^3 + \ldots\ldots\ldots + ar^n$ .....(ii)

Subtracting (ii) from (i), we get

$$(1-r)S_n = a - ar^n \Rightarrow S_n = \frac{a\left(1 - r^n\right)}{1 - r}$$

For convenience, we write $S_n = \dfrac{a\left(1 - r^n\right)}{1 - r}$ if $|r| > 1$.

$$\therefore \qquad S_n = \begin{cases} \dfrac{a(1-r^n)}{1-r} & if \qquad -1 \le r < 1 \\ na & if \qquad r = 1 \\ \dfrac{a(r^n-1)}{r-1} & if \quad |r| > 1 \, i.e., r < -1 \, or \, r > 1 \end{cases}$$

What happens if r = –1 ?

In this case the G.P. becomes a, $-a$, a, $-a$, ...

So the sum depends on the number of terms.

$S_n = 0$ if n = 2, 4, 6, ...*i.e.*, when n is odd.

The formula $S_n = \dfrac{a(1-r^n)}{1-r}$ holds in this case.

Let us apply the S_n formula for the series with

a = 10 and $r = -\dfrac{1}{2}$

$$\therefore S_n = \frac{a(1-r^n)}{1-r} = \frac{(10)\left[1 - \left(-\frac{1}{2}\right)^n\right]}{1 - \left(-\frac{1}{2}\right)}$$

The quantity $\left(-\dfrac{1}{2}\right)^n$ becomes nearly zero if n is very large.

$$\therefore S_\infty = \frac{10(1-0)}{1+\frac{1}{2}} = \frac{20}{3}$$

So, the sum of infinite number of terms is finite.

Therefore, a G.P. can be summed till infinity if $|r|<|$.

(because, as $n \to \infty$, $r^n \to 0$)

Hence, $S_\infty = \frac{a}{1-r}$, if $|r| < 1$ *i.e.*, $-1 < r < 1$.

Illustration 7: Find the least value of n for which $1 + 3 + 3^2 + ...+ 3^{n-1} > 1000$.

Solution : Given, $1 + 3 + 3^2 + ...+ 3^{n-1} > 1000$

$$\text{or } 1\left(\frac{3^n-1}{3-1}\right) > 1000$$

or $3^n - 1 > 2000$ or $3^n > 2001$ ....(i)

Now $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, $3^5 = 243$, $3^6 = 729$, $3^7 = 2187$, clearly $2187 > 2001$,

$\therefore$ least value of n for which $3^n > 2001$ is 7.

Illustration 8: If s is the sum to infinity of a G.P. whose first term is a then prove that the sum of the first n terms is $\left[1-\left(1\frac{a}{s}\right)^n\right]s$.

Solution : If r be the common ratio, we have $s = \frac{a}{1-r}$

$$\therefore r = 1-\frac{a}{s}$$

$$S_n = a\left[\frac{1-r^n}{1-r}\right] = a\left[\frac{1-\left(1-\frac{a}{s}\right)^n}{1-\left(1-\frac{a}{s}\right)}\right] = \frac{a\left[1-\left(1-\frac{a}{s}\right)^n\right]}{\frac{a}{s}}$$

$$S_n = \left[1-\left(1-\frac{a}{s}\right)^n\right]s$$

2.2 SOME RESULTS FOR G.P

1. If a, b, c are in G.P., then for k ($\neq 0$)

   ak, bk, ck are also in G.P. $a/k, b/k, c/k$ are also in G.P.

2. If a_1, a_2, a_3, ... and b_1, b_2, b_3; ..... are two G.P.'s, then the sequence a_1b_1, a_2b_2, a_3b_3; ..... is also in G.P.

3. Three terms in a G.P. are taken as $\frac{a}{r}$, a, ar.

4. Four terms in a G.P. are taken as $\frac{a}{r^3}, \frac{a}{r}, a, ar^3$.

5. If a, b, c and d are in G.P., then $\frac{a}{b}=\frac{b}{c}=\frac{c}{d} \Rightarrow b^2 = ac, c^2 = bd, ad = bc$.

6. Given $a_1, a_2, a_3, \ldots\ldots\ldots\ldots$ ($a_i > 0 \ \forall \ i$) are G.P. then $\log a_1, \log a_2, \log a_3, \ldots\ldots\ldots\ldots$ are in A.P. an vice versa.

Illustration 9: If the product of three number in G.P. be 216 and their sum is 19, find the numbers.

Solution : Let the three numbers be $\frac{a}{r}$, a, ar

Given, $\frac{a}{r} . a . ar = 216$

or $a^3 = 6^3 \quad \therefore \ a = 6$

Also $\frac{a}{r} + a + ar = 19$

$\therefore \frac{6}{r} + 6 + 6r = 19$

or $6r^2 - 13r + 6 = 0$ or $6r^2 - 9r - 4r + 6 = 0$

or $3r(2r - 3) - 2(2r - 3) = 0$ or $(3r - 2)(2r - 3) = 0$

$\therefore r = \frac{2}{3}, \frac{3}{2}$

When $r = \frac{2}{3}$, numbers will be 9, 6 and 4

When $r = \frac{3}{2}$, numbers will be 4, 6, 9. Hence the number are 4, 6, 9.

3. HARMONIC PROGRESSION

The sequence $a_1, a_2, a_3, \ldots\ldots\ldots, a_n$ is said to be a Harmonic progression (H.P.) if the sequence formed by the reciprocals, $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots\ldots \frac{1}{a_n}$ is an A.P.

Common difference of this A.P. $= \frac{1}{a_2} - \frac{1}{a_1} = \frac{a_1 - a_2}{a_2 a_1} = d$

and t_n of this A.P. $= \frac{1}{a_1} + (n-1)d$

Hence the nth term of the H.P. $= \frac{1}{a + (n-1)d}$, where $a = \frac{1}{a_1}$

Illustration 10: If *a, b, c* are in H.P., then show that $\frac{a}{b+c},\frac{b}{c+a},\frac{c}{a+b}$ are in H.P.

Solution : *a*, *b*, *c* are in H.P. $\Rightarrow \frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P.

$\Rightarrow \frac{a+b+c}{a},\frac{a+b+c}{b},\frac{a+b+c}{c}$ are in A.P. (Multiplying by $a + b + c$)

$\Rightarrow 1+\frac{b+c}{a},1+\frac{b+c}{b},1+\frac{b+c}{c}$ are in A.P.

$\Rightarrow \frac{b+c}{a},\frac{b+c}{b},\frac{b+c}{c}$ are in A.P. (Subtracting 1 from each)

$\Rightarrow \frac{a}{b+c},\frac{b}{c+a},\frac{c}{a+b}$ are in H.P.

Illustration 11: If the 7th term of a H.P. is 8 and the 8th term is 7, them find its 15th term.

Solution : t_7 of H.P. = 8 and t_8 of H.P. = 7

$\Rightarrow \frac{1}{a+6d}=8,\frac{1}{a+7d}=7$

Solving , $d=\frac{1}{56}=a$

$\therefore\ T_{15}$ of H.P. is $\frac{1}{a+14d}=\frac{56}{15}$

Can you find the 10th term of an H.P. having first two terms as $\frac{1}{5}$ and $\frac{1}{8}$?

Student Task

The following questions deal with the basic concepts of this section. Answer the following briefly . Go to the next section only if your score is greater than 80% . Do not consult the study material while attempting the questions.

1. Find the 15th term of sequence $x - 7, x - 2, x + 3, \ldots\ldots\ldots\ldots$
2. Find the sum to 15 terms of the sequence $x - 2, x + 3, x + 8, \ldots\ldots\ldots$ at $x = 10$.
3. Three numbers in A.P. have sum 24. The middle term is .....
4. The common ration of a G.P. having 10th term and 1st term equal to 1536 and –3 is .....
5. The sum to 10 terms of this G.P. is .....
6. Any geometric series can be summed till infinite number of terms by the formula $S_\infty=\frac{a}{1-r}$.

   (True / False)
7. If $a_1, a_2, \ldots\ldots\ldots\ldots a_n$ is an A.P. then *log* a_1. *log* a_2, ... *log* a_n is a G.P. (True / False)
8. The common ratio of a G.P. whose product of four consecutive terms is 81, third of these consecutive terms being 27, is 9. (True / False)

9. The first term of H.P., having k^{th} term equal to $\frac{1}{a+(k-1)d}$ is a. (True / False)

10 If a, b, c, d are in H.P., then $ab + bc + cd = 3ad$. (True / False)

Answer Key :

1. $x + 63$, 2. 645, 3. 8, 4. –2, 5. 1023, 6. False, 7. False 8. False,
9. False, 10. True

4. ARITHMETIC MEANS

If we are given two numbers, say a and b, then the number A which makes the progression a, A, b an arithmetic progression is called the Arithmetic mean of a and b.

Clearly, $A - a = b - A$

$$\Rightarrow A = \frac{a+b}{2}$$

Here we have inserted only one number between a and b. We can also insert n numbers (say A_1, A_2, ...A_n) between a and b such that the progression a, A_1, A_2, ..., A_n, b becomes arithmetic . There values can be obtained in terms of a and b as follows:

Consider the progression

a, A_1, A_2, ..., A_n, b

Here, first term = a, last term = b and number of terms = $n + 2$

Let d = common difference of A.P.

$\therefore\ b = a + (n + 2 - 1)d$

$$\Rightarrow d = \frac{b-a}{n+1}$$

So, $A_1 = a + d$ $\quad = a + \frac{b-a}{n+1}$

$A_2 = a + 2d$ $\quad = a + 2\frac{b-a}{n+1}$

$A_n = a + nd$ $\quad = a + n\frac{b-a}{n+1}$

So, the kth arithmetic mean is given by

$$A_k = a + \frac{k(b-a)}{n+1} \qquad \text{where } k = 1, 2, ..., n$$

e.g., the 5th of the 7 arithmetic means between the numbers 18 and 90 is given by

$$A_5 = 18 + \frac{5(90-18)}{7+1} = 63$$

If we have n numbers a_1, a_2, ... a_n then arithmetic mean of these numbers is defined as

$$A = \frac{a_1 + a_2 + ... + a_n}{n}$$

Note: The sum of n AM'S between two given quantites is equal to n times the AM between then. Sum of n AM'S between a and b $A_1 + A_2 + ... + A_n$

$= \frac{n}{2}(A_1 + A_n)$

$= \frac{n}{2}(a+b)$

Illustration 12: If n arithmetic means are inserted between 20 and 80 such that first mean : last mean = 1: 3, find n.

Solution : Given $A_1 : A_n = 1 : 3$

$$\Rightarrow \frac{20 + \frac{(1)(80-20)}{n+1}}{20 + \frac{(n)(80-20)}{n+1}} = \frac{1}{3} \qquad \Rightarrow \frac{20(n+1)+60}{20(n+1)+60n} = \frac{1}{3}$$

$$\Rightarrow \frac{n+4}{4n+1} = \frac{1}{3} \qquad \Rightarrow n = 11$$

Student Task

(a) If 11 AM'S are inserted between 28 and 10 then find the three middle terms in the series

(b) There are n AM'S between 3 and 54 such that 8th mean is to $(n-2)^{th}$ mean as 3 to 5, Find n.

5. GEOMETRIC MEANS

The geometric mean (G.M.) between two numbers a and b is the number G such that a, G, b are in G.P.

Hence $\frac{G}{a} = \frac{b}{G}$ or $G^2 = ab$ or $G = \sqrt{ab}$

The geometric mean of n positive numbers $a_1, a_2, ..., a_n$ is given by $G = \sqrt[n]{a_1 a_2 .... a_n}$

If $G_1, G_2, G_3, ... G_n$ be n geometric means (G.M.s) between a and b, then $a, G_1, G_2, ...G_n, b$ are in G.P.

If r be the common ratio of this sequence, then $b = t_{n+2} = ar^{n+1}$.

Hence $r^{n+1} = \frac{b}{a}$ and $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

The kth geometric mean is given as $G_k = a\left(\frac{b}{a}\right)^{\frac{k}{n+1}}$

This enables us to find $G_1, G_2, ...G_n$.

Note: The product of n geometric means between a and b is equal to the nth power of the geometric mean between a and b.

Illustration 13: Show that the product of *n* geometric means inserted between two positive quantites is equal to the nth power of the single geometric mean inserted between them.

Solution : Let $x_1, x_2, \ldots\ldots x_n$ be the *n* G.M.'s between two given positive quantities *a* and *b*, then

$$x_1 = a\left(\frac{b}{a}\right)^{\frac{1}{n+1}}, x_2 = a\left(\frac{b}{a}\right)^{\frac{2}{n+1}}, x_3 = a\left(\frac{b}{a}\right)^{\frac{3}{n+1}}, x_n = a\left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

$$\text{Now } x_1 x_2 \ldots\ldots\ldots\ldots x_n = a^n . \left(\frac{b}{a}\right)^{\frac{1}{n+1}+\frac{2}{n+1}+\ldots\ldots\ldots+\frac{n}{n+1}}$$

$$= a^n \left(\frac{b}{a}\right)^{\frac{1+2+\ldots\ldots+n}{n+1}} \qquad = a^n . \left(\frac{b}{a}\right)^{\frac{n(n+1)}{2(n+1)}}$$

$$= a^n . \left(\frac{b}{a}\right)^{\frac{n}{2}} = a^{n-\frac{n}{a}} (b)^{\frac{n}{2}} = (ab)^{\frac{n}{2}} = \left(\sqrt{ab}\right)^n$$

= (G.M. of *a* and *b*)^n

[$\because$ G.M. of a and b = $\sqrt{ab}$]

6. HARMONIC MEANS

Numbers $H_1, H_2, \ldots\ldots, H_n$ are said to be the *n* H.M.'s between two numbers *a* and *b* if $a, H_1, H_2, \ldots\ldots, H_n$ are in H.P.

For example $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ are the H.M.'s between 1 and $\frac{1}{5}$ because $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ are in H.P.

6.1 SINGLE HARMONIC MEAN BETWEEN TWO GIVEN QUANTITIES

Let *a* and *b* be the two given quantities and *H* be the H.M. between them, then *a, H*, *b* will be in H.P.

$\Rightarrow \frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ will be in A.P. $\Rightarrow \frac{1}{H} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H}$ *or* $\frac{2}{H} = \frac{1}{a} + \frac{1}{b}$

or $\frac{2}{H} = \frac{b+a}{ab} \qquad \because H = \frac{2ab}{a+b}$

Hence H.M. of a and b $= \frac{2ab}{a+b}$

6.2 n HARMONIC MEANS BETWEEN TWO GIVEN QUANTITIES

Let $H_1, H_2, H_3, \ldots\ldots\ldots\ldots H_n$ be the *n* H.M.'s between two given quantities *a* and *b*. *d* be the *c*, *d* of the corresponding A.P.

Then , $a, H_1, H_2, H_3, \ldots\ldots\ldots H_n$, *b* will be in H.P.

$\therefore \frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \ldots\ldots\ldots \frac{1}{H_n}$, will be in A.P.

Now $\frac{1}{b} = (n+2)^n$ term of A.P.

$$= \frac{1}{a} + (n+2-1)d \qquad [\because\ d = c.\ d.\ \text{of}\ A.P.]$$

$$\therefore d = \frac{\frac{1}{b} - \frac{1}{a}}{n+1} = \frac{a-b}{ab(n+1)}$$

Now, $\frac{1}{H_1} = 2^{nd}$ term of A.P. $= \frac{1}{a} + d$

$$= \frac{1}{a} + \frac{a-b}{ab(n+1)} = \frac{b(n+1)+a-b}{ab(n+1)} = \frac{bn+a}{ab(n+1)}$$

$$\therefore H_1 = \frac{ab(n+1)}{bn+a}$$

$$\frac{1}{H_2} = \frac{1}{a} + 2d = \frac{1}{a} + \frac{2(a-b)}{ab(n+1)} = \frac{2a+(n-1)b}{ab(n+1)}$$

$$\therefore H_2 = \frac{ab(n+1)}{2a+(n+1)b}$$

.....

$$\frac{1}{H_n} = \frac{1}{a} + nd = \frac{1}{a} + \frac{n(a-b)}{ab(n+1)} = \frac{na+b}{ab(n+1)}$$

$$\therefore H_n = \frac{ab(n+1)}{na+b}$$

Thus the n H.M.'s between a and b are

$$\frac{ab(n+1)}{a+nb}, \frac{ab(n+1)}{2a+(n-1)b}, \frac{ab(n+1)}{3a+(n-2)b}, \ldots\ldots, \frac{ab(n+1)}{na+b}$$

Illustration 14: Insert 6 harmonic means between 3 and $\frac{6}{23}$.

Solution : Let $x_1, x_2, x_3, \ldots\ldots x_6$ be the six H.M.'s between 3 and $\frac{6}{23}$

Then , 3 $x_1, x_2, \ldots\ldots x_6, \frac{6}{23}$ will be in H.P.

$$\therefore \frac{1}{3}, \frac{1}{x_1}, \frac{1}{x_2}, \ldots\ldots \frac{1}{x_6}, \frac{23}{6} \text{ are in A.P.}$$

Now $\frac{23}{6} = 8^{th}$ term of A.P. $\frac{1}{3} + 7d$, where $d = c.d.$ of A.P.

$$\therefore d = \frac{1}{2}$$

Now $\frac{1}{x_r} = (r+1)^{th}$ term of A.P.

$= \frac{1}{3} + rd = \frac{1}{3} + \frac{r}{2}\left[\because d = \frac{1}{2}\right]$

$\therefore x_r = \frac{6}{2+3r}$

Putting r = 1, 2, 3 ..... 6, we get

$x_1 = \frac{6}{5}, x_2 = \frac{6}{8}, x_3 = \frac{6}{11}, x_4 = \frac{6}{14}, x_5 = \frac{6}{17}, x_6 = \frac{6}{20}$

Hence the six H.M.'s are $\frac{6}{5}, \frac{3}{4}, \frac{6}{11}, \frac{3}{7}, \frac{6}{17}, \frac{3}{10}$

What is the harmonic mean of 0 and 5 ?

7. RELATION BETWEEN A.M., G.M. AND H.M. BETWEEN TWO REAL AND UNEQUAL QUANTITIES

Let a and b be two real positive and unequal quantities and A, G and H be the single A.M., G.M. and H.M. respectively between them.

Then $A = \frac{a+b}{2}, G = \sqrt{ab}, H = \frac{2ab}{a+b}$

Now $AH = \frac{a+b}{2}.\frac{2ab}{a+b} = ab = G^2$ $\therefore \frac{G}{A} = \frac{H}{G}$ .....(i)

Hence A, G, H are in G.P.

Again, $A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2}$

$= \frac{\left(\sqrt{a}\right)^2 + \left(\sqrt{b}\right)^2 - 2\sqrt{a}.\sqrt{b}}{2}$

$= \frac{\left(\sqrt{a} - \sqrt{b}\right)^2}{2} > 0$ $[\because a \neq b]$

Thus $A - G > 0$ or $A > G$ .....(ii)

From (i) $\frac{H}{G} = \frac{G}{H}$ .....(iii)

From (ii) , $\frac{G}{A} < 1 \quad \therefore \frac{H}{G} < 1 \Rightarrow H < G \Rightarrow G > H$ .....(iv)

From (i) and (iv), we get $A>G>H$

Thus A.M., G.M. and H.M. between two real positive and unequal quantities are in G.P. and *A.M. > G.M. > H.M.*

Note :

1. $A-G=\frac{1}{2}\left(\sqrt{a}-\sqrt{b}\right)^2=0$ $\quad\therefore A=G \Leftrightarrow a=b$

   Hence $A=G=H \Leftrightarrow a=b$

2. The relation $A \geq G \geq H$ holds for n positive real number also.

Illustration 15: The arithmetic mean of two numbers exceed their geometric mean by $\frac{3}{2}$ and the geometric mean exceed their harmonic mean by $\frac{6}{5}$, find the numbers.

Solution : Let the two numbers be a and b. A, G, H be the A.M., G.M., and H.M. respectively between a and b.

Then , $A=\frac{a+b}{2}; G=\sqrt{ab}$ and $H=\frac{2ab}{a+b}$ ....(i)

Given , $A=G+\frac{3}{2}$ and $\frac{S_3(1+8S_1)}{S_2^2}$ or $H=G-\frac{6}{5}$ ....(ii)

$\because$ $A.H. = G^2 \therefore \left(G+\frac{3}{2}\right)\left(G-\frac{6}{5}\right)=G^2$

or $(2G+3)(5G-6)=10G^2$ or $3G=18$ $\quad\therefore G=6$

From (ii), $A=6+\frac{3}{2}=\frac{15}{2}$

$\therefore \frac{a+b}{2}=\frac{15}{2}$ or $a+b=15$ ....(iii)

Also $G=6$ $\therefore \sqrt{ab}=6$ or $ab=36$ ....(iv)

Now $(a-b)^2=(a+b)^2-4ab=15^2-4.36=81$

$\therefore a-b=\pm 9$ .....(v)

from (iii) and (v), $a=12, b=3$ or $a=3, b=12$

Hence the number are 3 and 12.

Illustration 16: If $x>0$, prove that $x+\frac{1}{x}\geq 2$.

Solution $\because$ A.M. $\geq$ G.M.

$\therefore \frac{x+\frac{1}{x}}{2}\geq\sqrt{x.\frac{1}{x}} \Rightarrow \frac{x+\frac{1}{x}}{2}\geq 1 \Rightarrow x+\frac{1}{x}\geq 2$

Note : The least value of the sum of a positive number and its reciprocal is 2.

Illustration 17: If a, b, c, d be four distinct positive quantities in H.P., show that

(i) $a+d>b+c$ (ii) $ad>bc$

Solution : Since a, b, c, d are in H.P.

$\therefore$ a, b, c are in H.P. and b, c, d are also in H.P.

$\therefore$ b = H.M. of a and c

....(i)

and c = H.M. of b and d

....(ii)

Now A.M. of a and $c = \frac{a+c}{2}$

....(iii)

and G.M. of a and $c = \sqrt{ac}$

...(iv)

A.M. of b and $d = \frac{b+d}{2}$ (v) and G.M. of b and $d = \sqrt{bd}$

$\because$ A.M. > H.M. $\therefore \frac{a+c}{2} > b$ [From (i) and (iii)]

or $a + c > 2b$

From (ii) and (v) $\frac{b+d}{2} > c$, or $b + d > 2c$

Adding, we get $a + b + c + d > 2b + 2c$ or $a + d > b + c$

since G.M. > H.M.

$\therefore \sqrt{ac} > b$ or $ac > b^2$

Similarly. $\sqrt{bd} > c$ or $bd > c^2$

Multiplying we get $abcd > b^2c^2$ or $ad > bc$

Student Task

The following questions deal with the basic concepts of this section. Answer the following briefly . Go to the next section only if your score is greater than 80%. Do not consult the study material while attempting the question.

1. Insert 5 A.M.s between 10 and 100.
2. Insert 4 G.M.s between 5 and 160.
3. Insert 6 H.M.s between 3 and $\frac{6}{23}$.
4. Find the A.M., G.M. and H.M. of the numbers 2, 5, 1, 3.
5. Find the minimum value of $\tan\theta + \cot\theta$, $0 < \theta < \frac{\pi}{2}$.
6. Prove that for positive numbers a and b, $(a + b)\left(\frac{1}{a}+\frac{1}{b}\right) \geq 4$.
7. Prove that $\sqrt[3]{xyz}\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right) \geq 3$ where $x, y, z, \in \boldsymbol{R}^+$.

8. The A.M. of two numbers exceeds their G.M. by 2 and the G.M. exceeds their H.M. by $\frac{8}{5}$, find the numbers.

9. Show that the sum of the arithmetic means between two given numbers, equidistant from beginning and the end is constant.

10. If A.M. > G.M. > H.M. , then the numbers are not all equal. (True / False)

Answer Key :

1. 25, 40, 55, 70, 85, 2. 10, 20, 40, 80, 3. $\frac{6}{5},\frac{3}{4},\frac{6}{11},\frac{3}{7},\frac{6}{17},\frac{3}{10}$, 4. $\frac{11}{4},\sqrt[4]{30},\frac{120}{61}$

5. 2 8. 16, 4, 10. True

8. ARITHMETICO-GEOMETRIC SERIES (A.G.S.)

Suppose $a_1, a_2, a_3,.....a_n$, is an A.P. and $b_1, b_2, b_3, ...b_n$ is a G.P., then the sequence $a_1b_1, a_2b_2, a_3b_3,.....a_nb_n$ is called an arithmetico-geometric sequence.

The standard form of the sequence is

$a, (a + d)r, (a + 2d)r^2, ..., \{a + (n - 1)d\}r^{n-1}$

SUM OF *n* TERMS OF AN A.G.P.

Let $S_n = a + (a + d) r + (a + 2d) r^2 +.....+ \{a + (n - 1) d\} r^{n-1}$

Multiply both sides by r and subtract the second equation from the first,

$r . S_n = ar + (a + d) r^2 + ... + \{ a + (n - 2)d\} r^{n-1} + \{a + (n - 1)d\}r^n$

$S_n (1 - r) = a + dr + dr^2 + ... + dr^{n-1} - \{a + (n - 1)d\} r^n$

$$= a + dr\frac{(1-r^{n-1})}{(1-r)} - \{a+(n-1)d\}r^n$$

$$\therefore S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{\{a+(n-1)d\}r^n}{1-r}$$

$$= \frac{a}{1-r} + \frac{dr}{(1-r)^2} - \frac{dr^n}{(1-r)^2} - \frac{\{a+(n-1)d\}r^n}{1-r}$$

If $|r| < 1$, $r^n \to 0$ as $n \to \infty$ and therefore

$$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

Illustration 18: Find $1 + 2.2 + 3.2^2 + 4.2^3 + ... + 100.2^{99}$.

Solution : Here $a = 1$, $d = 1$, $r = 2$ and $n = 100$

$$\therefore S_{100} = \frac{1}{1-2} + \frac{(1)(2)[1-(2)^{100-1}]}{(1-2)^2} - \frac{\{1+(100-1)(1)\}}{1-2}$$

$= -1 + 2 - 2^{100} + 2^{100} + 99.2^{100}$

$= 99.2^{100} + 1.$

9. SUMMATION OF SERIES

9.1 SUM OF THE FIRST N NATURAL NUMBERS

$1 + 2 + 3 + \ldots\ldots\ldots\ldots + n = \sum_{r=1}^{n} r = \frac{n(n+1)}{2}$. This is evident as the numbers are in A.P.

9.2 SUM OF THE SQUARES OF THE FIRST N NATURAL NUMBERS

$1^2 + 2^2 + 3^2 + \ldots\ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Proof : Consider the identity $(x + 1)^3 - x^3 = 3x^2 + 3x + 1$

Putting $x = 1, 2, 3, \ldots\ldots\ldots\ldots\ldots, n$, we have

$(n + 1)^3 - n^3 = 3n^2 + 3n + 1$

$n^3 - (n - 1)^3 = 3(n - 1)^2 + 3(n - 1) + 1$

.....

$2^3 - 1^3 = 3.1^2 + 3.1 + 1$

Adding $(n + 1)^3 - 1^3 = 3\sum_{r}^{n} r^2 + 3\sum_{1}^{n} r + n$

Hence $3\sum_{r}^{n} r^2 = (n^3 + 3n^2 + 3n) - \frac{3n(n+1)}{2} - n$

$\therefore \sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$

9.3 SUM OF THE CUBES OF THE FIRST N NATURAL NUMBERS

$1^3 + 2^3 + \ldots\ldots\ldots + n^3 = \sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$

This can be proved by using the identity $(x + 1)^4 - x^4 \equiv 4x^3 + 6x^2 + 4x + 1$

Note : $\sum n^3 = \left(\sum n\right)^2 = \frac{n^2(n+1)^2}{4}$

Certain series can be summed up if each term of the series can be expressed as the difference of two consecutive terms of another series known as auxiliary series. This is illustrated in the solved examples ahead.

Illustration 19: Find the sum of n terms of the series whose nth term is $12n^2 - 6n + 5$.

Solution : Let $t_n = 12n^2 - 6n + 5$

$$\therefore S_n = \sum_{n=1}^{n} t_n = \sum_{n=1}^{n} \left(12n^2 - 6n + 5\right) = \sum_{n=1}^{n} 12n^2 - \sum_{n=1}^{n} 6n + \sum_{n=1}^{n} 5 = 12\sum_{n=1}^{n} n^2 - 6\sum_{n=1}^{n} n + 5n$$

$$= 12\frac{n(n+1)(2n+1)}{6} - \frac{6n(n+1)}{2} + 5n$$

$= 2n\,(2n^2 + 3n + 1) - 3n\,(n + 1) + 5n$

$= 4n^3 + 3n^2 + 4n$

Illustration 20: Find the sum of the series $1.n + 2.(n-1) + 3.(n-2) + \ldots\ldots + n.1$.

Solution : Here n occurs in each term, therefore, for general term we will find the rth term

Since r^{th} term of the sequence 1, 2, 3, ... $= 1 + (r-1).1$

and r^{th} term of the sequence $n, n-1, n-2, \ldots\ldots = n + (r-1)(-1)$

Now r^{th} term $t_r = \{1 + (r-1).1\}\{n + (r-1)(-1)\}$

$= r(n - r + 1) = nr - r^2 + r$

$$\therefore S_n = \sum_{r=1}^{n} t_r = \sum_{r=1}^{n} \left(nr - r^2 + r\right)$$

$$= n\sum_{r=1}^{n} r - \sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r = \frac{n(n+1)}{2}\left(n - \frac{2n+1}{3} + 1\right)$$

$$= \frac{n(n+1)}{2} \cdot \frac{(3n - 2n - 1 + 3)}{3} = \frac{n(n+1)(n+2)}{6}$$

Student Task

Can you find the 1000th term of the sequence 1, 2, 2, 3, 3, 3, 4, 4, 4, 4,.....?

Illustration 21: Find the nth term and the sum to n terms of the series

$1 + 6 + 23 + 58 + 117 + 206 + \ldots\ldots$

Solution : The terms of the series 1

6 23 58 117 206 .....

u_1 u_2 u_3 u_4 u_5 u_1 .....

The 1st order of $(u_2 - u_1)$ $(u_3 - u_2)$ $(u_4 - u_3)$ $(u_5 - u_4)$ $(u_6 - u_5)$.....

of difference

5 17 35 59 89 .....

v_2 v_2 v_3 v_4 v_5

The 2nd order of $(v_2 - v_1)$ $(v_3 - v_2)$ $(v_4 - v_3)$ $(v_5 - v_4)$

differences

12 18 24 30

The 3rd order of 6 6 6

difference

It may be noted that the terms of the successive order of difference series are obtained from the immediately preceding series by taking the difference of two consecutive terms.

If at any stage , in finding the successive order of difference series terms, all the terms reduce to the same number (as in this problem, this happens at the 3rd order of differences), then the nth term u_n of the given series is a polynomial in n of degree equal to that order of difference series whose terms are all the same. Thus in this problem u_n is of degree 3 and hence u_n can be taken as either $an^3 + bn^2 + cn + d$ or conveniently as $u_n = An(n-1)(n-2) + Bn(n-1) + Cn + D$

The constant A, B, C, D are determined as follows

put $n = 1$, $u_1 = 1$ and $u_1 = C + D$ $\therefore$ $C + D = 1$

put $n = 2$, $u_2 = 6$ and $u_2 = 2B + 2C + D$

$\therefore$ $2B + 2C + D = 6$

put $n = 3$, $u_2 = 23$ and $u_3 = 6A + 6B + 3C + D$

$\therefore$ $6A + 6B + 3C + D = 23$

put $n = 4$, $u_4 = 58$ and $u_4 = 24A + 12B + 4C + D$ $\therefore$ $24A + 12B + 4C + D = 58$

Solving for A, B, C, D, we get $A = 1$, $B = 3$, $C = -1$, $D = 2$

Hence $u_n = n(n-1)(n-2) + 3n(n-1) - n + 2$

$= n_3 - 2n + 2$

$$\therefore \sum_{n=1}^{n} u_n = \sum n^3 - 2\sum n + 2\sum 1$$

$$= \frac{n^2(n+1)^2}{4} - \frac{2n(n+1)}{2} + 2n$$

$$= \frac{n}{4}\left\{n^3 + 2n^2 - 3n + 4\right\}$$

Illustration 22: Sum to n terms the series 1 + 4 + 11 + 26 + 57 + 120.

Solution : 1 4 11 26 57 120

First order of differences

3 7 15 31 63

Second order of differences

4 8 16 32

The 2nd order of differences is a G.P. of common ratio 2.

In this case, $u_n = A$ (common ratio)^n + (first degree polynomial in n since the 2nd order difference is a G.P.)

i.e., $u_n = A.2^{n-1} + Bn + C$

obtain A, B and C using u_1, u_2 and u_3.

then $S_n = \sum_{n=1}^{n} u_n$

$= A(2^0 + 2^1 + 2^2 + \ldots\ldots 2^{n-1}) + B\sum n + C\sum(1)$

$$= A(2^n - 1) + \frac{Bn(n+1)}{2} + Cn$$

If $t_n = U_n - U_{n-1}$, what is the value of $\sum_{n=1}^{m} t_n$

Student Task

The following questions deal with the basic concepts of this section. Answer the following briefly. Go to the next section only if your score is greater than 80%. Do not consult the study material while attempting the questions.

1. Find the 10th term of the sequence 1, 2.2, 3.2^2, ...
2. Obtain the sum to *n* terms of the corresponding series in the above sequence.
3. Can the series $1+2\left(\frac{1}{3}\right)+3\left(\frac{1}{3}\right)^2+4\left(\frac{1}{3}\right)^3+$..... be summed till infinity. If so, what is the sum? If not , why ?
4. The sum to *n* terms of the series whose k^{th} term is $12k^2 - 6k + 5$ is $4n^3 + 3n^2 + 4n$ ...(True / False)
5. Sum to 10 terms of the series 1.2.3 + 2.3.4 + 3. 4.5 + ... is .....
6. If $t_k = k(k!)$, then $\sum_{k=1}^{n} t_k$ is equal to .....
7. The series $\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+$..... can be summed till infinity .(True / False)

Answer Key :

1. $(10)2^9$, 2. $1 - (1 - n)2^n$, 3. Yes, 9/4, 4. True, 5. 4290 6. $(n + 1)! - 1$
7. True

11. APPLICATION TO PROBLEMS OF MAXIMA

(i) If $a_1 + a_2 + a_3 + ... + a_n = k$ (constant) the value of $a_1a_2a_3.....a_n$ is greatest when $a_1 = a_2 = ... = a_n$ so that the greatest value of $a_1a_2....a_n$ is $\left(\frac{k}{n}\right)^n$

(ii) If $a_1a_2a_3.....a_n = k$ (Constant) the value of $a_1 + a_2 + a_3 + ... + a_n$ is least when $a_1 = a_2 = a_3 = ... = a_n$ So that the least of $a_1 + a_2 + a_3 + ... + a_n$ is $n(k)^{\frac{1}{n}}$

An Important Result:

(a) **If** $a_i > 0, i = 1,2,3,....n$ which are not identical then $\frac{a_1^m + a_2^m ..... + a_n^m}{n} > \left(\frac{a_1 + a_2 + ... + a_n}{n}\right)^m$ it m < o or m > 1

(b) $\frac{a_1^m + a_2^m ..... + a_n^m}{n} < \left(\frac{a_1 + a_2 + ... + a_n}{n}\right)^m$ if $0 < m < 1$

MISCELLANEOUS EXAMPLES

Example 1: The fourth term of the sequence 3, $\frac{3}{2}$, 1,..... is

(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{2}{3}$

Solution : The given sequence is $\frac{3}{1},\frac{3}{2},\frac{3}{3}$, ..., therefore, the fourth term is $\frac{3}{4}$ (by symmetry).

Alternatively , the given sequence is an H.P.; the corresponding A.P. being

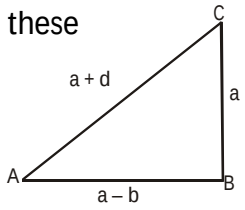
$\frac{1}{3}, \frac{2}{3}, 1$,..... whose C.D. = $\frac{1}{3}$

$\therefore$ Fourth term of this A.P. = $\frac{1}{3}+(4-1)\frac{1}{3}=\frac{4}{3}$ $\Rightarrow$ Fourth term of the given H.P. = $\frac{3}{4}$.

$\therefore$ (a)

Example 2 : If the sides of a right triangle are in A.P., then the sum of the sines of the two acute angles is

(a) $\frac{7}{5}$ (b) $\frac{4}{\sqrt{3}}$ (c) $\sqrt{\frac{\sqrt{5}-1}{2}}+\sqrt{\frac{\sqrt{5}+1}{2}}$ (d) none of these



Solution : Let the sides be $a - d$, a $a + d$ as shown ; $d > 0$,

then, $(a + d)^2 = a^2 + (a - d)^2$

$\Rightarrow a^2 = 4ad$ $\Rightarrow a = 4d$.

Hence, the three sides are $4d - d$, $4d$, $4d + d$ i.e. $3d$, $4d$, $5d$.

$\therefore \sin A + \sin C = \frac{a}{a+d}+\frac{a-d}{a+d}=\frac{4d}{5d}+\frac{3d}{5d}=\frac{7}{5}$.

$\therefore$ (a)

Example 3 : Sum of n terms of the series $\frac{1}{\sqrt{2}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{8}}+\frac{1}{\sqrt{8}+\sqrt{11}}+$ ..... is

(a) $\sqrt{3n+2}-\sqrt{2}$ (b) $\frac{1}{3}\left(\sqrt{3n+2}-\sqrt{2}\right)$ (c) $\sqrt{3n+2}+\sqrt{2}$ (d) None of these

Solution : Required sum $=\frac{1}{\sqrt{2}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{8}}+\frac{1}{\sqrt{8}+\sqrt{11}}+.....+\frac{1}{\sqrt{3n-1}+\sqrt{3n+2}}$

($\because$ n^{th} term of the sequence 2, 5, 8, ... is $3n - 1$ and that of 5, 8, 11, ... is $3n + 2$)

$=\frac{\sqrt{5}-\sqrt{2}}{5-2}+\frac{\sqrt{8}-\sqrt{5}}{8-5}+\frac{\sqrt{11}-\sqrt{8}}{11-8}+.....+\frac{\sqrt{3n+2}-\sqrt{3n-2}}{3n+2-(3n-1)}$

$\frac{1}{3}\left\{\sqrt{5}-\sqrt{2}+\sqrt{8}-\sqrt{5}+\sqrt{11}-\sqrt{8}+.....+\sqrt{3n+2}-\sqrt{3n-1}\right\}=\frac{1}{3}\left[\sqrt{3n+2}-\sqrt{2}\right]$

Example 4: If $|x| < 1$, then $1 + 3x + 5x^2 + 7x^2 +$ ..... to ∞ is equal to

(a) $\frac{1+x}{(1-x)^2}$ (b) $\frac{2+x}{(1-x)^2}$ (c) $\frac{x}{(1-x)^2}$ (d) None of these

Solution : Let

$S = 1 + 3x + 5x^2 + 7x^3 +$ ....(i)

then $x\,S = 1x + 3x^2 + 5x^3 +$ ....(ii)

Subtracting (ii) from (i), we get

$S - xs = 1 + 2x + 2x^2 + 2x^3 +$ ..... to ∞ $\Rightarrow S(1 - x) = 1\,(2x + 2x^2 + 2x^3 +$ ..... to ∞

$$\Rightarrow S(1-x) = 1 + \frac{2x}{1-x} = \frac{1-x+2x}{1-x} \Rightarrow S = \frac{1+x}{(1-x)^2}$$

$\therefore$ (a)

Example 5: The eleventh term of the sequence1, 1, 2, 3, 5, 8, 13, 21, 34, ... is

(a)89 (b) 66 (c) 72 (d) None of these

Solution : Each term, beginning with 3rd, is the sum of two pervious terms *i.e.*,

$T_3 = T_1 + T_2$, $T_4 = T_2 + T_3$, ... *etc*. $\therefore$ 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89.....

$\therefore$ (a)

Example 6 : If $a_1, a_2, a_3, ..., a_n$ are in A.P., where $a_i > 0$ for all i, show that

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + ... + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$$

Solution : $a_2 - a_1 = a_3 - a_2 = ... = a_n - a_{n-1} = d$ .....(i)

$$\text{L.H.S} = \frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + ... + \frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}$$

$$= \frac{1}{d}\left\{\sqrt{a_n}-\sqrt{a_1}\right\}$$ since the denominators are all equal to d.

$$\frac{1}{d}\left(\frac{a_n-a_1}{\sqrt{a_n}+\sqrt{a_1}}\right)$$ .....(ii)

But $a_n = a_1 + (n-1)\,d \;\therefore d = \left(\frac{a_n-a_1}{n-1}\right)$ .....(iii)

Substituting (iii) in (ii), we get L.H.S. $= \frac{n-1}{\sqrt{a_n}+\sqrt{a_1}}$ = R.H.S

Example 7: The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9), ... and so on. Show that the sum of the numbers in the nth group is $n^3 + (n-1)^3$.

Solution : The groups are (1), (2, 3, 4), (5, 6, 7, 8, 9), ...

The number of the terms in the groups are 1, 3, 5, ...

$\therefore$ the number of terms in the nth group = $(2n-1)$

The last term of the $(n-1)^{th}$ group = $1 + 3 + 5 + ... + (2n-3)$

$$= \left(\frac{n-1}{2}\right)\{2+(n-2)2\} = (n-1)^2 = n^2 - 2n + 1$$

$\therefore$ numbers in the last group are $n^2 - 2n + 2$, $n^2 - 2n + 3$,..... upto $(2n-1)$ terms.

$$\text{Sum of the numbers in the last group} = \left(\frac{2n-1}{2}\right)\left\{2(n^2-2n+2)+(2n-2)1\right\}$$

$= (2n-1)(n^2 - n + 1)$ $= 2n^3 - 3n^2 + 3n - 1 = n^3 + (n-1)^3$

Example 8: The sum of four numbers in G.P. is 60 and the arithmetic mean of the first and last numbers is 18. Find the numbers.

Solution : Let the four number be a, ar , ar^2, ar^3

$a(1 + r + r^2 + r^3) = 60$ ....(i)

$$\frac{a+ar^3}{2}=18 \quad \text{....(ii)}$$

$\therefore$ $a(1 + r)(1 + r^2)= 60$ and $a(1 + r)(1 – r + r^2) = 36$

Division gives $\frac{1+r^2}{1-r+r^2}=\frac{5}{3}$

$\therefore$ $2r^2 – 5r + 2 = 0$ giving $r = 2$ or ½

Putting $r = 2$ in equation (i), we get $a = 4$

$\therefore$ the numbers are 4, 8, 16, 32.

r = ½ gives the same four numbers .

Example 9: Sum to n terms the series : $(x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^3) +$.....

Solution : Let $S_n = (x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^2) +$ ..... upto n terms

Observe that multiplying both sides by $(x – y)$, the terms on R.H.S get reduced to

$x^2 – y^2$, $x^3 – y^3$, $x^4 – y^4$, ...

$\therefore (x – y)(S_n) = (x^2 – y^2) + (x^3 – y^3) + (x^4 + y^4) +....+ x^{n+1} – y^{n+1})$

$= (x^2 + x^3 + ...+ x^{n+1}) – (y^2 + y^3 + ...+ y^{n+1})$

$$= x^2\frac{(1-x^n)}{1-x}y^2\frac{(1-y^n)}{1-y} \qquad \therefore S_n=\frac{1}{x-y}\left[\frac{x^2(1-x^n)}{1-x}y^2\frac{(1-y^n)}{1-y}\right]$$

Example 10: Find the sum of an infinitely decreasing G.P. whose first term is equal to $(b + 2)$ and the common ratio is $2c^{-1}$ if b is the least value of the product of the roots of the equation $(m^2 + 1)x^2 – 3x + (m^2 + 1)^2 = 0$ and c is the greatest value of the sum of its roots.

Solution : b = least value of product of the roots of the given equation

= least value of $\frac{(m^2+1)^2}{m^2+1}$ = least value of $m^2 + 1$. when m is variable = 1

c = greatest value of sum of the roots

= greatest value of $\frac{3}{m^2+1}=3$

$\therefore$ the first term of the G.P. = 3 and the common ratio is 2/3

$$\therefore S_\infty=\frac{a}{1-r}=\frac{3}{1-\frac{2}{3}}=9$$

Example 11 : Four positive numbers form a G.P. The product of the first number and the fourth one equal to the greater root of the equation $x^{2+\log_{10}x}=(0.001)^{-2\frac{2}{3}}$ and the sum of the square of the second and square of the second of the third number is equal to 250. Find the numbers.

Solution : Let the four numbers be a, ar, ar^3, ar^3.

Now $x^{2+\log_{10} x} = (0.001)^{-8/3} = (10^{-3})^{-8/3} = 10^8$

Taking logarithm of both sides to the base 10,

$(2 + \log_{10} x) \log_{10} x = 8$

$\therefore\ y^2 + 2y - 8 = 0$ where $y = \log_{10} x$

$(y + 4)(y - 2) = 0\ \therefore \log_{10} x = -4$ or 2

$\therefore\ x = 10^{-4}$ or $x = 10^2$ and $10^2 > 10^{-4}$.

$x = 10^2$ is accepted since 10^2 is the greater root.

Now $a.\ ar^3 = 10^2$ or $a^2r^3 = 100$ .....(i)

and $(ar^2) + (ar^2)^2 = 250$ or $a^2r^2 + a^2r^4 = 250$ .....(ii)

Dividing one by the other , $\frac{1+r^2}{r} = \frac{5}{2}$

$2r^2 - 5r + 2 = 0$ giving $r = 2$ or $\frac{1}{2}$

When $r = 2$, $a^2 = \frac{100}{8} = \frac{25}{2}$ and the four numbers are $\sqrt{\frac{25}{2}}, \sqrt{50}, 10\sqrt{2} and 20\sqrt{2}$

When $r = ½$, we get the same four numbers.

Example 12: Prove that the numbers 49, 4489, 444889, ... obtained by inserting 48 into the middle of the preceding number are squares of integers.

Solution : $49 = 40 + 8 + 1 = 4 \times 10 + 8 + 1$

$4489 = 4000 + 400 + 80 + 8 + 1 = 4(10^3 + 10^2) + 8(10 + 1) + 1$

$444889 = 4(10^5 + 10^4 + 10^3) + 8(10^2 + 10 + 1) + 1$

$= 4 \times 10^3(1 + 10 + 10^2) + 8(1 + 10 + 10^2) + 1$

$$= (4 \times 10^3 + 8)\frac{\left(10^3 - 1\right)}{10 - 1} + 1 = \frac{4 \times 10^6 + 4 \times 10^3 + 1}{9} = \left(\frac{2 \times 10^3 + 1}{3}\right)^2$$

In the general case, $\underbrace{444....4}_{n digits}\underbrace{888....89}_{n digits} = \underbrace{44....4}_{n digits}\underbrace{88....8}_{n digits} + 1$

$= 4 \times 10^n(1 + 10 + 10^2 + ... + 10^{n-2}) + 8(1 + 10 + ...+10^{n-1}) + 1$

$$= (4 \times 10^n + 8)\left(\frac{10^n - 1}{10 - 1}\right) + 1 = \frac{4 \times 10^{2n} + 4 \times 10^n + 1}{9} = \left(\frac{2 \times 10^n + 1}{3}\right)^2$$

and $\left(\frac{2 \times 10^n + 1}{3}\right)$ is always an integer.

Example 13: If between any two numbers a and b, two arithmetic means a_1 and a_2, two geometric means g_1 and g_2 and two harmonic means h_1 and h_2 are inserted, show that

$$\frac{g_1 g_2}{a_1 + a_2} = \frac{h_1 h_2}{h_1 + h_2} ..$$

Solution : a, a_1, a_2, b are in A.P. $\Rightarrow a_1 + a_2 = a + b$

a, g_1, g_2, b are in G.P. $\Rightarrow g_1 g_2 = ab$

a, h_1, h_2, b are in H.P. $\Rightarrow \frac{1}{a}, \frac{1}{h_1}, \frac{1}{h_2}, \frac{1}{b}$ are in A.P.

$$\Rightarrow \frac{1}{h_1}+\frac{1}{h_2}=\frac{1}{a}+\frac{1}{b} \Rightarrow \frac{h_1+h_2}{h_1h_2}=\frac{a+b}{ab}=\frac{a_1+a_2}{g_1g_2} \Rightarrow \frac{g_1g_2}{a_1+a_2}=\frac{h_1h_2}{h_1+h_2}$$

Example 14: Find the sum of n terms of the series $1+5\left(\frac{4n+1}{4n-3}\right)+9\left(\frac{4n+1}{4n-3}\right)^2+13\left(\frac{4n+1}{4n-3}\right)^3+.....$

Solution : Let $\frac{4n+1}{4n-3}=x$. Then the required sum is

$S = 1 + 5x + 9x^2 + 13x^3 + ...+ (4n - 3)x^{n-1}$, (coefficients being in A.P.)

$xS = x + 5x^2 + 9x^3 + ... + (4n - 7)x^{n-1} + (4n - 3)x^n$

Subtracting

$S(1 - x) = 1 + 4x + 4x^2 + ...+ 4x^{n-1} - (4n - 3)x^n$

$$=1+4x\left\{\frac{1-x^{n-1}}{1-x}\right\}-(4n-3)x^n=1+\frac{4x}{1-x}-\frac{4x^n}{1-x}-(4n-3)x^n$$

$$\therefore S=\frac{1+3x}{(1-x)^2}-\frac{4x^n}{(1-x)^2}-\frac{(4x-3)x^n}{1-x}$$ = $4n^2 - 3n$ on replacing x by $\left(\frac{4n+1}{4n-3}\right)$

Example 15: If a, b, c are unequal positive numbers in H.P., then prove that $\frac{a+b}{2a-b}+\frac{c+b}{2c-b}>4$.

Solution : a, b, c are in H.P., $\frac{2}{b}=\frac{1}{a}+\frac{1}{c}$ ....(i)

$$\text{L.H.S.} = \frac{\frac{1}{b}+\frac{1}{a}}{\frac{2}{b}-\frac{1}{a}}+\frac{\frac{1}{b}+\frac{1}{c}}{\frac{2}{b}-\frac{1}{c}}=\frac{\frac{1}{b}+\frac{1}{a}}{\frac{1}{c}}+\frac{\frac{1}{b}+\frac{1}{c}}{\frac{1}{a}} \text{ using (i)} \qquad .....(ii)$$

$$=\frac{c}{b}+\frac{c}{a}+\frac{a}{b}+\frac{a}{c}=\frac{a+c}{b}+\left(\frac{c}{a}+\frac{a}{c}\right) \qquad .....(iii)$$

Since $A > G$, $\frac{\frac{a}{c}+\frac{c}{a}}{2}>\sqrt{\frac{a}{c}.\frac{c}{a}}$ $\qquad \therefore \frac{a}{c}+\frac{c}{a}>2$ .....(iv)

Since $A > H$, $\frac{a+c}{2}>\frac{2ac}{a+c}=b$ $\qquad \therefore \frac{a+c}{b}>2$ .....(v)

$\therefore$ L.H.S = $\left(\frac{a+c}{b}\right)+\left(\frac{c}{a}+\frac{a}{c}\right)$ > 2 + 2 = 4 in view of (iv) and (v) .

☺ **CONCEPTS**

Arithmetic Progression (A.P.)

1. General term : $t_k = a + (k-1)d$

2. Sum to *n* terms : $Sn = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

3. Means :

(i) A.M. of a and b; $A = \frac{a+b}{2}$

(ii) k^{th} A.M. out of n A.M.s between a and b; $A_k = a + k\left(\frac{b-a}{n+1}\right)$

(iii) A.M. Of $a_1, a_2, ..., a_n$ $A = \frac{a_1 + a_2 + ...a_n}{n}$

Geometric Progression (G.P.)

1. General term : $t_k = a.r^{k-1}$

2. Sum to *n* terms: $S_n = \frac{a(r^n - 1)}{r-1}, r \neq 1 = na, r = 1$

3. Sum till infinity : $S_\infty = \frac{a}{1-r}$, $|r| < 1$

4. Means:

(i) G.M. of a and b : $G = \sqrt{ab}$

(ii) k^{th} G.M. out of n G.M.'s between a and b : $G_k = a\left(\frac{b}{a}\right)^{\frac{k}{n+1}}$

(iii) G.M. of $a_1, a_2, ...a_n$ $G = (a_1 a_2 .....a_n)^{1/n}$

Harmonic Progression (H.P.)

1. General term : $t_k = \frac{1}{a+(k-1)d}$ with first term as $\frac{1}{a}$.

2. Means :

(i) H.M. of a and b : $H = \frac{2ab}{a+b}$

(ii) k^{th} H.M. out of n H.Ms. between a and b : $H_k = \frac{ab(n+1)}{k(a-b)+b(n+1)}$

(iii) H.M. of $a_1, a_2, ..., a_n$ $H = \dfrac{n}{\dfrac{1}{a_1}+\dfrac{1}{a_2}+....+\dfrac{1}{a_n}}$

Sequences And Series

Natural number sequences :

1. $1 + 2 + ... + n = \sum_{r=1}^{n} r = \dfrac{n(n+1)}{2}$

2. $1^2 + 2^2 + ... + n^2 = \sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$

3. $1^3 + 2^3 + ... + n^3 = \sum_{r=1}^{n} r^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

Arithmetico - Geometric Progession (A.G.P.)

1. General term : $t_k = [a + (k-1)d]br^{k-1}$
2. Sum of to n terms :

$$S_n = \frac{ab}{1-r} + \frac{dbr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]br^n}{1-r} \quad r \neq 1$$

3. Sum till infinity : $S_\infty = \dfrac{ab}{1-r} + \dfrac{dbr}{(1-r)^2}$, $|r| < 1$

Relations between Means :

1. For two numbers : $G^2 = AH$
2. For any number of positive numbers : $A \geq G \geq H$. Equality holds if numbers are equal .

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic:1 Arithmetic Progression

Q.1 The sum of the first 100 terms common to the series 17, 21, 25, 29, 33,... and 16, 21, 26, 31, 36....is
(a) 100101 (b) 111000
(c) 101100 (d) 110010

Q.2 For a certain A.P. and for a certain pair of natural numbers m and n, $\frac{S_m}{S_n}=\frac{m^2}{n^2}$, then $\frac{t_m}{t_n}$ for the given A.P. is
(a) $\frac{2m-1}{2n-1}$ (b) $\frac{2m+1}{2n+1}$
(c) $\frac{m+1}{n+1}$ (d) $\frac{m-1}{n-1}$

Q.3 The sum of three numbers in G.P. is 56. If we substract 1, 7, 21 from these numbers in that order, we obtain an A.P. the three numbers are
(a) 10, 18, 26 (b) 8, 16, 32
(c) 9, 16, 23 (d) 5, 8, 11

Q.4 If x, y, z and w are non-zero real numbers and $x^2 + 5y^2 + 5z^2 + 4w^2 - 4xy - 4yz - 4zw = 0$, then x, y, z, w are in
(a) A.P. (b) G.P.
(c) H.P. (d) A.G.P.

Q.5 If $x^{18} = y^{21} = z^{28}$, then $3\log_y z, 3\log_z y, 7\log_x z$ are
(a) A.P. (b) G.P.
(c) H.P. (d) A.G.P.

Q.6 If $a=\frac{1}{2^2-1}+\frac{1}{4^2-1}+\frac{1}{6^2-1}+\ldots\ \frac{1}{20^2-1}$,

$b=\frac{1}{2+\sqrt{2}}+\frac{1}{2\sqrt{3}+3\sqrt{2}}+\frac{1}{6+4\sqrt{3}}+\ldots$ upto 120 terms, then $\frac{10}{32}\left(\frac{1}{a}+\frac{1}{b}\right)$ equals
(a) 1 (b) 2 (c) 3 (d) 4

Q.7 If $a_1 = 1$ and $a_{n+1} = 2a_n + 5$, $n = 1, 2, \ldots$ then a_{100} equals
(a) $6 \times 2^{99} + 5$ (b) $6 \times 2^{99} - 5$
(c) $5 \times 2^{99} + 6$ (d) $5 \times 2^{99} - 6$

Q.8 If 1, $\log_9(3^{1-x} + 2)$ and $\log_3(4.3^x - 1)$ are in A.P., then x is
(a) $\log_3 4$ (b) $1 - \log_3 4$
(c) $\log_3 0.25$ (d) $\log_4 3$

Q.9 If the sides of a right-angled triangle are in A.P., then the sines of the acute angles are
(a) $\frac{3}{5}, \frac{4}{5}$ (b) $\sqrt{3}, \frac{1}{\sqrt{3}}$
(c) $\sqrt{\frac{\sqrt{5}-1}{2}}, \sqrt{\frac{\sqrt{5}+1}{2}}$ (d) $\sqrt{\frac{\sqrt{3}-1}{2}}, \sqrt{\frac{\sqrt{3}+1}{2}}$

Topic: 2 Geometric Progression

Q.10 Consider the following statements :
S_1 : Fourth term of a G.P. is 4 and tenth term is 1/16 then the value of sixth term is 1.
S_2 : If the sum of n terms of a G.P. is 1, 2, 2^2, 2^3, ... is S, then $S + 1 - 2^n$ is equal to zero.
S_3 : If $x \in \left(0, \frac{\pi}{2}\right)$ then the sum $1 + \sin x + \sin^2 x + \sin^3 x + \ldots$ upto ∞ is $\sec^2 x + \tan x \sec x$.
Which of only following is trure
(a) S_1 only (b) S_2 only
(c) S_3 only (d) S_1, S_2, S_3

Q.11 If the value of $4 + 44 + 444 + 4444 + \ldots\ldots$ upto n terms is $k\left[\frac{10^{n+1}-9n-10}{9}\right]$ then the value of k is
(a) 4/3 (b) 9/4 (c) 4/9 (d) 1/9

Q.12 If the sixth term of a GP be 2, then the product of first elevent terms is
(a) 1024 (b) 2047 (c) 2048 (d) 1023

Q.13 Let a_n be the nth term of the G.P. of positive numbers. Let $\sum_{n=1}^{100} a_{2n} = \alpha$ and $\sum_{n=1}^{100} a_{2n-1} = \beta$ such that $\alpha \neq \beta$, such that $\alpha \neq \beta$, then the common ratio is
(a) $\frac{\alpha}{\beta}$ (b) $\frac{\beta}{\alpha}$
(c) $\sqrt{\frac{\alpha}{\beta}}$ (d) $\sqrt{\frac{\beta}{\alpha}}$

Q.14 The series

$\frac{2x}{x+3}+\left(\frac{2x}{x+3}\right)^2+\left(\frac{2x}{x+3}\right)^3+\ldots\infty$ will have a definite sum when

(a) $-1 < x < 3$ (b) $0 < x < 1$
(c) $x = 0$ (d) $x = -3$

Q.15 The largest term of the sequence $\frac{1}{503},\frac{4}{524},\frac{9}{581},\frac{16}{692},\frac{25}{875},\ldots.$ is

(a) $\frac{16}{692}$ (b) $\frac{4}{524}$

(c) $\frac{49}{1529}$ (d) $\frac{49}{1234}$

Q.16 The value of x in $(-\pi, \pi)$ which satisfies the equation $8^{1+|\cos x|+|\cos^2 x|+|\cos^3 x|+\ldots\infty}=4^3$ is

(a) $\pi/3$ (b) $-\pi/3$
(c) $2\pi/3$ (d) All of these

Q.17 Let a, b be the roots of the equation $x^2 - 4x + k_1 = 0$ and c, d the roots of the equation $x^2 - 36x + k_2 = 0$. If $a < b < c < d$ and a, b, c, d are in G.P., then the product k_1k_2 equals

(a) 81 (b) 729
(c) 256 (d) 64

Q.18 If a, b, c are three distinct real numbers and they are in a G.P. If $a + b + c = xb$, then

(a) $x \le -1$ or $x \ge 3$ (b) $x \le -3$ or $x \ge 2$
(c) $x \le -4$ or $x \ge 3$ (d) $x \le 2$ or $x \ge 5$

Q.19 The product $2^{1/4}.4^{1/8}.8^{1/16}.....$ to ∞ equal to

(a) 4 (b) $\sqrt{2}$ (c) 8 (d) 2

Q.20 The three sides of a right angled triangle are in G.P. The tangents of two acute angles are

(a) $\frac{\sqrt{5}+1}{2}$ and $\frac{2}{\sqrt{5}+1}$

(b) $\sqrt{\frac{\sqrt{5}-2}{2}}$ and $\sqrt{\frac{2}{1+\sqrt{5}}}$

(c) $\sqrt{\frac{1+\sqrt{5}}{2}}$ and $\sqrt{\frac{2}{1+\sqrt{5}}}$

(d) $\frac{\sqrt{5}+1}{4}$ and $\frac{4}{\sqrt{5}+1}$

Q.21 The product $\prod_{n=2}^{\infty}\frac{n^3-1}{n^3+1}$ equals

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{12}$

Q.22 If $S = 1! + 2 \times 2! + 3 \times 3! + ... 10 \times 10!$, then $S + 2$, when divided by 11! leaves a remainder of

(a)1 (b)0 (c)7 (d)11

Q.23 Let $SC(0, \pi)$ denotes the set of values of x satisfying the equation $8^{1+|\cos x|+\cos^2 x|+\cos^3 x|+....}$ to $\infty = 4^3$, then S is

(a) $\left\{\frac{\pi}{3}\right\}$ (b) $\left\{\frac{\pi}{3}, -\frac{2\pi}{3}\right\}$

(c) $\left\{-\frac{\pi}{3}, \frac{2\pi}{3}\right\}$ (d) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$

Topic:3 Harmonic Progression

Q.24 Which of the following is false?

(a) If in a triangle PQR sin P, sin Q, sin R in A.P. then altitudes are in H.P.

(b) If three positive numbers a, b, c are in H.P. then $a^n + c^n \ge 2b^{n/2}$

(c) If a, b, c are in H.P. then $\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{d}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)=\frac{4}{ac}-\frac{3}{b^2}$

(d) $1+\frac{1}{1+2}+\frac{1}{1+2+3}+\ldots..$ upto $\infty = 1$

Q.25 If $b - c$, $2b - x$ and $b - a$ are in H.P. Then $a-\left(\frac{x}{2}\right), b-\left(\frac{x}{2}\right)$ and $\left(\frac{x}{2}\right)$ are in

(a) A.P. (b) G.P.
(c) H.P. (d) None of these

Topic: 4 Arithmetic Means, Geometric Means, Harmonic Means

Q.26 Four geometric means are inserted between the numbers $2^{11} - 1$ and $2^{11} + 1$. The product of these geometric means is

(a) $2^{22} - 2^{11} + 1$ (b) $2^{44} - 2^{33} + 1$
(c) $2^{44} - 2^{22} + 1$ (d) $2^{22} - 2^{12} + 1$

Q.27 If one geometric mean G and two A.Ms p and q be inserted between two given quantities then $(2p - q)(2q - p)$ equals

(a) G (b) G^2 (c) $2G^2$ (d) $2G$

Q.28 If $f(x) = x^2 - (a + b)x + ab$ and A and H be the A.M. and H.M. between two quantities a and b then

(a) $Af(A) = Hf(H)$
(b) $Af(H) = Hf(A)$
(c) $A + f(A) = H + f(A)$
(d) $f(A) + H = f(H) + A$

Q.29 The value of $x + y + z$ is 15 if a, x is 15 if a, x, y, z, b are in A.P. while the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is $\frac{5}{3}$ if a, x, y, z, b are in H.P. the value of a and b are
(a)9, 1 (b)7, 4 (c)8, 2 (d)–1, 3

Q.30 If a, b, c are in A.P., p, q, r are in H.P. and ap, bq, cr are in G.P., then
(a) $\frac{p}{r}+\frac{r}{p}=\frac{c}{a}+\frac{a}{c}$ (b) $\frac{p}{q}+\frac{q}{p}=\frac{c}{a}+\frac{a}{c}$
(c) $\frac{p}{r}+\frac{r}{p}=\frac{b}{c}+\frac{c}{b}$ (d) $\frac{p}{q}-\frac{q}{p}=\frac{c}{a}-\frac{a}{c}$

Q.31 If a, b, c satisfy $\frac{1}{a}+\frac{1}{c}+\frac{1}{a-b}+\frac{1}{c-b}=0$, then
(a) a, b, c are in H.P. or $a, b/2, c$ are in A.P.
(b) a, b, c arein A.P. or $a, b/2, c$ are in H.P.
(c) $a, b/2, c$ are in H.P. or a, b, c are in G.P.
(d) $a, b/2, c$ are in G.P. or a, b, c are in H.P.

Topic: 5 Arithmetic Geometric Series

Q.32 The sum of the series $1 + 2.2 + 3.2^2 + 4.2^2 + 4.2^2 + 5.2^4 + ... + 100.2^{99}$ is
(a) $99.2^{100} – 1$ (b) $100.2^{100} – 1$
(c) 99.2^{100} (d) $99.2^{100} + 1$

Q.33 If $3 + \frac{1}{4}(3 + p) + \frac{1}{4^2}(3 + 2p) + ...$ to $\infty = 8$ then p equals
(a)1 (b)5 (c)5 (d)9

Q.34 Let S denotes the infinite sum $2 + 5x + 9x^2 + 14x^3 + 20x^4 + ...$, where $|x| < 1$. Then S equals
(a) $\frac{2-x}{(1+x)^3}$ (b) $\frac{2-x}{(1-x)^3}$
(c) $\frac{2+x}{(1-x)^3}$ (d) $\frac{2+x}{(1+x)^3}$

Q.35 The sum to 100 terms of the series 1.2.3. + 2.3.4. + 3.4.5. + ... + $n(n + 1)(n + 2)$ + ... is integral multiple of
(a) 2525 (b) 2526
(c) 2527 (d) 2528

Q.36 The sum of the first n terms of the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + ...$ is $\frac{n(n+1)^2}{2}$ when n is even. Then the sum of n is odd, is
(a) $\frac{n(n^2+1)}{3}$ (b) $n(n + 1)^2$
(c) $\frac{n^2(n+1)}{2}$ (d) $\frac{n(n^2+1)}{2}$

Topic: 6 Application to Problems of Maxima and Minima

Q.37 If x, y, z are positive reals satisfying $4xy + 6yz + 8zx = 9$, then the greatest possible value of the product xyz is
(a) 0.125 (b) 0.375
(c) 0.500 (d) 0.625

Q.38 If $a + b + c = 3$ and $a > 0, b > 0, c > 0$, then the greatest value of $a^2b^3c^2$ is
(a) $\frac{3^9.2^4}{7^7}$ (b) $\frac{3^{10}.2^4}{7^7}$ (c) $\frac{3^8.2^4}{7^7}$ (d) $\frac{3^7.2^4}{7^7}$

Q.39 Let T be the set of integers {3, 11, 19, 27, ..., 451, 459, 467} and S be a subset of T such that the sum of no two eleemnts of S is 470. The maximum possible number of elements in S is
(a) 32 (b) 28 (c) 29 (d) 30

Topic: 7 Miscellaneous

Q.40 $\lim_{n\to\infty}(1+3^{-1})(1+3^{-2})(1+3^{-4})(1+3^{-8})$ ... $(1+3^{-2^n})$ =
(a) 1 (b) 1/2 (c) 3/2 (d) –1/2

Q.41 The coefficient of x^{101} in the expansion of $(1 – x)(1 – 2x)(1 – 2^2x)....(1 – 2^{101}x)$ is
(a) $2^{4950} – 2^{5050}$ (b) $2^{5051} – 2^{5152}$
(c) $2^{4950} – 2^{5051}$ (d) $2^{5050} – 2^{5152}$

Q.42 Let C be a circle with centre P_0 and AB be a diameter of C. Suppose P_1 is the midpoint of line segment P_0B, P_2 the midpoint of line segment P_1B and so on. Let C_1, C_2, C_3 be circles with diameters P_0P_1, P_1P_2, P_2P_3,..... respectively. Suppose the circles C_1, C_2, C_3,.... are all shaded. The ratio of the area of the unshaded portion of C to that of the original circle is
(a)8 : 9 (b)9 : 10 (c)10 : 11 (d)11 : 12

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ $(a, b, c \in R^+)$ are in A.P. $(a \equiv c)$ then product of roots of the equation $x^2 + \mu x + (2b^{51} – a^{51} – c^{51}) = 0$, $\mu \in R$ is negative.

Statement – 2 : $\frac{a^n + c^n}{2} > b^n$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** If *a, b, c, d* are positive and distinct numbers in H.P., then $a + d > b + c$.

Statement-2: If *a, b, c, d* are in H.P., $\frac{a+d}{ad} = \frac{b+c}{bc}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** For $n \in N$, $n > 1$, $2^n > 1 + n$, $2^{(n-1)/2}$.

Statements-2: A.M. of distinct positive number is greater than GM.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The sum of *n* terms of two arithmetic progressions are in A.P. in the ratio $(7n + 1) : (4n + 17)$, then the ratio of their n^{th} terms is 7 : 4.

Statement–2 : If $S_n = ax^2 + bx + c$, then $T_n = S_n - S_{n-1}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** Let $F_1(n) = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n}$, then $\sum_{r=1}^{n} F_1(r) = (n+1)(n+1)F_1(n) - n$.

Statement–2 : $\frac{1^{-1} + 2^{-1} + 3^{-1} + ... + n^{-1}}{n} > \left(\frac{1 + 2 + 3 + ...n}{n}\right)^{-1}$ or $\left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n}\right) > \frac{2n}{n+1}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.6 **Statement–1 :** If the angle of a convex polygon are in A.P. 120°, 125°, 130°, ... then it has 16 sides.

Statement–2 : The sum of the angles of a polygon of *x* sides is $(n - 2)180^o$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.7 **Statement–1 :** If the infinite A.G.P., 1, $\sqrt{3}$, 2, *x*,.... has a finite sum, then $x = 2$.

Statement–2 : The infinte A.G.P. *a*, $(a + d)r$, $(a + 2d)r^2$ ... has a finite sum only if $|r| < 1$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.8 Let (α, β) are the roots of $x^2 + 2x + mx + 2m = 0$; (α, γ) are the roots of $x^2 + 2x - mx - 2m = 0$ and (β, γ) are the roots of

$\frac{1}{x+p}+\frac{1}{x+q}=\frac{1}{r}$ $(m, p, q \neq 0)$.

Statement–1 : If $\alpha \neq \beta \neq \gamma$ then p, q, r in A.P.

Statement–2 : $\alpha + \beta + \gamma = -2$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.9 **Statement–1 :** The sum to n terms of the series $1^3 + 3.2^2 + 3^3 + 3.4^3 + 5^3$..... is $\frac{n^2(2n^2+6n+5)}{4}$, when n is even.

Statement–2 : The sum to infinity of $\frac{1}{7}+\frac{2}{7^2}+\frac{1}{7^3}+\frac{2}{7^4}$..... is $\frac{3}{16}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.10 Let $a_i \in R - \{0\}$, i=1, 2, 3,....n and all a_i's are distinct such that $f(x)$ =

$$\left(\sum_{i=1}^{n-1} a_i^2\right)x^2 - 2\left(\sum_{i=1}^{n-1} a_i a_{i+1}\right)x + \sum_{i=1}^{n} a_i^2$$

Statement–1 : If $f(x) \leq 0$ for some $x \in R$ then $a_1, a_2, ..., a_n$ are in G.P.

Statement–2 : For real a, b, c, $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.11 **Statement–1 :** If $a^x = b^y = c^z$; where x, y, z are unequal positive numbers and a, b, c are in G.P., then $x^3 + z^3 > 2y^3$.

Statement–2 : If a, b, c are in H.P. $a^3 + c^3 \geq 2b^3$, where a, b, c are positive real numbers.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 If x_1, x_2, x_3 and y_1, y_2, y_3 are both in GP with the same common ratio, then the points (x_1, y_1), (x_2, y_2) and (x_3, y_3):

(a) lie on a straight line

(b) lie on an elipse

(c) lie on a circle

(d) are vertices of a triangle.

Q.2 Let T_r be the rth term of an AP whose first term is a and common difference is d. If for some positive integers m & n, $m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then a – d equals :

(a) 0 (b) 1

(c) $\frac{1}{mn}$ (d) $\frac{1}{m}+\frac{1}{n}$

Q.3 If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a,b,c are in AP and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x,y,z are in

(a) HP

(b) Arithmetico–Geometric Progression

(c) AP

(d) GP

Q.4 If in a DABC, the altiudes from the vertices A, B, C on opposite sides are in H.P., then sin A, sin B, sin C are in-

(a) G.P.

(b) A.P.

(c) Arithmetico-Geometirc progression

(d) H.P.

Q.5 Let a_1, a_2, a_3,..... be terms of an AP. If $\frac{a_1+a_2+.....+a_p}{a_1+a_2+.....+a_q} = \frac{p^2}{q^2}$, $p \neq q$, then $\frac{a_6}{a_{21}}$ equals :

(a) $\frac{7}{2}$ (b) $\frac{2}{7}$ (c) $\frac{11}{41}$ (d) $\frac{41}{11}$

Q.6 If $a_1, a_2, ..., a_n$ are in HP, then the expression $a_1a_2 + a_2a_3 + ...+ a_{n-1}a_n$ is equal to :

(a) $(n-1)(a_1-a_n)$ (b) na_1a_n
(c) $(n-1)a_1a_n$ (d) $n(a_1-a_n)$

Q.7 In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals

(a) $\frac{1}{2}(1-\sqrt{5})$ (b) $\frac{1}{2}\sqrt{5}$

(c) $\sqrt{5}$ (d) $\frac{1}{2}(\sqrt{5}-1)$

Q.8 A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the nth minute.If $a_1 = a_2 = ...= a_{10} = 150$ and $a_{10}, a_{11},....$are in an AP with common difference –2, then the time taken by him to count all notes is
(a) 34 minutes (b) 125 minutes
(c) 135 minutes (d) 24 minutes

Q.9 A man saves Rs. 200 in each of the first three months of his service. In each of the subsequent months his saving increases by Rs. 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs. 11040 after :
(a) 18 months (b) 19 months
(c) 20 months (d) 21 months

Q.10 Let a_n be the nth term of an A.P. If $\sum_{r=1}^{100} a_{2r} = a$ and $\sum_{r=1}^{100} a_{2r-1} = b$, then the common difference of the A.P. is :

(a) a – b (b) $\frac{\alpha-\beta}{100}$ (c) b – a (d) $\frac{\alpha-\beta}{200}$

Q.11 The sum of first 20 terms of the sequence 0.7, 0.77, 0.777,....., is

(a) $\frac{7}{81}(179-10^{-20})$ (b) $\frac{7}{9}(99-10^{-20})$

(c) $\frac{7}{81}(179+10^{-20})$ (d) $\frac{7}{9}(99+10^{-20})$

Q.12 The sum of first 9 terms of the series $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+....$ is
(a) 142 (b) 192
(c)71 (d) 96

Q.13 If m is A.M of two distinct real numbers l and n $(l,n>1)$ and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4+2G_2^4+G_3^4$ equals

(a) $4lmn^2$ (b) $4l^2m^2n^2$

(c) $4l^2mn$ (d) $4lm^2n$

Q.14 If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is
(a) 8/5 (b) 4/3 (c) 1 (d) 7/4

Q.15 Let $a_1, a_2, a_3.....a_n,.....$ be in A.P. If $a_3+a_7+a_{11}+a_{15}=72$, then the sum of its first 17 terms is eqqual to
(a) 306 (b) 204
(c) 153 (d) 612

Q.16 For any three positive real numbers a, b and c , $9(25a^2+b^2)+25(c^2-3ac)=15b(3a+c)$.
Then
(a) b,c and a are in A.P.
(b) a, b and c are in A.P.
(c) a, b and c are in G.P
(d) b,c and a are in G.P.

Q.17 Let a, b, c, $c \in R$. If $f(x)=ax^2+bx+c$ is such that $a+b+c=3$ and $f(x+y)$ = f(x)+f(y) + xy, $\forall x,y \in R$, then $\sum_{n=1}^{10} f(n)$ is equal to
(a) 165 (b) 190
(c) 255 (d) 330

Q.18 If the sum of the first n terms of the series $\sqrt{3}+\sqrt{75}+\sqrt{243}+\sqrt{507}+...$ is $435\sqrt{3}$, then n equals
(a) 29 (b)18 (c) 15 (d) 13

Q.19 If the arithmetic mean of two numbers a and b, $a>b>0$, is five times their geometric mean, then $\frac{a+b}{a-b}$ is equal to

(a) $\frac{\sqrt{6}}{2}$ (b) $\frac{3\sqrt{2}}{4}$ (c) $\frac{5\sqrt{6}}{12}$ (d) $\frac{7\sqrt{3}}{12}$

Q.20 Let $S_n = \frac{1}{1^3}+\frac{1+2}{1^3+2^3}+\frac{1+2+3}{1^3+2^3+3^3}+...$ $+\frac{1+2+.....+n}{1^3+2^3+.....+n^3}$. If 100 $s_n = n$, then n is equal to
(a) 99 (b) 19 (c) 200 (d) 199

Q.21 If three positive number a, b and c are in A.P. such that abc=8, then the minimum possible value of b is
(a) $4^{2/3}$ (b) $4^{1/3}$ (c) 4 (d)2

Q.22 Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2+2.2^2+3^2+2.4^2+5^2+2.6^2+\ldots\ldots$ If $B-2A=100\lambda$, then λ is equal to:
(a) 248 (b) 464 (c) 496 (d) 232

Q.23 Let $a_1, a_2, a_3, \ldots\ldots, a_{49}$ be in A.P. Such that

$\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$.

If $a_1^2 + a_2^2 + \ldots\ldots + a_{17}^2 = 140m$, then m is equal to
(a) 68 (b) 34
(c) 33 (d) 66

Section-D
(Board pattern)

Q.1 Find the 19th and 20th terms of the sequence of defined by T_n =

$$\begin{cases} n^2, & \text{when } n \text{ is even} \\ n^2+1, & \text{when } n \text{ is odd} \end{cases}.$$

Q.2 Find the 19th term of the sequence 5, 2, –1, –4,.....

Q.3 Which term of the sequence 18, –12, 8,.... is $\frac{512}{729}$?

Q.4 The second term of a G.P. is $\frac{25}{4}$ and the 8th term is $\frac{16}{625}$, find the G.P.

Q.5 If the 9th term of an AP is zero. Prove that 29th term is double the 19th term.

Q.6 In an AP, $S_n = 5$, $a = \frac{5}{7}$ and $d = -\frac{1}{21}$. Find n.

Q.7 Solve for x, the equation $2 + 7 + 12 + \ldots. + x = 156$.

Q.8 Find the sum of all three digit numbers which leaves the remainder 2 when divided by 5.

Q.9 Sum to n terms. 6 + 66 + 666 + ...

Q.10 Find the three numbers in a GP. Whose sum is 19 and product is 216.

Q.11 Find the sum of the following series, $10^3 + 11^3 + 12^3 + \ldots. + 20^3$.

Q.12 Find the sum 0 to n terms of the series whose n^{th} term is $n^2 + 2n + 2 + 2^n$.

Q.13 Find the sum of n term of the series

(i) $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}$

(ii) $\sqrt{7}+\sqrt{21}+3\sqrt{7}+\ldots\ldots$

Q.14 Insert 4 geometric means between $\frac{4}{9}$ and $\frac{27}{8}$.

Q.15 Between 1 and 31, m A.M.'s have been inserted in such a way that the ratio of 7th and $(m-1)^{th}$ A.M.'s is 5 : 9. Find the value of m.

Q.16 If $\frac{a+bx}{a-bx}=\frac{b+cx}{b-cx}=\frac{c+dx}{c-dx}(x\neq 0)$ then show that a, b, c and d are in GP.

Q.17 The pth, q^{th} and r^{th} terms of an A.P. are a, b, c respectively. Show that $(q-r)a + (r-p)b + (p-q)c = 0$.

Q.18 If a, b, c are in A.P., b, c, d are in G.P. and $\frac{1}{c}, \frac{1}{d}, \frac{1}{e}$ are in A.P., prove that a, c, e are in G.P.

Q.19 A farmer buys a used tractor for Rs. 12000. He pays 6000 cash and agrees to pay the balance in annual installment of Rs. 500 plus 10% interest on unpaid amount. How much will the tractor cost him?

Q.20 20th terms of the series. $2 \times 4 + 4 \times 6 + 6 \times 8 + \ldots..$

CHAPTER 4

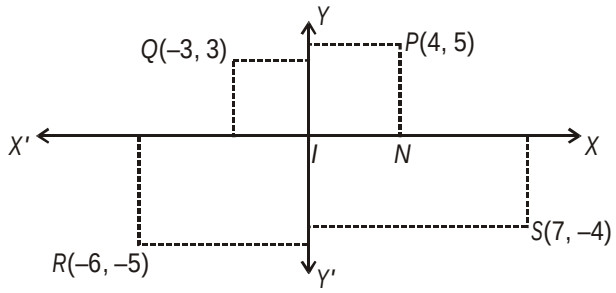
STRAIGHT LINE

1 PRELIMNARIES

The system of Geometry in which a point is specified by means of an ordered number-pair is known as Coordinate Geometry. It enables us to solve geometrical problems by algebraic methods.

1.1 Coordinates of point

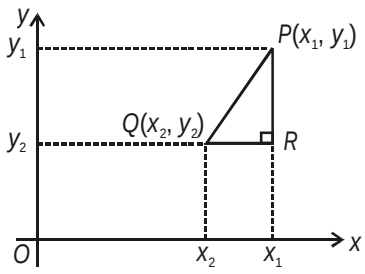
Let *P* be a point in a plane; draw two perpendicular lines *X'OY, Y'OY* in the plane, and draw *PM*, *PN* parallel to *OX, OY* respectively. *X'OX, Y'OY* are known as the coordinate axes. *MP* is the *x*-coordinate, called the abscissa, and *NP* is the *y*-coordinate called the ordinate of *P*. These two distances *MP* and *NP* together fix the position of *P* in the plane.



If *MP* = 4, *NP* = 5, then the position of *P* is denoted by (4, 5) the *x*-coordinate being written first. *Q* is the point (–3, 3), for the *x*-coordinate, is measured to the left of the *y*-axis, and is negative in sign; *R* is the point (–6, –5) for both coordinates are negative in sign. Similarly *S* is the point (7, –4). It should be observed that *O*, the origin is the point (0, 0) and the *y*-coordinate of any point on the *x*-axis is zero while the *x*-coordinate of any point on the *y*-axis is zero.

Every point in th eplane of the axes of coordiantes is represented correspondingly by an ordered pair (*x, y*) the first one, namely, *x* representing the *x*-coordinate of the oint and the second one, namely *y*, representing the *y*-coordinate of the point. Conversely to every ordered pair there corresponds only one point. Thus this method is representation of points in a plane by an ordered pair of real number is, what is usually called one-one ant onto by which we mean that for every point there is only one ordered pair of real numbers andfor every ordered pair of real numbers there is only one point, and neithre any point nor any ordered pair is left out without its associated ordered pair of real numbers or a point as the case may be.

1.2 To find the distance between two given points in the cartesian system



Let the coordiantes of the given points *P* and *Q* be(x_1, y_1) and (x_2, y_2) respectively.

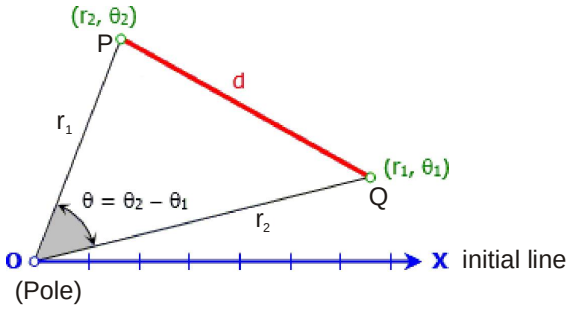
From the figure

$$PQ = \sqrt{QR^2 + PR^2}$$

i.e., $PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Distance blw points in pol.ar co-ordinates

Let O be the pole and OX be the initial lines let P and Q be two given points whose polar co-ordiantes are (r_1, θ_1) and (r_2, θ_2) respectively



$OP = r_1, OQ = r_2 \angle POX = \theta_1 \angle QOX = \theta_2 \angle POQ = \theta_1 - \theta_2$

Student Task

What is the distance of the point (cos α, sin α) from origin?

Illustration 1 : Show that the points (2, 3)(1, 5) (–2, 0) and (–1, –2) are vertices of a parallelogram.

Solution : Let the points be denoted by *A, b, C, D* in order

$AB^2 = (2 - 1)^2 + (3 - 5)^2$

$= 1 + 4 = 5$

$BC^2 = (1 + 2)^2 + (5 - 1)^2$

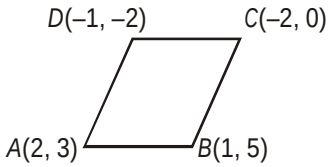
$= 9 + 25 = 34$

$CD^2 = (-2 + 1)^2 + (0 + 2)^2$

$= 1 + 4 = 5$

$DA^2 = (-1 - 2)^2 + (-2 - 3)^2$

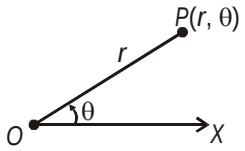
$= 9 + 25 = 34$



Since the opposite sides are equal, the figure drawn is a parallelogram.

1.3 Polar coordinates

Let *OX* be a given line and *P* be any point in a given plane. Let θ be the angle through which a line rotates in moving from *OX* to the position *OP*. Then, if *r* is the lenth of *OP*, the position of *P* is known when *r* and θ are known.



r is called the radius vector, q the vectorial angle and (*r*, θ) is the polar coordinates of the point *P*, which is briefly referred to as the point (*r*, θ). *OX* is called the initial line and *O* the pole.

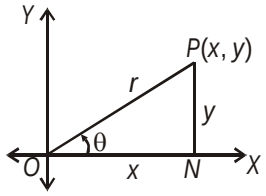
1.4 Relation between Cartesian and Polar coordinates

Let *P* be the point (*x, y*) with reference to rectangular axes *OX, OY* and the point (r, θ) with reference to the pole *O* and initial line *OX*. i.e., the *X*-axis.

$\therefore$ We have $x = r\cos\theta$; $y = r\sin\theta$

Hence $\tan\theta = \frac{y}{x}$ and $x^2 + y^2 = r^2$

These relations enable us to change from one system of coordinates to the other.



$$(r\cos\theta, r\sin\theta) = (x, y) \quad \text{.....(i)}$$

$$\left(\sqrt{x^2 + y^2}, \tan^{-1}\left(\frac{y}{x}\right)\right) = (r, \theta) \quad \text{.....(ii)}$$

If r and θ are known then we can find (x,y) from equation (i) and if x and y known then we can find (r, θ)from Equation (ii)

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\alpha = \tan^{-1}\left|\frac{y}{x}\right|$$

The values of θ in four quadrants

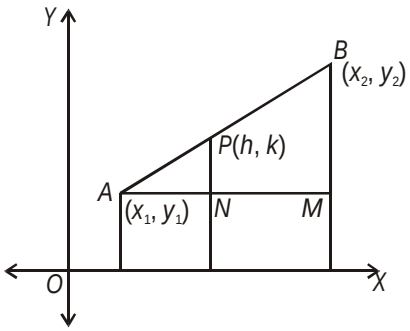
Quadrant	I	II	III	IV
θ	α	$\pi - \alpha$	$-\pi + \alpha$	$-\alpha$

2. SECTION FORMULA

Let *A, B* are the points (x_1, y_1), (x_2, y_2), here we will find the coordinates of the point *P* on *AB* such that *AP* : *PB* = ℓ : *m*.

Let the coordinates of *P* be (*h*, *k*)

In the figure, triangles *ANP and AMP* are similar.



$\therefore \quad \frac{PN}{BM} = \frac{AP}{AB} = \frac{\ell}{\ell + m}$ since $\frac{AP}{PB} = \frac{\ell}{m}$

i.e. $\frac{k - y_1}{y_2 - y_1} = \frac{\ell}{\ell + m}$

$\therefore \quad (\ell + m)k = (\ell + m)y_1 + \ell(y_2 - y_1)$

$= \ell y_2 + m y_1$

$\therefore \quad k = \frac{\ell y_2 + m y_1}{\ell + m}$

Similarly, $k = \frac{\ell x_2 + m x_1}{\ell + m}$

Note : Putting $\ell = m$, we find that the mid-point of the line joining the points (x_1, y_1), (x_2, y_2), i.e. $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Note : When *P* lies outside *AB* i.e., external to *AB* such that *AP* : *BP* = ℓ : *m.*

It has been assumed that l and *m* are positive numbers; if, however, we take *AP* : *BP* to be positive in the former case and negative in the latter case, we have in both cases the formula

$$h = \frac{\ell x_2 + m x_1}{\ell + m}, k = \frac{\ell y_2 + m y_1}{\ell + m}.$$

Illustration 2: Find the cooordinates of the points *L, M* which divide internally and externally in the ratio 2 : 3 the line joining the points *A*(8, 10), *B*(18, 20), and show that $TL \cdot TM = TB^2$, where *T* is the midpoint of *AB*.

Solution :



Since *AL* : *LB* = 2 : 3, the coordintes of *L* are $\left(\frac{2 \times 18 + 3 \times 8}{2 + 3}, \frac{2 \times 20 + 3 \times 10}{2 + 3}\right)$

i.e., (12, 14)

Since *M* divides *AB* externally in the ratio 2 : 3, abscissa = $\frac{2x18 - 3x8}{2 - 3} = -12$

Ordinate = $\frac{2x20 - 3x10}{2 - 3} = -10$

Since *T* is the mid - point of *AB*, the coordinates of *T* are $\left(\frac{8 + 18}{2}, \frac{10 + 20}{2}\right)$

i.e., (13, 15)

T (13, 15), *L*(12, 14), *M*(–12, –10), *B*(18, 20)

TL = $\sqrt{(13 - 12)^2 + (15 - 14)^2} = \sqrt{2}$

TM = $\sqrt{(13 - 12)^2 + (15 - 10)^2} = 25\sqrt{2}$

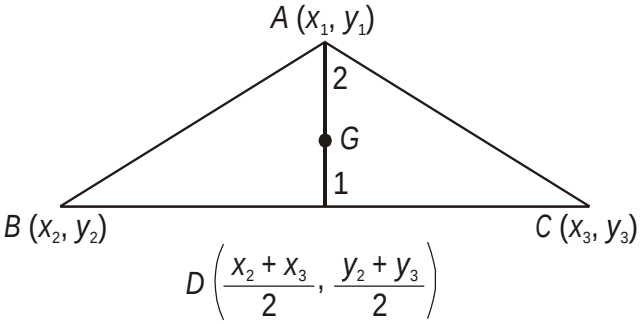
$\therefore$ $TL \cdot TM = 50$

$TB^2 = (13 - 18)^2 + (15 - 20)^2 = 50$

$\therefore$ $TL \cdot TM = TB^2$

Illustration 3: *A, B, C* are the points (x_1, y_1), (x_2, y_2), (x_3, y_3). *D* is the mid point of *BC*, and *G* divides *AD* in the ratio 2 : 1, find the coordinates of *G* and show that the medians of triangle *ABC* are concurrent.

Solution : Since $BD = DC$, *D* is the point $\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}$



Since $AG : GD = 2 : 1$, the coordinates of *G* are

$$\left(\frac{1.x_1 + 2\left(\frac{x_2 + x_3}{2}\right)}{1+2}, \frac{1.y_1 + 2\left(\frac{y_2 + y_3}{2}\right)}{1+2} \right)$$

i.e., $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$

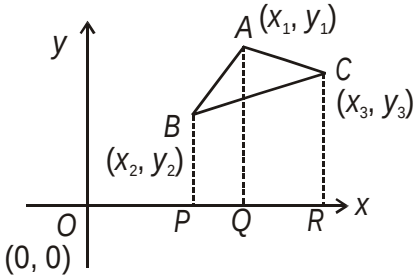
The symmetry of this result in x_1, x_2, x_3 and in y_1, y_2, y_3 shows that *G* lies on each median.

Hence the medians of triangle *ABC* are concurrent.

3. AREA OF A TRIANGLE

Let the vertices of the triangle be

$A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$



We have considered the vertices of the triangle in the first quadrant for the ease of calculations. Drops perpendiculars form *A*, *B* and *C* on the *x*-axis.

Area Of ΔABC = Area of trapezium *ABPQ*

+ Area of trapezium $AQRC$

– Area of trapezium $BCRP$

$$= \frac{1}{2}(BP + AQ)(PQ) + \frac{1}{2}(AQ + CR)(QR) - \frac{1}{2}(BP + CR)(PR)$$

$$= \frac{1}{2}(y_2 + y_1)(x_1 - x_2) + \frac{1}{2}(y_1 + y_3)(x_3 - x_1) - \frac{1}{2}(y_2 + y_3)(x_3 - x_2)$$

$$= \frac{1}{2}\{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\}$$

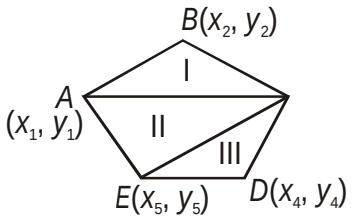
expressed in the determinant form,

$$\Rightarrow \Delta = \text{absolute value of } \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$$

The area of a triangle is considered positive when its vertices are named in counter-clockwise order, and negative when named in clockwise order.

If three points are collinear, we have the area of the $\Delta = 0$.

Similarly to calculate the area of pentagon whose vertices are given, we can divide the pentagon into a number of triangle and then sum up the areas. e.q.



Area of pentagon $ABCDE$ = Area of I + Area of II + Area of III

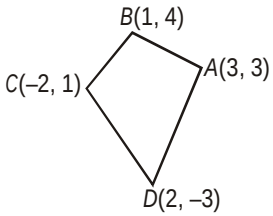
$$= \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \\ x_5 & y_5 & 1 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} x_3 & y_3 & 1 \\ x_4 & y_4 & 1 \\ x_5 & y_5 & 1 \end{vmatrix}$$

Student Task

What is the area of triangle formed by the point (–6, –3), (1, 4) and (5, 8) ?

Illustration 4: Find the area of the quadrilateral with vertices (3, 3), (1, 4), (–2, 1), (2, –3).

Solution : Let the given vertices be A, B, C, D respectively.



Join AC

Area of *ABCD* = area of triangle *ABC* + Area of triangle *ACD*

$$= \frac{1}{2}[(3(4-1)+1(1-3)-2(3-4)] + \frac{1}{2}[3(1+3)-2(-3-3)+2(3-1)]$$

$$= \frac{1}{2}(9-2+2+2+12+12+14) = \frac{37}{2}$$

4. EQUATION OF LOCUS OF A POINT

The locus of a point, as it moves in accordance with a given geometrical condition, is the path traced out by the moving point. In geometry we mean, by locus, the curve itself. We say that the locus is a straight line (or) the locus is a circle. On the contrary in coordiante geometry the moving point can be represented by the ordered pair (*x*, *y*). The geometrical condition imposed on the moving point gets transformed into an algebraic relation connecting (*x*, *y*). This relation, between *x* and *y*(the coordiantes of the moving point), is called the equation to the curve described by points.

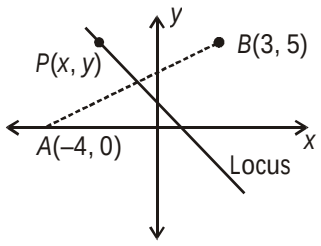
For example, if *P*(*x, y*) is any point equidistant from *A*(–4, 0) and *B*(3, 5),

we have $PA^2 = PB^2$ and this can be expressed in terms of coordiantes of *A, P* and *B* as

$(x + 4)^2 + y^2 = (x - 3)^2 + (y - 5)^2$

i.e., $14x + 10y - 18 = 9$

i.e., $7x + 5y - 9 = 0$



Moreover, if *P*(*x, y*) is any point whose coordiantes satisfy the equation $7x + 5y - 9 = 0$,

We have $PA^2 = PB^2$

i.e. *P* is equidistant from *A* and *B*. Hence the relation $7x + 5y - 9 = 0$, is called the equation of the locus of points equidistant from *A* and *B*.

It is said that the equation $7x + 5y - 9 = 0$, represents the perpendiuclar bisector of *AB*.

Coordiante Geometry is based on the concept of locus of a point. Large number of problems will involve the idea of the locus of a point.

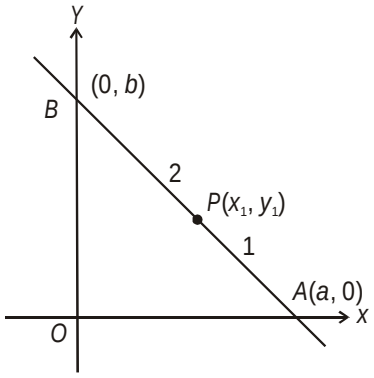
To find locus of a point *P*, proceed as follows :

1. Assume co-ordinate of *P* as (*h, k*); *h* and *k* are unknown quantities.
2. Translate the given problem into equations, which will contain coordiantes of *P*, some given quantities and some unknown quantities.
3. Suppose two unknown quantities are introduced in the above method. Then one must get three equations (or more) in step 2.
4. Eliminate unknowns to get an equation which should be generalized by replacing *h* by *x, k* by *y*, to get the required locus.

Illustration 5: A straight line segment of length '*p*' moves with its ends on two mutually perpendicular lines. Find the locus of the point which divides the line in the ratio 1 : 2.

Solution : Choose the two mutually perpendicular lines as axes of coordinates.

The straight line segment *AB* of constant length '*p*' slides so that *A* and *B* move along *OX* and *OY* respectively. Let $P(x_1, y_2)$ be a point of *AB* such that *AP* : *BP* = 1 : 2. It is required to find the locus of *P*.



At any position of *AB*, let the intercepts *OA, OB* be *a, b* respectively, so that

$a^2 + b^2 = p^2$

Since *AP* : *PB* = 1 : 2, we have

$X_1 = \frac{2a}{3}, y_1 = \frac{b}{3}$

$\therefore a = \frac{3x_1}{2}$, b = $3y_1$

Using (i) $\left(\frac{3x_1}{2}\right)^2 + (3y_1)^2 = p^2$, and this implies that

$P(x_1, y_1)$ should lie on the curve $\left(\frac{3x_1}{2}\right)^2 + (3y_1)^2 = p^2$

i.e., the required locus is $\frac{x^2}{4} + \frac{y^2}{1} = \frac{p^2}{9}$.

5. SHIFTING OF AXES

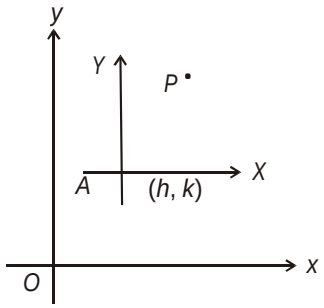
5.1 Translation of Coordinate Axes

When origin is shifted to a new position *A*(*h*, *k*) without changing the directions of the axes, then coordinates of all points in the plane are obtained by the formulae:

$x = X + h \qquad y = Y + k$

where, *P* (*x*, *y*) referred to *ox*, *oy*

and P (*X*, *Y*) referred to *AX, AY*



5.2 Rotation of Coordinate Axes

When, coordinate axes are rotated in anti-clockwise direction through an angle θ, about the origin, then coordiantes of a point *P* in the plane are obtained by the formulae.

$$\left.\begin{aligned} x &= X\cos\theta - Y\sin\theta \\ y &= X\sin\theta + Y\cos\theta \end{aligned}\right\} \text{ and } \left\{\begin{aligned} X &= x\cos\theta + y\sin\theta \\ Y &= y\cos\theta - x\sin\theta \end{aligned}\right.$$

Where *P* is (x, y) referred to coordinate system *oxy* and *P* is (X, Y) referred to new coordiante system *oXY*.

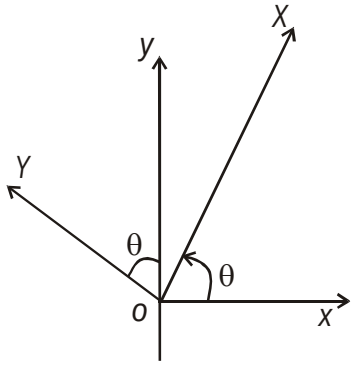


Illustration 6: Find the equation of the curve $x^2 + 2y^2 + 3x - 4x + 7 = 0$, when the origin is transferred to the point $(1, -2)$ without changing the direction of axes.

Solution : If $P(x, y)$ be any point on the curve and (x_1, y_1) be the coordinates of *P*. *w*. *r*. to new axes then $x_1 = x - \alpha = x - 1$ and $y_1 = y - \beta = y + 2$

$\therefore\ x = x_1 + 1,\ y = y_1 - 2$

Hence new equation will be

$(x_1 + 1)^2 + 2(y_1 - 2)^2 + 3(x_1 + 1) - 4(y_1 - 2) + 7 = 0$

or $x_1^2 + 2x_1 + 1 + 2(y_1^2 - 4y_1 + 4) + 3x_1 + 3 - 4y_1 + 8 + 7 = 0$

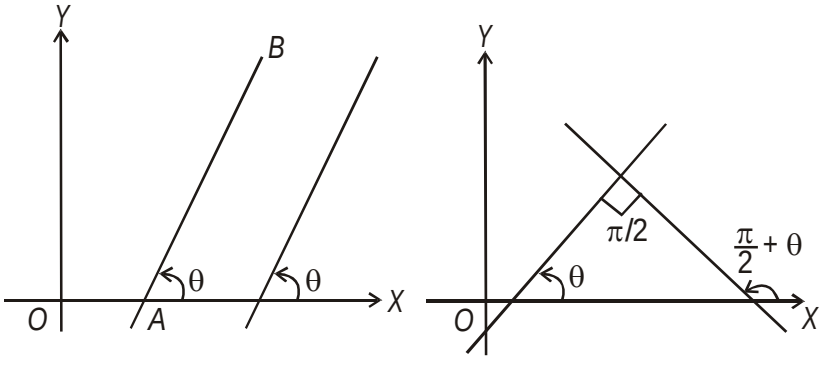
or $x_1^2 + 2y_1^2 + 5x_1 - 12y_1 + 27 = 0$

Thus new equation of the curve will be

$x^2 + 2y^2 + 5x - 12y + 27 = 0$

Note :- New equation of the curve can be directly obtained by putting (x + 1) in place of x and (y –2) in place of y in the given equation of the curve.

6. THE STRAIGHT LINE



If a straight line *AB* makes an angle θ with the positive direction of the *x*-axis, tan θ is called the slope or gradient of the straight line and is usually denoted by the letter '*m*'.

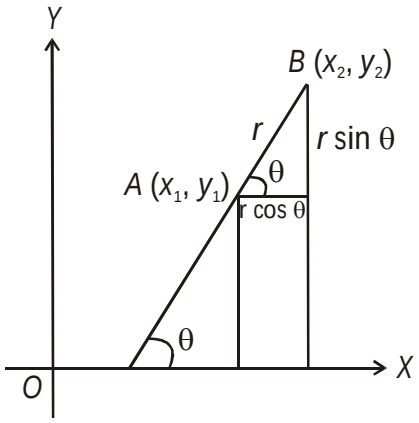
It follows that

(i) If two lines are parallel, their slopes are equal, for the lines must be equally inclined to the positive direction of the *x*-axis.

(ii) If two lines are perpendicular the product of their slopes is –1, for if one line is Inclined at an angle θ to the *x*-axis, the other must be inclined at $\frac{\pi}{2}+\theta$, hence their slopes are tan θ and $\tan\left(\frac{\pi}{2}+\theta\right)$, i.e., tan θ and –cot θ.

$\therefore$ the product is –1.

6.1 To Find The Slope of Line Joining Any Two Points (x_1, y_1) And (x_2, y_2)



Let the segment joining of the points be of length *r* and let the line be inclined to the *x*-axis at angle θ.

$\therefore$ $r\cos\theta = x_2 - x_1$

$r\sin\theta = y_2 - y_1$

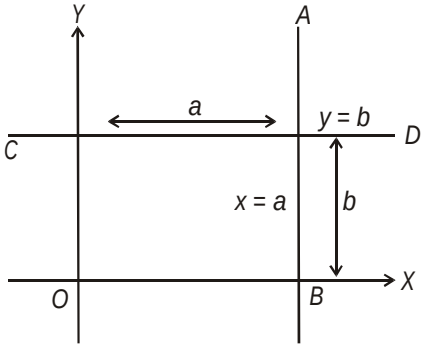
and therefore $\tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$

and tan θ is the required slope. The expression for the slope is, therefore,

$$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{\text{difference of the ordinates of the two points}}{\text{difference of the abscissa}}$$

6.2 Lines Parallel To The Co Ordinate Axes

Let the line *AB* be parallel to the *Y*-axis and at a distance '*a*' from it. Every point on *AB* will have its abscissa '*a*', and hence the equation of *AB* is $x = a$. By putting $a = 0$, we deduce that the equation of the *y*-axis is $x = 0$.



Similarly, the equation of the straight line *CD* parallel to the *X*-axis and at a distance '*b*' from it is $y = b$. By putting b = 0 we deduce that the equation to the *X*-axis is $y = 0$

6.3 Angle Between Two Given Straight Lines

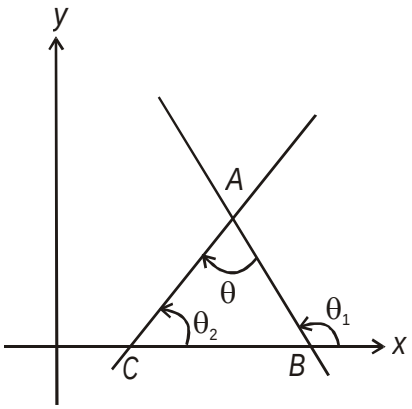
Let *AB, AC* have slopes m_1, m_2 and be inclined to the *X*-axis at θ_1, θ_2.

Then $\angle = \theta_1 - \theta_2 = \theta$

$$\therefore \tan \angle \boldsymbol{CAB} = \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1 \tan\theta_2}$$

i.e., $$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

where θ is the angle between *AB* and *AC*



Note :- It is customary to take θ, as the acute angle between the two lines, and hence mostly one can take the above formula as tan θ = $\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$

If the lines are parallel,

$\tan\theta = 0$, since $\theta = 0$

$\therefore\ m_1 - m_2 = 0$

$\therefore\ m_1 = m_2$

and if the lines are perpendicular, $\tan\theta$ is not defined, since $\theta = \frac{\pi}{2}$, and

therefore $m_1 m_2 + 1 = 0$

$\therefore\ m_1 m_2 = -1$.

Student Task

If $m_1 m_2 = -1$, the two lines are perpendicular. Is the converse also true?

Illustration 7 : Find the acute angle between the two lines with slopes $\frac{1}{5}$ and $\frac{3}{2}$.

solution : If the angle beyween the lines is θ,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\frac{1}{5} - \frac{3}{2}}{1 + \left(\frac{1}{5}\right)\times\left(\frac{3}{2}\right)}\right| = |-1| = 1$$

Therefore the angle is 45°.

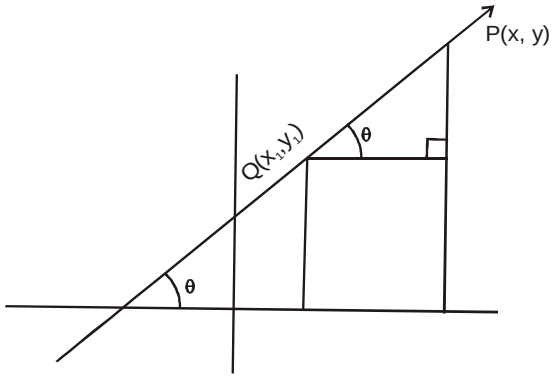
7. VARIOUS FORMS OF THE EQUATION OF A STRAIGHT LINE

It was said earlier that every curve, in coordinate geometery, is represented by an equation which is a relation connecting *x* and *y*, the coordinates of any point which lies on that curve. In particular, any straight line has also an equation to represent it. It may be seen that to represent -a straight line we need to be given two independent information; and depending upon the two, the equation also changes.

We have therefore the following forms of equations to straight lines:

7.1 Slope one Point form

Given that *a* line has a slope ($m = \tan\theta$) which gives the direction it may be noted that '*m*' alone does not give the equation of the line and with the same slope there can be any number of straight lines all of which are parallel. Given that it passes through a given point $y - y_1 = m(x - x_1)$



$$m = \frac{y - y_1}{x - x_1} \qquad y - y_1 = m(x - x_1)$$

Illustration 8 : If a line has a slope $= \frac{1}{2}$ and passes through (–1, 2); find its equation.

Solution : The equation of the line is $y - 2 = \frac{1}{2}(x - (-1))$

i.e., $2y - 4 = x + 1$

i.e., $x - 2y + 5 = 0$

7.2 Y - Intercept Form

Given the slope '*m*' and the length '*c*' (called the intercept) cut off on the *y*-axis by the line. In this case the form of the equation is $y = mx + c$

Illustration 9: If a line has a slope $\frac{1}{2}$ and cuts off long the positive *y*-axis of length $\frac{5}{2}$ find the equation of the line.

Solution : $y = \frac{1}{2}x + \frac{5}{2}$

i.e., $2y = x + 5$

i.e., $x - 2y + 5 = 0$

7.3 Two Points Form

Equation of a line passing through two points (x_1, y_1) and (x_2, y_2).
In this case the equation to the line is of the form

$$\frac{y-y_1}{x-x_1} = \frac{y_1-y_2}{x_1-x_2}$$

Where $x_1 \neq x_2$ and $\left(\frac{y_1-y_2}{x_1-x_2}\right)$ is the slope (m) of the line.

Let AB be a line which passes through two points $Q(x_1y_1)$ and $R(x_2, y_2)$ let $P(x,y)$ be any point an He line AB

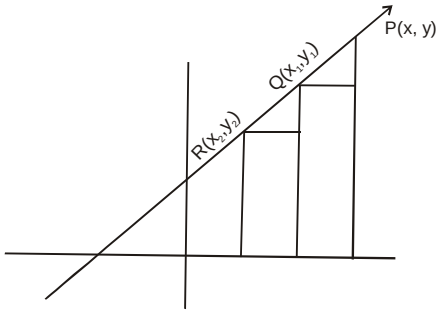


Illustration 10: If a line passes through two points (1, 5) and (3, 7), find its equation.

Solution : $\frac{y-5}{x-1} = \frac{-7+5}{-3+1} = \frac{-2}{-2} = 1$

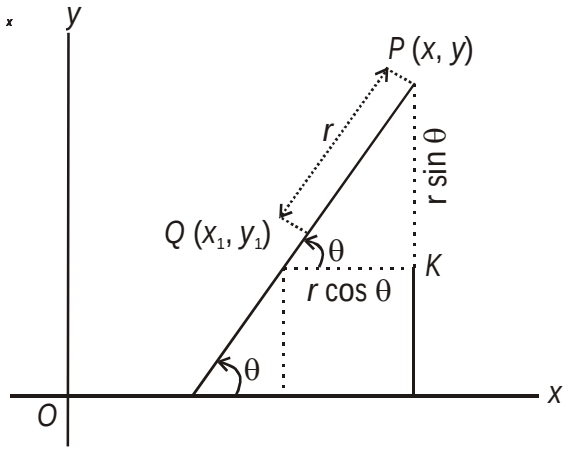
i.e., $y - 5 = x - 1$

i.e., $x - y + 4 = 0$

7.3 Symmetric Form or distance form or Parametric form of a line

If (x_1, y_1) is a given point on a straight line, (x, y) any point on the line, θ the inclination of the line and r the distance between the points (x, y) and (x_1, y_1) then $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta}$

Let $P(x, y)$ represents any point on the line. $Q(x_1, y_1)$ is a given point, such that $PQ = r$.
Draw lines through Q and P parallel to QK parallel to OX.
From the triangle QKP.
$QK = PQ \cos\theta$



or $x - x_1 = r\cos\theta$ and $\frac{x-x_1}{\cos\theta} = r$

Also $PK = PQ \sin\theta$ or

$y - y_1 = r\sin\theta$ and $\frac{y-y_1}{\sin\theta} = r$

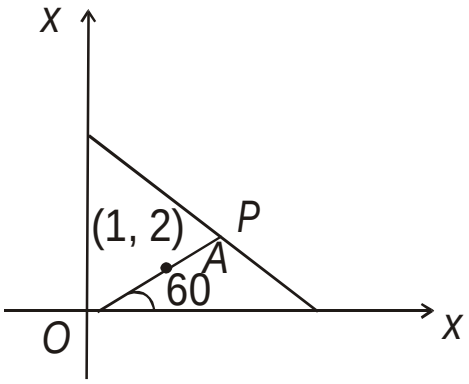
$\therefore \frac{x-x_1}{\cos\theta}\frac{y-y_1}{\sin\theta} = r$

This is called the symmetric form of the equation to a line (This will be found very useful in cases where we have to deal with the length (r) of any portion of a line.)

Illustration 11: A straight line passes through a point A (1, 2) and makes an angle 60° with the x-axis. This line intersects the line $x + y = 6$ at the point P. Find AP.

Solution : Let required line be

$$\frac{x-1}{\cos 60^o} = \frac{y-2}{\sin 6^o} = r \ (\equiv AP)$$



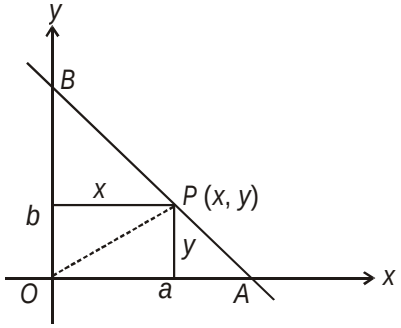
$\therefore$ P is $(1 + r\cos 60^o , 2 + r \sin 60^o)$ which lies on $x + y = 6$

The condition is $\left(1+\frac{r}{2}\right)+\left(2+r\frac{\sqrt{3}}{2}\right) = 6$ or $r\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\right)=3$

$$\therefore r = \frac{6}{1+\sqrt{3}} = \frac{6\left(1-\sqrt{3}\right)}{\left(1+\sqrt{3}\right)\left(1-\sqrt{3}\right)} = \frac{6\left(1-\sqrt{3}\right)}{\left(1+\sqrt{3}\right)\left(1-\sqrt{3}\right)} = \frac{6\left(1-\sqrt{3}\right)}{2} = 3\left(\sqrt{3}-1\right)$$

7.5 Intercept Form

Let the line make intercepts a and b on the axes of x and y respectively.
Let $P(x, y)$ represent any point on the line. Join OP.



We have,

Area of triangle *OAB* = area of triangle *OAP* + area of triangle *OPB*

i.e., $\frac{1}{2}ab = \frac{1}{2}ay + \frac{1}{2}bx$

or $\frac{x}{a} + \frac{y}{b} = 1$

This is called the intercept form of the equation of straight line.

Illustration 12: Find the equation of the straight line , which passes through the point (3, 4) and whose intercept on *y*-axis is twice that on *x*-axis.

Solution :

Let the equation of the line be

$\frac{x}{a} + \frac{y}{b} = 1$ .....(i)

According to the question $b = 2a$

$\therefore$ from (i) equation of line will become

$\frac{x}{a} + \frac{y}{b} = 1$ or $2x + y = 2a$ .....(ii)

Since line (ii) passes through the point (3, 4)

$\therefore$ $2 \times 3 + 4 = 2a$ $\therefore$ $a = 5$

$\therefore$ from (ii), equation of required line will be

$2x + y = 10$.

7.6 Normal Form

If *P* is the length of the perpendicular from the origin upon a straight line, and that α is the angle the perpendicular makes with the axis of *x*, equation of the straight line can be obtained as follows :

Let *PQ* be the straight line, *OQ* (= *p*) the perpendicular drawn to it from the origin *O*, and $\angle QOX = \alpha$

Draw the ordinate *PN* also draw *NR* perpendicular to *OQ*, and *PM* perpendicular to *RN*

We have $\angle PNM = 90^\circ - \angle RNO = \alpha$

$\therefore$ $p = OQ = OR + RQ = OR + PM$

$= ON \cos\alpha + PN \sin\alpha$

or $p = x \cos\alpha + y \sin\alpha$

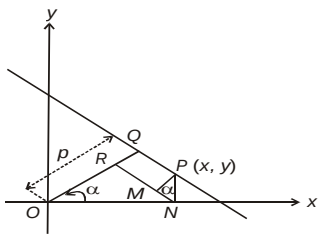
$\therefore$ $x \cos\alpha + y \sin\alpha = p$ is the required equation.

This is called the perpendicular form.

Alternatively,

Suppose $\frac{x}{a} + \frac{y}{b} = 1$ is the equation of the straight line

In this case, $a = \frac{p}{\cos\alpha}, b = \frac{p}{\sin\alpha}$



$\therefore$ by substitution, the equation becomes

$$\frac{x\cos\alpha}{p}+\frac{y\sin\alpha}{p}=1 \text{ or } x\cos\alpha + y\sin\alpha = p$$

This is known as the normal form of the equation to a straight line.

Illustration 13: Find the equation of the straight line upon which the length of perpendicular from origin is $3\sqrt{2}$ units and this perpendicular makes an angle of 75° with the positive direction of x-axis.

Solution :

Let *AB* be the required line and *OL* be perpendicular to it.

Given *OL* = $3\sqrt{2}$

And $\angle LOA = 75^o$

$\therefore$ equation of line *AB* will be

$x\cos 75^o + y\sin 75^o = 3\sqrt{2}$

[Normal Form] ....(i)

Now $\cos 75^o = \cos(30^o + 40^o)$

$$= \frac{\sqrt{3}}{2}.\frac{1}{\sqrt{2}}-\frac{1}{2}.\frac{1}{\sqrt{2}}=\frac{\sqrt{3}-1}{2\sqrt{2}}$$

and $\sin 75^o = \sin(30^o + 45^o) = \sin 30^o \cos 45^o + \cos 30^o \sin 45^o$

$$= \frac{1}{2}.\frac{1}{\sqrt{2}}-\frac{\sqrt{3}}{2}.\frac{1}{\sqrt{2}}=\frac{\sqrt{3}+1}{2\sqrt{2}}$$

$\therefore$ from (i) equation of line *AB* is

$$x\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)+y\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)=3\sqrt{2}$$

or $\left(\sqrt{3}-1\right)x+\left(\sqrt{3}+1\right)y-12=0$

7.7 Reduction of General Equation to Standard Form

Let $Ax+By+C=0$ be the general equation of a straight line where A and B are not both zero.

(i) Reduction of Slope intercept form

$Ax+By+C=0$

$By=-Ax-C$

$$y=\left(-\frac{A}{B}\right)+\left(\frac{C}{B}\right)$$

$$\text{Slope (m)} = \frac{-A}{B}=-\frac{\text{coefficient of x}}{\text{coefficient of y}}$$

$$\alpha\text{intercept}=\frac{-C}{\beta}=-\frac{\text{Constant term}}{\text{Coeffcient of y}}$$

(ii) Reduction to intercept form

$Ax+By+C=0$

$Ax+By=-C$

$$\frac{A}{-C}x+\frac{By}{-C}=1$$

$$\frac{x}{\left(\frac{-C}{A}\right)}+\frac{y}{\left(\frac{-C}{B}\right)}=1$$

$$X-\text{intercept}=\frac{-C}{A}=\frac{\text{Constant term}}{\text{Coefficient of x}}$$

$$Y-\text{intercpt}=\frac{-C}{B}=\frac{\text{Constant term}}{\text{Coefficient of y}}$$

(iii) Reduction to Normal form

General Equation is $Ax+By+C=0$ **Let its** normal form be $x\cos\alpha+y\sin\alpha=p$ the equations $Ax+By+C=0$ and $x\cos\alpha+y\sin\alpha=p$ represent the same line

$$\frac{A}{\cos\alpha}=\frac{B}{\sin\alpha}=\frac{C}{-P}$$

$$\cos\alpha=\frac{-AP}{C},\sin\alpha=\frac{-BP}{C}$$

$$\cos^2\alpha+\sin^2\alpha=1$$

$$\left(\frac{-AP}{C}\right)^2+\left(\frac{-BP}{C}\right)^2=1$$

$$p^2=\frac{C^2}{A^2+B^2}$$

$$P=\frac{|C|}{\sqrt{A^2+B^2}}$$

Hance $\cos\alpha=-\frac{|C|}{C}\frac{A}{\sqrt{A^2+B^2}}$

Illustration 14: Find the value of *k* so that the straight line $2x + 3y + 4 + k(6x – y + 12) = 0$ and $7x + 5y – 4 = 0$ are perpendicular to each other.

Solution : Given lines are

$(2 + 6k)\,x + (3 – k)y + 4 + 12k = 0$ .....(i)

and $7x + 5y – 4 = 0$

slope of line (i) $m_1 = -\frac{2+6k}{3-k}=\frac{2+6k}{k-3}$

and slope of line (ii) $m_2 = -\frac{7}{5}$

Since line (i) is perpendicular to line (ii)

$\therefore\left(\frac{2+6k}{k-3}\right)\left(-\frac{7}{5}\right)=-1$ or $(2 + 6k)\,7 = 5\,(k – 3)$ or $14 + 42k = 5k – 15$ or $37k = –29$ or k

$=-\frac{29}{37}$

Student Task

Q1. Let A (6, –1), B (1, 3) and C (λ, 8) be three points such that $AB = BC$, find the value (s) of λ.

Q2. The coordinates of the point which divides the line joining (5, –2) and (9, 6) externally in the ration 3 : 1 are (8, 4).(True / False)

Q3. The area of a triangle haing vertices (λ, $\lambda - 2$), ($\lambda + 3$, λ) and ($\lambda + 2$, $\lambda + 2$), is independent of λ.(True /false)

Q4. If $P\left(t+\frac{1}{t}, t-\frac{1}{t}\right)$ is a variable point, find its locus.

Q.5 The equation of line joining the points (-1, 3) and (4, –2) is .....

Q.6 The equation of line whose intercept on x-axis and y-axis are respectively twice and thrice of those by the line $3x + 4y = 12$ is .....

Q.7 Write the parametic form of the line passing through (–1, 2) and having slope $-\frac{1}{\sqrt{3}}$.

Q8. Find the y – intercept of the line $3x – 2y + 7 = 0$.

Q9. The parameter α in the normal form of the line $\sqrt{3}x + y + 4 = 0$ has the value $\frac{\pi}{6}$.

Q10. The acute angle between the lines $y = 4x – 1$and $x = 4y + 1$ is .....

Answer Key :

1. (–3, 5), 2. False 3. True 4. $x^2 – y^2 = 4$ 5. $x + y – 2 = 0$

6. $9x + 8y = 72$ 7. $\frac{x+1}{\cos\left(\frac{5\pi}{6}\right)} = \frac{y-2}{\sin\left(\frac{5\pi}{6}\right)} = r$ 8. 7/2 9. Flase 10. $\tan^{-1}\frac{15}{8}$

7.8 Equations of Lines Parallel and Perpendicualr to a Given Line

Theorem :1 The equation of line parallel to $ax + by + c = 0$ is $ax + by + \lambda = 0$ where λ is some corstant.

Theorem :2 The equation of the line perpendicular to the line $ax + by + c = 0$ is $bx – ay + \lambda = 0$ where λ is some constant.

8. POINT OF INTERSECTION OF TWO LINES

Consider two non-parallel lines : $ax + by + c = 0$ and $a'x + b'y + c' = 0$

i.e. $-\frac{a}{b} \neq \frac{a'}{b} \Rightarrow ab' - a'b' \neq 0$

Their point of intersection can be obtained by solving their equations simultaneously which can be done through substitution, elimination or cross-multiplication rule. The point of intersection is given by

$$x = \frac{bc' - bc}{ab' - a'b}, y = \frac{a'c - ac'}{ab' - a'b}$$

Illustration 15: Show that the lines $2x - y - 12 = 0$ and $3x + y - 8 = 0$ intersect at a point which is equidistant from both the coordinate areas.

Solution : $2x - y - 12 = 0$

$3x + y - 8 = 0$

Using cross multiplication rule, we have

$$\frac{x}{(-1)(-8)-(1)(-12)} = \frac{-y}{(2)(-8)-(3)(-12)} = \frac{1}{(2)(1)-(3)(-1)}$$

$$\Rightarrow \frac{x}{20} = \frac{-y}{20} = \frac{1}{5} \Rightarrow x = \frac{20}{5} = 4 \text{ and } y = -\frac{20}{5} = -4$$

So, the point is (4, –4) which is at a distance 4 units from both the coordinate axes.

9. AREA OF TRIANGLE WHEN EQUATION OF SIDES ARE GIVEN

Let equation of sides be

$a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

$a_3x + b_3y + c_3 = 0$

Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$\therefore$ area of triangle $ABC = \dfrac{\Delta^2}{2|C_1C_2C_3|}$,

where C_1, C_2, C_3 are cofactors of c_1, c_2, c_3 in the determinant Δ.

Illustration 16: Find the area of triangle formed by the lines $x - y + 1 = 0$, $2x + y + 4 = 0$ and $x + 3 = 0$.

Solution : $$\text{Area} = \frac{1}{2|C_1C_2C_3|}\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 4 \\ 1 & 0 & 3 \end{vmatrix}^2$$

where $C_1 = (2)(0) - (1)(1) = -1$

$C_2 = -[(1)(0) - (1)(-1) = -1$

$C_3 = (1)(1) - (2)(-1) = 3$

Putting these values, we get

$$\text{Area} = \frac{1}{2(3)}\left[1(-4-1)+3(1+2)\right]^2 \text{ (expanding along last row)} = \frac{8}{3} \text{ sq. units}$$

10. CONDITION OF COLLINEARITY

Three points A, B and C or more are said to be collinear when they lie on the same line.

Collinearity can be cheaked in two simple ways.

(i) If Slope of AB = slope of BC, points A, B, C are collinear.

(ii) If Area of $\Delta ABC = 0$, points A, B, C are collinear.

(iii) Collinear points must satisfy section formulae.

Illustration 17 : Prove that the points (1, 2), (–3, λ) and (4, 0)are collinear if λ is equal to $\frac{14}{3}$.

Solution : Let *A* (1, 2), *B*(–3, λ) and *C* (4, 0) to be the three points.

Slope of *AB* = slope of *AC*

$$\Rightarrow \frac{\lambda-2}{-3-1}=\frac{2-0}{1-4}\Rightarrow \frac{\lambda-2}{-4}=-\frac{2}{3}$$

$$\Rightarrow \lambda=2+\frac{8}{3}=\frac{14}{3}$$

Alternatively,

$$\text{Area } (\Delta ABC) = \frac{1}{2}\begin{vmatrix}1 & 2 & 1\\ -3 & \lambda & 1\\ 4 & 0 & 1\end{vmatrix}=0$$

$\Rightarrow 4(2 – \lambda) + (\lambda + 6) = 0$

$\Rightarrow –4\lambda + \lambda + 8 + 6 = 0$ $\Rightarrow \lambda = \frac{14}{3}$

Student Task

How will you check collinearity of four points?

11. CONDITION OF CONCURRENCY

Three or more lines are said to be concurrent if they all pass through the same points.

If the lines

$a_1 x + b_1 y + c_1 = 0$

$a_2 x + b_2 y + c_2 = 0$

$a_3 x + b_3 y + c_3 = 0$

are concurrent, the area of triangle formed by them is zero.

$$\therefore \begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$$

Illustration 18: The line x + ly – 4 = 0 passes through the point of intersection of $4x – y + 1 = 0$ and $x + y + 1 = 0$. Find the values of λ.

Solution :

$$\Rightarrow \begin{vmatrix}1 & \lambda & -4\\ 4 & -1 & 1\\ 1 & 1 & 1\end{vmatrix}=0 \Rightarrow -2 – 3\lambda – 20 = 0 \quad \Rightarrow \lambda = -\frac{22}{3}$$

12. FAMILY OF LINES

Family means a group of lines having a particular characterstic e.g., family of lines parallel to *x* - axis are of the from $y = k$, where *k* is any arbitrary real numer. Family of lines perpendicular to $2x – y + 4 = 0$ is $x + 2y + k = 0$ since product of slopes has to be –1. so, we can say

(i) Family of lines parallel to $ax + by + c = 0$ is of the form $ax + by + k = 0$

(ii) Family of lines perpendicular to $ax + by + c = 0$ is the form $bx - ay + k = 0$

(iii) Family of lines through the point of intersection of the lines

$ax + by + c = 0$ ....(i)

and $a'x + b'y + c' = 0$ ....(ii)

can be written as

$(ax + by + c) + k(a'x + b'y + c') = 0$ .....(iii)

Equation (i) represent a straight line, for it is of the first degree.

Let (x_1, y_1) be the coordinates of the common points of (i) and (ii)

The points (x_1, y_1) lies on (ii);

$\therefore\ ax_1 + by_1 + c = 0$ .....(iv)

The points (x_1, y_1) lies on (ii)

$\therefore\ a' x_1 + b' y_1 + c' = 0$ .....(v)

Multiplying (v) by k, and adding to (iv),we get

$ax_1 + by_1 + c + k(a' x_1 + b'y_1 + c') = 0$ .....(iv)

From this we see that the values x_1, y_1 satisfy the equation (iii)

$\therefore$ the straight line represented by (iii) passes through (x_1, y_1) the common point of (i) and (ii).

Student Task

What is the family of lines passing through origin ?

Illustration 19: Find the equation of a line parallel to $x + 2y = 3$ and passing through the points (3, 4).

Solution : Any line parallel to $x + 2y = 3$ is of the form $x + 2y = k$.

The point (3, 4) satisfies this equation

$\Rightarrow 3 + 2(4) = k \Rightarrow k = 11$

So, required line is $x + 2y = 11$

Illustration 20: Find the equation of line perpendicular to $2x - 3y = 5$ and cutting off an intercept 1 on the x-axis

Solution : Any line perpendicular to $2x - 3y = 5$ is of the form $3x + 2y = k$

Putting $y = 0$, we get $x = \frac{k}{3}$, the x-intercept.

Since, $\frac{k}{3} = 1 \Rightarrow k = 3$

The required line is $3x + 2y - 3 = 0$

Student Task

What is the equation of line perpendicular to $x = 5$ and passing through origin?

Illustration 21: Find the equation of the straight line passing through (2, –9) and the point of intersection of lines $2x + 5y - 8 = 0$, $3x - 4y - 35 = 0$.

Solution : Any line through teh point of intersection of given lines is $2x + 5y - 8 + k(3x - 4y - 35) = 0$ Point (2, –9) lies on it, implies $k = 7$.

$\therefore$ The required line becomes $x - y - 11 = 0$

Illustration 22: Find the equation of straight line passing through the point of intersection of lines $3x - 4y + 1 = 0$, $5x + y - 1 = 0$ and cutting off equal intercept from coordiate axes.

Solution : Any line passing through the point of intersection of given lines is

$3x - 4y + 1 + k(5x + y - 1) = 0$

or $(3 + 5k)\,x + (k - 4)\,y = k - 1$

Writing it in intercept form as $\frac{x}{\frac{k-1}{3+5k}} + \frac{y}{\frac{k-1}{k-4}} = 1, if\, k \neq 1$

For equal intercepts: $\frac{k-1}{k-4} = \frac{k-1}{3+5k}$ which gives $k = \frac{-7}{4}$

Hence the required equation is $23x + 23y = 11$

13. DISTANCE OF A POINT FROM A LINE

The perpendicular distance of a point from a line can be obtained when the equation of the line and the coordinates of the point are given.

Case I :

Let us first derive the formula for this purpose when the equation of the line is given in normal form.

Let the equation of the line l in normal form be

$x \cos \alpha + y \sin \alpha = p$,

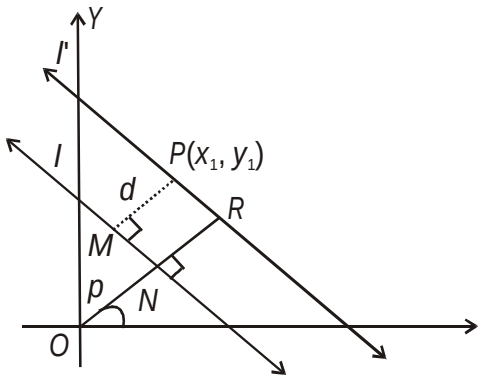
when α is the angle made by the perpendicular from origin to the line with positive direction of x-axis and p is the length of this perpendicular. Let $P(x_1, y_1)$ be the point, not on the line l. Let the perpendicular drawn from the point P to the line l be PM and $PM = d$. Point P is assumed to lie on opposite side of the line l from the origin O. Draw a line l' parallel to the line l through the point P. Let ON be perpendicular from the origin to the line l which meets the line l' in points R.

Obviously

$ON = p$ and $\angle XON = \alpha$

Also, from the figure, we note that

$OR = ON + NR = p + MP$



Therefore, length of the perpendicular from the origin to the line l' is

$OR = p + d$

and the angle made by the perpendicular OR with positive direction of x-axis is α. Hence, the equation of the line l' in normal form is

$x \cos \alpha + y \sin \alpha = p + d$

Since the line l' passes through the point P, the coordinates (x_1, y_1) of the point P should satisfy the equation of the line l' giving

$x_1 \cos \alpha + y_1 \sin \alpha = p + d$,

or $\quad d = x_1 \cos \alpha + y_1 \sin \alpha - p$.

The length of a segment is always non - negative. Therefore, we take the absolute value of the RHS, i.e.,

$d = |\, x_1 \cos\alpha + y_1 \sin\alpha - p \,|$.

Thus, the length of the perpendicular is the absolute value of the result obtained by substituting the coordinates of point P in the expression $x \cos\alpha + y \sin\alpha - p$

Case II :

Let the equation of the line be

$Ax + By + C = 0$

Reducing the general equation to the normal form, we have

$$\pm\frac{A}{\sqrt{A^2+B^2}}x \pm \frac{B}{\sqrt{A^2+B^2}}y = \pm\frac{C}{\sqrt{A^2+B^2}}$$

Where sign is taken + or – so that the RHS is positive.

(a) When $C < 0$

In this case the normal form of the equation of the line / becomes

$$\frac{A}{\sqrt{A^2+B^2}}x + \frac{B}{\sqrt{A^2+B^2}}y = -\frac{C}{\sqrt{A^2+B^2}}$$

Now, from the result of case (I), the length of the perpendicular segment drawn from the point $P(x_1, y_1)$ to the line (II) is

$$d = \left|\frac{A}{\sqrt{A^2+B^2}}x_1 + \frac{B}{\sqrt{A^2+B^2}}y_1 = +\frac{C}{\sqrt{A^2+B^2}}\right| \quad \text{i.e., } d = \left|\frac{Ax_1+By_1+C}{\sqrt{A^2+B^2}}\right|$$

(b) When $C > 0$

In this case the normal form of the equation (II) becomes

$$-\frac{A}{\sqrt{A^2+B^2}}x - \frac{B}{\sqrt{A^2+B^2}}y = \frac{C}{\sqrt{A^2+B^2}}$$

Again, by the result of case (I), the distance d is

$$d = \left|\frac{-Ax_1-By_1-C}{\sqrt{A^2+B^2}}\right| \qquad \text{or} \quad d = \left|\frac{Ax_1+By_1+C}{\sqrt{A^2+B^2}}\right|$$

The perpendicular distance of (x_1, y_1) from the line $ax + by + c = 0$ is given by

$$d = \left|\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|$$

So, Perpendicular sidtance from origin is $\frac{|c|}{\sqrt{a^2+b^2}}$.

Illustration 23: The equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$. Find the length of side of the triangle.

Solution : Equation of side BC is

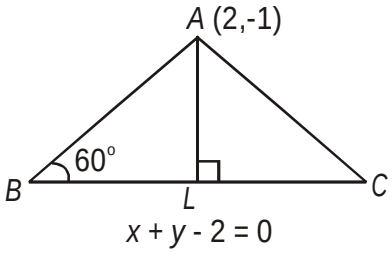
$x + y - 2 = 0$ ...(i)

$A \equiv (2, -1)$

Now $AL = \left|\frac{2-1-2}{\sqrt{1^2+1^2}}\right| = \frac{1}{\sqrt{2}}$

From $\Delta\, ABL$, $\sin 60^\circ = \frac{AL}{AB}$

$$\therefore \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{2}AB} \text{ or } AB = \sqrt{\frac{2}{3}}$$



14. DISTANCE BETWEEN PARALLEL LINES

The distance between $ax + by + c = 0$ and $ax + by + c' = 0$

is given by, $d = \dfrac{|c - c'|}{\sqrt{a^2 + b^2}}$

Make sure that coefficients of x and y are same in both the lines before applying the formula.

Illustration 24: Find the distance between the lines $5x + 12y + 40 = 0$ and $10x + 24y - 25 = 0$.

Solution : Here, coefficients of x and y are not the same in both the equations. So, we write them as

$5x + 12y + 40 = 0$

$$5x + 12y - \frac{25}{2} = 0$$

Now, distance between them $= \dfrac{\left|40 - \left(-\dfrac{25}{2}\right)\right|}{\sqrt{(5)^2 + (12)^2}} = \dfrac{105}{2(13)} = \dfrac{105}{26}$

14.1 AREA OF PARALLELOGRAM

Therom:1 Area of parallelogram ABCD where sides AB, BC, CD and DA are represented by

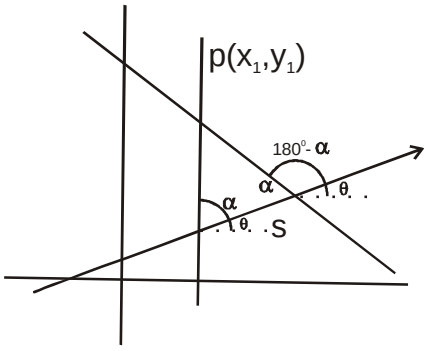
$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0$ $a_1x + b_1y + d_1 = 0, a_2x + b_2y + d_2 = 0$ is

$$\frac{P_1P_2}{\sin\theta} \text{ or } \frac{|C_1 - d_1||c_2 - d_2|}{\left|\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}\right|}$$

Where P_1 and P_2 are the distances between parallel sides and θ is the angle between two adjaeent sites

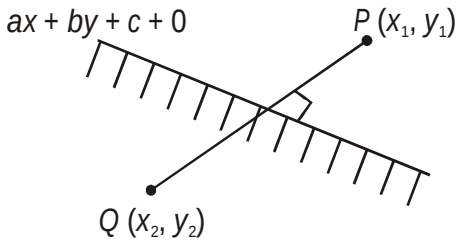
14. 2 Equation of straight lines passing thrrough a given point and making a given angle with a given line

He equations of the Straight lines which pass through a given point $(x_1\ y_1)$ and make a given the given straight line $y = mx + c$ are $y - y_1 = \tan(\theta \pm \alpha)(x - x_1)$ where $m = \tan\theta$



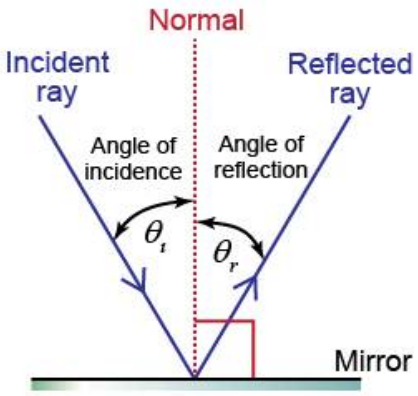
15. IMAGE OF A POINT IN A LINE

The point Q situated at the same distance behind the line mirror $ax + by + c = 0$ as the point P in front of it, is called the image of P.



15.1 Reflection of Light

When a ray of light falls an a smoth polished surface (mirror separating two mediam, a portion of its is reflected back into the first medium .



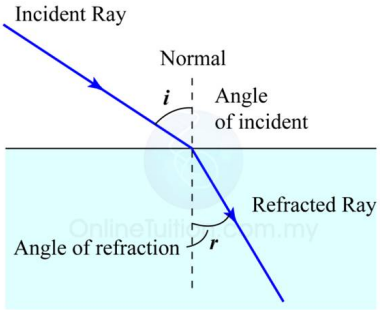
Laws of reflectional

(i) The incident ray, normal and the reflected ray to a surface at the point of incidence all lie in the same plane

(ii) the angle of incidence = angle of reflection $\angle\theta_i = \angle\theta_r$

Refraction of light

When a ray of light falls on the boundary seprating the two trans parent media there is a change in the direction of the ray



$$1\mu_2 = \frac{\sin i}{\sin r}$$

Illustration 25: If P is (1, 2) and the line mirror is $2x - y + 4 = 0$, find coordinates of its image (*i.e.*, Q).

Solution : Let the coordinates of Q be (x_2, y_2).

The mid - point of PQ lies on $2x - y + 4 = 0$

$$\Rightarrow 2\left(\frac{x_2+1}{2}\right) - \left(\frac{y_2+2}{2}\right) + 4 = 0$$

$\Rightarrow 2x_2 + 2 - y_2 - 2 + 8 = 0$

$\Rightarrow 2x_2 - y_2 + 8 = 0$

.....(i)

Also the line PQ is perpendicular to the line mirror.

$\Rightarrow 2y_2 - 4 + x_2 - 1 = 0$

$\Rightarrow x_2 + 2y_2 - 5 = 0$

.....(ii)

Solving (i) and (ii), we get

$$\frac{x_2}{5-16} = \frac{-y_2}{-10-8} = \frac{1}{4+1}$$

$$\Rightarrow x_2 = -\frac{11}{5}, y_2 = \frac{18}{5}$$

$\therefore$ *i.e.* $\left(-\frac{11}{5}, \frac{18}{5}\right)$

Student Task

Q1. The lines $x - y + 1 = 0$ and $2x + 5y - 4 = 0$ intersect at the point (4, 5). (True / False)

Q2. The area of triangle formed by the lines $x + y + 1 = 0$, $x = 5$ and $y = -3$ is .....

Q3. Points (1, 2), (2, –3) and (–3, 4) are collinear. (True / False)

Q4. The lines $x = 0$, $y = 0$ and $2x - 3y = 0$ are concurrent. (True / False)

Q5. Line parallel to $x - 2y + 4 = 0$ and at a distance 4 form it is (are) .....

Q6. The line $x - y + \lambda y = 0$ where λ is variable passes through a fixed point (True / False)

Q7. The distance of $3x - 4y + 7 = 0$ from origin is .....

Q8. The distance between the lines $3x - 4y + 7 = 0$ and $6x - 8y + 7 = 0$ is .....

Q9. The image of origin in the line $x + 4y - 1 = 0$ is .....

Q10. Line perpendicular to $2x + 5y - 1 = 0$ and passing through the point (4, 2) is .....

Answer Key :

1. False 2. 9/2 3. False 4. True 5. $x - 2y + 4\left(1 \pm \sqrt{5}\right) = 0$

6. True 7. 7/5 8. 7/10 9. $\left(\frac{2}{17}, \frac{8}{17}\right)$ 10. $5x - 2y - 12 = 0$

16. POSITION OF A POINT WITH RESPECT TO A LINE

Two points (x_1, y_1) and (x_2, y_2) are located on the same or opposite sides of the line $ax + by + c = 0$ according as the sign of $ax_1 + by_1 + c = 0$ and $ax_2 + by_2 + c$ is same or opposite.

e.g., the points (2, –3) and (4, 2) are on the same side of the line $6x - y + 2 = 0$ as $6(2) - (-3) + 2 = 17$ and $6(4) - (2) + 2 = 24$ are of same sign (positive here).

On these lines, we can say that a point (x_1, y_1) lies on the origin side of $ax + by + c = 0$ (i.e. the side in which (0, 0) lies) if $ax_1 + by_1 + c$ is of the same sign as c.

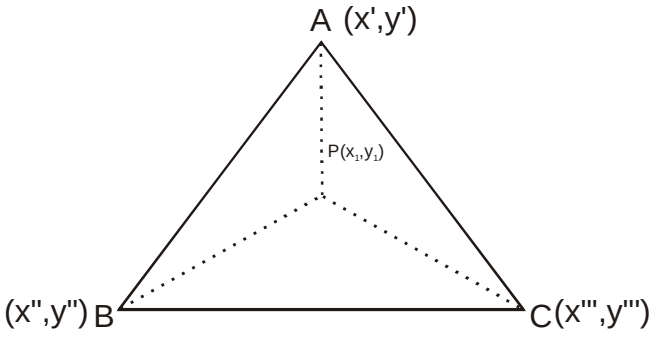
Position of a point which lies inside a triangle

Let $P(x_1, y_1)$ be the point and equations of the sides of a triangle are

$BC: a_1x + b_1y + C_1 = 0$

$CA: a_2x + b_2y + C_2 = 0$

$AB: a_3x + b_3y + C_3 = 0$



First find the co-ordinates of A,B and C says

$A = (x', y'), B = (x'', y'')$ and $C = (X''', y''')$ and if co-ordinates of A,B,C are given, then find equation of BC, CA and AB.

If $P(x_1, y_1)$ lies inside the triangle, then P and A must be on the same side of BC, P and B must be on the same side of AC, P and C must be on the same side of AB

Then

$$\frac{a_1x_1 + b_1y_1 + c_1}{a_1x' + b_1y' + c_1} > 0 \quad \text{.....(i)}$$

$$\frac{a_2x_1 + b_2y_1 + c_2}{a_2x'' + b_2y'' + c_2} > 0 \quad \text{.....(ii)}$$

$$\frac{a_3x_1 + b_3y_1 + c_3}{a_3x''' + b_3y''' + c_3} > 0 \quad \text{.....(iii)}$$

The required values of $P(x_1, y_1)$ must be intersection of there interection of there inequvalities .

Illustration 26: Find the values of λ for which the point $(2 - \lambda, 1 + 2\lambda)$ lies on the non-origin side of the line $4x - y - 2 = 0$.

Solution : Since 'c' is negative, the expression obtained by putting the coordinates of the point in the equation of line must be positive for the point to lie on non-origin side of the line.

$\therefore \quad 4(2 - \lambda) - (1 + 2\lambda) - 2 > 0 \Rightarrow 5 - 6\lambda > 0 \Rightarrow \lambda < 5/6$

Illustration 26: Determine all values of a for which the point the point (a, a^2) lies inside the triangle formed by the lines $x - y + 1 = 0$; $x + y - 3 = 0$ and $y + 2x = 0$.

Solution : ABC is the triangle formed by the three lines and $P(a, a^2)$ lies inside the triangle ABC.

Refer to the figure

1. For $x + y - 3 = 0$; (0, 0) and (a, a^2) lie on the same side.

$$a^2 + a - 3 < 0$$

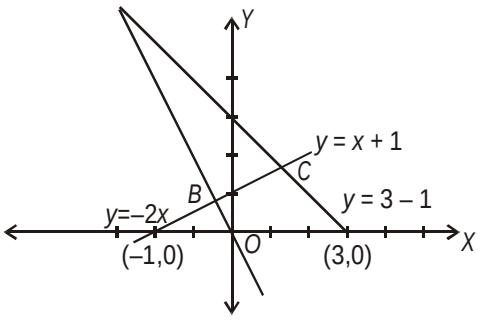
i.e., $\frac{-1-\sqrt{13}}{2} < a < \frac{-1+\sqrt{13}}{2}$ ....(i)

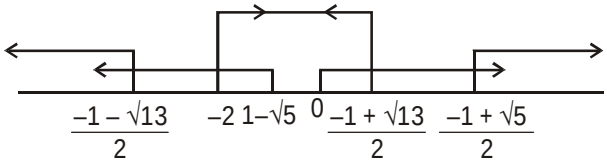
2. For $x - y + 1 = 0$; (0, 0) and (a, a^2) lie on the opposite sides.

$$a - a^2 + 1 < 0 \quad \therefore \quad a < \frac{1-\sqrt{5}}{2} (\text{or}) \, a > \frac{1+\sqrt{5}}{2}$$ ....(ii)

3. For the line $y + 2x = 0$, we have $2a + a^2 > 0$

i.e., $a > 0$ (or) $a < -2$ ....(iii)





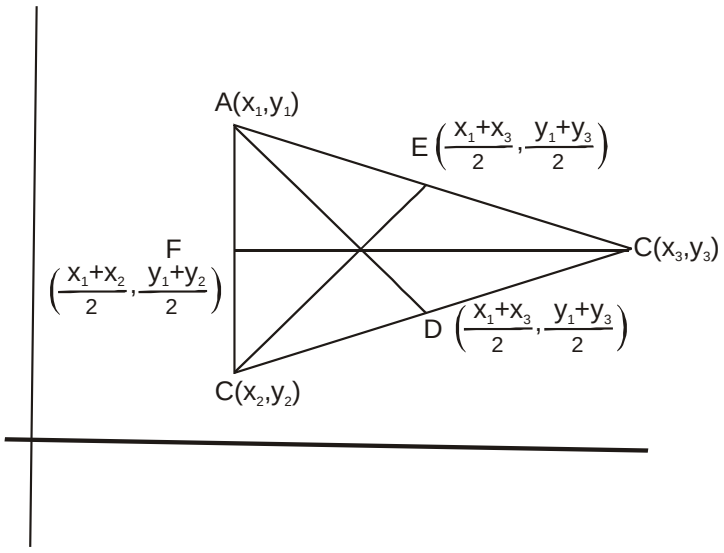
17. SOME POINTS AND THEIR COORDAINTES RELATED TO A TRIANGLE

For a triangle having vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, the sides opposite to these vertices being a, b and c respectively, we state the following results :

17.1 Centroid of a Triangle

Definition : the point of interaction of the medians of a triangle is called the centroid of the triangle and it divides the median internally in the ratio 2:1

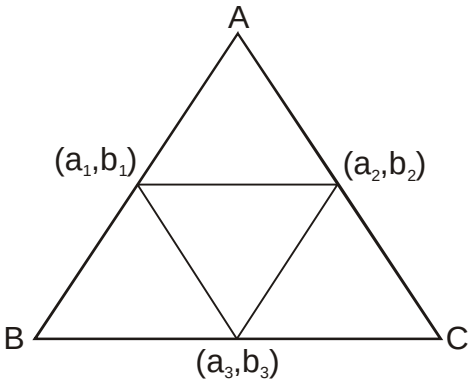
Theorem: The Co-ordinates of the centroid of the triangle whose vertices are (x_1, y_1), (x_2, y_2) and (x_3, y_3) are $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$



Note: Centroid of the triangle obtained by joining the middle points of the sides of a triange is hte same as the centroid of the original triangle

OR

If (a_1, b_1), (a_2, b_2) and (a_3, b_3) are the mid-points of the siter of a triangle. Then its centroid is given by $\left(\frac{a_1+a_2+a_3}{3}, \frac{b_1+b_2+b_3}{3}\right)$



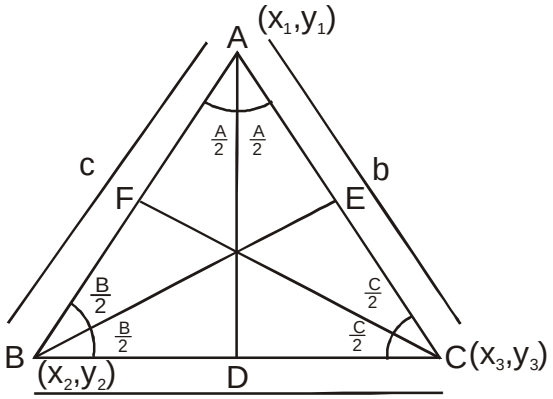
Centroid, G : Point of intersection of medians *G* is $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

17.2 Incentre

Definition: The point of intersection fo internal angle bisectors of triangle is called the incentre of the triangle.

Theorem: The co-ordinates of the incentre of a triangle whose vertices are $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ are $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$

where a,b,c are the lengths of sides BC, CA and AB respectively

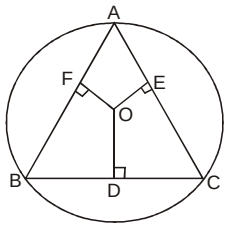


17.3 Excentres

Excentres I_1, I_2, I_3 : Point of intersection of two exterior angle bisectors and the remaining internal angle bisector. Excentre opposite angle *A* is $\left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$.

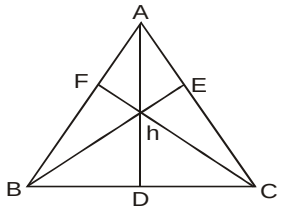
17.4 Circumcentre

The circumcentre of a triangle is the point of intersection of the perpendicular bisectors of the sides of a triangle



17.5 Orhtocentre of triangle

Let A (x_1,y_1),B(x_2,y_2) and C(x_3,y_3) be the vertices of ΔABC and it angles of ΔABC are given, then cordinates of orthocentre are.



$$\left(\frac{x_1\tan A + x_2\tan B + x_3\tan C}{\tan A+\tan B+\tan C}, \frac{y_1\tan A + y_2\tan B + y_3\tan C}{\tan A+\tan B+\tan C}\right)$$

CONCEPTS

Think about the possibilities of all/some of the above mentioned points to lie on or outside the triangle.

Illustration 28: Find the incentre I of ΔABC if *A* is (4, –2), *B* is (–2, 4) and *C* is (5, 5).

Solution :

$$BC \equiv a = \sqrt{(5+2)^2+(5-4)^2} = 5\sqrt{2}$$

$$CA \equiv b = \sqrt{(5-4)^2+(5+2)^2} = 5\sqrt{2}$$

$$AB \equiv c = \sqrt{(-2-4)^2+(4+2)^2} = 6\sqrt{2}$$

If incentre I is $(\bar{x},\bar{y})$, then by the formula,

$$\bar{x} = \frac{ax_1+bx_2+cx_3}{a+b+c} = \frac{20\sqrt{2}-10\sqrt{2}+30}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}} = \frac{5}{2}$$

$$\bar{y} = \left(\frac{ay_1+by_2+cy_3}{a+b+c}\right) = \frac{-10\sqrt{2}+20\sqrt{2}+30\sqrt{2}}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}} = \frac{5}{2}$$

$\therefore$ Incentre is $\left(\frac{5}{2},\frac{5}{2}\right)$.

Illustration 29: Find the coordinates of the orthocentre of the triangle whose vertices are (0, 1), (2, –1) and (–1, 3).

Solution : Let $A \equiv (0, 0)$, $B \equiv (2, -1)$ and $C \equiv (-1, 3)$

$$\left[\because \text{Slope of } BC = \frac{-1-3}{2+1}\right]$$

Since $AL \perp BC$, therefore equation of line AL is

$$y - 0 = \frac{3}{4}(x - 0)$$

or $4y = 3x$ or $3x - 4y = 0$ ....(i)

Since $BM \perp AC$, therefore, equation of line BM is

$$y + 1 = \frac{1}{3}(x - 2) \text{ or } 3(y + 1) = x - 2$$

or $x - 3y = 5$ ....(ii)

Solving equations (i) and (ii), we get $x = -4$, $y = -3$

$\therefore$ Orthocentre $(-4, -3)$.

18. ANGLE BISECTORS

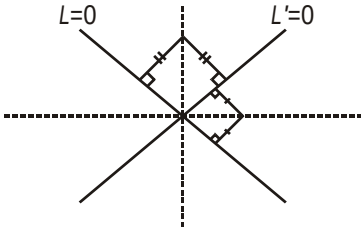
The locus of a point which moves such that its distance from two intersecting lines is same lies on the angle bisector of the two lines.

Let the two lines be

$L : ax + by + c = 0$

$L' : a'x + b'y + c' = 0$

$$\therefore \quad \frac{|ax + by + c|}{\sqrt{a^2 + b^2}} = \frac{|a'x + b'y + c'|}{\sqrt{a'^2 + b'^2}}$$



Clearly, there are two angle bisectors.

Methods to distinguish between acute angle bisector and obtuse angle bisector.

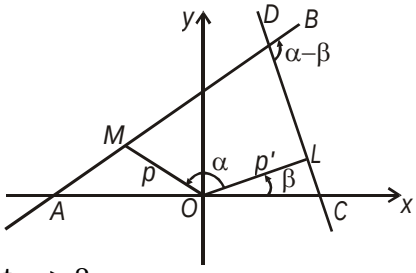
(i) Take any bisector and take any line. Find the acute angle between them. If its is less than 45°, the bisector taken lies in the acute angle region otherwise in the obtuse angle region.

(ii) The two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ divide the xy plane (the coordiante axes plane) into acute angle region and obtuse angle region. It is proposed to ascertain that, in which of these two regions, the origin is situated.

Without loss of generality, the two equations may be so adjusted that both c and c' are positive.

Let AB and CD be teh two given lines intersecting at E. p and p' are the lengths of the perpendiculars from O on AB and CD respectively.

$$\therefore \quad p = \frac{c}{\sqrt{a^2 + b^2}} \quad \text{and} \quad p' = \frac{c'}{\sqrt{a'^2 + b'^2}}$$



Let α and β be the angles made by p and p' with positive OX. Let $\alpha > \beta$.

Then $\alpha - \beta$ is the angle contained between p and p' and it is easy to see that $\angle CEB = \alpha - \beta$ (since $OLEM$ is cyclic) and this angle $\alpha - \beta$ is the supplement of the angle between the given lines in which the origin is situated. In other words, the origin is situated in that region of the angle between the two given lines, this angle being $180^o - (\alpha - \beta)$.

Case-I If $\alpha - \beta$ is acute, the origin is situated in the obtuse angle region and in that case $\cos(\alpha - \beta)$ is positive.

Case-II

If $\alpha - \beta$ is obtuse, then the origin is situated in the acute angle region and in this case $\cos(\alpha - \beta)$ is negative.

The first equation $ax + by + c = 0$ may be written as $\frac{a}{\sqrt{a^2+b^2}}x + \frac{b}{\sqrt{a^2+b^2}}y + \frac{c}{\sqrt{a^2+b^2}} = 0$ and

identifying this with $-x\cos\alpha - y\sin\alpha + p = 0$ $\left(\text{Note } p = \frac{c}{\sqrt{a^2+b^2}}\right)$

we have, $\cos\alpha = \frac{-a}{\sqrt{a^2+b^2}}$

$\sin\alpha = \frac{-b}{\sqrt{a^2+b^2}}$

Simiilarly we have, $\cos\beta = \frac{-a'}{\sqrt{a'^2+b'^2}}$

$\sin\beta = \frac{-b'}{\sqrt{a'^2+b'^2}}$

and $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \frac{aa'+bb'}{\sqrt{(a^2+b^2)(a'^2+b'^2)}}$

Again, consider two cases,

$\therefore$ In the case(i) referred to above $\cos(\alpha - \beta) > 0$

$\therefore$ $aa' + bb' > 0$

and in this case origin is in the obtuse angle region.

In case (ii) $\cos(\alpha - \beta) < 0$

$\therefore$ $aa' + b'' < 0$

and in this case origin lies in the acute angle region, c is assumed to be positive and hence $ax + by + c > 0$ for the origin region. Similarly $a'x + b'y + c' > 0$ for the region in which the origin is contained.

The bisector of the angle in which the origin is situated is

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}} \quad \text{....(i)}$$

and this is the acute angle bisector if $aa' + bb' < 0$; and is the obtuse angle bisector if $aa' + bb' > 0$; and the bisector of the angle in which the origin is not situated is

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = -\frac{a'x+b'y+c'}{\sqrt{a^{12}+b^{12}}} \quad \text{....(ii)}$$

and this is the acute angle bisector if $aa' + bb' > 0$; and is the obtuse angle bisector if $aa' + bb' < 0$.

Student Task

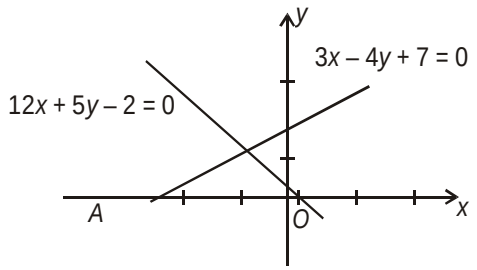
How can you find the angle between the two bisectors?

Illustration 30: Find the equation of the bisector of the acute angle between the lines $3x - 3y + 7 = 0$, $12x + 5y - 2 = 0$. Draw a rough sketch.

Solution : The bisectors of the angle between the given lines are given by

$$\frac{3x-4y+4}{\sqrt{9+16}} = \pm\frac{12x+5y-2}{\sqrt{144+25}}$$

i.e., $13(3x - 4y + 7) = \pm 5(12x + 5y - 2)$



From the figure, the bisector of the acute angle between the lines makes acute angle with the direction *OX* and hence its gradient is positive.

$\therefore$ the required bisector is given by

$13(3x - 4y + 7) = -5(12x + 5y - 2)$

i.e., $99x - 27y + 81 = 0$

i.e., $11x - 3y + 9 = 0$

Note : The other equation gives a line of negative gradient, and it represents the bisector of obtuse angle between the lines.

Alternatively, the equations of the lines may be written as

$3x - 4y + 7 = 0$

$-12x - 5y + 2 = 0$ (keeping the constant term positive)

Here, we have $a_1a_2 + b_1b_2 = -36 + 20 = -16$

i.e., $a_1a_2 + b_1b_2 < 0$, and hence the origin is situated in one of the acute angle regions.

$\therefore$ the required bisector is given by $\frac{3x-4y+7}{5} = \frac{-12x-5y+2}{13}$

i.e., $13(3x - 4y + 7) = 5(-12x - 5y + 2)$

i.e., $99x - 27y + 81 = 0$ or $11x - 3y + 9 = 0$

Again alternatively, the equations of the bisectors of the angles between the straight lines

$3x - 4y + 7 = 0$, $12x + 5y - 2 = 0$

$$\frac{3x-4y+7}{5} = \pm\frac{12x+5y-2}{13}$$

i.e., $21x + 77y - 101 = 0$ and $11x - 3y + 9 = 0$

The following rule may be used to distinguish between these two lines.

Consider one of the given lines and one of the bisectors. We may take $3x - 4y + 7 = 0$ and $11x - 3y + 9 = 0$. Determine the tangent of the angle between these two lines.

The slope of the line $3x - 4y + 7 = 0$ is $\frac{3}{4}$

and the slope of the line $11x - 3y + 9 = 0$ is $\frac{11}{3}$

Let θ be the angle between these lines.

Then $\tan\theta = \dfrac{\frac{11}{3} - \frac{3}{4}}{1 + \frac{11}{3}.\frac{3}{4}} = \dfrac{35}{45} = \dfrac{7}{9}$

i.e., $\tan\theta < 1$, and hence $\theta < 45^o$

$\therefore$ we conclude that $11x - 3y + 9 = 0$ is the equation of the bisector of the acute angle between the given lines.

Note : The other bisector has therefore equation $21x + 77y - 101 = 0$

Illustration 31: The sides of a rhombus *ABCD* are parallel to the lines $x - y + 2 = 0$ and $7x - y - 3 = 0$. The diagonals intersect at (2, 3). The vertex *A* is on the *y*-axis. Find the possible positions of *A* on the *y*-axis.

Solution : Let *A* be $(0, x_1)$

The sides of the rhombus are parallel to the lines $x - y + 2 = 0$ and $7x - y - 3 = 0$ and hence the diagonals of the rhombus are parallel to the bisectors of the angle between these lines. The bisectors have their slopes equal to the slopes of the diagonals.

The bisectors are $\dfrac{x - y + 2}{\sqrt{2}} = \pm\dfrac{7x - y - 3}{\sqrt{50}}$

i.e., $5x - 5y + 10 = 7x - y - 3$ i.e., $2x + 4y - 13 = 0$

or $5x - 6y + 10 = -7x + y + 3$ i.e., $12x - 6y + 7 = 0$

The slope of a diagonal joining $(0, x_1)$ and (2, 3) is $\dfrac{3 - x_1}{2 - 0}$ and this is $= -\dfrac{1}{2}$ or 2

$3 - x_1 = -1$ (or) $3 - x_1 = 5$

$x_1 = 4$ $x_1 = -1$

The possible positions o *A* are (0, 4) or (0, –1).

19. HOMOGENEOUS EQUATION OF THE 2ND DEGREE

The equation $ax^2 + 2hxy + by^2 = 0$, homogeneous and of the second degree always represents a pair of straight lines through the origin. If the lines are taken as $y = mx$ and $y = m'x$ then it is identical to $(y - mx)(y - m'x) = 0$.

Copmaring,

$$m + m' = \frac{-2h}{b} \text{ and } mm' = \frac{a}{b}$$

Therefore the angle between the lines is given by

$$\tan^{-1}\left|\frac{2\sqrt{h^2-ab}}{a+b}\right| \text{ and}$$

(i) when $h^2 = ab$, the two lines coincide and

(ii) when $a + b = 0$, the lines are perpendicular.

20. NON-HOMOGENEOUS 2ND DEGREE EQUATION

The general equation of 2nd degree

$ax^2 + 2hxy + 2gx + 2fy + c = 0$

represents a pair of straight lines only

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

or equivalently in the determinant form $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$

In that case, when representing two straight lines, the equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

may be written as $(\ell x + my + n)(\ell' x + m'y + n') = 0$

so that the individual equations to the straight lines are

$\ell x + my + n = 0,\ \ell' x + m'y + n' = 0$

Moreover the lines are real if $h^2 \geq ab$.

The two equations compared : When $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, with $\Delta = 0$, represents a pair of straight lines; then $ax^2 + 2hxy + by^2 = 0$ represents two straight lines through the origin and respectively parallel to the other two lines.

It is, therefore, that when

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c \equiv (\ell x + my + n)(\ell' x + m'y + n') = 0$

represents two straight lines through a point say (x_1, y_1); then

$ax^2 + 2hxy + by^2 \equiv (\ell x + my)(\ell' x + m'y) = 0$

represents two straight lines through the origin respectively parallel to the first pair.

Note : The angle formula in Article 19 holds here also.

Illustration 32 : Find straight lines represented by $6x^2 + 13xy + 6y^2 + 8x + 7y + 2 = 0$ and also find the point of intersection.

Solution : $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$\equiv 6(6)2 + 2\left(\frac{7}{2}\right)4\left(\frac{13}{2}\right) - 6\left(\frac{7}{2}\right)^2 - 6(4)^2 - 2\left(\frac{13}{2}\right)^2 = 254 - 254 = 0$$

Hence given equation represents a pair of lines.

To find the lines, we rewrite the given equation as $6x^2 + (13y + 8)x + (6y^2 + 7y + 2) = 0$,

(a quadratic in x)

Solve for x :

$$x = \frac{-(13y+8) \pm \sqrt{(13y+8)^2 - 24(6y^2+7y+2)}}{12}$$

$$= \frac{-(13y+8) \pm (5y+4)}{12}$$

$$= \frac{-(2y+1)}{3}, \frac{-(3y+2)}{2}$$

Hence lines are $3x + 2y + 1 = 0$

....(i)

$2x + 3y + 2 = 0$

....(ii)

Solve equations (i) and (ii) for the point of intesection, we get

$$x = \frac{1}{5}, y = -\frac{4}{5}$$

Thus lines intersect at point.

Illustration 33: Show that $bx^2 - 2hxy + ay^2 = 0$ represent a pair of straight lines, which are at right angles to the pair of lines given by $ax^2 + 2hxy + by^2 = 0$.

Solution : Let $ax^2 + 2hxy + by^2 = 0$ represent the lines $y - m_1x = 0$ and $y - m_2x = 0$

$\therefore \quad ax^2 + 2hxy + by^2 \equiv (y - m_1x)(y - m_2x)$

$\Rightarrow \quad m_1m_2x^2 - (m_1 + m_2)xy + y^2 \equiv ax^2 + 2hxy + by^2$

Comparing the coefficients on the both sides,

$$\frac{m_1m_2}{a} = \frac{m_1 + m_2}{-2h} = \frac{1}{b}$$

or $m_1 + m_2 = -\frac{2h}{b}, m_1m_2 = \frac{a}{b}$ ....(i)

(In other words, m_1 and m_2 are roots of the quadratic equation $bm^2 + 2hm + a = 0$)

Since required lines are perpendicular to the given lines.

$$y = -\frac{1}{m_1}x \text{ and } y = -\frac{1}{m_2}x$$

Required pair of lines is $\left(y + \frac{x}{m}\right)\left(y + \frac{1}{m_2}x\right) = 0$

or $(m_1y + x)(x_2y + x) = 0$

$m_1m_2y^2 + (m_1 + m_2)xy + x^2 = 0$

Using (i) ;

$\frac{a}{b}y^2 - \frac{2h}{b}xy + x^2 = 0$ or $ay^2 - 2hxy + bx^2 = 0$

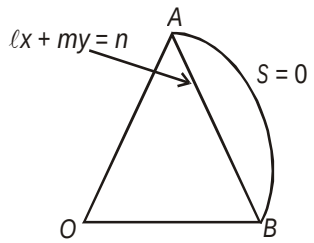
21. HOMOGENISATION

Let $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ be a non-homogeneous second-degree equation. It generally represents a curve. [It may be a pair of striaght lines also]

The equation of a straight line be $\ell x + my = n$ [First degree equation].

Now $S = 0$ and the above line generally meet at two points, say A and B.

Let O be the origin.



We get the combined equation of OA and OB by homogenizing

$ax^2 + 2hy + by^2 + 2gx + 2fy + c = 0$

with the help of $\frac{\ell x + my}{n} = 1$

This is because the equation to a pair of straight lines through the origin, is a homogeneous equation.

Homogenisation is done as

$$ax^2 + 2hxy + byh^2 + 2gx\left(\frac{\ell x + my}{n}\right) + 2fy\left(\frac{\ell x + my}{n}\right) + c\left(\frac{\ell m + my}{n}\right)^2 = 0$$

Illustration 34: Prove that the straight lines joining the origin to the points of intersection of $x - y = 2$ and the curve $5x^2 - 12xy - 8y^2 + 8y^2 + 8x - 4y + 12 = 0$ make equal angles with the axes.

Solution : Using $x - y = 2$, homogenise the second degree equation to give

$(5x^2 + 12xy - 8y^2)(2)(2) + (8x - 4y)(2)(x - y) + 12(x - y)(x - y) = 0$

This simplified gives

$x^2(12 + 16 + 12) + xy(46 - 16 - 8 - 24) + y^2(-32 + 8 + 12) = 0$

$48x^2 - 12y^2 = 0$ i.e., $4x^2 - y^2 = 0$

This represents the two lines $y = 2x$(slope = 2) and $y = -2x$ (slope = –2). These equally inclined to the axes.

Student Task

Q.1 The point (–4, 1) lies on the non-origin side of the line $x - 4y + 2 = 0$. (True/False).

Q.2 The two points (–1, 2) and (0, 4) lie on the same side of the line $3x - y + 4 = 0$. (True/False).

Q.3 The centroid of the triangle whose sides are $x + y = 3$, $y - x = 3$ and $y = 0$ is at _____.

Q.4 The excentre opposite to vertex B of the triangle having vertices $A(-5, 0)$, $B(5, 0)$, $C(0, -12)$ is at _____.

Q.5 The circumcentre of the triangle having vertices $A(1, 2)$, $B(1, -5)$, $C(-3, -10)$ lies outside the triangle. (True/False).

Q.6 The orthocentre of triangle whose two sides are along coordinate axes and the third side lies along the line $x + 10y - 4 = 0$ is at _____.

Q.7 The obtuse angle bisector of the line $x - 2y + 4 = 0$ and $4x - 3y + 2 = 0$ is _____.

Q.8 The angle between lines represented by the equation $4x^2 - 4xy - y^2 = 0$ is _____.

Q.9 The equation $\ell x^2 - 2xy - y^2 + 8x - 4y + 1 = 0$ represents pair of straight lines if _____.

Q.10 The acute angle between the lines joining the origin to the points of intersection of the curve $x^2 + y^2 = 8$ and $x + y = 2$ is _____.

Answer Key :

1. True, 2. False 3. (0, 1) 4. (–13, –12) 5. True 6. Origin

7. $x(4 - \sqrt{5}) + y(2\sqrt{5} - 3) + (2 - 4\sqrt{5}) = 0$ 8. $\tan^{-1}\frac{4\sqrt{2}}{3}$ 9. $\frac{31}{5}$ 10. 60°

IMPORTANT FORMULAE

Distance Formula : $D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Section formula : $x = \frac{mx_2 + nx_1}{m+n}, y = \frac{my_2 + ny_1}{m+n}$

Area of Triangle : $\Delta = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

slope : m = $\tan\theta = \frac{y_2 - y_1}{x_2 - x_1}, x_1 \neq x_2$

$0 \leq \theta < \pi$

A, *B* & *C* are collinear if slope *AB* = Slope *AC* or Area (Δ *ABC*) = 0

Acute Angle between lines having slope m_1 and m_2:

$$\theta = \tan^{-1}\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

Parallel if $m_1 = m_2$

Perpendicular if $m_1 m_2 = -1$

Family Of Lines :

Parallel to $ax + by + c = 0$ is $ax + by + k = 0$

Perpendicular to $ax + by + c = 0$ is $bx - ay + k = 0$

Passing through point of intersection of $L = 0$ & $L' = 0$ is $L + k L' = 0$

Distance of (x_1, y_1)

From $ax + by + c = 0$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Distance between parallel lines

$$d = \frac{|c - c'|}{\sqrt{a^2 + b^2}}$$

(x_1, y_1) & (x_2, y_2) are on same/opposite of $ax + by + c$ if and $ax_1 + by_1 + c$ are of same/opposite sign.

(x_1, y_1) & (x_2, y_2) are on origin/non-origin side of

$ax + by + c = 0$ if $ax_1 + by_1 + c$ is of same /opposite sign to that of c.

Points related to a triangle:

Centroid (G) : $\left(\frac{\sum x_1}{3}, \frac{\sum y_1}{3}\right)$

Excentre $I_1 = \left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$ etc.

Circumcentre (S) : Intersection Pt. of Perpendicular bisectors

Incentre (I) : $\left(\frac{\sum ax_1}{\sum a}, \frac{\sum ay_1}{\sum a}\right)$

Orthocenter (H) : Intersection Pt. of altitudes G,H,S are collinear & $HG : GS = 2 : 1$ except in equilateral triangle.

Angle Bisectors :

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = \pm\frac{a'x+b'x+c'}{\sqrt{a'^2+b'^2}}$$

If $aa' + bb' < 0$: '+' sign gives acute angle bisector, if $aa' + bb' > 0$: '–' sign gives acute angle bisector, provided c & c' are positive.

Forms of Straight Lines:

Slope one point form : $y - y_1 = m(x - x_1)$

Two point form : $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Intercept form : $\frac{x}{a} + \frac{y}{b} = 1$

y-intercept form : $y = mx + c$

Normal form : $x\cos\theta + y\sin\theta = p$

Parametric form : $x = x_1 + r\cos\theta$

$y = y_1 + r\sin\theta$

$|r|$ = distance between (x, y) and (x_1, y_1)

Genral form : $ax + by + c = 0$

Area of triangle formed by

$a_i x + b_i y + c_i = 0, i = 1, 2, 3$

$$\Delta = \frac{1}{|2C_1C_2C_3|}\begin{vmatrix} a_1 & h & g \\ a_2 & b & f \\ a_3 & f & c \end{vmatrix} = 0, h^2 \geq ab$$

Acute Angle between them : $\theta = \tan^{-1}\left|\frac{2\sqrt{h^2-ab}}{a+b}\right|$

Homogenisation :

Joint equation of lines joining origin to points of intersection of $lx + my + n = 0$ with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is given by $ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{lx+my}{-n}\right) + c\left(\frac{lx+my}{-n}\right) = 0$.

MISCELLANEOUS EXAMPLES

Example 1 :

If a, b, c are all distinct, then the equations

$(b-c)x + (c-a)y + a - b = 0$ and $(b^3 - c^3)x + (c^3 - a^3)y + a^3 - b^3 = 0$ represent the same line if

(a) $a + b + c \neq 0$ (b) $a + b + c = 0$

(c) $a + b = 0$ or $b + c = 0$ (d) None of these

Solution :

The given equation represents the same line if

$$\frac{b^3 - c^3}{b-c} = \frac{c^3 - a^3}{c-a} = \frac{a^3 - b^3}{a-b} = \lambda \text{ (say)}$$

i.e., if $b^2 + c^2 + bc = c^2 + a^2 + ca = a^2 + b^2 + ab = \lambda$

i.e., if $b^2 - a^2 + bc - ca = 0$

and $c^2 - b^2 + ca - ab = 0$

i.e., if $(b-a)(b + a + c) = 0$ and $(c-b)(c + b + a) = 0$

i.e., $a + b + c = 0$ $(\because\ a \neq b \neq c)$

Example 2 :

If the pair of lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ are such that each pair bisects the angle between the other pair, then pq equals

(a) –1 (b) 0 (c) 1 (d) 2

Solution :

The second pair must be identical with $\frac{x^2 - y^2}{1-(-1)} = \frac{xy}{-p}$

i.e., with $x^2 + \frac{2}{p}xy - y^2 = 0$. Consequently, $\frac{2}{p} = -2q$

$\Rightarrow$ $pq = -1$.

Example 3 :

Angles made with x-axis by the two lines through the point (1, 2) and cutting the line $x + y = 4$ at a distance $\frac{1}{3}\sqrt{6}$ from the point (1, 2) are

(a) $\frac{\pi}{6}$ and $\frac{\pi}{3}$ (b) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (c) $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ (d) None of these

Solution :

Any line through (1, 2) can be written as $\frac{x-1}{\cos\theta} = \frac{y-2}{\sin\theta} = r$

where θ is the angle which the line makes with +ve direction of x-axis. Any point on this line is $(r\cos\theta + 1, r\sin\theta + 2)$ when $|r| = \frac{1}{3}\sqrt{6}$, this point lies on the line $x + y = 4$

i.e., $r\cos\theta + 1 + r\sin\theta + 2 = 4$, $|r| = \frac{1}{3}\sqrt{6}$

$\Rightarrow r(\cos\theta + \sin\theta) = 1$, $|r| = \frac{1}{3}\sqrt{6} \Rightarrow r^2(1 + 2\sin\theta\cos\theta) = 1$, $r^2 = \frac{6}{9}$

$\Rightarrow 1 + \sin2\theta = \frac{1}{r^2} = \frac{9}{6}$ Þ $\sin2\theta = \frac{1}{2} \Rightarrow 2\theta = \frac{\pi}{6}$ or $\frac{5\pi}{6} \Rightarrow \theta = \frac{\pi}{12}$ or $\frac{5\pi}{12}$

Example 4 :

If the algebraic sum of the perpendicular distances of a variable line from the points (0, 2), (2, 0) and (1, 1) is zero, then the line always passes through the point

(a) (0, 0) (b) (0, 1) (c) (1, 0) (d) (1, 1)

Solution :

If the line be $ax + by + c = 0$, then $\frac{0+2b+c}{\sqrt{a^2+b^2}} + \frac{2a+c}{\sqrt{a^2+b^2}} + \frac{a+b+c}{\sqrt{a^2+b^2}} = 0$

$\Rightarrow 3a + 3b + 3c = 0 \Rightarrow a + b + c = 0$

$\Rightarrow$ the line $ax + by + c = 0$ always passes through the point (1, 1).

Example 5 :

Let $P \equiv (-1, 0)$, $Q \equiv (0, 0)$ and $R \equiv (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of angle PQR is

(a) $\frac{\sqrt{3}}{2}x + y = 0$ (b) $x + \sqrt{3}\,y = 0$ (c) $\sqrt{3}\,x + y = 0$ (d) $x + \frac{\sqrt{3}}{2}y = 0$

Solution :

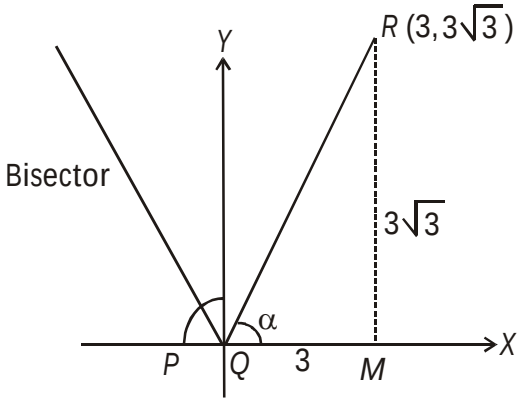
In the figure observe that $\alpha = \frac{3\sqrt{3}}{3} = \sqrt{3}$

$\Rightarrow \alpha = 60^o$ $\angle PQR = 120^o$

Hence, the bisector makes an angle of $(60^o + \alpha)$

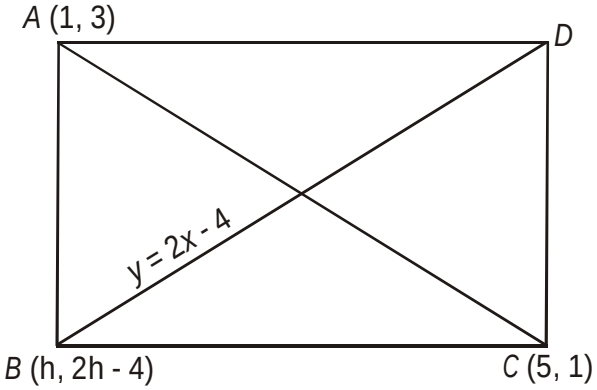
i.e., $60^o + 60^o = 120^o$ with positive direction of x-axis.

Hence its equation is $y - 0 = (\tan 120^o)(x - 0)$.



Example 6 :

The point (1, 3) and (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c and the remaining vertices.



Solution :

Using the Figure the midpoint of *AC* lies on *BD*, i.e., the point (3, 2) lies on $y = 2x + c$.

$\therefore$ $2 = 6 + c$ or $c = -4$, and therefore *BD* has equation $y = 2x - 4$.

The slope of *AC* is $\frac{3-1}{1-5} = -\frac{1}{2}$ and the slope of *BD* = 2.

Since their product is –1, the diagonals cut at right angles, and hence the rectangle becomes a square.

B lies on $y = 2x - 4$, and $\therefore$ its coordinates may be taken $(h, 2h - 4)$.

We have $2\,AB^2 = AC^2$ *i.e.*, $2\,[(1-h)^2 + (7-2h)^2] = (1-5)^2 + (3-1)^2$

i.e., $2\,[5h^2 - 30h + 50] = 16 + 4 = 20$.

$\therefore$ $h^2 - 6h + 10 = 2$ or $h^2 - 6h + 8 = 0$

$\therefore$ $(h-2)(h-4) = 0$ *i.e.*, $h = 2$ or 4.

$\therefore$ the vertex *B* is the point (2, 0) and the vertex *D* is the point (4, 4).

Note :- It is not absolutely necessary to use the idea of a square.

Example 7 :

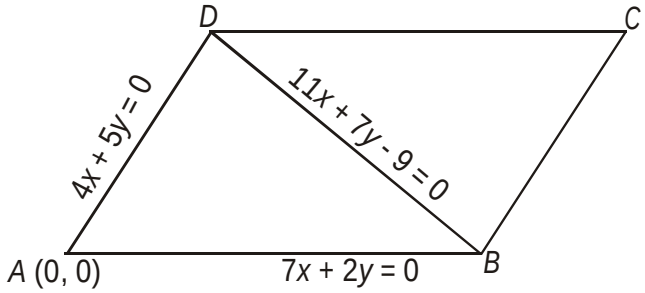
The consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x + 2y = 0$. If the equation of one diagonal be $11x + 7y = 9$, Find the equation of the other diagonal.

Solution :

A(0, 0) does not lie on $11x + 7y = 9$ and $11x + 7y = 9$ represents the diagonal *BD*.

Now $11x + 7y - 9 = 0$ can be written as $7x + 2y + 1(4x + 5y - 9) = 0$ and again $4x + 5y + 1\,(7x + 2y - 9) = 0$. But $7x + 2y + 1\,(4x + 5y - 9) = 0$ represents a straight line passing through the point of intersection of $7x + 2y = 0$ and $4x + 5y - 9 = 0$.

It follows that *B* is a point commom to $11x + 7y - 9 = 0$, $7x + 2y = 0$ and $4x + 5y - 9 = 0$.



Also $4x + 5y - 9 = 0$ is clearly parallel to $4x + 5y = 0$. Hence $4x + 5y - 9 = 0$ represents the equation of *BC*. Similarly $7x + 2y - 9 = 0$ represents the equation of *DC*.

$\therefore$ the equation of *AC* is of the form $4x + 5y - 9 + k\,(7x + 2y - 9) = 0$.

Since *AC* passes through the origin, we have, $-9 - 9k = 0$, $\therefore$ $k = -1$.

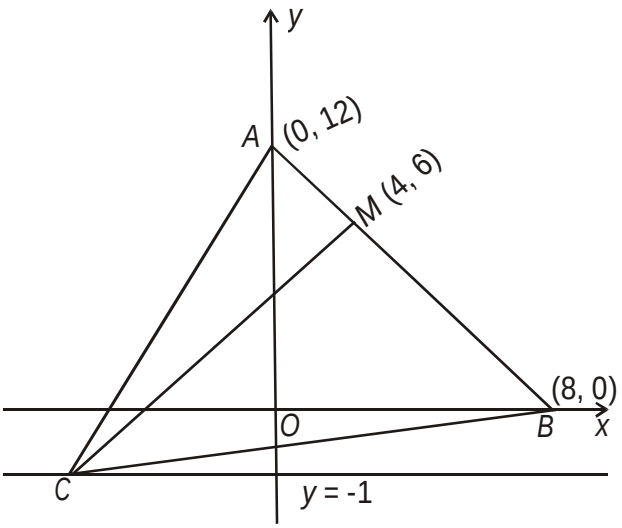
Substituting the value of *k*, *AC* is given by

$4x + 5y - 9 - (7x + 2y - 9) = 0$. *i.e.*, $-3x + 3y = 0$; *i.e.*, $y = x$

Note :- This example can also be done by finding the cooddinates of the points B and D.

Example 8 :

The line $3x + 2y = 24$ meets the *y*-axis at *A* and the *x*-axis at *B*. The perpendicular bisector of *AB* meets the line through (0, –1) parallel to the *x*-axis at *C*. Find the area fo the triangle *ABC*.



Solution :

Let *M* be the mid point of *AB* whose slope is –3/2.

$\therefore$ the perpendicular bisector of *AB* has for its equation $y - 6 = \left(\frac{2}{3}\right)(x-4)$

i.e., $2x - 3y + 10 = 0$

Now the ordinate of *C* is –1.

$\therefore$ its abscissa is $-\frac{13}{2}$

$\therefore$ *CM* = perpendicular length from $\left(-\frac{13}{2}, -1\right)$ upon $3x + 2y - 24 = 0 = \left|\frac{-\frac{39}{2} - 2 - 24}{\sqrt{9+4}}\right| = \frac{91}{2.\sqrt{3}}$

$AB = \sqrt{144+64} = \sqrt{208} = 4\sqrt{3}$

$\therefore$ the area of triangle *ABC* = $\frac{1}{2}AB.CM = \frac{1}{2}.4\sqrt{3}\frac{91}{2\sqrt{13}} = 91$ square units.

Example 9 :

A variable straight line, drawn through the point of intersection of the straight lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{a}+\frac{y}{b}=1$ meets the coordinate axes in *A* and *B*. Show that the locus of the mid-point of *AB* is the curve $2(a + b)\, xy = ab\,(x + y)$.

Solution :

Any line through the point of intersection of the given lines has equation of the form $bx + ay - ab + k\,(ax + by - ab) = 0$, where this line meets the *x*-axis ($y = 0$), we have $(b + ak)\, x - (k + 1)\, ab$ *i.e.*,

the coordinates of A are $\left(\frac{(k+1)ab}{b+ak}, 0\right)$

Similarly, the coordinates of B are $\left(0, \frac{(k+1)ab}{a+bk}\right)$

$\therefore$ if (x_1, y_1) be the mid-point of AB, we have

$$2x_1 = \frac{(k+1)ab}{b+ak} \text{ and } 2y_1 = \frac{(k+1)ab}{a+bk} \quad \therefore \quad \frac{(k+1)ab}{2x_1} = b+ak \text{ and } \frac{(k+1)ab}{2y_1} = a+bk.$$

By addition, $(k+1)\ ab\left[\frac{1}{2x_1} + \frac{1}{2y_1}\right] = (k+1)(a+b)$.

$k + 1 \neq 0$ for in that case both A and B coincide at (0, 0).

$\therefore$ cancelling the common factor and simplifying, $ab\,(x_1 + y_1) = (a + b)\,2x_1y_1$

i.e., (x_1, y_1) lies on the curve $ab(x + y) = 2\,(a + b)\,xy$, which is the required locus.

Alternatively

The two lines, by symmetry considerations, intersect at the point P with coordinates $x + y = \frac{ab}{a+b}$

A line through this point P with intercepts $2x_1$, $2y_1$ along the axes is $\frac{x}{2x_1} + \frac{y}{2y_1} = 1$, and this line passes through P.

$$\therefore \frac{ab}{a+b}\left(\frac{x_1+y_1}{2x_1y_1}\right) = 1 \qquad \text{...(i)}$$

Now A is $(2x_1, 0)$; B is $(0, 2y_1)$

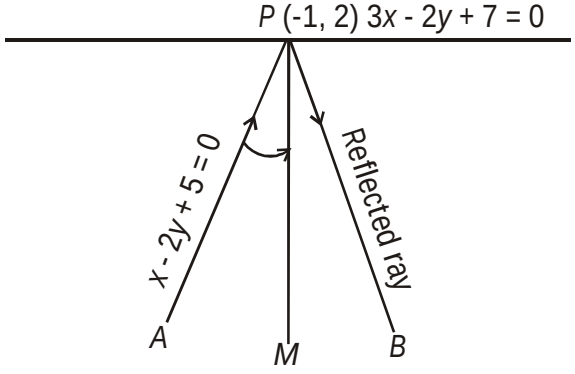
$\therefore$ mid-point of AB is (x_1, y_1)

The locus of the mid-point (dropping the suffixes in (1)) is

$\frac{ab}{a+b}\left(\frac{x+y}{2xy}\right) = 1$ *i.e.*, $2xy\,(a + b) = ab\,(x + y)$

Example 10 :

A ray of light is sent along the line $x - 2y + 5 = 0$. Upon reaching the line $3x - 2y + 7 = 0$, the ray is reflected from it. Find the equation of the line containing the reflected ray.



Solution :

The reflected ray *PB* passes through (–1, 2)

Let its slope be *m*, slope of *PM* = –2/3

($\because$ *PM* is perpendicular to $3x - 2y + 7 = 0$)

$$\therefore \tan\theta = \frac{\frac{1}{2}+\frac{2}{3}}{1-\frac{1}{2}.\frac{2}{3}} = +\frac{7}{4}$$

$\angle MPB = \theta$, and this is the angle between *PM* whose slope is 2/3 and *PB* whose slope in m

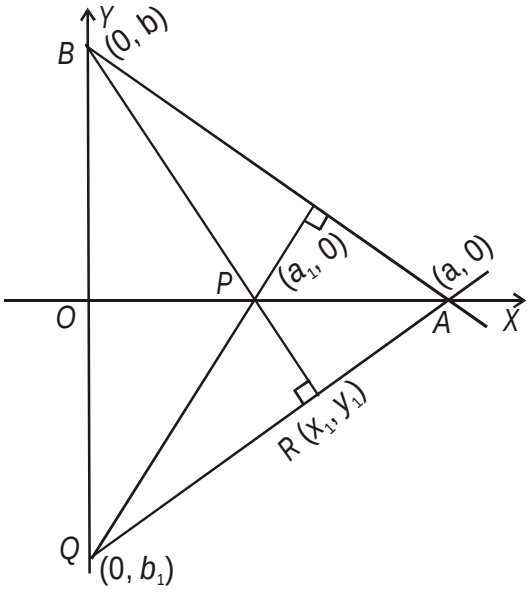
$$\therefore \frac{\frac{-2}{3}-m}{1-\frac{2}{3}m} = +\frac{7}{4} \Rightarrow 8 + 12m = -21 + 14m \Rightarrow m = \frac{29}{2}$$

$\therefore$ equation to *PB*, the reflected ray is $y - 2 = = \frac{29}{2}(x+1) = \frac{29}{2}(x+1)$

i.e., $2y - 4 = 29x + 29$, *i.e.*, $29x - 2y + 33 = 0$.

Example 11 :

The straight line $\frac{x}{a}+\frac{y}{b}=1$ cuts the axes of coordinates in *A* and *B*. *A* line perpendicular to *AB* cuts the axes in *P* and *Q*. Find the locus of the point of intersection of *AQ* and *BP*.



Solution :

Let $(a_1, 0)$ be the coordinates of *P* and $(0, b_1)$ the coordinates of *Q*.

The slope of *AB* is $-\frac{b}{a}$ and the slope of *PQ* is $-\frac{b_1}{a_1}$

But *AB* and *PQ* are at right angles, and hence $\left(-\frac{b}{a}\right)\left(-\frac{b_1}{a_1}\right) = -1$

or $\frac{bb_1}{aa_1} = -1$ .....(i)

Let AQ and BP meet at $R(x_1, y_1)$.

We have $\frac{y_1 - b}{x_1} = \frac{b}{a_1}$ (each being the slope of BP)

$\frac{y_1}{x_1 - a} = \frac{b_1}{a}$ (each being the slope of AQ)

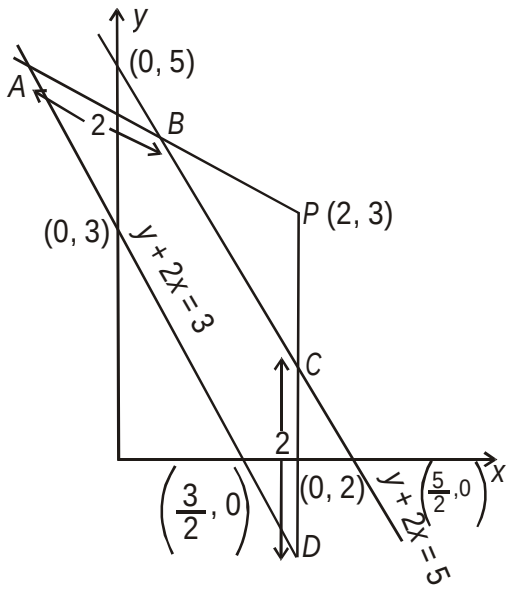
$\therefore$ by multiplication, $\frac{y_1 - b}{x_1} \cdot \frac{y_1}{x_1 - a} = \left(-\frac{b}{a_1}\right)\left(-\frac{b_1}{a}\right) = \frac{bb_1}{aa_1} = -1$, using (i)

$\therefore$ rearranging we have $x_1(x_1 - a) + y_1(y_1 - b) = 0$

Hence the required locus is $x\left(x - a\right) + y(y - b) = 0$ or $x^2 + y^2 - ax - by = 0$.

Example 12 :

Find the equation of the line through (2, 3) and making an intercept at length 2 units from the lines $y + 2x = 3$ and $y + 2x = 5$.



Solution :

A line through (2, 3) is of the form $\frac{x - 2}{\cos\theta} = \frac{y - 3}{\sin\theta} = r$

For $r = r_1$, let the point $(2 + r_1 \cos\theta, 3 + r_1 \sin\theta)$ on this line lie on $y + 2x = 3$.

$\therefore$ $3 + r_1 \sin\theta + 2(2 + r_1 \cos\theta) = 3$

$2r_1 \cos\theta + r_1 \sin\theta = -4$ (1)

Similarly for $r = r^2$, let the point $(2 + r_2 \cos\theta, 3 + r_2 \sin\theta)$ on this line lie on $y + 2x = 5$.

$\therefore$ $3 + r_2 \sin\theta + 2(2 + r_2 \cos\theta) = 5$

$2r_2 \cos\theta + r_2 \sin\theta = -2$ (2)

(1) – (2) gives

$2(r_1 - r_2)\cos\theta + (r_1 - r_2)\sin\theta = -2$

By question $r_1 - r_2 = AB = 2$

$\therefore\ 2\cos\theta + \sin\theta = -1$

Dividng by $\cos\theta$ and squaring

$(2 + \tan\theta)^2 = \sec^2\theta = 1 + \tan^2\theta$ .....(3)

$\therefore\ 3 + 4\tan\theta = 0 \Rightarrow \tan\theta = -\frac{3}{4}$

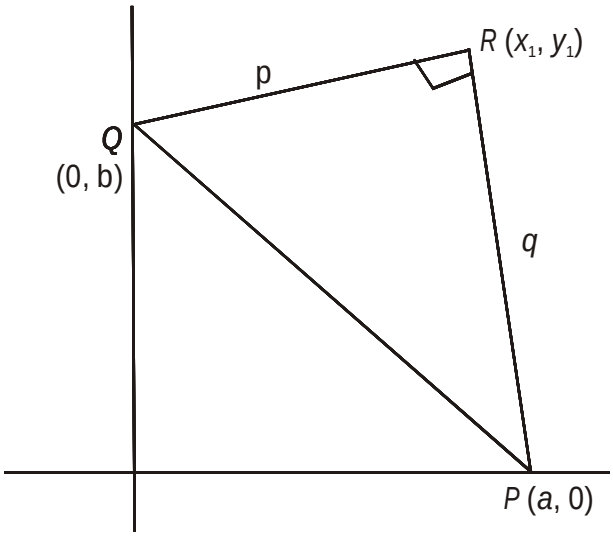
The required line is therefore *AB*, through (2, 3) and with a slope = – 3/4, and hence its equation is

$y - 3 = -\frac{3}{4}(x - 2)$ i.e., $3x + 4y - 18 = 0$

Note :- Usually step (3) gives a quadratic in tan θ, so that there are always two lines satisfying the conditions of the problem. In this case the quadratic has become linear (quadratic with one infinite root) and hence the second line in this case is CD parallel to the y-axis and whose equation is x – 2 = 0

Example 13:

A triangle *PQR* is right angled at *R* with *RP* = *q* and *RQ* = *p*; and moves such that the angular points *P* and *Q* slide along the *x*-axis and *y*-axis respectively. Find the locus of *R*.



Solution :

Let *R* be (x_1, y_1) ; *P* $(a, 0)$; *Q* $(0, b)$

Then we have

$(x_1 - a)^2 + y_1^2 = q^2$ ....(i)

$x_1^2 + (y_1 - b)^2 = p^2$ ....(ii)

$p^2 + q^2 = a^2 + b^2$ (= PQ^2) ....(iii)

From (i), $a = x_1 \pm \sqrt{q^2 - y_1^2}$

From (ii), $b = y_1 \pm \sqrt{p^2 - x_1^2}$

Substituting in (iii)

$\left(x_1 \pm \sqrt{q^2 - y_1^2}\right)^2 + \left(y_1 \pm \sqrt{p^2 - x_1^2}\right)^2 = p^2 + q^2$

i.e., $x_1^2 + y_1^2 + (q^2 - y_1^2) + (p^2 - x_1^2) \pm 2x_1\sqrt{q^2 - y_1^2} \pm 2y_1\sqrt{p^2 - x_1^2} = p^2 + q^2$

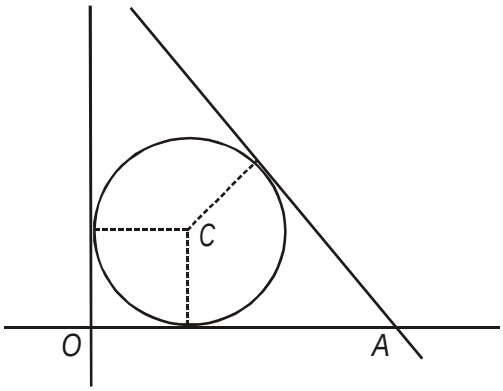
$\therefore \pm x_1\sqrt{q^2 - y_1^2} = \pm y_1\sqrt{p^2 - x_1^2}$

Squaring on both sides $x_1^2\left(q^2 - y_1^2\right) = y_1^2\left(p^2 - x_1^2\right) i.e., q^2x_1^2 - p^2y_1^2 = 0$

Locus of (x_1, y_1) is the pair of lines through the origin namely $qx + py = 0$, $qx - py = 0$.

Example 14 :

The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle, which has two of its sides along the coordinate axes. If the locus of the circumcentre is of the form $x + y - xy + k\sqrt{x^2 + y^2} = 0$ find k.



Solution :

Let (x_1, y_1) be the circumencentre of triangle OAB (B not shown in the figure). A is no the x-axis and b is on the y-axis.

Circumcentre is the midpoint of AB.

$\therefore$ A is $(2x_1, 0)$ and B is $(0, 2y_1)$.

Equation of AB is $\frac{x}{2x_1} + \frac{y}{2y_1} - 1 = 0$

Centre of the given circle is at C (2, 2) and radius 2 units. (2, 2) and (0, 0) lie on the same side of $\frac{x}{2x_1} + \frac{y}{2y_1} - 1 = 0$ (0, 0) makes $\frac{x}{2x_1} + \frac{y}{2y_1} - 1$ negative and hence

$\frac{2}{2x_1} + \frac{2}{2y_1} - 1$ is also negative.

Perpendicular distance from (2, 2) on $\frac{x}{2x_1} + \frac{y}{2y_1} - 1 = 0$ is $\frac{\left|\frac{2}{2x_1} + \frac{2}{2y_1} - 1\right|}{\sqrt{\frac{1}{4x_1^2} + \frac{1}{4y_1^2}}}$ = radius

= 2 since AB is a tangent to the circle.

$$2\left(\sqrt{\frac{1}{4x_1^2}+\frac{1}{4y_1^2}}\right)=1-\frac{1}{x_1}-\frac{1}{y_1},\left(\text{since } 1-\frac{1}{x_1}-\frac{1}{y_1}>0\right)$$

$$\frac{\sqrt{x_1^2+y_1^2}}{x_1y_1}=\frac{x_1y_1-x_1-y_1}{x_1y_1}$$

Locus of (x_1, y_1) is $x + y - xy + \sqrt{x^2+y^2}=0$; and this compared with $x + y - xy + k\sqrt{x^2+y^2}=0$ gives the value $k = 1$.

Example 15 :

Show that all chords of $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin pass through a fixed point; also find the fixed point.

Solution :

Let $lx + my = n$ be one such chord. Using this first degree equation, if we homogenise

$3x^2 - y^2 - 2x + 4y = 0$ we get

$(3x^2 - y^2)\ (n) + (-2x + 4y)(lx + my) = 0$

These two lines, by question, are perpendicular.

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$3n - n - 2l + 4m = 0$

$l - 2m = n$

But $l - 2m = n$ is the condition that the point $(1, -2)$ lies on $lx + my = n$. In other words the line $lx + my = n$ always passes through the fixed point $(1, -2)$.

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic:1 Straight Line

Q.1 If the point $(k, 2-2k)$, $(-k+1, 2k)$ and $(-4-k, 6-2k)$ are collinear, then k is equal to
(a) 11 (b) 1/2
(c) 1 (d) –1/2

Q.2 A line passing through (2, 2) is perpendicular to the line $3x + y = 3$. Its y intercept is
(a) 1/3 (b) 2/3
(c) 1 (d) 4/3

Q.3 The equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$ is
(a) $21x + 7y - 101 = 0$
(b) $11x + 3y + 20 = 0$
(c) $21x - 7y + 3 = 0$
(d) $11x - 3y + 9 = 0$

Q.4 P and Q are the points on the line joining $A(-2, 5)$ *ad* $B(3, 1)$ such that $AP = AQ = QB$. Then the mid point of PQ is
(a) (1/2, 3) (b) (–1/2, 4)
(c) (2, 3) (d) (–1, 4)

Q.5 If the line $6x - y + 2 + k(2x + 3y + 13) = 0$ is parallel to x-axis, the value of k is
(a) –1/3 (b) 1/3
(c) –3 (d) 3

Q.6 The equation of the straight line which is perpendicular to $y = x$ and passes through (3, 2) will be given by
(a) $x - y = 5$ (b) $x + y = 5$
(c) $x + y = 1$ (d) $x - y = 1$

Q.7 The nearest point on the line $3x - 4y = 25$ from the origin is
(a) (–4, 5) (b) (3, –4)
(c) (3, 4) (d) (3, 5)

Q.8 The line $(p + 2q)x + (p - 3q)y = p - q$ for different values of p and q passes through the point
(a) $\left(\frac{3}{2}, \frac{5}{2}\right)$ (b) $\left(\frac{2}{3}, \frac{2}{5}\right)$
(c) $\left(\frac{3}{5}, \frac{3}{5}\right)$ (d) $\left(\frac{2}{5}, \frac{3}{5}\right)$

Q.9 The angle between the lines $2x + 3 = 0$ and $3y = 5$ is
(a) 0° (b) 30°
(c) 60° (d) 90°

Q.10 If the line $lx + my + n = 0$ and $px + qy + r = 0$ are perpendicular, then
(a) $lp - mq = 0$ (b) $lp + mq = 0$
(c) $lm = pq$ (d) $lm + pq = 0$

Q.11 The equation to the pair of straight lines through the origin and perpendicular to $2x^2 - 4xy - 7y^2 = 0$ is
(a) $7x^2 - 4xy - 2y^2 = 0$
(b) $7x^2 + 4xy - 2y^2 = 0$
(c) $7x^2 + 4xy + 2y^2 = 0$
(d) $-7x^2 + 4xy - 2y^2 = 0$

Q.12 Given the system of straight lines $a(2x + y - 3) + b(3x + 2y - 5) = 0$, the line of the system situated farthest from the point (4, –3) has the equation
(a) $4x + 11y - 15 = 0$ (b) $7x + y - 8 = 0$
(c) $4x + 3y - 7 = 0$ (d) $3x - 4y + 1 = 0$

Q.13 The parametric equation of a line is given by $x = -2 + \frac{r}{\sqrt{10}}$ and $y = 1 + 3\frac{r}{\sqrt{10}}$. Then, for the line
(a) Intercept on the x-axis = 7/3
(b) Intercept on the y-axis = –7
(c) Slope of the line = 1/3
(d) Slope of the line = 3

Q.14 The area of the triangle with vertices at the point $(a, b + c)$, $(b, c + a)$, $(c, a + b)$ is
(a) 0 (b) $a + b + c$
(c) $ab + bc + ca$ (d) None of these

Q.15 The line L has intercept a and b on the co-ordinate axes. When keeping the origin fixed, the coordinate axes are rotated through a fixed angle, then the same line has intercepts p and q on the roatated axes, then
(a) $a^2 + b^2 = p^2 + q^2$
(b) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
(c) $a^2 + p^2 = b^2 + q^2$
(d) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$

Q.16 The co-ordinates of foot of the perpendicular from the point (2, 4) on the line $x + y = 1$ are

(a) $\left(\frac{1}{2},\frac{3}{2}\right)$ (b) $\left(-\frac{1}{2},\frac{3}{2}\right)$

(c) $\left(\frac{4}{3},\frac{1}{2}\right)$ (d) $\left(\frac{3}{4},-\frac{1}{2}\right)$

Q.17 The area of a triangle is 5 and two of its vertices are A (2, 1) and B (3, –2). The third vertex which lies on theline $y = x + 3$ is

(a) $\left(-\frac{3}{2},\frac{3}{2}\right)$ (b) $\left(\frac{5}{2},\frac{11}{2}\right)$

(c) $\left(-\frac{7}{2},-\frac{13}{2}\right)$ (d) (0, 0)

Q.18 The equation of the lines joining the origin to the points of trisection of the portion of the line $3x + y = 12$ intercepted between the axes are

(a) $y=\frac{1}{2}x$ and $y = x$ (b) $y = x$ and $y = –x$

(c) $y=\frac{3}{2}x$ and $y = 6x$ (d) none of these

Q.19 The area of the quadrilateral whose sides are given by $|x| + |y| = 1$ is

(a) 1 sq. unit (b) 2 sq. units
(c) 4 sq. units (d) None of these

Q.20 The equation of bisector of that angle between the lines $x + y + 1 = 0$ and $2x – 3y –$ $= 0$ which contains the point (10, – 20) is

(a) $x\left(\sqrt{13}+2\sqrt{2}\right)+y\left(\sqrt{13}-3\sqrt{2}\right)+\left(\sqrt{13}-5\sqrt{2}\right)=0$

(b) $x\left(\sqrt{13}-2\sqrt{2}\right)+y\left(\sqrt{13}+3\sqrt{2}\right)+\left(\sqrt{13}+5\sqrt{2}\right)=0$

(c) $x\left(\sqrt{13}+2\sqrt{2}\right)+y\left(\sqrt{13}+3\sqrt{2}\right)+$

$\left(\sqrt{13}+5\sqrt{2}\right)=0$

(d) None of these

Q.21 Three lines $px + qy + r = 0$, $qx + ry + p = 0$ and $rx + py + q = 0$ are concurrent if

(a) $p + q – r = 0$
(b) $p^2 – q^2 + r^2 = pq + qr + rp$
(c) $p^3 + q^3 + r^3 = 3pqr$
(d) None of these

Q.22 The straight lines $x + y = 0$, $3x + y – 4 = 0$ and $x + 3y – 4 = 0$ form a triangle, which is

(a) Isosceles (b) Equilateral
(c) Right-angled (d) None of these

Q.23 If $\frac{x}{c}+\frac{y}{d}=1$ be any line through the intersection of lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$, then

(a) $\frac{1}{a}+\frac{1}{d}=\frac{1}{b}+\frac{1}{c}$ (b) $\frac{1}{b}+\frac{1}{d}=\frac{1}{c}+\frac{1}{a}$

(c) $\frac{1}{c}+\frac{1}{d}=\frac{1}{a}+\frac{1}{b}$ (d) None of these

Q.24 A straight line passes through (x_1, y_1) and its segment between the axes is bisected at this point. Then its equation is given by

(a) $\frac{x}{x_1}+\frac{y}{y_1}=2$ (b)$2(xy_1 + yx_1) = x_1 y_1$

(c) $xy_1 + yx_1 = x_1 y_1$ (d) None of these

Q.25 The rquation of the diagonal through the origin of the quadrilateral formed by the lines $x = 0$, $y = 0$, $x + y – 1 = 0$ and $6x + y – 3 = 0$ is

(a) $4x –3y = 0$ (b) $3x – 2y = 0$
(c) $x – y = 0$ (d) $x + y = 0$

Q.26 A ray of light coming from the point (1, 2) is reflected at a point A on the x-axis and then passes through the point (5, 3). The coordinates of the point A is

(a) $\left(\frac{13}{5},0\right)$ (b) $\left(\frac{5}{13},0\right)$

(c) (–7, 0) (d) None of these

Q.27 In a rhomus $ABCD$ the diagonals AC and BD intersect at the point (3, 4). If the point A is (1, 2) the diagonal BD has the equation

(a) $x – y – 1 = 0$ (b) $x + y – 1 = 0$
(c) $x – y + 1 = 0$ (d) $x + y – 7 = 0$

Q.28 If the points (–2, 0), $\left(-1,\frac{1}{\sqrt{3}}\right)$ and $(\cos\theta,$ $\sin\theta)$ are collinear, then the number of values of $\theta\in[0,2\pi]$ is

(a) 0 (b) 1
(c) 2 (d) Infinite

Q.29 If $bx + cy = a$, where a, b, c are of the same sign, be a line such that the area enclosed by the line and the axes of reference is 1/8 unit^2, then

(a) b, a, c are in G.P.
(b) $2b$, $2a$, c are in G.P.

(c) $b, \frac{a}{2}, c$ are in A.P.

(d) $b, -2a, c$ are in G.P.

Q.30 If (x, y) be a variable point on the line $y = 2x$ lying between the lines $2(x + 1) + y = 0$ and $x + 3(y - 1) = 0$, then

(a) $x \in \left(-\frac{1}{2}, \frac{6}{7}\right)$ (b) $x \in \left(\frac{1}{2}, \frac{3}{7}\right)$

(c) $y \in \left(-1, \frac{3}{7}\right)$ (d) $y \in \left(-1, \frac{6}{7}\right)$

Q.31 The point (4, 1) undergoes the following three transformations successively

I. Reflection about the line $y = x$

II. Translation through a distance 2 units along the positive direction of x-axis

III. Rotation through an angle $\pi/4$ about the origin in the anticlockwise direction. The final position of the point is given by the co-ordinates

(a) $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (b) $\left(-\sqrt{2}, 7\sqrt{2}\right)$

(c) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (d) $\left(\sqrt{2}, 7\sqrt{2}\right)$

Q.32 If one of the diagonals of a square is along the line $x = 2y$ and one of its vertices is (3, 0), then the sides through this vertex are given by

(a) $y - 3x + 9 = 0, 3y + x - 3 = 0$
(b) $y + 3x + 9 = 0, 3y + x - 3 = 0$
(c) $y - 3x + 9 = 0, 3y - x + 3 = 0$
(d) $y - 3x + 3 = 0, 3y + x + 9 = 0$

Q.33 The equation of the line through (–5, 4) such that its segment intercepted by the lines $x + 2y + 1 = 0$ and $x + 2y - 1 = 0$ is of length $\frac{2}{\sqrt{5}}$ is

(a) $2x - y + 14 = 0$ (b) $2x - y - 14 = 0$
(c) $2x - y + 1 = 0$ (d) $2x - y + 13 = 0$

Topic:2 Pair of Straight line

Q.34 If $x^2 - 3xy + \lambda y^2 + 3x - 5y + 2 = 0$, represents a pair of straight lines, then the value of λ is

(a) 1 (b) 4
(c) 3 (d) 2

Q.35 The equation : $\lambda y^2 + (1 - \lambda^2)xy - \lambda x^2 = 0$ represents a pair of straight lines. Then $3x + y = 12$ intercepted between the axes are

(a) $\lambda = 2$ necessarily
(b) $\lambda = -2$ necessarily
(c) λ may be any real number
(d) None of these

Q.36 The equation to a pair of opposite sides of a parallelogram are $x^2 - 5x + 6 = 0$ and $y^2 - 6y + 5 = 0$. The equations to its diagonals are

(a) $x + 4y = 13$ and $y = 4x - 7$
(b) $4x + y = 13$ and $4y = x - 7$
(c) $4x + y = 13$ and $y = 4x - 7$
(d) $y - 4x = 13$ and $y + 4x = 7$

Topic:3 Equation of Locus of a Point

Q.37 The locus of a point which the line segment joining the given points (α, β) and (γ, δ) subtends a right angle is

(a) Straight line (b) Ellipse
(c) Parabola (d) Circle

Q.38 A point moves so that sum of square of its distances from the four sides of a square is constant. This point always lie on a

(a) Straight line (b) Circle
(c) Ellipse (d) Parabola

Q.39 The locus of the mid-point of the portion intercepted between the axes by the line $x \cos \alpha + y \sin \alpha = p$, where p is a constant is

(a) $x^2 + y^2 = 4p^2$ (b) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

(c) $x^2 + y^2 = \frac{4}{p^2}$ (d) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$

Topic:4 Miscellaneous

Q.40 Area of the triangle with vertices (a, b), (x_1, y_1) and (x_2, y_2) where a, x_1, x_2 are in G.P. with common ratio r and b, y_1, y_2 are in G.P. with common ratio s is

(a) $ab(r - 1)(s - 1)(s - r)$
(b) $ab(r + 1)(s + 1)(r - s)$

(c) $\frac{1}{2}ab(r - 1)(s - 1)(s - r)$

(d) $\frac{1}{2}ab(r + 1)(s + 1)(s - r)$

Q.41 A (4, 2), B (7, 6) and C (x, y) are three points such that the angle BAC is a right angle and the area of the triangle ABC is 7 square units. Then the number of such points C is

(a) 0 (b) 1
(c) 2 (d) 4

Q.42 The equation of the base of an equilateral triangle is $x + y - 2 = 0$ and one vertex is (2,

–1). then the length of the side of the triangle is

(a) $\sqrt{\frac{1}{2}}$ (b) $\sqrt{\frac{3}{2}}$

(c) $\sqrt{\frac{2}{3}}$ (d) $\sqrt{2}$

Q.43 The orthocentre, circumcentre, centroid and incenter of the triangle formed by the lines $x = 0$, $y = 0$ and $x + y = a$ lie on

(a) The straight line $y = x$

(b) $y = 2x$

(c) $y = 3x$

(d) None of these

Q.44 If O be the origin and if $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are two points, the $OP_1(OP_2)COS \angle P_1OP_2$ is equal to

(a) $x_1y_2 + x_2y_1$

(b) $x_1x_2 + y_1y_2$

(c) $\left(x_1^2 + y_1^2\right)\left(x_2^2 + y_2^2\right)$

(d) $(x_1 - x_2)^2 + (y_1 - y_2)^2$

Q.45 If (α, α^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 3 = 0$ and $5x - 6y - 1 = 0$, then

(a) $2\alpha + 3\alpha^2 - 1 > 0$ (b) $\alpha + 3\alpha^2 - 3 < 0$

(c) $\alpha + 2\alpha^2 - 3 > 0$ (d) None of these

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** Centroid and incenter of a triangle always lie inside the triangle.

Statement – 2 : Lines which divides the two different internal angle of triangle, will meet inside the trianlge.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** $(\lambda L_1 + \mu L_2 = aL_3 + bL_4) \Leftrightarrow$ lines L_1, L_2, L_3 and L_4 are concurrent. $(a, b, \lambda, \mu \in R)$

Statement-2: $L_1 + kL_2 = 0$ represents family of lines passing through point of intersection of L_1 and L_2.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** If $ah + bk + c = 0$, then the point (h, k) lies above $ax + by + c = 0$. $(a, b, c$ are non zero real numbers)

Statements-2: It $x = h$, $y = \frac{-(ah + c)}{b}$ which is less than k (for $b > 0$) i.e., $\frac{-(ah + c)}{b} < k$, hence (h, k) lies above the line.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** A homogeneous equation in x and y of degree n represents family of lines (not more than n), all intersecting at origin.

Statement–2 : Substituting $\frac{y}{x} = m$ gives a polynomial in m of degree n, which can have maximum n real roots of the form $y - m_1x = 0$, $y - m_2x = 0$, ...

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** Two mutually perpendicular lines intersect with co-ordinate axes, then the angle which one makes with positive x axis anticlockwise is equal to angle that other one makes with positive y-axis anticlockwise.

Statement–2 : All quadrilaterals that can be forned by points of intersection of pair of lines with co-ordinate axes are concyclic.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Years Question)

Q.1 If the equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)\,x + (b_1 - b_2)\,y + c = 0$, then the value of 'c' is :

(a) $\frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$

(b) $a_1^2 - a_2^2 + b_1^2 - b_2^2$

(c) $\frac{1}{2}(a_1^2 + a_2^2 + b_1^2 + b_2^2)$

(d) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$

Q.2 Locus of centroid of the triangle whose vertices are (a cos t, a sin t), (b sin t, – b cos t) and (1, 0), where t is a parameter is :

(a) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(b) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$

(c) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

(d) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$

Q.3 If the pair of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then :

Q.4 A square of side 'a' lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle a $\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x-axis. The equation of its diagonal not passing through the origin is :

(a) y (cos a – sin a) – x (sin a – cos a) = a

(b) y (cos a + sin a) + x (sin a – cos a) = a

(c) y (cos a + sin a) + x (sin a + cos a) = a

(d) y (cos a + sin a) + x (cos a – sin a) = a

Q.5 Let A(2,–3) and B(–2,1) be vertices of a triangle ABC. If the centroid of this triangle moves on the line 2x + 3y = 1, then the locus of the vertex C is the line :

(a) 2x + 3y = 9 (b) 2x – 3y = 7

(c) 3x + 2y = 5 (d) 3x – 2y = 3

Q.6 The equation of the straight line passing through the point (4,3) and making intercepts on the co–ordinate axes whose sum is –1, is :

(a) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(b) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(c) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

(d) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

Q.7 If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value:

(a) 1 (b) –1 (c) 2 (d) –2

Q.8 If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals :

(a) 1 (b) –1 (c) 3 (d) –3

Q.9 The line parallel to the x–axis and passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0, where (a,b) ¹ (0,0) is :

(a) above the x–axis at a distance of (2/3) from it

(b) above the x–axis at a distance of (3/2) from it

(c) below the x–axis at a distance of (2/3) from it

(d) below the x–axis at a distance of (3/2) from it

Q.10 If non–zero numbers a,b,c are in HP, then the straight line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ always passes through a fixed point. That point is :

(a) $\left(1, -\frac{1}{2}\right)$ (b) (1, – 2)

(c) (–1, – 2) (d) (–1, 2)

Q.11 If a vertex of a triangle is (1,1) and the mid–points of two sides through this vertex are (–1,2) and (3,2), then the centroid of the triangle is :

(a) $\left(\frac{1}{3}, \frac{7}{3}\right)$ (b) $\left(1, \frac{7}{3}\right)$

(c) $\left(-\frac{1}{3}, \frac{7}{3}\right)$ (d) $\left(-1, \frac{7}{3}\right)$

Q.12 If the pair of lines $ax^2 + 2(a + b)xy + by^2 = 0$ lie along diameter of a circle and divide the circle into four sectors such that the area of one of the sectors is thrice the area of another sector, then

(a) $3a^2 + 2ab + 3b^2 = 0$
(b) $3a^2 + 10ab + 3b^2 = 0$
(c) $3a^2 - 2ab + 3b^2 = 0$
(d) $3a^2 - 10ab + 3b^2 = 0$

Q.13 A straight line through the point A (3, 4) is such that its intercept between the axes is bisected at A. Its equation is :
(a) $3x - 4y + 7 = 0$ (b) $4x + 3y = 24$
(c) $3x + 4y = 25$ (d) $x + y = 7$

Q.14 If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then 'a' belongs to :

(a) $(3, \infty)$ (b) $\left(\frac{1}{2}, 3\right)$

(c) $\left(-3, -\frac{1}{2}\right)$ (d) $\left(0, \frac{1}{2}\right)$

Q.15 Let A(h, k), B(1, 1) and C(2, 1) be the vertices of a right angled triangle with AC as its hypotenuse. If the area of triangle is 1, then the set of values which 'k' can take is given by
(a) {1, 3} (b) {0, 2}
(c) {–1, 3} (d) {–3, – 2}

Q.16 Let P = (–1, 0) Q = (0, 0) and R = $(3, 3\sqrt{3})$ be three points. The equation of the bisector of the $\angle PQR$ is

(a) $\sqrt{3}\ x + y = 0$ (b) $x + \frac{\sqrt{3}}{2}\ y = 0$

(c) $\frac{\sqrt{3}}{2}\ x + y = 0$ (d) $x + \sqrt{3}\ y = 0$

Q.17 If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is

(a) $-\frac{1}{2}$ (b) – 2
(c) ± 1 (d) 2

Q.18 The perpendicular bisector of the line segment joining P(1, 4) and Q(k, 3) has y-intercept – 4. Then a possible value of k is
(a) – 4 (b) 1
(c) 2 (d) – 2

Q.19 The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2 x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for:
(a) exactly one value of p
(b) exactly two values of p
(c) more than two values of p
(d) no value of p

Q.20 Three distinct points A, B and C are given in the 2-dimensional coordinate plane such that the ratio of the distance of any one of them from the point (1, 0) to the distance from the point (–1, 0) is equal to $\frac{1}{3}$. Then the circumcentre of the triangle ABC is at the point :

(a) $\left(\frac{5}{4}, 0\right)$ (b) $\left(\frac{5}{2}, 0\right)$

(c) $\left(\frac{5}{3}, 0\right)$ (d) 0, 0

Q.21. The line L given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point (13, 32). The line K is parallel to L and has the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L and K is

(a) $\sqrt{17}$ (b) $\frac{17}{\sqrt{15}}$

(c) $\frac{23}{\sqrt{17}}$ (d) $\frac{23}{\sqrt{15}}$

Q.22 The line $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement-1 : The ratio PR : RQ equals $2\sqrt{2} : \sqrt{5}$

Statement-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

(a) Statement-1 is true, Statement-2 is true ; Statement-2 is correct explanation for Statement-1
(b) Statement-1 is true, Statement-2 is true ; Statement-2 is **not** a correct explanation for Statement-1
(c) Statement-1 is true, Statement-2 is false
(d) Statement-1 is false, Statement-2 is true

Q.23 The lines $x + y = |a|$ and $ax - y = 1$ intersect each other in the first quadrant. Then the set of all possible values of a is the interval :
(a) $(0, \infty)$ (b) $[1, \infty)$
(c) $(-1, \infty)$ (d) $(-1, 1]$

Q.24 If A(2, –3) and B(–2, 1) are two vertices of a triangle and third vertex moves on the line

$2x + 3y = 9$, then the locus of the centroid of the triangle is :

(a) $x - y = 1$ (b) $2x + 3y = 1$

(c) $2x + 3y = 3$ (d) $2x - 3y = 1$

Q.25 If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals :

(a) $\frac{29}{5}$ (b) 5

(c) 6 (d) $\frac{11}{5}$

Q.26 A line is drawn through the point (1, 2) to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. if the area of the triangle OPQ is least, then the slope of the line PQ is :

(a) $-\frac{1}{4}$ (b) -4

(c) -2 (d) $-\frac{1}{2}$

Q.27 A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is

(a) $y = x + \sqrt{3}$ (b) $\sqrt{3}y = x - \sqrt{3}$

(c) $y = \sqrt{3}x - \sqrt{3}$ (d) $\sqrt{3}y = x - 1$

Q.28 The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1) (1, 1) and (1, 0) is :

(a) $2 + \sqrt{2}$ (b) $2 - \sqrt{2}$

(c) $1 + \sqrt{2}$ (d) $1 - \sqrt{2}$

Q.29 Let a, b,c and d be non-zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes, then

(a) $2bc + 3ad = 0$ (b) $3bc - 2ad = 0$

(c) $3bc + 2ad = 0$ (d) $2bc - 3ad = 0$

Q30. Let PS be the median of the triangle whose vertices P(2, 2), Q(6,-1)and R(7, 3). The equation of the line passing through (1, -1) and parallel to PS is

(a) $2x + 9y + 7 = 0$ (b) $4x + 7y + 3 = 0$

(c) $2x - 9y - 11 = 0$ (d) $4x - 7y - 11 = 0$

Q.31 Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its digonals intersect at $(-1,-2)$, then which one of the following is a vertex of this rhombus?

(a) (-3,-9) (b) (-3, -8)

(c) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (d) $\left(-\frac{10}{3}, \frac{7}{3}\right)$

Q.32 If a variable line drawn through the intersection of the lines $\frac{x}{3} + \frac{y}{4} = 1$ and $\frac{x}{4} + \frac{y}{3} = 1$, meets the coordinate axes at A and B, $(A \neq B)$, then the locus of the mid-point of AB is

(a) $7xy = 6(x + y)$

(b) $4(x + y)^2 - 28(x + y) + 49 = 0$

(c) $6xy = 7(x + y)$

(d) $14(x + y)^2 - 97(x + y) + (x + y) + 168 = 0$

Q.33 The point (2,1) is translated parallel to the line L: $x - y = 4$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the line passing throguh Q and perpendicualr to L is

(a) $x + y = 2 - \sqrt{6}$ (b) $2x + 2y = 1 - \sqrt{6}$

(c) $x + y = 3 - 3\sqrt{6}$ (d) $x + y = 3 - 2\sqrt{6}$

Q.34 A straight line through origin O meets the lines $3y = 10 - 4x$ and $8x + 6y + 5 = 0$ at points A and B respectively. Then O divides the segment AB in the ratio

(a) 2 : 3 (b) 1 : 2

(c) 4 : 1 (d) 3 : 4

Q.35 A ray of light is incident along along a line which meets another line, $7x - y + 1 = 0$, at the point (0, 1). the ray is then reflected from this point along the line, $y + 2x = 1$. Then the equation of the line of incidence of the ray of light is

(a) $41x - 25y + 25 = 0$ (b) $41x + 25y - 25 = 0$

(c) $41x - 38y + 38 = 0$ (d) $41x + 38y - 38 = 0$

Q.36 Let k be an integer such that triangle with vertices $(k,-3k),(5,k)$ and $(-k,2)$ has area 28 sq. units. Then the orthocentre of this triangle is at the point

(a) $\left(1,\frac{3}{4}\right)$ (b) $\left(1,-\frac{3}{4}\right)$

(c) $\left(2\frac{1}{2}\right)$ (d) $\left(2,-\frac{1}{2}\right)$

Q.37 A square, of each side 2, lies above hte x-axis and has one vertex at the origin. If one of the sides passing through the origin makes and angle 30^0 with the positive direction of the x-axis, then the sum of the x-coordinates of the vertices of the square is

(a) $\sqrt{3}-2$ (b) $2\sqrt{3}-1$

(c) $\sqrt{3}-1$ (d) $2\sqrt{3}-2$

Q.38 Let the orthocentre and centroid of a triangle be A(-3, 5) and B(3,3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is:

(a) $2\sqrt{10}$ (b) $3\sqrt{\frac{5}{2}}$

(c) $\frac{3\sqrt{5}}{2}$ (d) $\sqrt{10}$

Section-D
(Board pattern)

Q.1 Find a point on the line through the point $A(1, 3)$ and $B(-5, 4)$ that is twice as far from A as from B.

Q.2 Find the area of the quadrilateral whose vertices, taken in order, are (–1, 6), (1, 2), (5, 4) and (3, 8).

Q.3 Prove analytically that the sum of squares of diagonals of a parallelogram is equal to the sum of squares of its sides.

Q.4 Find the equation to the locus of a point which moves so that its distance from the point $(ak, 0)$ is $k(> 0, \neq 1)$ times its distance from the point $\left(\frac{a}{k}, 0\right)$.

Q.5 A straight line is such that the portion of it intercepted between the axes is bisected at the point (x_1, y_1), prove that its equation is $\frac{x}{x_1}+\frac{y}{y_1}=2$.

Q.6 Find the equation of the internal bisector of the angle A of the triangle ABC formed by the points $A(-36, 7)$, $B(20, 7)$ and $C(0, -8)$.

Q.7 If p is the length of perpendicular from the origin on a line whose intercepts on the axes are a and b, then show that $\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}$.

Q.8 Which of the two lines $2x + y - 3 = 0$ and $4x + y + 5 = 0$ is farther from the origin?

Q.9 Show that the lines $2x - 3y = 7$, $3x - 4y = 13$ and $8x - 11y = 33$ are concurrent. Find the point at which they meet.

Q.10 Prove analytically that the altitudes of a triangle are concurrent.

Q.11 Find the equations of the two lines which can be drawn through the point (2, 2) to make an angle of 45° with the line $x + y = 2$.

Q.12 Find the foot of operpendicular drawn from the point (1, –2) on the line $4x - 3y - 5 = 0$.

Q.13 Find the orthocentre of the triangle formed by the points (1, 0), (2, –4) and (–5, –2).

Q.14 Find the coordinates of the circumcentre of the triangle whose vertices are (1, 2), (–1, 6) and (5, 4)

Q.15 Find the area of the triangle formed by the lines $y = x$, $y = 2x$ and $y = 3x + 4$.

Q.16 Two opposite vertices of a square are (1, 2) and (3, 8). Find the remaining two vertices.

Q.17 Find the coordinates of the incentre of the triangle formed by the lines $y = 15$, $5x - 12y = 0$, $3x + 4y = 0$.

Q.18 Find the direction in which a line be drawn through the point (1, 2) so as to meet the line $x + y = 4$ at a distance $\frac{1}{3}\sqrt{6}$ from the point (1, 2).

Q.19 Find the equation to the locus of the point P which satisfies the condition $|PA| + |PB| = 2a$ where A and B are the points $(ae, 0)$ and $(-ae, 0)$ respectively, $0 < e < 1$ and $a > 0$.

Q.20 If A and B are respectively the points (–1, 6) and (5, 4), find the point P such that $|PA| = |PB|$ and area $\Delta PAB = 10$ units.

ANSWER KEY

CHAPTER-01 (BINOMIAL THEOREM)

SECTION-A (Straight objective type questions)

1. (a)	2. (b)	3. (d)	4. (b)	5. (c)	6. (b)	7. (b)
8. (c)	9. (d)	10. (a)	11. (b)	12. (c)	13. (a)	14. (d)
15. (c)	16. (b)	17. (b)	18. (a)	19. (b)	20. (c)	21. (c)
22. (c)	23. (c)	24. (b)	25. (a)	26. (d)	27. (d)	28. (c)
29. (a)	30. (c)	31. (b)	32. (c)	33. (b)	34. (a)	35. (a)
36. (c)	37. (a)	38. (a)	39. (b)	40. (a)	41. (b)	42. (b)
43. (a)						

SECTION-B (Assertion-Reason type questions)

1. (b)	2. (d)	3. (b)	4. (d)	5. (d)	6. (a)	7. (d)
8. (b)	9. (d)	10. (c)				

SECTION-C (Previous Year Question)

1. (b)	2. (a)	3. (b)	4. (a)	5. (c)	6. (b)	7. (d)
8. (c)	9. (a)	10. (b)	11. (c)	12. (d)	13. (c)	14. (c)
15. (a)	16. (c)	17. (b)	18. (c)	19. (c)	20. (c)	

CHAPTER-02 (PROBABILITY)

SECTION-A (Straight objective type questions)

1. (d)	2. (a)	3. (a)	4. (c)	5. (a)	6. (a)	7. (b)
8. (a)	9. (b)	10. (a)	11. (b)	12. (b)	13. (a)	14. (c)
15. (b)	16. (b)	17. (b)	18. (d)	19. (d)	20. (d)	21. (d)
22. (c)	23. (b)	24. (a)	25. (c)	26. (b)	27. (c)	28. (a)
29. (d)	30. (c)	31. (a)	32. (a)	33. (a)	34. (a)	35. (d)
36. (d)	37. (a)	38. (b)	39. (a)	40. (a)	41. (b)	42. (c)
43. (b)	44. (a)	45. (a)				

SECTION-B (Assertion-Reason type questions)

1. (b)	2. (b)	3. (c)	4. (c)	5. (d)	6. (d)	7. (b)
8. (a)	9. (b)	10. (b)				

SECTION-C (Previous Year Question)

1. (b)	2. (c)	3. (a)	4. (b)	5. (c)	6. (c)	7. (c)
8. (b)	9. (c)	10. (d)	11. (a)	12. (a)	13. (c)	14. (c)
15. (c)	16. (Bonus)	17. (a)	18. (a)	19. (d)	20. (a)	21. (a)

CHAPTER-03 (SEQUENCE AND SERIES)

SECTION-A (Straight objective type questions)

1. (c)	2. (a)	3. (b)	4. (d)	5. (a)	6. (a)	7. (b)
8. (b)	9. (a)	10. (d)	11. (c)	12. (c)	13. (a)	14. (a)
15. (c)	16. (d)	17. (b)	18. (a)	19. (d)	20. (d)	21. (c)

22. (a) 23. (d) 24. (d) 25. (b) 26. (b) 27. (b) 28. (b)
29. (a) 30. (a) 31. (a) 32. (d) 33. (d) 34. (b) 35. (a)
36. (c) 37. (b) 38. (b) 39. (d) 40. (c) 41. (d) 42. (d)

SECTION-B (Assertion-Reason type questions)

1. (a) 2. (b) 3. (a) 4. (d) 5. (b) 6. (d) 7. (a)
8. (a) 9. (b) 10. (a) 11. (a)

SECTION-C (Previous Years Question)

1. (a) 2. (a) 3. (a) 4. (b) 5. (c) 6. (c) 7. (d)
8. (a) 9. (d) 10. (b) 11. (c) 12. (d) 13. (d) 14. (b)
15. (a) 16. (a) 17. (d) 18. (c) 19. (c) 20. (d) 21. (d)
22. (a) 23. (b)

CHAPTER-04 (STRAIGHT LINE)

SECTION-A (Straight objective type questions)

1. (b) 2. (d) 3. (d) 4. (a) 5. (c) 6. (b) 7. (b)
8. (d) 9. (d) 10. (b) 11. (a) 12. (d) 13. (d) 14. (a)
15. (b) 16. (b) 17. (a) 18. (c) 19. (b) 20. (a) 21. (c)
22. (a) 23. (c) 24. (a) 25. (b) 26. (a) 27. (d) 28. (b)
29. (d) 30. (d) 31. (c) 32. (a) 33. (a) 34. (d) 35. (c)
36. (c) 37. (d) 38. (b) 39. (b) 40. (c) 41. (c) 42. (c)
43. (c) 44. (b) 45. (a)

SECTION-B (Assertion-Reason type questions)

1. (a) 2. (d) 3. (d) 4. (a) 5. (c)

SECTION-C (Previous Years Question)

1. (a) 2. (b) 3. (d) 4. (d) 5. (a) 6. (d) 7. (c) 8. (d)
9. (d) 10. (b) 11. (b) 12. (a) 13. (b) 14. (b) 15. (c) 16. (a)
17. (c) 18. (a) 19. (a) 20. (a) 21. (c) 22. (c) 23. (b) 24. (b)
25. (c) 26. (c) 27. (b) 28. (b) 29. (b) 30. (a) 31. (c) 32. (a)
33. (d) 34. (c) 35. (c) 36. (c) 37. (d) 38. (b)